

101

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| 1. | 8 box gauge nail penetrate a 4" dry timber? | $5 - 3\frac{1}{2}$ |
| 2. | 7 penny nail penetrate a $3\frac{1}{2}$ " green mast? | $2\frac{1}{4} - 3\frac{1}{16}$ |
| 3. | 6 common gauge nail penetrate a $4\frac{1}{2}$ " dry plank? | $4 - 4$ |
| 4. | 16 penny nail penetrate a 2" green rafter? | $3\frac{1}{2} - 1\frac{9}{16}$ |
| 5. | 4 common gauge nail penetrate a 2" green girder? | $5 - 1\frac{9}{16}$ |
| 6. | 2 common gauge nail penetrate a $\frac{5}{8}$ " dry rafter? | $6 - \frac{9}{16}$ |
| 7. | 8 box gauge nail penetrate a $4\frac{1}{2}$ " green dimension? | $5 - 4\frac{1}{16}$ |
| 8. | 8 box gauge nail penetrate a $2\frac{1}{2}$ " dry mast? | $5 - 2$ |
| 9. | $10\frac{1}{2}$ box gauge nail penetrate a $4\frac{1}{2}$ " green pole? | $3 - 4\frac{1}{16}$ |
| 10. | 3 penny nail penetrate a 3" dry beam? | $1\frac{1}{4} - 2\frac{1}{2}$ |
| 11. | $15\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " green pole? | $1 - 3\frac{1}{16}$ |
| 12. | $12\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " dry plank? | $1\frac{1}{2} - \frac{5}{16}$ |
| 13. | $12\frac{1}{2}$ box gauge nail penetrate a $2\frac{1}{2}$ " green frame? | $2 - 2\frac{1}{16}$ |
| 14. | 8 box gauge nail penetrate a $1\frac{1}{2}$ " green beam? | $5 - 1\frac{9}{32}$ |
| 15. | 30 penny nail penetrate a $3\frac{1}{2}$ " green board? | $4\frac{1}{2} - 3\frac{1}{16}$ |
| 16. | $11\frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " green plank? | $2 - 1\frac{9}{32}$ |

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| 1. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 2. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 3. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 4. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 5. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 6. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 7. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 8. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 9. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 10. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 11. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 12. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 13. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 14. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 15. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 16. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |

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|-----|---|----------------|
| 1. | 30 penny nail penetrate a 2 1/2" green girder? | 4 1/2 — 2 1/16 |
| 2. | 10 1/2 box gauge nail penetrate a 3 1/2" dry dimension? | 3 — 3 |
| 3. | 10 penny nail penetrate a 3/4" dry board? | 3 — 5/8 |
| 4. | 6 penny nail penetrate a 5/8" dry dimension? | 2 — 9/16 |
| 5. | 3 common gauge nail penetrate a 1/2" green mast? | 5 1/2 — 15/32 |
| 6. | 10 1/4 common gauge nail penetrate a 3 1/2" dry girder? | 2 1/2 — 3 |
| 7. | 9 common gauge nail penetrate a 4" green beam? | 3 — 3 9/16 |
| 8. | 10 1/2 box gauge nail penetrate a 3/4" dry girder? | 3 — 5/8 |
| 9. | 10 1/4 common gauge nail penetrate a 3" dry board? | 2 1/2 — 2 1/2 |
| 10. | 8 penny nail penetrate a 4 1/2" green pole? | 2 1/2 — 4 1/16 |
| 11. | 15 1/2 box gauge nail penetrate a 4 1/2" green timber? | 1 — 4 1/16 |
| 12. | 9 box gauge nail penetrate a 2 1/2" dry timber? | 4 — 2 |
| 13. | 20 penny nail penetrate a 3/8" green rafter? | 4 — 11/32 |
| 14. | 9 box gauge nail penetrate a 1 1/4" green frame? | 4 — 1 1/32 |
| 15. | 12 penny nail penetrate a 3 1/2" dry mast? | 3 1/4 — 3 |
| 16. | 11 1/2 common gauge nail penetrate a 1 1/2" dry frame? | 2 — 1 1/4 |

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| 1. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 2. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 3. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 4. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 5. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 6. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 7. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 8. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 9. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 10. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 11. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 12. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 13. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 14. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 15. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 16. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |

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| 1. | 12 ½ box gauge nail penetrate a 3/8" green timber? | 2 — 11/32 |
| 2. | 20 penny nail penetrate a 2" dry board? | 4 — 1 1/2 |
| 3. | 2 penny nail penetrate a 3/8" green board? | 1 — 11/32 |
| 4. | 6 penny nail penetrate a 2" green plank? | 2 — 19/16 |
| 5. | 16 penny nail penetrate a 2 ½" green board? | 3 ½ — 2 1/16 |
| 6. | 5 penny nail penetrate a 2 ½" green frame? | 1 ¾ — 2 1/16 |
| 7. | 14 box gauge nail penetrate a 3/4" dry timber? | 1 ½ — 5/8 |
| 8. | 12 penny nail penetrate a 3 ½" dry girder? | 3 ¼ — 3 |
| 9. | 9 penny nail penetrate a 4" green timber? | 2 ¾ — 3 9/16 |
| 10. | 10 ¼ common gauge nail penetrate a 5/8" dry pole? | 2 ½ — 9/16 |
| 11. | 20 penny nail penetrate a ½" green mast? | 4 — 15/32 |
| 12. | 16 penny nail penetrate a 2 ½" green girder? | 3 ½ — 2 1/16 |
| 13. | 14 box gauge nail penetrate a 2 ½" dry beam? | 1 ½ — 2 |
| 14. | 5 penny nail penetrate a 1 ½" dry mast? | 1 ¾ — 1 ¼ |
| 15. | 60 penny nail penetrate a 5/8" dry timber? | 6 — 9/16 |
| 16. | 5 common gauge nail penetrate a 5/8" dry beam? | 4 ½ — 9/16 |

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| 1. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 2. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 3. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 4. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 5. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 6. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 7. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 8. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 9. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 10. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 11. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 12. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 13. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 14. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 15. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 16. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |

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|-----|--|--------------------------------|
| 1. | 14 box gauge nail penetrate a $1\frac{1}{4}$ " green frame? | $1\frac{1}{2} - 1\frac{1}{32}$ |
| 2. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry girder? | $2\frac{1}{2} - \frac{5}{16}$ |
| 3. | 10 $\frac{1}{4}$ common gauge nail penetrate a $4\frac{1}{2}$ " green plank? | $2\frac{1}{2} - 4\frac{1}{16}$ |
| 4. | 5 common gauge nail penetrate a $\frac{3}{4}$ " green board? | $4\frac{1}{2} - 1\frac{1}{16}$ |
| 5. | 8 penny nail penetrate a $2\frac{1}{2}$ " green beam? | $2\frac{1}{2} - 2\frac{1}{16}$ |
| 6. | 3 penny nail penetrate a $4\frac{1}{2}$ " green mast? | $1\frac{1}{4} - 4\frac{1}{16}$ |
| 7. | 5 penny nail penetrate a $\frac{3}{4}$ " dry girder? | $1\frac{3}{4} - \frac{5}{8}$ |
| 8. | 60 penny nail penetrate a $2\frac{1}{2}$ " dry timber? | $6 - 2$ |
| 9. | 40 penny nail penetrate a $\frac{3}{8}$ " dry board? | $5 - \frac{5}{16}$ |
| 10. | 3 common gauge nail penetrate a $1\frac{1}{4}$ " dry board? | $5\frac{1}{2} - 1$ |
| 11. | 30 penny nail penetrate a $4\frac{1}{2}$ " dry board? | $4\frac{1}{2} - 4$ |
| 12. | 6 penny nail penetrate a $1\frac{1}{2}$ " green beam? | $2 - 1\frac{9}{32}$ |
| 13. | 9 box gauge nail penetrate a 2" dry beam? | $4 - 1\frac{1}{2}$ |
| 14. | 10 penny nail penetrate a $2\frac{1}{2}$ " green dimension? | $3 - 2\frac{1}{16}$ |
| 15. | 5 common gauge nail penetrate a $\frac{1}{2}$ " dry frame? | $4\frac{1}{2} - \frac{7}{16}$ |
| 16. | 12 $\frac{1}{2}$ box gauge nail penetrate a $2\frac{1}{2}$ " dry dimension? | $2 - 2$ |

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|---|-------------------------------|
| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 3. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 4. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 5. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 6. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 7. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 8. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 9. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 10. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 11. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 12. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 13. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 14. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 15. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 16. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |

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| 1. | 10 ½ box gauge nail penetrate a 5/8" dry mast? | 3 — 9/16 |
| 2. | 2 common gauge nail penetrate a 3" green dimension? | 6 — 29/16 |
| 3. | 2 penny nail penetrate a 1 ½" green beam? | 1 — 19/32 |
| 4. | 9 common gauge nail penetrate a 3" green joist? | 3 — 29/16 |
| 5. | 9 box gauge nail penetrate a 2" dry board? | 4 — 1 ½ |
| 6. | 60 penny nail penetrate a 1 ¼" green joist? | 6 — 11/32 |
| 7. | 10 ¼ common gauge nail penetrate a 2 ½" dry joist? | 2 ½ — 2 |
| 8. | 4 penny nail penetrate a 3/8" green frame? | 1 ½ — 11/32 |
| 9. | 4 common gauge nail penetrate a 1 ½" dry pole? | 5 — 1 ¼ |
| 10. | 9 penny nail penetrate a 1" green plank? | 2 ¾ — 25/32 |
| 11. | 11 ½ common gauge nail penetrate a 3" dry frame? | 2 — 2 ½ |
| 12. | 12 ½ box gauge nail penetrate a ½" green plank? | 2 — 15/32 |
| 13. | 14 box gauge nail penetrate a 3" dry timber? | 1 ½ — 2 ½ |
| 14. | 2 penny nail penetrate a 1" green dimension? | 1 — 25/32 |
| 15. | 15 common gauge nail penetrate a 4" green board? | 1 — 39/16 |
| 16. | 4 penny nail penetrate a ½" green rafter? | 1 ½ — 15/32 |

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| 1. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 2. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 3. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 4. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 5. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 6. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 7. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 9. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 10. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 11. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 12. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 13. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 14. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 15. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 16. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |

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| 1. | 14 common gauge nail penetrate a $5/8$ " green rafter? | $1 \frac{1}{4} - \frac{19}{32}$ |
| 2. | 10 $\frac{1}{2}$ box gauge nail penetrate a 3" dry frame? | $3 - 2 \frac{1}{2}$ |
| 3. | 60 penny nail penetrate a $4 \frac{1}{2}$ " dry frame? | $6 - 4$ |
| 4. | 15 $\frac{1}{2}$ box gauge nail penetrate a 2" green timber? | $1 - \frac{19}{16}$ |
| 5. | 9 common gauge nail penetrate a $3/4$ " dry beam? | $3 - \frac{5}{8}$ |
| 6. | 12 $\frac{1}{2}$ common gauge nail penetrate a $3 \frac{1}{2}$ " dry board? | $1 \frac{1}{2} - 3$ |
| 7. | 50 penny nail penetrate a 4" green board? | $5 \frac{1}{2} - \frac{39}{16}$ |
| 8. | 12 penny nail penetrate a 2" green beam? | $3 \frac{1}{4} - \frac{19}{16}$ |
| 9. | 11 $\frac{1}{2}$ common gauge nail penetrate a 4" green plank? | $2 - \frac{39}{16}$ |
| 10. | 9 penny nail penetrate a $1 \frac{1}{2}$ " green girder? | $2 \frac{3}{4} - \frac{19}{32}$ |
| 11. | 16 penny nail penetrate a $3/4$ " dry joist? | $3 \frac{1}{2} - \frac{5}{8}$ |
| 12. | 8 box gauge nail penetrate a $3/8$ " green pole? | $5 - \frac{11}{32}$ |
| 13. | 4 penny nail penetrate a $2 \frac{1}{2}$ " dry pole? | $1 \frac{1}{2} - 2$ |
| 14. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " green pole? | $2 \frac{1}{2} - \frac{15}{32}$ |
| 15. | 11 $\frac{1}{2}$ common gauge nail penetrate a $3/8$ " dry mast? | $2 - \frac{5}{16}$ |
| 16. | 20 penny nail penetrate a $\frac{1}{2}$ " dry pole? | $4 - \frac{7}{16}$ |

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| 1. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 2. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 3. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 4. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 5. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 6. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 7. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 8. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 9. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 10. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 11. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 12. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 13. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 14. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 15. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 16. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |

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|-----|---|----------------|
| 1. | 7 penny nail penetrate a 2 1/2" green mast? | 2 1/4 — 2 1/16 |
| 2. | 4 penny nail penetrate a 3 1/2" dry mast? | 1 1/2 — 3 |
| 3. | 5 penny nail penetrate a 4" dry mast? | 1 3/4 — 3 1/2 |
| 4. | 12 1/2 common gauge nail penetrate a 2 1/2" green beam? | 1 1/2 — 2 1/16 |
| 5. | 11 1/2 box gauge nail penetrate a 1/2" green joist? | 2 1/2 — 15/32 |
| 6. | 10 penny nail penetrate a 4" green pole? | 3 — 3 9/16 |
| 7. | 12 penny nail penetrate a 5/8" green rafter? | 3 1/4 — 19/32 |
| 8. | 15 common gauge nail penetrate a 2 1/2" green rafter? | 1 — 2 1/16 |
| 9. | 4 penny nail penetrate a 3/4" dry timber? | 1 1/2 — 5/8 |
| 10. | 4 common gauge nail penetrate a 3 1/2" dry rafter? | 5 — 3 |
| 11. | 2 penny nail penetrate a 3 1/2" dry mast? | 1 — 3 |
| 12. | 9 penny nail penetrate a 1 1/4" green plank? | 2 3/4 — 1 1/32 |
| 13. | 40 penny nail penetrate a 2 1/2" dry beam? | 5 — 2 |
| 14. | 30 penny nail penetrate a 1 1/4" dry pole? | 4 1/2 — 1 |
| 15. | 8 box gauge nail penetrate a 1 1/4" dry board? | 5 — 1 |
| 16. | 15 common gauge nail penetrate a 3" dry pole? | 1 — 2 1/2 |

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| 1. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 2. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 3. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 4. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 5. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 6. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 7. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 8. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 9. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 10. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 11. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 12. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 13. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 14. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 15. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 16. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |

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|-----|---|---------------------------------|
| 1. | 50 penny nail penetrate a 1" dry dimension? | $5 \frac{1}{2} - \frac{3}{4}$ |
| 2. | 6 penny nail penetrate a $1 \frac{1}{2}$ " dry plank? | $2 - 1 \frac{1}{4}$ |
| 3. | 9 common gauge nail penetrate a $1 \frac{1}{2}$ " green rafter? | $3 - 1\frac{9}{32}$ |
| 4. | 8 penny nail penetrate a $3 \frac{1}{2}$ " green beam? | $2 \frac{1}{2} - 3\frac{1}{16}$ |
| 5. | 10 penny nail penetrate a $3 \frac{1}{2}$ " dry mast? | $3 - 3$ |
| 6. | 14 common gauge nail penetrate a $\frac{5}{8}$ " green girder? | $1 \frac{1}{4} - 1\frac{9}{32}$ |
| 7. | 6 common gauge nail penetrate a 1" dry mast? | $4 - \frac{3}{4}$ |
| 8. | 12 penny nail penetrate a $1 \frac{1}{4}$ " green timber? | $3 \frac{1}{4} - 1\frac{1}{32}$ |
| 9. | 4 penny nail penetrate a 2" green rafter? | $1 \frac{1}{2} - 1\frac{9}{16}$ |
| 10. | 3 penny nail penetrate a 1" dry frame? | $1 \frac{1}{4} - \frac{3}{4}$ |
| 11. | 2 penny nail penetrate a $\frac{3}{8}$ " dry plank? | $1 - \frac{5}{16}$ |
| 12. | $14 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " green timber? | $1 \frac{1}{4} - 1\frac{1}{32}$ |
| 13. | 60 penny nail penetrate a $\frac{1}{2}$ " green mast? | $6 - 1\frac{15}{32}$ |
| 14. | $14 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry mast? | $1 \frac{1}{4} - 1$ |
| 15. | 15 common gauge nail penetrate a $4 \frac{1}{2}$ " green girder? | $1 - 4\frac{1}{16}$ |
| 16. | 6 penny nail penetrate a $1 \frac{1}{2}$ " green beam? | $2 - 1\frac{9}{32}$ |

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|---|--------------------------|
| 1. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 2. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 3. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 4. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 5. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 6. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 7. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 8. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 9. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 10. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 11. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 12. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 13. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 14. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 15. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 16. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |

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|-----|---|----------------|
| 1. | 4 penny nail penetrate a 3 1/2" green board? | 1 1/2 — 3 1/16 |
| 2. | 20 penny nail penetrate a 1/2" green rafter? | 4 — 15/32 |
| 3. | 10 1/4 common gauge nail penetrate a 1" dry joist? | 2 1/2 — 3/4 |
| 4. | 10 1/2 box gauge nail penetrate a 3/8" dry rafter? | 3 — 5/16 |
| 5. | 5 penny nail penetrate a 3/8" green board? | 1 3/4 — 11/32 |
| 6. | 20 penny nail penetrate a 5/8" green plank? | 4 — 19/32 |
| 7. | 8 box gauge nail penetrate a 1/2" dry timber? | 5 — 7/16 |
| 8. | 9 box gauge nail penetrate a 2 1/2" dry beam? | 4 — 2 |
| 9. | 5 penny nail penetrate a 1 1/2" dry joist? | 1 3/4 — 1 1/4 |
| 10. | 7 penny nail penetrate a 2 1/2" green timber? | 2 1/4 — 2 1/16 |
| 11. | 4 common gauge nail penetrate a 2" green dimension? | 5 — 19/16 |
| 12. | 16 penny nail penetrate a 1/2" green plank? | 3 1/2 — 15/32 |
| 13. | 14 box gauge nail penetrate a 5/8" green plank? | 1 1/2 — 19/32 |
| 14. | 30 penny nail penetrate a 3 1/2" dry mast? | 4 1/2 — 3 |
| 15. | 10 penny nail penetrate a 1/2" green joist? | 3 — 15/32 |
| 16. | 9 penny nail penetrate a 1 1/4" green beam? | 2 3/4 — 1 1/32 |

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|---|--------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 3. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 4. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 5. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 6. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 7. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 8. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 9. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 10. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 11. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 12. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 13. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 14. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 15. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 16. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |

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|-----|--|--------------------------------|
| 1. | 14 box gauge nail penetrate a $\frac{1}{2}$ " dry rafter? | $1\frac{1}{2} - \frac{7}{16}$ |
| 2. | 10 penny nail penetrate a 3" green board? | $3 - 2\frac{9}{16}$ |
| 3. | 50 penny nail penetrate a $2\frac{1}{2}$ " green beam? | $5\frac{1}{2} - 2\frac{1}{16}$ |
| 4. | 9 box gauge nail penetrate a $4\frac{1}{2}$ " green joist? | $4 - 4\frac{1}{16}$ |
| 5. | 4 penny nail penetrate a 2" dry rafter? | $1\frac{1}{2} - 1\frac{1}{2}$ |
| 6. | 8 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $2\frac{1}{2} - \frac{15}{32}$ |
| 7. | 3 penny nail penetrate a $\frac{3}{4}$ " green beam? | $1\frac{1}{4} - \frac{11}{16}$ |
| 8. | 7 penny nail penetrate a $2\frac{1}{2}$ " dry frame? | $2\frac{1}{4} - 2$ |
| 9. | 50 penny nail penetrate a $\frac{3}{4}$ " dry dimension? | $5\frac{1}{2} - \frac{5}{8}$ |
| 10. | 9 box gauge nail penetrate a 3" green board? | $4 - 2\frac{9}{16}$ |
| 11. | 14 box gauge nail penetrate a $4\frac{1}{2}$ " green girder? | $1\frac{1}{2} - 4\frac{1}{16}$ |
| 12. | 40 penny nail penetrate a $\frac{1}{2}$ " dry frame? | $5 - \frac{7}{16}$ |
| 13. | 7 penny nail penetrate a $4\frac{1}{2}$ " green board? | $2\frac{1}{4} - 4\frac{1}{16}$ |
| 14. | 30 penny nail penetrate a $2\frac{1}{2}$ " dry frame? | $4\frac{1}{2} - 2$ |
| 15. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry frame? | $2\frac{1}{2} - \frac{5}{16}$ |
| 16. | 30 penny nail penetrate a 3" dry plank? | $4\frac{1}{2} - 2\frac{1}{2}$ |

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|--|---|
| 1. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 4. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 5. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 6. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 7. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 8. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 11. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 12. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 13. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 14. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 15. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 16. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |

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|-----|--|---------------------------------------|
| 1. | 20 penny nail penetrate a 2 ½" green joist? | 4 — 2 ¹ / ₁₆ |
| 2. | 15 ½ box gauge nail penetrate a 3/8" green timber? | 1 — 11 ¹ / ₃₂ |
| 3. | 60 penny nail penetrate a ½" green girder? | 6 — 15 ¹ / ₃₂ |
| 4. | 6 common gauge nail penetrate a 3/8" green board? | 4 — 11 ¹ / ₃₂ |
| 5. | 9 box gauge nail penetrate a 2 ½" dry board? | 4 — 2 |
| 6. | 11 ½ box gauge nail penetrate a 3/4" dry mast? | 2 ½ — 5 ¹ / ₈ |
| 7. | 8 box gauge nail penetrate a 2 ½" dry beam? | 5 — 2 |
| 8. | 4 penny nail penetrate a 5/8" dry mast? | 1 ½ — 9 ¹ / ₁₆ |
| 9. | 10 ½ box gauge nail penetrate a 3/4" green joist? | 3 — 11 ¹ / ₁₆ |
| 10. | 40 penny nail penetrate a 4" green timber? | 5 — 3 ⁹ / ₁₆ |
| 11. | 9 common gauge nail penetrate a 4" dry joist? | 3 — 3 ½ |
| 12. | 30 penny nail penetrate a 2" green rafter? | 4 ½ — 1 ⁹ / ₁₆ |
| 13. | 8 penny nail penetrate a 1" green dimension? | 2 ½ — 25 ¹ / ₃₂ |
| 14. | 8 common gauge nail penetrate a 1 ¼" dry beam? | 3 ½ — 1 |
| 15. | 9 common gauge nail penetrate a 1" green girder? | 3 — 25 ¹ / ₃₂ |
| 16. | 14 box gauge nail penetrate a 1 ½" dry board? | 1 ½ — 1 ¼ |

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|--|-------------------------------|
| 1. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 2. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 3. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 4. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 5. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 6. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 7. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 8. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 9. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 10. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 11. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 12. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 13. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 14. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 15. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 16. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |

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|-----|---|---------------------------------|
| 1. | 9 box gauge nail penetrate a $\frac{3}{8}$ " green pole? | $4 - \frac{11}{32}$ |
| 2. | 20 penny nail penetrate a $4 \frac{1}{2}$ " green plank? | $4 - 4\frac{1}{16}$ |
| 3. | $12 \frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " green pole? | $2 - \frac{15}{32}$ |
| 4. | 8 box gauge nail penetrate a $4 \frac{1}{2}$ " dry dimension? | $5 - 4$ |
| 5. | 14 common gauge nail penetrate a 3" dry beam? | $1 \frac{1}{4} - 2 \frac{1}{2}$ |
| 6. | $12 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green board? | $2 - \frac{11}{16}$ |
| 7. | 9 box gauge nail penetrate a $3 \frac{1}{2}$ " green beam? | $4 - 3\frac{1}{16}$ |
| 8. | 7 penny nail penetrate a $\frac{1}{2}$ " green frame? | $2 \frac{1}{4} - \frac{15}{32}$ |
| 9. | 20 penny nail penetrate a 2" green frame? | $4 - 1\frac{9}{16}$ |
| 10. | 60 penny nail penetrate a 4" green board? | $6 - 3\frac{9}{16}$ |
| 11. | 6 penny nail penetrate a $1 \frac{1}{2}$ " green frame? | $2 - 1\frac{9}{32}$ |
| 12. | 4 common gauge nail penetrate a 4" green beam? | $5 - 3\frac{9}{16}$ |
| 13. | 16 penny nail penetrate a $2 \frac{1}{2}$ " dry joist? | $3 \frac{1}{2} - 2$ |
| 14. | 60 penny nail penetrate a $1 \frac{1}{2}$ " green board? | $6 - 1\frac{9}{32}$ |
| 15. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " dry board? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 16. | 8 box gauge nail penetrate a $3 \frac{1}{2}$ " green joist? | $5 - 3\frac{1}{16}$ |

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|---|--|
| 1. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 2. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 3. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 4. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 5. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 6. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 7. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 8. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 9. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 10. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 11. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 12. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 13. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 14. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 15. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 16. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |

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|-----|---|----------------------------------|
| 1. | 4 common gauge nail penetrate a 1" green plank? | $5 - \frac{25}{32}$ |
| 2. | 9 penny nail penetrate a 2 ½" dry frame? | $2 \frac{3}{4} - 2$ |
| 3. | 40 penny nail penetrate a 2" dry rafter? | $5 - 1 \frac{1}{2}$ |
| 4. | 14 box gauge nail penetrate a 5/8" green rafter? | $1 \frac{1}{2} - \frac{19}{32}$ |
| 5. | 15 ½ box gauge nail penetrate a 5/8" green joist? | $1 - \frac{19}{32}$ |
| 6. | 8 penny nail penetrate a 4 ½" green pole? | $2 \frac{1}{2} - 4 \frac{1}{16}$ |
| 7. | 11 ½ box gauge nail penetrate a 1 ½" dry pole? | $2 \frac{1}{2} - 1 \frac{1}{4}$ |
| 8. | 50 penny nail penetrate a 3/4" green frame? | $5 \frac{1}{2} - \frac{11}{16}$ |
| 9. | 8 penny nail penetrate a 3/8" dry plank? | $2 \frac{1}{2} - \frac{5}{16}$ |
| 10. | 20 penny nail penetrate a 2 ½" green frame? | $4 - 2 \frac{1}{16}$ |
| 11. | 50 penny nail penetrate a 5/8" green joist? | $5 \frac{1}{2} - \frac{19}{32}$ |
| 12. | 3 penny nail penetrate a 2" dry plank? | $1 \frac{1}{4} - 1 \frac{1}{2}$ |
| 13. | 50 penny nail penetrate a 4" green board? | $5 \frac{1}{2} - 3 \frac{9}{16}$ |
| 14. | 10 penny nail penetrate a 4" green dimension? | $3 - 3 \frac{9}{16}$ |
| 15. | 16 penny nail penetrate a 1 ¼" dry frame? | $3 \frac{1}{2} - 1$ |
| 16. | 9 penny nail penetrate a 3/4" green girder? | $2 \frac{3}{4} - \frac{11}{16}$ |

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|---|--|
| 1. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 2. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 3. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 4. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 5. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 6. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 7. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 8. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 9. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 10. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 11. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 13. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 14. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 15. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |

114

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|-----|--|--------------|
| 1. | 8 penny nail penetrate a 1 ½" dry mast? | 2 ½ — 1 ¼ |
| 2. | 40 penny nail penetrate a ½" green timber? | 5 — 15/32 |
| 3. | 10 penny nail penetrate a 3" green girder? | 3 — 29/16 |
| 4. | 40 penny nail penetrate a ½" dry beam? | 5 — 7/16 |
| 5. | 2 penny nail penetrate a 4" dry dimension? | 1 — 3 ½ |
| 6. | 7 penny nail penetrate a 1 ½" green frame? | 2 ¼ — 19/32 |
| 7. | 9 common gauge nail penetrate a 3/8" green timber? | 3 — 11/32 |
| 8. | 20 penny nail penetrate a 2" green rafter? | 4 — 19/16 |
| 9. | 12 ½ common gauge nail penetrate a 5/8" dry frame? | 1 ½ — 9/16 |
| 10. | 50 penny nail penetrate a 5/8" green timber? | 5 ½ — 19/32 |
| 11. | 8 penny nail penetrate a 2 ½" green mast? | 2 ½ — 2 1/16 |
| 12. | 2 penny nail penetrate a 2" dry plank? | 1 — 1 ½ |
| 13. | 9 penny nail penetrate a 2" dry dimension? | 2 ¾ — 1 ½ |
| 14. | 16 penny nail penetrate a 1 ¼" dry joist? | 3 ½ — 1 |
| 15. | 12 penny nail penetrate a 3/4" dry mast? | 3 ¼ — 5/8 |
| 16. | 5 penny nail penetrate a 3 ½" dry joist? | 1 ¾ — 3 |

1. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$
2. clearance hole for a 20 gauge screw?	Q $21/64 = 21/64$
3. countersink hole for a 24 gauge screw?	$1/2 - 3/8 + 1/8 = 1/2$
4. pilot hole for a 2 gauge screw?	67 $3/32 - 1/16 = 1/32$
5. countersink hole for a 20 gauge screw?	$29/64 - 21/64 + 1/8 = 29/64$
6. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
7. countersink hole for a 4 gauge screw?	B $7/64 + 1/8 = 15/64$
8. countersink hole for a 7 gauge screw?	L $5/32 + 1/8 = 9/32$
9. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
10. clearance hole for a 18 gauge screw?	N $19/64 = 19/64$
11. clearance hole for a 18 gauge screw?	N $19/64 = 19/64$
12. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
13. countersink hole for a 14 gauge screw?	V $1/4 + 1/8 = 3/8$
14. pilot hole for a 14 gauge screw?	12 $1/4 - 1/16 = 3/16$
15. countersink hole for a 11 gauge screw?	Q $13/64 + 1/8 = 21/64$
16. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$

115

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|-----|---|---------------------------------|
| 1. | 4 penny nail penetrate a 4" dry dimension? | $1\frac{1}{2} - 3\frac{1}{2}$ |
| 2. | 5 common gauge nail penetrate a 2" dry timber? | $4\frac{1}{2} - 1\frac{1}{2}$ |
| 3. | 6 penny nail penetrate a 2" dry rafter? | $2 - 1\frac{1}{2}$ |
| 4. | 16 penny nail penetrate a $1\frac{1}{2}$ " green pole? | $3\frac{1}{2} - 1\frac{9}{32}$ |
| 5. | 14 common gauge nail penetrate a 4" dry board? | $1\frac{1}{4} - 3\frac{1}{2}$ |
| 6. | 4 common gauge nail penetrate a $3\frac{1}{2}$ " green mast? | $5 - 3\frac{1}{16}$ |
| 7. | $12\frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " dry timber? | $1\frac{1}{2} - \frac{7}{16}$ |
| 8. | $14\frac{1}{2}$ box gauge nail penetrate a 2" dry rafter? | $1\frac{1}{4} - 1\frac{1}{2}$ |
| 9. | 3 common gauge nail penetrate a $\frac{3}{4}$ " green pole? | $5\frac{1}{2} - 1\frac{11}{16}$ |
| 10. | 16 penny nail penetrate a $\frac{1}{2}$ " green plank? | $3\frac{1}{2} - 1\frac{15}{32}$ |
| 11. | 7 penny nail penetrate a 3" green timber? | $2\frac{1}{4} - 2\frac{9}{16}$ |
| 12. | $11\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry girder? | $2\frac{1}{2} - \frac{9}{16}$ |
| 13. | $11\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry pole? | $2\frac{1}{2} - \frac{7}{16}$ |
| 14. | 40 penny nail penetrate a $\frac{1}{2}$ " dry mast? | $5 - \frac{7}{16}$ |
| 15. | 3 penny nail penetrate a 3" dry mast? | $1\frac{1}{4} - 2\frac{1}{2}$ |
| 16. | 2 penny nail penetrate a $3\frac{1}{2}$ " green frame? | $1 - 3\frac{1}{16}$ |

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|---|--|
| 1. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 2. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 3. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 4. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 5. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 6. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 7. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 8. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 9. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 10. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 11. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 12. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 13. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 15. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 16. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |

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|-----|--|--------------------------------------|
| 1. | 12 ½ common gauge nail penetrate a 4 ½" dry pole? | 1 ½ — 4 |
| 2. | 4 common gauge nail penetrate a 4 ½" green frame? | 5 — 4 ¹ / ₁₆ |
| 3. | 4 common gauge nail penetrate a ½" green rafter? | 5 — 15 ¹ / ₃₂ |
| 4. | 14 common gauge nail penetrate a 1 ½" green pole? | 1 ¼ — 1 ⁹ / ₃₂ |
| 5. | 20 penny nail penetrate a 3/8" green timber? | 4 — 11 ¹ / ₃₂ |
| 6. | 16 penny nail penetrate a 3" dry pole? | 3 ½ — 2 ½ |
| 7. | 12 ½ common gauge nail penetrate a 4" dry beam? | 1 ½ — 3 ½ |
| 8. | 9 box gauge nail penetrate a 4" dry dimension? | 4 — 3 ½ |
| 9. | 11 ½ common gauge nail penetrate a 4" green plank? | 2 — 3 ⁹ / ₁₆ |
| 10. | 7 penny nail penetrate a 2 ½" dry pole? | 2 ¼ — 2 |
| 11. | 10 ½ box gauge nail penetrate a 4" dry board? | 3 — 3 ½ |
| 12. | 9 box gauge nail penetrate a 1 ¼" dry plank? | 4 — 1 |
| 13. | 8 box gauge nail penetrate a 3/8" green girder? | 5 — 11 ¹ / ₃₂ |
| 14. | 15 common gauge nail penetrate a 4 ½" dry board? | 1 — 4 |
| 15. | 12 ½ box gauge nail penetrate a 1 ½" green joist? | 2 — 1 ⁹ / ₃₂ |
| 16. | 11 ½ box gauge nail penetrate a 2" green beam? | 2 ½ — 1 ⁹ / ₁₆ |

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|---|-----------------------------------|
| 1. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 2. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 4. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 5. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 6. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 7. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 8. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 9. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 10. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 11. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 12. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 13. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 14. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 15. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 16. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |

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|-----|--|-----------------------------------|
| 1. | 6 penny nail penetrate a 4" dry board? | $2 - 3 \frac{1}{2}$ |
| 2. | 9 penny nail penetrate a $\frac{3}{8}$ " green beam? | $2 \frac{3}{4} - 11 \frac{1}{32}$ |
| 3. | 4 common gauge nail penetrate a $\frac{3}{8}$ " dry joist? | $5 - \frac{5}{16}$ |
| 4. | 50 penny nail penetrate a $1 \frac{1}{4}$ " green plank? | $5 \frac{1}{2} - 11 \frac{1}{32}$ |
| 5. | 9 penny nail penetrate a 3" green beam? | $2 \frac{3}{4} - 2 \frac{9}{16}$ |
| 6. | 16 penny nail penetrate a $\frac{3}{4}$ " green board? | $3 \frac{1}{2} - 11 \frac{1}{16}$ |
| 7. | 9 penny nail penetrate a 1" dry rafter? | $2 \frac{3}{4} - \frac{3}{4}$ |
| 8. | 3 common gauge nail penetrate a 4" dry joist? | $5 \frac{1}{2} - 3 \frac{1}{2}$ |
| 9. | $10 \frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " green beam? | $2 \frac{1}{2} - 11 \frac{1}{32}$ |
| 10. | 7 penny nail penetrate a 3" green frame? | $2 \frac{1}{4} - 2 \frac{9}{16}$ |
| 11. | 5 penny nail penetrate a $4 \frac{1}{2}$ " green beam? | $1 \frac{3}{4} - 4 \frac{1}{16}$ |
| 12. | $10 \frac{1}{2}$ box gauge nail penetrate a 3" dry joist? | $3 - 2 \frac{1}{2}$ |
| 13. | 6 penny nail penetrate a $4 \frac{1}{2}$ " dry plank? | $2 - 4$ |
| 14. | 14 box gauge nail penetrate a $4 \frac{1}{2}$ " green plank? | $1 \frac{1}{2} - 4 \frac{1}{16}$ |
| 15. | 16 penny nail penetrate a 4" green dimension? | $3 \frac{1}{2} - 3 \frac{9}{16}$ |
| 16. | 10 penny nail penetrate a $1 \frac{1}{2}$ " dry board? | $3 - 1 \frac{1}{4}$ |

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|--|-------------------------------|
| 1. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 2. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 3. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 4. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 5. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 6. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 7. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 8. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 9. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 10. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 11. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 12. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 13. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 14. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 15. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 16. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |

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|-----|--|--|
| 1. | 12 ½ common gauge nail penetrate a 3 ½" green plank? | 1 ½ — 3 ¹ / ₁₆ |
| 2. | 40 penny nail penetrate a ¾" dry girder? | 5 — 5 ⁵ / ₈ |
| 3. | 4 common gauge nail penetrate a 1 ¼" green girder? | 5 — 1 ¹ / ₃₂ |
| 4. | 15 common gauge nail penetrate a 1 ¼" dry girder? | 1 — 1 |
| 5. | 12 ½ box gauge nail penetrate a 4 ½" dry rafter? | 2 — 4 |
| 6. | 10 ½ box gauge nail penetrate a 1" green mast? | 3 — 25 ⁵ / ₃₂ |
| 7. | 14 box gauge nail penetrate a 5/8" green joist? | 1 ½ — 19 ¹⁹ / ₃₂ |
| 8. | 9 common gauge nail penetrate a 3/8" green girder? | 3 — 11 ¹¹ / ₃₂ |
| 9. | 5 penny nail penetrate a 5/8" green mast? | 1 ¾ — 19 ¹⁹ / ₃₂ |
| 10. | 8 box gauge nail penetrate a 1 ½" green girder? | 5 — 19 ¹⁹ / ₃₂ |
| 11. | 3 penny nail penetrate a 2" green plank? | 1 ¼ — 19 ⁹ / ₁₆ |
| 12. | 30 penny nail penetrate a 1" dry frame? | 4 ½ — ¾ |
| 13. | 10 ¼ common gauge nail penetrate a ¾" dry rafter? | 2 ½ — 5 ⁵ / ₈ |
| 14. | 12 ½ common gauge nail penetrate a 1" dry plank? | 1 ½ — ¾ |
| 15. | 10 ½ box gauge nail penetrate a 4 ½" dry timber? | 3 — 4 |
| 16. | 4 penny nail penetrate a 3" green dimension? | 1 ½ — 29 ⁹ / ₁₆ |

1. countersink hole for a 2 gauge screw?	2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$
2. countersink hole for a 6 gauge screw?	H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$
3. clearance hole for a 12 gauge screw?	2 $\frac{7}{32} = \frac{7}{32}$
4. countersink hole for a 7 gauge screw?	L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$
5. clearance hole for a 4 gauge screw?	35 $\frac{7}{64} = \frac{7}{64}$
6. pilot hole for a 1 gauge screw?	78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$
7. countersink hole for a 7 gauge screw?	L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$
8. pilot hole for a 8 gauge screw?	35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$
9. clearance hole for a 10 gauge screw?	12 $\frac{3}{16} = \frac{3}{16}$
10. clearance hole for a 10 gauge screw?	12 $\frac{3}{16} = \frac{3}{16}$
11. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
12. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$
13. pilot hole for a 20 gauge screw?	H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$
14. clearance hole for a 7 gauge screw?	22 $\frac{5}{32} = \frac{5}{32}$
15. countersink hole for a 9 gauge screw?	N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$
16. clearance hole for a 8 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$

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|-----|--|-----------------------------------|
| 1. | 3 penny nail penetrate a 3" green plank? | $1 \frac{1}{4} - 2 \frac{9}{16}$ |
| 2. | 9 penny nail penetrate a 3" green dimension? | $2 \frac{3}{4} - 2 \frac{9}{16}$ |
| 3. | 12 penny nail penetrate a 2" dry frame? | $3 \frac{1}{4} - 1 \frac{1}{2}$ |
| 4. | 12 $\frac{1}{2}$ common gauge nail penetrate a 2" green rafter? | $1 \frac{1}{2} - 1 \frac{9}{16}$ |
| 5. | 9 common gauge nail penetrate a 1" green rafter? | $3 - 2 \frac{5}{32}$ |
| 6. | 10 penny nail penetrate a 1" green dimension? | $3 - 2 \frac{5}{32}$ |
| 7. | 9 box gauge nail penetrate a 3 $\frac{1}{2}$ " dry pole? | $4 - 3$ |
| 8. | 3 penny nail penetrate a 1 $\frac{1}{4}$ " dry board? | $1 \frac{1}{4} - 1$ |
| 9. | 50 penny nail penetrate a 2" green board? | $5 \frac{1}{2} - 1 \frac{9}{16}$ |
| 10. | 11 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{4}$ " dry mast? | $2 \frac{1}{2} - 1$ |
| 11. | 16 penny nail penetrate a $\frac{3}{8}$ " dry girder? | $3 \frac{1}{2} - \frac{5}{16}$ |
| 12. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " green mast? | $1 \frac{1}{2} - 1 \frac{9}{32}$ |
| 13. | 60 penny nail penetrate a $\frac{3}{4}$ " green pole? | $6 - 1 \frac{1}{16}$ |
| 14. | 3 penny nail penetrate a $\frac{1}{2}$ " green pole? | $1 \frac{1}{4} - 1 \frac{15}{32}$ |
| 15. | 20 penny nail penetrate a $\frac{3}{8}$ " green plank? | $4 - 1 \frac{11}{32}$ |
| 16. | 3 penny nail penetrate a 2 $\frac{1}{2}$ " dry rafter? | $1 \frac{1}{4} - 2$ |

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|---|---|
| 1. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 2. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 3. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 4. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 5. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 6. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 7. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 8. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 9. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 11. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 12. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 13. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 14. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 15. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 16. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |

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|-----|--|----------------------------------|
| 1. | 4 penny nail penetrate a $\frac{1}{2}$ " green beam? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 2. | 3 penny nail penetrate a $\frac{3}{8}$ " dry pole? | $1 \frac{1}{4} - \frac{5}{16}$ |
| 3. | 3 penny nail penetrate a 1" dry timber? | $1 \frac{1}{4} - \frac{3}{4}$ |
| 4. | 30 penny nail penetrate a $1 \frac{1}{2}$ " dry joist? | $4 \frac{1}{2} - 1 \frac{1}{4}$ |
| 5. | 8 penny nail penetrate a 1" dry beam? | $2 \frac{1}{2} - \frac{3}{4}$ |
| 6. | 2 penny nail penetrate a $\frac{3}{4}$ " dry dimension? | $1 - \frac{5}{8}$ |
| 7. | 3 penny nail penetrate a $1 \frac{1}{2}$ " green frame? | $1 \frac{1}{4} - \frac{19}{32}$ |
| 8. | 16 penny nail penetrate a $\frac{5}{8}$ " dry dimension? | $3 \frac{1}{2} - \frac{9}{16}$ |
| 9. | 9 common gauge nail penetrate a $\frac{3}{4}$ " dry timber? | $3 - \frac{5}{8}$ |
| 10. | $11 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green board? | $2 - \frac{11}{16}$ |
| 11. | 12 penny nail penetrate a $\frac{3}{8}$ " green rafter? | $3 \frac{1}{4} - \frac{11}{32}$ |
| 12. | 2 common gauge nail penetrate a $1 \frac{1}{4}$ " dry frame? | $6 - 1$ |
| 13. | 14 box gauge nail penetrate a $2 \frac{1}{2}$ " green rafter? | $1 \frac{1}{2} - 2 \frac{1}{16}$ |
| 14. | 3 penny nail penetrate a $4 \frac{1}{2}$ " dry timber? | $1 \frac{1}{4} - 4$ |
| 15. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " green rafter? | $1 \frac{1}{2} - \frac{11}{32}$ |
| 16. | 12 penny nail penetrate a $\frac{5}{8}$ " dry timber? | $3 \frac{1}{4} - \frac{9}{16}$ |

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|---|--|
| 1. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 2. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 3. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 4. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 5. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 6. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 7. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 8. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 9. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 12. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 13. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 14. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 15. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 16. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |

121

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|-----|--|--------------|
| 1. | 10 ½ box gauge nail penetrate a 3" dry board? | 3 — 2 ½ |
| 2. | 10 ½ box gauge nail penetrate a 1" green mast? | 3 — 25/32 |
| 3. | 60 penny nail penetrate a 3" dry plank? | 6 — 2 ½ |
| 4. | 60 penny nail penetrate a 3 ½" green rafter? | 6 — 3 1/16 |
| 5. | 11 ½ common gauge nail penetrate a 4" dry pole? | 2 — 3 ½ |
| 6. | 14 ½ box gauge nail penetrate a 3/8" dry board? | 1 ¼ — 5/16 |
| 7. | 7 penny nail penetrate a 4" dry beam? | 2 ¼ — 3 ½ |
| 8. | 12 ½ common gauge nail penetrate a 2 ½" green plank? | 1 ½ — 2 1/16 |
| 9. | 9 box gauge nail penetrate a 3/8" green timber? | 4 — 11/32 |
| 10. | 6 common gauge nail penetrate a 4 ½" dry dimension? | 4 — 4 |
| 11. | 4 penny nail penetrate a 3/8" dry pole? | 1 ½ — 5/16 |
| 12. | 9 penny nail penetrate a 3/4" green board? | 2 ¾ — 11/16 |
| 13. | 5 penny nail penetrate a 1" dry rafter? | 1 ¾ — ¾ |
| 14. | 10 ¼ common gauge nail penetrate a 3/8" dry joist? | 2 ½ — 5/16 |
| 15. | 30 penny nail penetrate a 3 ½" dry timber? | 4 ½ — 3 |
| 16. | 12 ½ box gauge nail penetrate a 1 ¼" green joist? | 2 — 1 1/32 |

1. pilot hole for a 1 gauge screw?	78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$
2. pilot hole for a 9 gauge screw?	35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$
3. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
4. pilot hole for a 20 gauge screw?	H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$
5. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
6. pilot hole for a 5 gauge screw?	52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$
7. pilot hole for a 24 gauge screw?	O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$
8. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
9. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
10. clearance hole for a 5 gauge screw?	30 $\frac{1}{8} = \frac{1}{8}$
11. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$
12. clearance hole for a 20 gauge screw?	Q $\frac{21}{64} = \frac{21}{64}$
13. pilot hole for a 12 gauge screw?	22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$
14. pilot hole for a 20 gauge screw?	H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$
15. countersink hole for a 5 gauge screw?	E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$
16. pilot hole for a 9 gauge screw?	35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$

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|-----|--|----------------------------------|
| 1. | 4 common gauge nail penetrate a $\frac{3}{4}$ " dry dimension? | $5 - \frac{5}{8}$ |
| 2. | 9 common gauge nail penetrate a $\frac{5}{8}$ " green mast? | $3 - \frac{19}{32}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a 2" green plank? | $3 - \frac{19}{16}$ |
| 4. | 11 $\frac{1}{2}$ box gauge nail penetrate a 2 $\frac{1}{2}$ " green board? | $2 \frac{1}{2} - 2 \frac{1}{16}$ |
| 5. | 50 penny nail penetrate a 1 $\frac{1}{4}$ " dry plank? | $5 \frac{1}{2} - 1$ |
| 6. | 5 penny nail penetrate a 1 $\frac{1}{4}$ " green mast? | $1 \frac{3}{4} - 1 \frac{1}{32}$ |
| 7. | 50 penny nail penetrate a 1 $\frac{1}{4}$ " dry plank? | $5 \frac{1}{2} - 1$ |
| 8. | 40 penny nail penetrate a $\frac{1}{2}$ " dry dimension? | $5 - \frac{7}{16}$ |
| 9. | 8 box gauge nail penetrate a 2" dry mast? | $5 - 1 \frac{1}{2}$ |
| 10. | 10 $\frac{1}{2}$ box gauge nail penetrate a 3 $\frac{1}{2}$ " green beam? | $3 - 3 \frac{1}{16}$ |
| 11. | 8 penny nail penetrate a 3" dry pole? | $2 \frac{1}{2} - 2 \frac{1}{2}$ |
| 12. | 5 penny nail penetrate a $\frac{3}{8}$ " green dimension? | $1 \frac{3}{4} - 1 \frac{1}{32}$ |
| 13. | 5 common gauge nail penetrate a 4" green plank? | $4 \frac{1}{2} - 3 \frac{9}{16}$ |
| 14. | 30 penny nail penetrate a 1 $\frac{1}{4}$ " green rafter? | $4 \frac{1}{2} - 1 \frac{1}{32}$ |
| 15. | 7 penny nail penetrate a 3 $\frac{1}{2}$ " dry pole? | $2 \frac{1}{4} - 3$ |
| 16. | 6 common gauge nail penetrate a $\frac{1}{2}$ " dry mast? | $4 - \frac{7}{16}$ |

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|---|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 5. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 8. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 9. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 10. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 11. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 12. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 13. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 14. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 15. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 16. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |

123

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|-----|---|----------------|
| 1. | 50 penny nail penetrate a 1 1/2" green rafter? | 5 1/2 — 19/32 |
| 2. | 4 common gauge nail penetrate a 1 1/4" green rafter? | 5 — 11/32 |
| 3. | 11 1/2 box gauge nail penetrate a 3/8" green plank? | 2 1/2 — 11/32 |
| 4. | 12 1/2 common gauge nail penetrate a 1 1/2" dry beam? | 1 1/2 — 1 1/4 |
| 5. | 14 common gauge nail penetrate a 4 1/2" green plank? | 1 1/4 — 4 1/16 |
| 6. | 3 common gauge nail penetrate a 4 1/2" green timber? | 5 1/2 — 4 1/16 |
| 7. | 60 penny nail penetrate a 3/8" green girder? | 6 — 11/32 |
| 8. | 11 1/2 box gauge nail penetrate a 4" green girder? | 2 1/2 — 3 9/16 |
| 9. | 30 penny nail penetrate a 4 1/2" dry girder? | 4 1/2 — 4 |
| 10. | 40 penny nail penetrate a 1 1/4" dry rafter? | 5 — 1 |
| 11. | 50 penny nail penetrate a 1 1/2" dry pole? | 5 1/2 — 1 1/4 |
| 12. | 15 common gauge nail penetrate a 3/4" green pole? | 1 — 11/16 |
| 13. | 9 box gauge nail penetrate a 1/2" dry dimension? | 4 — 7/16 |
| 14. | 12 1/2 common gauge nail penetrate a 3" dry girder? | 1 1/2 — 2 1/2 |
| 15. | 30 penny nail penetrate a 1/2" green timber? | 4 1/2 — 15/32 |
| 16. | 8 penny nail penetrate a 3/4" green timber? | 2 1/2 — 11/16 |

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|---|---|
| 1. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 2. clearance hole for a 12 gauge screw? | $2 \quad \frac{7}{32} = \frac{7}{32}$ |
| 3. pilot hole for a 1 gauge screw? | $78 \quad \frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 4. pilot hole for a 14 gauge screw? | $12 \quad \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 5. clearance hole for a 1 gauge screw? | $47 \quad \frac{5}{64} = \frac{5}{64}$ |
| 6. countersink hole for a 4 gauge screw? | $B \quad \frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 7. pilot hole for a 7 gauge screw? | $41 \quad \frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 8. clearance hole for a 8 gauge screw? | $17 \quad \frac{11}{64} = \frac{11}{64}$ |
| 9. countersink hole for a 14 gauge screw? | $V \quad \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 10. pilot hole for a 2 gauge screw? | $67 \quad \frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 11. clearance hole for a 20 gauge screw? | $Q \quad \frac{21}{64} = \frac{21}{64}$ |
| 12. clearance hole for a 16 gauge screw? | $H \quad \frac{17}{64} = \frac{17}{64}$ |
| 13. clearance hole for a 2 gauge screw? | $41 \quad \frac{3}{32} = \frac{3}{32}$ |
| 14. clearance hole for a 7 gauge screw? | $22 \quad \frac{5}{32} = \frac{5}{32}$ |
| 15. clearance hole for a 10 gauge screw? | $12 \quad \frac{3}{16} = \frac{3}{16}$ |
| 16. pilot hole for a 11 gauge screw? | $27 \quad \frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |

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|-----|--|-------------------------|
| 1. | 4 penny nail penetrate a 1 ¼" green rafter? | 1 ½ — 1 ⅓ ₂ |
| 2. | 5 penny nail penetrate a 2" green beam? | 1 ¾ — 1 ⅑ ₁₆ |
| 3. | 12 ½ common gauge nail penetrate a 5/8" dry plank? | 1 ½ — ⅑ ₁₆ |
| 4. | 12 ½ common gauge nail penetrate a 3" dry pole? | 1 ½ — 2 ½ |
| 5. | 2 penny nail penetrate a 4" green joist? | 1 — 3 ⅑ ₁₆ |
| 6. | 50 penny nail penetrate a 4 ½" green rafter? | 5 ½ — 4 ⅑ ₁₆ |
| 7. | 8 box gauge nail penetrate a 3" dry pole? | 5 — 2 ½ |
| 8. | 60 penny nail penetrate a 5/8" dry beam? | 6 — ⅑ ₁₆ |
| 9. | 14 ½ box gauge nail penetrate a 3 ½" dry timber? | 1 ¼ — 3 |
| 10. | 7 penny nail penetrate a 4 ½" dry pole? | 2 ¼ — 4 |
| 11. | 20 penny nail penetrate a 4" dry plank? | 4 — 3 ½ |
| 12. | 9 box gauge nail penetrate a 1 ½" dry pole? | 4 — 1 ¼ |
| 13. | 10 ¼ common gauge nail penetrate a 3" dry beam? | 2 ½ — 2 ½ |
| 14. | 10 penny nail penetrate a ½" dry board? | 3 — 7/16 |
| 15. | 8 box gauge nail penetrate a 1 ¼" green frame? | 5 — 1 ⅓ ₂ |
| 16. | 12 penny nail penetrate a 2" green dimension? | 3 ¼ — 1 ⅑ ₁₆ |

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|--|---|
| 1. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 2. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 3. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 6. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 7. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 8. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 9. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 10. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 11. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 12. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 13. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 14. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 15. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 16. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |

125

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|-----|--|----------------|
| 1. | 4 penny nail penetrate a 3 1/2" green joist? | 1 1/2 — 3 1/16 |
| 2. | 40 penny nail penetrate a 3/4" dry beam? | 5 — 5/8 |
| 3. | 14 common gauge nail penetrate a 2 1/2" green board? | 1 1/4 — 2 1/16 |
| 4. | 5 penny nail penetrate a 3 1/2" green dimension? | 1 3/4 — 3 1/16 |
| 5. | 15 1/2 box gauge nail penetrate a 1 1/4" green timber? | 1 — 1 1/32 |
| 6. | 4 penny nail penetrate a 3" green dimension? | 1 1/2 — 2 9/16 |
| 7. | 15 1/2 box gauge nail penetrate a 1" green pole? | 1 — 25/32 |
| 8. | 12 1/2 common gauge nail penetrate a 5/8" dry timber? | 1 1/2 — 9/16 |
| 9. | 14 common gauge nail penetrate a 1/2" dry board? | 1 1/4 — 7/16 |
| 10. | 4 penny nail penetrate a 2 1/2" green joist? | 1 1/2 — 2 1/16 |
| 11. | 14 box gauge nail penetrate a 4" dry rafter? | 1 1/2 — 3 1/2 |
| 12. | 20 penny nail penetrate a 2" dry timber? | 4 — 1 1/2 |
| 13. | 12 1/2 box gauge nail penetrate a 1 1/4" green rafter? | 2 — 1 1/32 |
| 14. | 8 common gauge nail penetrate a 4 1/2" dry joist? | 3 1/2 — 4 |
| 15. | 5 penny nail penetrate a 3/8" dry pole? | 1 3/4 — 5/16 |
| 16. | 8 box gauge nail penetrate a 2" dry timber? | 5 — 1 1/2 |

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|---|--------------------------|
| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 3. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 4. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 5. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 6. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 7. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 8. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 9. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 10. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 11. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 12. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 13. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 14. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 15. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 16. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |

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|-----|---|--------------------------------|
| 1. | 8 penny nail penetrate a 3" dry board? | $2\frac{1}{2} - 2\frac{1}{2}$ |
| 2. | 8 box gauge nail penetrate a $1\frac{1}{2}$ " dry board? | $5 - 1\frac{1}{4}$ |
| 3. | 50 penny nail penetrate a $1\frac{1}{4}$ " green board? | $5\frac{1}{2} - 1\frac{1}{32}$ |
| 4. | 5 common gauge nail penetrate a $\frac{3}{8}$ " dry pole? | $4\frac{1}{2} - \frac{5}{16}$ |
| 5. | 20 penny nail penetrate a $1\frac{1}{4}$ " green rafter? | $4 - 1\frac{1}{32}$ |
| 6. | 10 penny nail penetrate a $\frac{5}{8}$ " dry joist? | $3 - \frac{9}{16}$ |
| 7. | $10\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green frame? | $3 - \frac{11}{16}$ |
| 8. | 50 penny nail penetrate a $\frac{3}{8}$ " green girder? | $5\frac{1}{2} - 1\frac{1}{32}$ |
| 9. | 30 penny nail penetrate a $\frac{3}{8}$ " dry frame? | $4\frac{1}{2} - \frac{5}{16}$ |
| 10. | 60 penny nail penetrate a 1" dry joist? | $6 - \frac{3}{4}$ |
| 11. | 7 penny nail penetrate a $1\frac{1}{4}$ " dry rafter? | $2\frac{1}{4} - 1$ |
| 12. | 8 box gauge nail penetrate a $\frac{3}{4}$ " dry beam? | $5 - \frac{5}{8}$ |
| 13. | 9 common gauge nail penetrate a $3\frac{1}{2}$ " dry mast? | $3 - 3$ |
| 14. | 8 penny nail penetrate a $\frac{3}{8}$ " green pole? | $2\frac{1}{2} - 1\frac{1}{32}$ |
| 15. | 60 penny nail penetrate a $\frac{3}{8}$ " green pole? | $6 - 1\frac{1}{32}$ |
| 16. | 14 common gauge nail penetrate a 1" green girder? | $1\frac{1}{4} - \frac{25}{32}$ |

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|--|-------------------------------|
| 1. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 2. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 3. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 4. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 5. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 6. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 7. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 8. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 9. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 10. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 11. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 12. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 13. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 15. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 16. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |

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|-----|--|----------------|
| 1. | 20 penny nail penetrate a 3/8" dry joist? | 4 — 5/16 |
| 2. | 12 penny nail penetrate a 1" green dimension? | 3 1/4 — 25/32 |
| 3. | 10 1/4 common gauge nail penetrate a 2 1/2" dry plank? | 2 1/2 — 2 |
| 4. | 40 penny nail penetrate a 1 1/2" green board? | 5 — 19/32 |
| 5. | 12 1/2 box gauge nail penetrate a 4" green joist? | 2 — 39/16 |
| 6. | 12 1/2 box gauge nail penetrate a 3 1/2" dry girder? | 2 — 3 |
| 7. | 30 penny nail penetrate a 2" green joist? | 4 1/2 — 19/16 |
| 8. | 10 1/4 common gauge nail penetrate a 1 1/2" dry frame? | 2 1/2 — 1 1/4 |
| 9. | 8 common gauge nail penetrate a 3 1/2" dry pole? | 3 1/2 — 3 |
| 10. | 3 penny nail penetrate a 3 1/2" green pole? | 1 1/4 — 3 1/16 |
| 11. | 5 penny nail penetrate a 4 1/2" green timber? | 1 3/4 — 4 1/16 |
| 12. | 20 penny nail penetrate a 5/8" dry joist? | 4 — 9/16 |
| 13. | 14 common gauge nail penetrate a 1" green beam? | 1 1/4 — 25/32 |
| 14. | 6 penny nail penetrate a 5/8" dry beam? | 2 — 9/16 |
| 15. | 4 penny nail penetrate a 3/4" green rafter? | 1 1/2 — 11/16 |
| 16. | 40 penny nail penetrate a 1 1/4" dry dimension? | 5 — 1 |

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|--|---|
| 1. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 2. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 3. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 4. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 5. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 6. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 7. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 8. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 9. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 10. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 11. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 14. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 15. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |

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|-----|---|-------------|
| 1. | 10 ½ box gauge nail penetrate a 3/4" green rafter? | 3 — 11/16 |
| 2. | 11 ½ box gauge nail penetrate a 4" green plank? | 2 ½ — 39/16 |
| 3. | 2 penny nail penetrate a 3" dry frame? | 1 — 2 ½ |
| 4. | 3 penny nail penetrate a 1 ¼" green plank? | 1 ¼ — 11/32 |
| 5. | 6 common gauge nail penetrate a 5/8" green dimension? | 4 — 19/32 |
| 6. | 15 ½ box gauge nail penetrate a 3 ½" dry girder? | 1 — 3 |
| 7. | 12 ½ common gauge nail penetrate a 3/8" dry timber? | 1 ½ — 5/16 |
| 8. | 4 penny nail penetrate a 1 ½" dry joist? | 1 ½ — 1 ¼ |
| 9. | 3 common gauge nail penetrate a 1 ½" green plank? | 5 ½ — 19/32 |
| 10. | 3 common gauge nail penetrate a 1 ¼" dry plank? | 5 ½ — 1 |
| 11. | 14 box gauge nail penetrate a 5/8" dry beam? | 1 ½ — 9/16 |
| 12. | 5 penny nail penetrate a 3 ½" dry mast? | 1 ¾ — 3 |
| 13. | 5 penny nail penetrate a ½" green plank? | 1 ¾ — 15/32 |
| 14. | 9 box gauge nail penetrate a 1 ¼" dry rafter? | 4 — 1 |
| 15. | 9 penny nail penetrate a 1" dry plank? | 2 ¾ — ¾ |
| 16. | 9 penny nail penetrate a 1 ½" green girder? | 2 ¾ — 19/32 |

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|--|--------------------------|
| 1. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 2. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 3. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 4. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 5. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 7. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 8. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 9. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 10. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 11. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 12. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 13. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 14. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 15. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 16. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |

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|-----|---|---------------------|
| 1. | 9 common gauge nail penetrate a $5/8$ " green frame? | $3 - 19/32$ |
| 2. | 40 penny nail penetrate a $2\ 1/2$ " green beam? | $5 - 2^{1/16}$ |
| 3. | 3 penny nail penetrate a $3/4$ " dry pole? | $1\ 1/4 - 5/8$ |
| 4. | 16 penny nail penetrate a $2\ 1/2$ " dry mast? | $3\ 1/2 - 2$ |
| 5. | $11\ 1/2$ box gauge nail penetrate a $1\ 1/4$ " green beam? | $2\ 1/2 - 1^{1/32}$ |
| 6. | 16 penny nail penetrate a $1\ 1/4$ " dry girder? | $3\ 1/2 - 1$ |
| 7. | $12\ 1/2$ box gauge nail penetrate a $3/4$ " dry girder? | $2 - 5/8$ |
| 8. | 4 penny nail penetrate a $1/2$ " green timber? | $1\ 1/2 - 15/32$ |
| 9. | 10 penny nail penetrate a $3/8$ " dry beam? | $3 - 5/16$ |
| 10. | 8 box gauge nail penetrate a 2" dry rafter? | $5 - 1\ 1/2$ |
| 11. | $11\ 1/2$ common gauge nail penetrate a $1\ 1/4$ " dry plank? | $2 - 1$ |
| 12. | 2 common gauge nail penetrate a 4" green mast? | $6 - 3^{9/16}$ |
| 13. | 14 box gauge nail penetrate a 1" green joist? | $1\ 1/2 - 25/32$ |
| 14. | 7 penny nail penetrate a $2\ 1/2$ " dry mast? | $2\ 1/4 - 2$ |
| 15. | 14 box gauge nail penetrate a 2" dry frame? | $1\ 1/2 - 1\ 1/2$ |
| 16. | 8 penny nail penetrate a $1/2$ " dry pole? | $2\ 1/2 - 7/16$ |

1. pilot hole for a 10 gauge screw?	30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$
2. countersink hole for a 14 gauge screw?	V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$
3. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
4. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
5. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
6. clearance hole for a 5 gauge screw?	30 $\frac{1}{8} = \frac{1}{8}$
7. countersink hole for a 6 gauge screw?	H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$
8. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
9. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$
10. countersink hole for a 14 gauge screw?	V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$
11. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$
12. pilot hole for a 10 gauge screw?	30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$
13. countersink hole for a 6 gauge screw?	H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$
14. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
15. clearance hole for a 6 gauge screw?	27 $\frac{9}{64} = \frac{9}{64}$
16. countersink hole for a 6 gauge screw?	H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$

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|-----|--|---------------------------------|
| 1. | 14 box gauge nail penetrate a $\frac{1}{2}$ " green rafter? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 2. | 9 penny nail penetrate a $1 \frac{1}{2}$ " green girder? | $2 \frac{3}{4} - \frac{19}{32}$ |
| 3. | 4 common gauge nail penetrate a 1" green beam? | $5 - \frac{25}{32}$ |
| 4. | 5 penny nail penetrate a $1 \frac{1}{2}$ " dry dimension? | $1 \frac{3}{4} - 1 \frac{1}{4}$ |
| 5. | 6 common gauge nail penetrate a $\frac{1}{2}$ " green frame? | $4 - \frac{15}{32}$ |
| 6. | 8 box gauge nail penetrate a 4" green frame? | $5 - 3\frac{9}{16}$ |
| 7. | 20 penny nail penetrate a 2" dry beam? | $4 - 1 \frac{1}{2}$ |
| 8. | 8 penny nail penetrate a $\frac{3}{4}$ " green frame? | $2 \frac{1}{2} - \frac{11}{16}$ |
| 9. | 12 penny nail penetrate a $4 \frac{1}{2}$ " dry girder? | $3 \frac{1}{4} - 4$ |
| 10. | 4 penny nail penetrate a $2 \frac{1}{2}$ " dry mast? | $1 \frac{1}{2} - 2$ |
| 11. | $10 \frac{1}{2}$ box gauge nail penetrate a 4" green girder? | $3 - 3\frac{9}{16}$ |
| 12. | 5 penny nail penetrate a $1 \frac{1}{4}$ " dry beam? | $1 \frac{3}{4} - 1$ |
| 13. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " dry beam? | $1 \frac{1}{2} - \frac{9}{16}$ |
| 14. | 10 penny nail penetrate a 2" dry joist? | $3 - 1 \frac{1}{2}$ |
| 15. | 14 box gauge nail penetrate a $\frac{3}{4}$ " green beam? | $1 \frac{1}{2} - \frac{11}{16}$ |
| 16. | 30 penny nail penetrate a $\frac{3}{8}$ " green plank? | $4 \frac{1}{2} - \frac{11}{32}$ |

1. pilot hole for a 18 gauge screw? B $19/64 - 1/16 = 15/64$
2. pilot hole for a 24 gauge screw? O $3/8 - 1/16 = 5/16$
3. clearance hole for a 8 gauge screw? 17 $11/64 = 11/64$
4. clearance hole for a 16 gauge screw? H $17/64 = 17/64$
5. pilot hole for a 6 gauge screw? 47 $9/64 - 1/16 = 5/64$
6. countersink hole for a 4 gauge screw? B $7/64 + 1/8 = 15/64$
7. countersink hole for a 8 gauge screw? N $11/64 + 1/8 = 19/64$
8. pilot hole for a 5 gauge screw? 52 $1/8 - 1/16 = 1/16$
9. countersink hole for a 18 gauge screw? $27/64$ $19/64 + 1/8 = 27/64$
10. pilot hole for a 1 gauge screw? 78 $5/64 - 1/16 = 1/64$
11. countersink hole for a 10 gauge screw? O $3/16 + 1/8 = 5/16$
12. clearance hole for a 14 gauge screw? E $1/4 = 1/4$
13. pilot hole for a 24 gauge screw? O $3/8 - 1/16 = 5/16$
14. clearance hole for a 3 gauge screw? 41 $3/32 = 3/32$
15. clearance hole for a 18 gauge screw? N $19/64 = 19/64$
16. clearance hole for a 5 gauge screw? 30 $1/8 = 1/8$

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|-----|---|--------------------------------|
| 1. | 12 penny nail penetrate a $\frac{3}{8}$ " dry board? | $3\frac{1}{4} - \frac{5}{16}$ |
| 2. | 9 box gauge nail penetrate a $2\frac{1}{2}$ " green beam? | $4 - 2\frac{1}{16}$ |
| 3. | 9 penny nail penetrate a $2\frac{1}{2}$ " green mast? | $2\frac{3}{4} - 2\frac{1}{16}$ |
| 4. | 11 $\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " dry dimension? | $2\frac{1}{2} - 3$ |
| 5. | 10 penny nail penetrate a 1" green mast? | $3 - \frac{25}{32}$ |
| 6. | 14 box gauge nail penetrate a $3\frac{1}{2}$ " green beam? | $1\frac{1}{2} - 3\frac{1}{16}$ |
| 7. | 8 penny nail penetrate a 2" green timber? | $2\frac{1}{2} - 1\frac{9}{16}$ |
| 8. | 8 common gauge nail penetrate a $4\frac{1}{2}$ " green frame? | $3\frac{1}{2} - 4\frac{1}{16}$ |
| 9. | 40 penny nail penetrate a $\frac{3}{4}$ " green mast? | $5 - \frac{11}{16}$ |
| 10. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green girder? | $2\frac{1}{2} - \frac{11}{16}$ |
| 11. | 30 penny nail penetrate a 1" dry girder? | $4\frac{1}{2} - \frac{3}{4}$ |
| 12. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green beam? | $1\frac{1}{2} - \frac{11}{16}$ |
| 13. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1" green pole? | $2 - \frac{25}{32}$ |
| 14. | 15 common gauge nail penetrate a $\frac{3}{8}$ " dry joist? | $1 - \frac{5}{16}$ |
| 15. | 50 penny nail penetrate a $2\frac{1}{2}$ " green plank? | $5\frac{1}{2} - 2\frac{1}{16}$ |
| 16. | 4 penny nail penetrate a $\frac{5}{8}$ " dry joist? | $1\frac{1}{2} - \frac{9}{16}$ |

1. pilot hole for a 18 gauge screw?	B $19/64 - 1/16 = 15/64$
2. pilot hole for a 5 gauge screw?	52 $1/8 - 1/16 = 1/16$
3. clearance hole for a 10 gauge screw?	12 $3/16 = 3/16$
4. countersink hole for a 11 gauge screw?	Q $13/64 + 1/8 = 21/64$
5. clearance hole for a 24 gauge screw?	V $3/8 = 3/8$
6. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
7. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
8. clearance hole for a 18 gauge screw?	N $19/64 = 19/64$
9. countersink hole for a 11 gauge screw?	Q $13/64 + 1/8 = 21/64$
10. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$
11. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$
12. countersink hole for a 14 gauge screw?	V $1/4 + 1/8 = 3/8$
13. pilot hole for a 14 gauge screw?	12 $1/4 - 1/16 = 3/16$
14. countersink hole for a 12 gauge screw?	S $7/32 + 1/8 = 11/32$
15. countersink hole for a 7 gauge screw?	L $5/32 + 1/8 = 9/32$
16. countersink hole for a 7 gauge screw?	L $5/32 + 1/8 = 9/32$

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|-----|---|----------------------------------|
| 1. | 8 box gauge nail penetrate a 1" dry pole? | $5 - \frac{3}{4}$ |
| 2. | 7 penny nail penetrate a 1" green board? | $2 \frac{1}{4} - \frac{25}{32}$ |
| 3. | 8 common gauge nail penetrate a $3 \frac{1}{2}$ " green girder? | $3 \frac{1}{2} - 3 \frac{1}{16}$ |
| 4. | 16 penny nail penetrate a $\frac{5}{8}$ " green pole? | $3 \frac{1}{2} - \frac{19}{32}$ |
| 5. | 50 penny nail penetrate a $2 \frac{1}{2}$ " dry frame? | $5 \frac{1}{2} - 2$ |
| 6. | 30 penny nail penetrate a $1 \frac{1}{2}$ " dry frame? | $4 \frac{1}{2} - 1 \frac{1}{4}$ |
| 7. | 12 penny nail penetrate a $\frac{1}{2}$ " dry rafter? | $3 \frac{1}{4} - \frac{7}{16}$ |
| 8. | 7 penny nail penetrate a $1 \frac{1}{4}$ " green pole? | $2 \frac{1}{4} - 1 \frac{1}{32}$ |
| 9. | 3 common gauge nail penetrate a $\frac{3}{4}$ " green timber? | $5 \frac{1}{2} - \frac{11}{16}$ |
| 10. | 8 common gauge nail penetrate a $\frac{5}{8}$ " green mast? | $3 \frac{1}{2} - \frac{19}{32}$ |
| 11. | 30 penny nail penetrate a $2 \frac{1}{2}$ " green frame? | $4 \frac{1}{2} - 2 \frac{1}{16}$ |
| 12. | $11 \frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " green board? | $2 \frac{1}{2} - 3 \frac{1}{16}$ |
| 13. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green frame? | $1 \frac{1}{2} - \frac{11}{16}$ |
| 14. | 8 common gauge nail penetrate a $3 \frac{1}{2}$ " dry girder? | $3 \frac{1}{2} - 3$ |
| 15. | 15 common gauge nail penetrate a $2 \frac{1}{2}$ " dry board? | $1 - 2$ |
| 16. | 4 common gauge nail penetrate a $2 \frac{1}{2}$ " dry mast? | $5 - 2$ |

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| 1. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 2. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 5. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 6. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 7. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 8. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 9. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 10. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 11. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 12. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 13. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 15. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 16. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |

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| 1. | 15 common gauge nail penetrate a 2" green girder? | $1 - 1\frac{9}{16}$ |
| 2. | 12 penny nail penetrate a $\frac{1}{2}$ " dry girder? | $3\frac{1}{4} - \frac{7}{16}$ |
| 3. | 30 penny nail penetrate a $1\frac{1}{2}$ " green beam? | $4\frac{1}{2} - 1\frac{9}{32}$ |
| 4. | $10\frac{1}{4}$ common gauge nail penetrate a 2" green dimension? | $2\frac{1}{2} - 1\frac{9}{16}$ |
| 5. | $14\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green mast? | $1\frac{1}{4} - 1\frac{11}{32}$ |
| 6. | 9 penny nail penetrate a $\frac{5}{8}$ " green plank? | $2\frac{3}{4} - 1\frac{19}{32}$ |
| 7. | 9 common gauge nail penetrate a 4" dry plank? | $3 - 3\frac{1}{2}$ |
| 8. | 10 penny nail penetrate a $\frac{3}{4}$ " green frame? | $3 - 1\frac{11}{16}$ |
| 9. | 9 penny nail penetrate a $\frac{3}{8}$ " dry girder? | $2\frac{3}{4} - \frac{5}{16}$ |
| 10. | 14 box gauge nail penetrate a 2" green board? | $1\frac{1}{2} - 1\frac{9}{16}$ |
| 11. | 4 penny nail penetrate a $4\frac{1}{2}$ " dry plank? | $1\frac{1}{2} - 4$ |
| 12. | $12\frac{1}{2}$ common gauge nail penetrate a $1\frac{1}{4}$ " green rafter? | $1\frac{1}{2} - 1\frac{1}{32}$ |
| 13. | 10 penny nail penetrate a $1\frac{1}{2}$ " green plank? | $3 - 1\frac{9}{32}$ |
| 14. | $12\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green rafter? | $1\frac{1}{2} - 1\frac{11}{16}$ |
| 15. | $10\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry mast? | $3 - \frac{5}{8}$ |
| 16. | $14\frac{1}{2}$ box gauge nail penetrate a $2\frac{1}{2}$ " green plank? | $1\frac{1}{4} - 2\frac{1}{16}$ |

1. countersink hole for a 12 gauge screw?	S $7/32 + 1/8 = 11/32$
2. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
3. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
4. countersink hole for a 14 gauge screw?	V $1/4 + 1/8 = 3/8$
5. clearance hole for a 16 gauge screw?	H $17/64 = 17/64$
6. clearance hole for a 8 gauge screw?	17 $11/64 = 11/64$
7. clearance hole for a 5 gauge screw?	30 $1/8 = 1/8$
8. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$
9. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$
10. pilot hole for a 18 gauge screw?	B $19/64 - 1/16 = 15/64$
11. clearance hole for a 7 gauge screw?	22 $5/32 = 5/32$
12. pilot hole for a 2 gauge screw?	67 $3/32 - 1/16 = 1/32$
13. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$
14. clearance hole for a 8 gauge screw?	17 $11/64 = 11/64$
15. clearance hole for a 9 gauge screw?	17 $11/64 = 11/64$
16. clearance hole for a 18 gauge screw?	N $19/64 = 19/64$

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| 1. | 3 common gauge nail penetrate a $\frac{3}{4}$ " dry rafter? | $5\frac{1}{2} - \frac{5}{8}$ |
| 2. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry plank? | $3 - \frac{5}{16}$ |
| 3. | 20 penny nail penetrate a $2\frac{1}{2}$ " green frame? | $4 - 2\frac{1}{16}$ |
| 4. | 5 common gauge nail penetrate a $\frac{1}{2}$ " dry pole? | $4\frac{1}{2} - \frac{7}{16}$ |
| 5. | 9 box gauge nail penetrate a 2" dry plank? | $4 - 1\frac{1}{2}$ |
| 6. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry timber? | $2\frac{1}{2} - \frac{5}{8}$ |
| 7. | 20 penny nail penetrate a 4" dry mast? | $4 - 3\frac{1}{2}$ |
| 8. | 50 penny nail penetrate a $\frac{3}{4}$ " dry rafter? | $5\frac{1}{2} - \frac{5}{8}$ |
| 9. | 6 penny nail penetrate a $\frac{3}{4}$ " green rafter? | $2 - \frac{11}{16}$ |
| 10. | 14 box gauge nail penetrate a $\frac{3}{4}$ " dry frame? | $1\frac{1}{2} - \frac{5}{8}$ |
| 11. | 7 penny nail penetrate a 2" green frame? | $2\frac{1}{4} - 1\frac{9}{16}$ |
| 12. | 12 $\frac{1}{2}$ common gauge nail penetrate a $1\frac{1}{2}$ " green frame? | $1\frac{1}{2} - 1\frac{9}{32}$ |
| 13. | 10 $\frac{1}{2}$ box gauge nail penetrate a 3" green plank? | $3 - 2\frac{9}{16}$ |
| 14. | 12 $\frac{1}{2}$ common gauge nail penetrate a 1" dry rafter? | $1\frac{1}{2} - \frac{3}{4}$ |
| 15. | 10 $\frac{1}{2}$ box gauge nail penetrate a 1" green beam? | $3 - 2\frac{25}{32}$ |
| 16. | 4 common gauge nail penetrate a 3" dry frame? | $5 - 2\frac{1}{2}$ |

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| 1. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 2. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 3. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 4. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 5. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 7. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 8. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 9. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 10. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 11. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 12. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 13. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 14. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 15. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 16. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |

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| 1. | 10 ½ box gauge nail penetrate a 4" dry frame? | 3 — 3 ½ |
| 2. | 9 box gauge nail penetrate a 2 ½" green rafter? | 4 — 2 ⅛ |
| 3. | 8 common gauge nail penetrate a 3" green girder? | 3 ½ — 2 ⅞ |
| 4. | 10 ½ box gauge nail penetrate a ½" green girder? | 3 — 15/32 |
| 5. | 12 ½ box gauge nail penetrate a 3" green dimension? | 2 — 2 ⅞ |
| 6. | 50 penny nail penetrate a ¾" dry beam? | 5 ½ — ⅝ |
| 7. | 8 penny nail penetrate a 4 ½" green frame? | 2 ½ — 4 ⅛ |
| 8. | 7 penny nail penetrate a ¾" green plank? | 2 ¼ — 11/16 |
| 9. | 10 penny nail penetrate a ¾" dry timber? | 3 — ⅝ |
| 10. | 2 penny nail penetrate a ½" dry plank? | 1 — 7/16 |
| 11. | 50 penny nail penetrate a 5/8" green joist? | 5 ½ — 19/32 |
| 12. | 9 box gauge nail penetrate a 2 ½" green board? | 4 — 2 ⅛ |
| 13. | 5 penny nail penetrate a 2 ½" green plank? | 1 ¾ — 2 ⅛ |
| 14. | 2 penny nail penetrate a 5/8" dry rafter? | 1 — 9/16 |
| 15. | 40 penny nail penetrate a 4" dry beam? | 5 — 3 ½ |
| 16. | 12 ½ common gauge nail penetrate a 3 ½" dry mast? | 1 ½ — 3 |

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| 1. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 2. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 3. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 4. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 5. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 6. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 7. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 8. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 9. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 10. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 11. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 12. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 13. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 14. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 15. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 16. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |

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|-----|---|----------------|
| 1. | 5 penny nail penetrate a 4 1/2" green board? | 1 3/4 — 4 1/16 |
| 2. | 14 box gauge nail penetrate a 3 1/2" dry frame? | 1 1/2 — 3 |
| 3. | 8 box gauge nail penetrate a 3/8" dry dimension? | 5 — 5/16 |
| 4. | 5 penny nail penetrate a 3" dry joist? | 1 3/4 — 2 1/2 |
| 5. | 9 box gauge nail penetrate a 1 1/4" green girder? | 4 — 1 1/32 |
| 6. | 9 penny nail penetrate a 1 1/4" green dimension? | 2 3/4 — 1 1/32 |
| 7. | 2 penny nail penetrate a 1/2" dry frame? | 1 — 7/16 |
| 8. | 12 1/2 common gauge nail penetrate a 1 1/2" dry timber? | 1 1/2 — 1 1/4 |
| 9. | 8 penny nail penetrate a 4 1/2" dry timber? | 2 1/2 — 4 |
| 10. | 16 penny nail penetrate a 2" green rafter? | 3 1/2 — 1 9/16 |
| 11. | 60 penny nail penetrate a 3/8" dry frame? | 6 — 5/16 |
| 12. | 16 penny nail penetrate a 3 1/2" green mast? | 3 1/2 — 3 1/16 |
| 13. | 14 common gauge nail penetrate a 1 1/4" green plank? | 1 1/4 — 1 1/32 |
| 14. | 30 penny nail penetrate a 3" green board? | 4 1/2 — 2 9/16 |
| 15. | 7 penny nail penetrate a 5/8" dry rafter? | 2 1/4 — 9/16 |
| 16. | 2 penny nail penetrate a 3/8" green girder? | 1 — 11/32 |

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|--|---|
| 1. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 2. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 3. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 4. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 5. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 6. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 7. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 8. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 9. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 10. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 14. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 15. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 16. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |

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|-----|---|---------------|
| 1. | 6 penny nail penetrate a 3/8" dry rafter? | 2 — 5/16 |
| 2. | 2 penny nail penetrate a 3/4" green mast? | 1 — 11/16 |
| 3. | 50 penny nail penetrate a 1/2" green joist? | 5 1/2 — 15/32 |
| 4. | 5 common gauge nail penetrate a 3/4" green frame? | 4 1/2 — 11/16 |
| 5. | 10 1/4 common gauge nail penetrate a 3" green mast? | 2 1/2 — 29/16 |
| 6. | 4 common gauge nail penetrate a 3 1/2" green joist? | 5 — 31/16 |
| 7. | 30 penny nail penetrate a 5/8" dry board? | 4 1/2 — 9/16 |
| 8. | 9 common gauge nail penetrate a 1" green dimension? | 3 — 25/32 |
| 9. | 10 1/4 common gauge nail penetrate a 3/4" dry plank? | 2 1/2 — 5/8 |
| 10. | 50 penny nail penetrate a 4" green pole? | 5 1/2 — 39/16 |
| 11. | 11 1/2 common gauge nail penetrate a 2" dry mast? | 2 — 1 1/2 |
| 12. | 11 1/2 common gauge nail penetrate a 3 1/2" green rafter? | 2 — 31/16 |
| 13. | 12 1/2 box gauge nail penetrate a 1" green girder? | 2 — 25/32 |
| 14. | 9 common gauge nail penetrate a 1 1/2" green plank? | 3 — 19/32 |
| 15. | 6 penny nail penetrate a 2 1/2" green frame? | 2 — 21/16 |
| 16. | 10 1/4 common gauge nail penetrate a 4 1/2" dry girder? | 2 1/2 — 4 |

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|--|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 3. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 4. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 5. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 6. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 7. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 8. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 9. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 10. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 11. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 12. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 14. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 15. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |

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| 1. | 8 box gauge nail penetrate a 1" green board? | $5 - \frac{25}{32}$ |
| 2. | 14 common gauge nail penetrate a 2" green rafter? | $1 \frac{1}{4} - 1 \frac{9}{16}$ |
| 3. | 30 penny nail penetrate a 3" dry joist? | $4 \frac{1}{2} - 2 \frac{1}{2}$ |
| 4. | 4 penny nail penetrate a 2" green board? | $1 \frac{1}{2} - 1 \frac{9}{16}$ |
| 5. | 6 penny nail penetrate a $\frac{3}{4}$ " dry pole? | $2 - \frac{5}{8}$ |
| 6. | 60 penny nail penetrate a $1 \frac{1}{4}$ " green board? | $6 - 1 \frac{1}{32}$ |
| 7. | 7 penny nail penetrate a $1 \frac{1}{2}$ " dry board? | $2 \frac{1}{4} - 1 \frac{1}{4}$ |
| 8. | 14 common gauge nail penetrate a $\frac{3}{4}$ " green mast? | $1 \frac{1}{4} - 1 \frac{1}{16}$ |
| 9. | 40 penny nail penetrate a 3" green girder? | $5 - 2 \frac{9}{16}$ |
| 10. | 2 penny nail penetrate a $4 \frac{1}{2}$ " green board? | $1 - 4 \frac{1}{16}$ |
| 11. | 12 penny nail penetrate a $1 \frac{1}{4}$ " green joist? | $3 \frac{1}{4} - 1 \frac{1}{32}$ |
| 12. | 8 penny nail penetrate a 4" green mast? | $2 \frac{1}{2} - 3 \frac{9}{16}$ |
| 13. | 30 penny nail penetrate a $\frac{3}{8}$ " dry timber? | $4 \frac{1}{2} - \frac{5}{16}$ |
| 14. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green board? | $3 - 1 \frac{1}{32}$ |
| 15. | 50 penny nail penetrate a $3 \frac{1}{2}$ " green dimension? | $5 \frac{1}{2} - 3 \frac{1}{16}$ |
| 16. | 9 box gauge nail penetrate a 1" green pole? | $4 - \frac{25}{32}$ |

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|--|--------------------------|
| 1. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 2. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 3. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 5. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 6. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 7. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 8. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 9. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 10. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 11. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 12. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 13. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 14. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 15. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 16. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |

139

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|-----|--|--------------------------------|
| 1. | 5 common gauge nail penetrate a $\frac{1}{2}$ " dry pole? | $4\frac{1}{2} - \frac{7}{16}$ |
| 2. | 30 penny nail penetrate a 2" dry dimension? | $4\frac{1}{2} - 1\frac{1}{2}$ |
| 3. | 15 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry frame? | $1 - \frac{5}{8}$ |
| 4. | 11 $\frac{1}{2}$ box gauge nail penetrate a 4" green timber? | $2\frac{1}{2} - 3\frac{9}{16}$ |
| 5. | 11 $\frac{1}{2}$ common gauge nail penetrate a 3" green girder? | $2 - 2\frac{9}{16}$ |
| 6. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry joist? | $3 - \frac{9}{16}$ |
| 7. | 7 penny nail penetrate a $1\frac{1}{4}$ " green pole? | $2\frac{1}{4} - 1\frac{1}{32}$ |
| 8. | 12 penny nail penetrate a 3" green board? | $3\frac{1}{4} - 2\frac{9}{16}$ |
| 9. | 2 penny nail penetrate a $\frac{5}{8}$ " dry girder? | $1 - \frac{9}{16}$ |
| 10. | 60 penny nail penetrate a $\frac{3}{8}$ " dry mast? | $6 - \frac{5}{16}$ |
| 11. | 9 box gauge nail penetrate a $2\frac{1}{2}$ " dry mast? | $4 - 2$ |
| 12. | 7 penny nail penetrate a $1\frac{1}{4}$ " green board? | $2\frac{1}{4} - 1\frac{1}{32}$ |
| 13. | 16 penny nail penetrate a $1\frac{1}{2}$ " green plank? | $3\frac{1}{2} - 1\frac{9}{32}$ |
| 14. | 6 common gauge nail penetrate a $1\frac{1}{4}$ " green beam? | $4 - 1\frac{1}{32}$ |
| 15. | 50 penny nail penetrate a $4\frac{1}{2}$ " dry plank? | $5\frac{1}{2} - 4$ |
| 16. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry board? | $2\frac{1}{2} - \frac{9}{16}$ |

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|---|---|
| 1. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 2. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 3. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 4. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 5. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 6. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 7. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 8. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 9. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 10. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 11. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 12. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 13. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 14. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 15. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 16. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |

140

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|-----|---|-------------|
| 1. | 12 ½ box gauge nail penetrate a 4" dry timber? | 2 — 3 ½ |
| 2. | 5 penny nail penetrate a 3" dry board? | 1 ¾ — 2 ½ |
| 3. | 8 penny nail penetrate a ¾" green mast? | 2 ½ — 11/16 |
| 4. | 12 ½ box gauge nail penetrate a 1 ¼" dry pole? | 2 — 1 |
| 5. | 15 common gauge nail penetrate a ¾" green plank? | 1 — 11/16 |
| 6. | 3 common gauge nail penetrate a 2" green pole? | 5 ½ — 19/16 |
| 7. | 14 common gauge nail penetrate a 4 ½" dry pole? | 1 ¼ — 4 |
| 8. | 6 penny nail penetrate a 5/8" dry dimension? | 2 — 9/16 |
| 9. | 7 penny nail penetrate a ½" dry joist? | 2 ¼ — 7/16 |
| 10. | 9 common gauge nail penetrate a ½" dry girder? | 3 — 7/16 |
| 11. | 40 penny nail penetrate a 5/8" dry timber? | 5 — 9/16 |
| 12. | 15 common gauge nail penetrate a 1" green girder? | 1 — 25/32 |
| 13. | 8 common gauge nail penetrate a 1 ½" dry frame? | 3 ½ — 1 ¼ |
| 14. | 7 penny nail penetrate a 4" green pole? | 2 ¼ — 39/16 |
| 15. | 15 common gauge nail penetrate a 2 ½" green rafter? | 1 — 21/16 |
| 16. | 16 penny nail penetrate a 4" dry rafter? | 3 ½ — 3 ½ |

1. pilot hole for a 5 gauge screw?	52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$
2. clearance hole for a 7 gauge screw?	22 $\frac{5}{32} = \frac{5}{32}$
3. clearance hole for a 14 gauge screw?	E $\frac{1}{4} = \frac{1}{4}$
4. clearance hole for a 16 gauge screw?	H $\frac{17}{64} = \frac{17}{64}$
5. clearance hole for a 10 gauge screw?	12 $\frac{3}{16} = \frac{3}{16}$
6. clearance hole for a 16 gauge screw?	H $\frac{17}{64} = \frac{17}{64}$
7. clearance hole for a 6 gauge screw?	27 $\frac{9}{64} = \frac{9}{64}$
8. pilot hole for a 10 gauge screw?	30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$
9. countersink hole for a 3 gauge screw?	2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$
10. pilot hole for a 6 gauge screw?	47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$
11. pilot hole for a 6 gauge screw?	47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$
12. clearance hole for a 18 gauge screw?	N $\frac{19}{64} = \frac{19}{64}$
13. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$
14. clearance hole for a 20 gauge screw?	Q $\frac{21}{64} = \frac{21}{64}$
15. clearance hole for a 8 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$
16. pilot hole for a 16 gauge screw?	6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$

141

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|-----|---|--------------------------------------|
| 1. | 3 penny nail penetrate a 1 ½" green mast? | 1 ¼ — 1 ⁹ / ₃₂ |
| 2. | 8 common gauge nail penetrate a 3 ½" dry pole? | 3 ½ — 3 |
| 3. | 8 penny nail penetrate a 3" green beam? | 2 ½ — 2 ⁹ / ₁₆ |
| 4. | 16 penny nail penetrate a 4" green joist? | 3 ½ — 3 ⁹ / ₁₆ |
| 5. | 16 penny nail penetrate a 1" green timber? | 3 ½ — 2 ⁵ / ₃₂ |
| 6. | 60 penny nail penetrate a 4" dry board? | 6 — 3 ½ |
| 7. | 2 common gauge nail penetrate a 1" green mast? | 6 — 2 ⁵ / ₃₂ |
| 8. | 10 ¼ common gauge nail penetrate a ¾" dry joist? | 2 ½ — ⅝ |
| 9. | 2 common gauge nail penetrate a 2 ½" green frame? | 6 — 2 ¹ / ₁₆ |
| 10. | 40 penny nail penetrate a 2" green rafter? | 5 — 1 ⁹ / ₁₆ |
| 11. | 2 common gauge nail penetrate a ¾" dry beam? | 6 — ⅝ |
| 12. | 11 ½ common gauge nail penetrate a ¾" dry joist? | 2 — ⅝ |
| 13. | 6 common gauge nail penetrate a 2" green girder? | 4 — 1 ⁹ / ₁₆ |
| 14. | 8 box gauge nail penetrate a 1" dry pole? | 5 — ¾ |
| 15. | 9 common gauge nail penetrate a 5/8" dry rafter? | 3 — 9/16 |
| 16. | 30 penny nail penetrate a ¾" green rafter? | 4 ½ — 11/16 |

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|---|--|
| 1. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 2. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 3. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 4. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 5. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 9. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 10. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 11. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 12. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 14. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 15. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 16. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |

142

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|-----|---|-----------------|
| 1. | 3 penny nail penetrate a 1 1/2" green joist? | 1 1/4 — 1 9/32 |
| 2. | 10 1/2 box gauge nail penetrate a 1/2" dry dimension? | 3 — 7/16 |
| 3. | 14 box gauge nail penetrate a 3 1/2" dry plank? | 1 1/2 — 3 |
| 4. | 9 penny nail penetrate a 1 1/2" green plank? | 2 3/4 — 1 9/32 |
| 5. | 2 penny nail penetrate a 3" dry pole? | 1 — 2 1/2 |
| 6. | 10 1/2 box gauge nail penetrate a 3 1/2" green frame? | 3 — 3 1/16 |
| 7. | 12 1/2 box gauge nail penetrate a 3 1/2" green mast? | 2 — 3 1/16 |
| 8. | 5 common gauge nail penetrate a 2" green rafter? | 4 1/2 — 1 9/16 |
| 9. | 12 1/2 box gauge nail penetrate a 1 1/2" dry plank? | 2 — 1 1/4 |
| 10. | 4 common gauge nail penetrate a 3/8" dry plank? | 5 — 5/16 |
| 11. | 9 penny nail penetrate a 2" dry rafter? | 2 3/4 — 1 1/2 |
| 12. | 40 penny nail penetrate a 1 1/4" dry mast? | 5 — 1 |
| 13. | 4 common gauge nail penetrate a 3/8" dry timber? | 5 — 5/16 |
| 14. | 14 box gauge nail penetrate a 3/4" green plank? | 1 1/2 — 1 11/16 |
| 15. | 3 penny nail penetrate a 5/8" dry dimension? | 1 1/4 — 9/16 |
| 16. | 30 penny nail penetrate a 1/2" green plank? | 4 1/2 — 1 5/32 |

1. clearance hole for a 1 gauge screw? 47 $5/64 = 5/64$
2. pilot hole for a 24 gauge screw? O $3/8 - 1/16 = 5/16$
3. clearance hole for a 12 gauge screw? 2 $7/32 = 7/32$
4. countersink hole for a 14 gauge screw? V $1/4 + 1/8 = 3/8$
5. pilot hole for a 9 gauge screw? 35 $11/64 - 1/16 = 7/64$
6. countersink hole for a 18 gauge screw? $27/64 - 19/64 + 1/8 = 27/64$
7. clearance hole for a 24 gauge screw? V $3/8 = 3/8$
8. pilot hole for a 4 gauge screw? 55 $7/64 - 1/16 = 3/64$
9. clearance hole for a 16 gauge screw? H $17/64 = 17/64$
10. pilot hole for a 3 gauge screw? 67 $3/32 - 1/16 = 1/32$
11. countersink hole for a 20 gauge screw? $29/64 - 21/64 + 1/8 = 29/64$
12. clearance hole for a 14 gauge screw? E $1/4 = 1/4$
13. pilot hole for a 7 gauge screw? 41 $5/32 - 1/16 = 3/32$
14. countersink hole for a 12 gauge screw? S $7/32 + 1/8 = 11/32$
15. clearance hole for a 18 gauge screw? N $19/64 = 19/64$
16. countersink hole for a 24 gauge screw? $1/2 - 3/8 + 1/8 = 1/2$

143

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|-----|---|----------------|
| 1. | 12 penny nail penetrate a 1 1/4" green girder? | 3 1/4 — 1 1/32 |
| 2. | 14 box gauge nail penetrate a 4 1/2" green girder? | 1 1/2 — 4 1/16 |
| 3. | 6 common gauge nail penetrate a 3/4" green pole? | 4 — 11/16 |
| 4. | 9 box gauge nail penetrate a 1" dry mast? | 4 — 3/4 |
| 5. | 10 penny nail penetrate a 5/8" green girder? | 3 — 19/32 |
| 6. | 14 1/2 box gauge nail penetrate a 2 1/2" dry board? | 1 1/4 — 2 |
| 7. | 14 common gauge nail penetrate a 2 1/2" dry plank? | 1 1/4 — 2 |
| 8. | 3 common gauge nail penetrate a 1/2" dry joist? | 5 1/2 — 7/16 |
| 9. | 40 penny nail penetrate a 5/8" dry joist? | 5 — 9/16 |
| 10. | 5 common gauge nail penetrate a 2" green rafter? | 4 1/2 — 19/16 |
| 11. | 9 penny nail penetrate a 1 1/4" green dimension? | 2 3/4 — 1 1/32 |
| 12. | 3 penny nail penetrate a 4" green joist? | 1 1/4 — 39/16 |
| 13. | 10 1/2 box gauge nail penetrate a 1" dry joist? | 3 — 3/4 |
| 14. | 14 common gauge nail penetrate a 3" dry board? | 1 1/4 — 2 1/2 |
| 15. | 10 1/2 box gauge nail penetrate a 2" green timber? | 3 — 19/16 |
| 16. | 20 penny nail penetrate a 4 1/2" green mast? | 4 — 4 1/16 |

1. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
2. pilot hole for a 3 gauge screw?	67 $3/32 - 1/16 = 1/32$
3. clearance hole for a 16 gauge screw?	H $17/64 = 17/64$
4. clearance hole for a 9 gauge screw?	17 $11/64 = 11/64$
5. pilot hole for a 20 gauge screw?	H $21/64 - 1/16 = 17/64$
6. clearance hole for a 6 gauge screw?	27 $9/64 = 9/64$
7. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
8. pilot hole for a 10 gauge screw?	30 $3/16 - 1/16 = 1/8$
9. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
10. countersink hole for a 3 gauge screw?	2 $3/32 + 1/8 = 7/32$
11. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$
12. pilot hole for a 11 gauge screw?	27 $13/64 - 1/16 = 9/64$
13. clearance hole for a 16 gauge screw?	H $17/64 = 17/64$
14. clearance hole for a 24 gauge screw?	V $3/8 = 3/8$
15. countersink hole for a 16 gauge screw?	X $17/64 + 1/8 = 25/64$
16. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$

144

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|-----|--|--------------------------------|
| 1. | 7 penny nail penetrate a $\frac{1}{2}$ " dry mast? | $2\frac{1}{4} - \frac{7}{16}$ |
| 2. | 11 $\frac{1}{2}$ common gauge nail penetrate a $1\frac{1}{2}$ " green frame? | $2 - 1\frac{9}{32}$ |
| 3. | 2 penny nail penetrate a $4\frac{1}{2}$ " dry board? | $1 - 4$ |
| 4. | 8 box gauge nail penetrate a $1\frac{1}{2}$ " green pole? | $5 - 1\frac{9}{32}$ |
| 5. | 8 box gauge nail penetrate a $\frac{3}{8}$ " dry rafter? | $5 - \frac{5}{16}$ |
| 6. | 6 common gauge nail penetrate a $\frac{5}{8}$ " green timber? | $4 - 1\frac{9}{32}$ |
| 7. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green beam? | $2\frac{1}{2} - 1\frac{9}{32}$ |
| 8. | 40 penny nail penetrate a $\frac{5}{8}$ " dry timber? | $5 - \frac{9}{16}$ |
| 9. | 8 box gauge nail penetrate a $2\frac{1}{2}$ " dry plank? | $5 - 2$ |
| 10. | 9 penny nail penetrate a $4\frac{1}{2}$ " green frame? | $2\frac{3}{4} - 4\frac{1}{16}$ |
| 11. | 20 penny nail penetrate a $1\frac{1}{4}$ " dry dimension? | $4 - 1$ |
| 12. | 8 common gauge nail penetrate a $2\frac{1}{2}$ " dry beam? | $3\frac{1}{2} - 2$ |
| 13. | 12 $\frac{1}{2}$ common gauge nail penetrate a $2\frac{1}{2}$ " dry board? | $1\frac{1}{2} - 2$ |
| 14. | 9 penny nail penetrate a $\frac{1}{2}$ " green beam? | $2\frac{3}{4} - 1\frac{5}{32}$ |
| 15. | 10 penny nail penetrate a 2" green joist? | $3 - 1\frac{9}{16}$ |
| 16. | 9 box gauge nail penetrate a $1\frac{1}{2}$ " dry board? | $4 - 1\frac{1}{4}$ |

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|--|-----------------------------------|
| 1. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 2. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 3. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 5. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 6. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 7. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 9. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 10. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 11. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 12. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 13. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 14. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 15. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 16. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |

145

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|-----|---|----------------------------------|
| 1. | 6 penny nail penetrate a 4" dry beam? | $2 - 3 \frac{1}{2}$ |
| 2. | 8 penny nail penetrate a $1 \frac{1}{4}$ " green girder? | $2 \frac{1}{2} - 1 \frac{1}{32}$ |
| 3. | $11 \frac{1}{2}$ box gauge nail penetrate a 3" dry girder? | $2 \frac{1}{2} - 2 \frac{1}{2}$ |
| 4. | 4 penny nail penetrate a $\frac{1}{2}$ " dry beam? | $1 \frac{1}{2} - \frac{7}{16}$ |
| 5. | 50 penny nail penetrate a $\frac{3}{4}$ " green rafter? | $5 \frac{1}{2} - 1 \frac{1}{16}$ |
| 6. | 60 penny nail penetrate a $\frac{3}{4}$ " dry girder? | $6 - \frac{5}{8}$ |
| 7. | 8 box gauge nail penetrate a $4 \frac{1}{2}$ " green dimension? | $5 - 4 \frac{1}{16}$ |
| 8. | $15 \frac{1}{2}$ box gauge nail penetrate a 3" dry dimension? | $1 - 2 \frac{1}{2}$ |
| 9. | 8 common gauge nail penetrate a 3" dry board? | $3 \frac{1}{2} - 2 \frac{1}{2}$ |
| 10. | 60 penny nail penetrate a $3 \frac{1}{2}$ " dry dimension? | $6 - 3$ |
| 11. | $11 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " green beam? | $2 \frac{1}{2} - 1 \frac{1}{32}$ |
| 12. | 8 box gauge nail penetrate a $\frac{1}{2}$ " dry plank? | $5 - \frac{7}{16}$ |
| 13. | 2 common gauge nail penetrate a 3" dry timber? | $6 - 2 \frac{1}{2}$ |
| 14. | 6 common gauge nail penetrate a $2 \frac{1}{2}$ " green rafter? | $4 - 2 \frac{1}{16}$ |
| 15. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry plank? | $3 - \frac{5}{8}$ |
| 16. | 8 penny nail penetrate a $3 \frac{1}{2}$ " green frame? | $2 \frac{1}{2} - 3 \frac{1}{16}$ |

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|---|--|
| 1. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 2. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 3. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 6. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 7. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 8. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 9. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 13. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 14. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 15. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |

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|-----|---|--------------|
| 1. | 15 ½ box gauge nail penetrate a ½" green frame? | 1 — 15/32 |
| 2. | 11 ½ common gauge nail penetrate a 2 ½" dry rafter? | 2 — 2 |
| 3. | 9 box gauge nail penetrate a 5/8" dry rafter? | 4 — 9/16 |
| 4. | 2 penny nail penetrate a 3/4" dry dimension? | 1 — 5/8 |
| 5. | 50 penny nail penetrate a 1 ¼" green girder? | 5 ½ — 1 1/32 |
| 6. | 8 penny nail penetrate a ½" green rafter? | 2 ½ — 15/32 |
| 7. | 11 ½ box gauge nail penetrate a 4" green pole? | 2 ½ — 3 9/16 |
| 8. | 8 box gauge nail penetrate a 1 ½" green beam? | 5 — 1 9/32 |
| 9. | 2 penny nail penetrate a 3 ½" green rafter? | 1 — 3 1/16 |
| 10. | 8 box gauge nail penetrate a 3" green beam? | 5 — 2 9/16 |
| 11. | 14 ½ box gauge nail penetrate a 3/8" dry pole? | 1 ¼ — 5/16 |
| 12. | 4 penny nail penetrate a 4 ½" dry girder? | 1 ½ — 4 |
| 13. | 12 ½ common gauge nail penetrate a 4 ½" dry pole? | 1 ½ — 4 |
| 14. | 4 common gauge nail penetrate a 3/4" dry plank? | 5 — 5/8 |
| 15. | 11 ½ common gauge nail penetrate a 3" green joist? | 2 — 2 9/16 |
| 16. | 9 penny nail penetrate a 1 ½" dry girder? | 2 ¾ — 1 ¼ |

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|--|--------------------------|
| 1. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 2. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 3. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 4. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 5. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 7. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 8. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 9. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 10. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 11. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 12. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 13. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 14. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 15. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 16. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |

147

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|-----|--|-------------|
| 1. | 12 ½ common gauge nail penetrate a ½" green joist? | 1 ½ — 15/32 |
| 2. | 6 common gauge nail penetrate a 4 ½" green timber? | 4 — 4 1/16 |
| 3. | 12 penny nail penetrate a 2" green board? | 3 ¼ — 19/16 |
| 4. | 10 ½ box gauge nail penetrate a 3/4" green girder? | 3 — 11/16 |
| 5. | 10 ½ box gauge nail penetrate a 1 ½" dry beam? | 3 — 1 ¼ |
| 6. | 10 ½ box gauge nail penetrate a 1 ¼" dry frame? | 3 — 1 |
| 7. | 8 common gauge nail penetrate a 5/8" green mast? | 3 ½ — 19/32 |
| 8. | 12 penny nail penetrate a 5/8" green mast? | 3 ¼ — 19/32 |
| 9. | 40 penny nail penetrate a 4" green rafter? | 5 — 39/16 |
| 10. | 8 box gauge nail penetrate a 3/4" dry beam? | 5 — 5/8 |
| 11. | 30 penny nail penetrate a ½" dry plank? | 4 ½ — 7/16 |
| 12. | 8 box gauge nail penetrate a 1" green frame? | 5 — 25/32 |
| 13. | 10 ¼ common gauge nail penetrate a 4 ½" dry dimension? | 2 ½ — 4 |
| 14. | 12 penny nail penetrate a 3/8" green timber? | 3 ¼ — 11/32 |
| 15. | 14 box gauge nail penetrate a 2" green girder? | 1 ½ — 19/16 |
| 16. | 2 penny nail penetrate a 1" green pole? | 1 — 25/32 |

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|--|---|
| 1. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 2. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 3. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 4. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 5. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 6. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 7. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 8. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 9. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 10. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 11. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 12. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 14. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 15. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 16. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |

148

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|-----|--|----------------|
| 1. | 20 penny nail penetrate a 1 1/2" green timber? | 4 — 19/32 |
| 2. | 14 box gauge nail penetrate a 1 1/4" dry frame? | 1 1/2 — 1 |
| 3. | 14 common gauge nail penetrate a 4" dry pole? | 1 1/4 — 3 1/2 |
| 4. | 9 box gauge nail penetrate a 2 1/2" dry frame? | 4 — 2 |
| 5. | 14 1/2 box gauge nail penetrate a 3/8" green board? | 1 1/4 — 11/32 |
| 6. | 10 1/2 box gauge nail penetrate a 4 1/2" dry girder? | 3 — 4 |
| 7. | 9 box gauge nail penetrate a 1 1/2" green pole? | 4 — 19/32 |
| 8. | 4 common gauge nail penetrate a 1" green plank? | 5 — 25/32 |
| 9. | 9 box gauge nail penetrate a 3/4" green board? | 4 — 11/16 |
| 10. | 10 1/2 box gauge nail penetrate a 1 1/4" green rafter? | 3 — 11/32 |
| 11. | 12 1/2 common gauge nail penetrate a 1/2" green pole? | 1 1/2 — 15/32 |
| 12. | 11 1/2 box gauge nail penetrate a 1/2" green rafter? | 2 1/2 — 15/32 |
| 13. | 5 common gauge nail penetrate a 3 1/2" green mast? | 4 1/2 — 3 1/16 |
| 14. | 2 common gauge nail penetrate a 1 1/2" green mast? | 6 — 19/32 |
| 15. | 5 penny nail penetrate a 2 1/2" green plank? | 1 3/4 — 2 1/16 |
| 16. | 6 penny nail penetrate a 3/8" green mast? | 2 — 11/32 |

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|--|--------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 3. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 4. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 5. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 6. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 7. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 8. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 9. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 10. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 11. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 12. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 13. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 14. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 15. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 16. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |

149

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|-----|--|--------------------------------|
| 1. | 40 penny nail penetrate a 2" dry plank? | $5 - 1\frac{1}{2}$ |
| 2. | 7 penny nail penetrate a 3" dry rafter? | $2\frac{1}{4} - 2\frac{1}{2}$ |
| 3. | 14 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry rafter? | $1\frac{1}{4} - \frac{5}{8}$ |
| 4. | 3 penny nail penetrate a 2" dry dimension? | $1\frac{1}{4} - 1\frac{1}{2}$ |
| 5. | 16 penny nail penetrate a 2" dry rafter? | $3\frac{1}{2} - 1\frac{1}{2}$ |
| 6. | 3 penny nail penetrate a $3\frac{1}{2}$ " green plank? | $1\frac{1}{4} - 3\frac{1}{16}$ |
| 7. | 9 penny nail penetrate a 3" dry plank? | $2\frac{3}{4} - 2\frac{1}{2}$ |
| 8. | 6 penny nail penetrate a $\frac{3}{8}$ " green joist? | $2 - \frac{11}{32}$ |
| 9. | 5 common gauge nail penetrate a $\frac{1}{2}$ " dry mast? | $4\frac{1}{2} - \frac{7}{16}$ |
| 10. | 12 $\frac{1}{2}$ box gauge nail penetrate a $4\frac{1}{2}$ " green mast? | $2 - 4\frac{1}{16}$ |
| 11. | 5 common gauge nail penetrate a 1" dry dimension? | $4\frac{1}{2} - \frac{3}{4}$ |
| 12. | 9 penny nail penetrate a 4" dry board? | $2\frac{3}{4} - 3\frac{1}{2}$ |
| 13. | 5 common gauge nail penetrate a $4\frac{1}{2}$ " dry timber? | $4\frac{1}{2} - 4$ |
| 14. | 2 penny nail penetrate a $\frac{3}{4}$ " green frame? | $1 - \frac{11}{16}$ |
| 15. | 5 penny nail penetrate a $1\frac{1}{4}$ " dry beam? | $1\frac{3}{4} - 1$ |
| 16. | 50 penny nail penetrate a 3" dry timber? | $5\frac{1}{2} - 2\frac{1}{2}$ |

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|--|---|
| 1. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 2. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 3. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 4. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 5. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 6. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 7. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 8. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 9. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 10. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 12. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 13. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 14. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 15. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 16. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

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|-----|--|--------------------------------|
| 1. | 20 penny nail penetrate a 1" dry beam? | $4 - \frac{3}{4}$ |
| 2. | 2 penny nail penetrate a 3" dry joist? | $1 - 2\frac{1}{2}$ |
| 3. | 9 penny nail penetrate a $1\frac{1}{4}$ " dry joist? | $2\frac{3}{4} - 1$ |
| 4. | 10 penny nail penetrate a 3" dry pole? | $3 - 2\frac{1}{2}$ |
| 5. | 60 penny nail penetrate a $\frac{3}{4}$ " dry rafter? | $6 - \frac{5}{8}$ |
| 6. | 14 common gauge nail penetrate a 2" dry plank? | $1\frac{1}{4} - 1\frac{1}{2}$ |
| 7. | 3 common gauge nail penetrate a 1" green timber? | $5\frac{1}{2} - \frac{25}{32}$ |
| 8. | 60 penny nail penetrate a $\frac{5}{8}$ " dry mast? | $6 - \frac{9}{16}$ |
| 9. | 9 penny nail penetrate a 1" dry beam? | $2\frac{3}{4} - \frac{3}{4}$ |
| 10. | 6 penny nail penetrate a 3" green rafter? | $2 - 2\frac{9}{16}$ |
| 11. | $10\frac{1}{2}$ box gauge nail penetrate a 2" dry joist? | $3 - 1\frac{1}{2}$ |
| 12. | 40 penny nail penetrate a $1\frac{1}{2}$ " dry rafter? | $5 - 1\frac{1}{4}$ |
| 13. | $15\frac{1}{2}$ box gauge nail penetrate a $2\frac{1}{2}$ " green joist? | $1 - 2\frac{1}{16}$ |
| 14. | 60 penny nail penetrate a $1\frac{1}{4}$ " green board? | $6 - 1\frac{1}{32}$ |
| 15. | 14 common gauge nail penetrate a $1\frac{1}{2}$ " dry joist? | $1\frac{1}{4} - 1\frac{1}{4}$ |
| 16. | 8 box gauge nail penetrate a 4" green timber? | $5 - 3\frac{9}{16}$ |

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|---|---|
| 1. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 2. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 3. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 4. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 5. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 6. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 7. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 8. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 10. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 11. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 12. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 14. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 15. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 16. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |

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|-----|---|---------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 2 ½" green rafter? | 3 — 2 ¹ / ₁₆ |
| 2. | 10 penny nail penetrate a 4 ½" green timber? | 3 — 4 ¹ / ₁₆ |
| 3. | 9 penny nail penetrate a 4 ½" green frame? | 2 ¾ — 4 ¹ / ₁₆ |
| 4. | 9 common gauge nail penetrate a 5/8" dry board? | 3 — 9 ¹ / ₁₆ |
| 5. | 20 penny nail penetrate a 4" green plank? | 4 — 3 ⁹ / ₁₆ |
| 6. | 12 ½ box gauge nail penetrate a 3/4" green rafter? | 2 — 11 ¹ / ₁₆ |
| 7. | 8 box gauge nail penetrate a 1 ¼" dry pole? | 5 — 1 |
| 8. | 10 ¼ common gauge nail penetrate a 1" green timber? | 2 ½ — 25 ¹ / ₃₂ |
| 9. | 9 common gauge nail penetrate a 3/8" dry joist? | 3 — 5 ¹ / ₁₆ |
| 10. | 4 common gauge nail penetrate a 4 ½" green beam? | 5 — 4 ¹ / ₁₆ |
| 11. | 40 penny nail penetrate a 3/8" dry timber? | 5 — 5 ¹ / ₁₆ |
| 12. | 9 penny nail penetrate a 4 ½" green beam? | 2 ¾ — 4 ¹ / ₁₆ |
| 13. | 10 ¼ common gauge nail penetrate a 3 ½" dry frame? | 2 ½ — 3 |
| 14. | 3 penny nail penetrate a 1" dry board? | 1 ¼ — ¾ |
| 15. | 14 common gauge nail penetrate a 2" green beam? | 1 ¼ — 1 ⁹ / ₁₆ |
| 16. | 10 ¼ common gauge nail penetrate a 1 ½" green mast? | 2 ½ — 1 ⁹ / ₃₂ |

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|---|-------------------------------|
| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 3. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 4. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 5. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 6. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 7. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 8. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 9. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 10. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 11. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 12. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 13. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 14. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 15. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 16. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |

152

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|-----|--|-----------------------------------|
| 1. | 30 penny nail penetrate a 4" green plank? | $4 \frac{1}{2} - 3 \frac{9}{16}$ |
| 2. | 8 box gauge nail penetrate a $\frac{3}{4}$ " green plank? | $5 - 1 \frac{11}{16}$ |
| 3. | 3 penny nail penetrate a $3 \frac{1}{2}$ " green beam? | $1 \frac{1}{4} - 3 \frac{1}{16}$ |
| 4. | 7 penny nail penetrate a $1 \frac{1}{4}$ " green timber? | $2 \frac{1}{4} - 1 \frac{1}{32}$ |
| 5. | 6 penny nail penetrate a $2 \frac{1}{2}$ " green beam? | $2 - 2 \frac{1}{16}$ |
| 6. | 5 penny nail penetrate a 4" dry plank? | $1 \frac{3}{4} - 3 \frac{1}{2}$ |
| 7. | $10 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{2}$ " dry plank? | $3 - 1 \frac{1}{4}$ |
| 8. | 60 penny nail penetrate a $3 \frac{1}{2}$ " dry plank? | $6 - 3$ |
| 9. | 5 penny nail penetrate a $1 \frac{1}{4}$ " green rafter? | $1 \frac{3}{4} - 1 \frac{1}{32}$ |
| 10. | 8 box gauge nail penetrate a $\frac{1}{2}$ " green beam? | $5 - 1 \frac{15}{32}$ |
| 11. | 6 penny nail penetrate a 2" green mast? | $2 - 1 \frac{9}{16}$ |
| 12. | 60 penny nail penetrate a $\frac{3}{8}$ " green mast? | $6 - 1 \frac{11}{32}$ |
| 13. | 10 penny nail penetrate a $4 \frac{1}{2}$ " green girder? | $3 - 4 \frac{1}{16}$ |
| 14. | 16 penny nail penetrate a $1 \frac{1}{2}$ " green frame? | $3 \frac{1}{2} - 1 \frac{9}{32}$ |
| 15. | 60 penny nail penetrate a 4" dry frame? | $6 - 3 \frac{1}{2}$ |
| 16. | 14 common gauge nail penetrate a $\frac{1}{2}$ " green joist? | $1 \frac{1}{4} - 1 \frac{15}{32}$ |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 3. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 4. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 5. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 6. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 7. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 8. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 9. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 10. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 11. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 12. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 13. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 14. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 15. countersink hole for a 24 gauge screw? | $\frac{1}{2} \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 16. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |

153

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|-----|---|-----------------------------------|
| 1. | 3 penny nail penetrate a $3/8$ " green mast? | $1 \frac{1}{4} - 1 \frac{11}{32}$ |
| 2. | 2 penny nail penetrate a $1 \frac{1}{4}$ " dry board? | $1 - 1$ |
| 3. | 2 common gauge nail penetrate a $5/8$ " green pole? | $6 - 1 \frac{19}{32}$ |
| 4. | 9 penny nail penetrate a $3 \frac{1}{2}$ " dry mast? | $2 \frac{3}{4} - 3$ |
| 5. | 4 common gauge nail penetrate a $3/8$ " green timber? | $5 - 1 \frac{11}{32}$ |
| 6. | 2 penny nail penetrate a $1 \frac{1}{2}$ " green timber? | $1 - 1 \frac{9}{32}$ |
| 7. | $10 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{2}$ " dry girder? | $3 - 1 \frac{1}{4}$ |
| 8. | $10 \frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " dry pole? | $3 - 2$ |
| 9. | 6 penny nail penetrate a 4" dry pole? | $2 - 3 \frac{1}{2}$ |
| 10. | 15 common gauge nail penetrate a 3" dry girder? | $1 - 2 \frac{1}{2}$ |
| 11. | 6 common gauge nail penetrate a $2 \frac{1}{2}$ " dry frame? | $4 - 2$ |
| 12. | 50 penny nail penetrate a $1 \frac{1}{2}$ " green mast? | $5 \frac{1}{2} - 1 \frac{9}{32}$ |
| 13. | 20 penny nail penetrate a $4 \frac{1}{2}$ " green dimension? | $4 - 4 \frac{1}{16}$ |
| 14. | 50 penny nail penetrate a $1 \frac{1}{2}$ " green rafter? | $5 \frac{1}{2} - 1 \frac{9}{32}$ |
| 15. | 4 common gauge nail penetrate a $3 \frac{1}{2}$ " dry joist? | $5 - 3$ |
| 16. | $10 \frac{1}{2}$ box gauge nail penetrate a $1/2$ " green pole? | $3 - 1 \frac{15}{32}$ |

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|--|---|
| 1. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 2. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 3. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 4. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 5. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 6. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 7. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 8. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 9. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 10. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 11. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 12. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 13. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 14. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 15. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 16. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |

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|-----|--|---------------------------------|
| 1. | 7 penny nail penetrate a $\frac{3}{8}$ " green timber? | $2\frac{1}{4} - 1\frac{11}{32}$ |
| 2. | 2 penny nail penetrate a 1" dry pole? | $1 - \frac{3}{4}$ |
| 3. | 4 penny nail penetrate a 3" dry pole? | $1\frac{1}{2} - 2\frac{1}{2}$ |
| 4. | 15 $\frac{1}{2}$ box gauge nail penetrate a 2" green mast? | $1 - 1\frac{9}{16}$ |
| 5. | 10 penny nail penetrate a 4" green dimension? | $3 - 3\frac{9}{16}$ |
| 6. | 8 common gauge nail penetrate a $\frac{3}{8}$ " green frame? | $3\frac{1}{2} - 1\frac{11}{32}$ |
| 7. | 11 $\frac{1}{2}$ common gauge nail penetrate a $2\frac{1}{2}$ " green plank? | $2 - 2\frac{1}{16}$ |
| 8. | 6 common gauge nail penetrate a 1" green board? | $4 - 2\frac{5}{32}$ |
| 9. | 3 penny nail penetrate a $2\frac{1}{2}$ " dry pole? | $1\frac{1}{4} - 2$ |
| 10. | 8 common gauge nail penetrate a $\frac{1}{2}$ " dry plank? | $3\frac{1}{2} - \frac{7}{16}$ |
| 11. | 10 penny nail penetrate a 2" green mast? | $3 - 1\frac{9}{16}$ |
| 12. | 6 common gauge nail penetrate a 4" dry board? | $4 - 3\frac{1}{2}$ |
| 13. | 50 penny nail penetrate a 2" dry timber? | $5\frac{1}{2} - 1\frac{1}{2}$ |
| 14. | 15 $\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " green frame? | $1 - 3\frac{1}{16}$ |
| 15. | 40 penny nail penetrate a $\frac{3}{8}$ " green board? | $5 - 1\frac{11}{32}$ |
| 16. | 2 common gauge nail penetrate a $1\frac{1}{4}$ " dry plank? | $6 - 1$ |

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|---|--|
| 1. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 2. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 3. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 4. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 5. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 6. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 7. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 8. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 9. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 10. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 12. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 13. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 14. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 15. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 16. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |

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|-----|--|----------------------------------|
| 1. | 12 penny nail penetrate a $5/8$ " green plank? | $3 \frac{1}{4} - \frac{19}{32}$ |
| 2. | 50 penny nail penetrate a $2 \frac{1}{2}$ " green board? | $5 \frac{1}{2} - 2 \frac{1}{16}$ |
| 3. | 5 penny nail penetrate a 4" dry frame? | $1 \frac{3}{4} - 3 \frac{1}{2}$ |
| 4. | 8 penny nail penetrate a 4" dry mast? | $2 \frac{1}{2} - 3 \frac{1}{2}$ |
| 5. | 40 penny nail penetrate a 1" dry mast? | $5 - \frac{3}{4}$ |
| 6. | 8 box gauge nail penetrate a $2 \frac{1}{2}$ " dry rafter? | $5 - 2$ |
| 7. | 5 penny nail penetrate a $3/8$ " dry rafter? | $1 \frac{3}{4} - \frac{5}{16}$ |
| 8. | $10 \frac{1}{2}$ box gauge nail penetrate a $3/4$ " dry plank? | $3 - \frac{5}{8}$ |
| 9. | 30 penny nail penetrate a $5/8$ " green girder? | $4 \frac{1}{2} - \frac{19}{32}$ |
| 10. | 14 box gauge nail penetrate a 1" green rafter? | $1 \frac{1}{2} - \frac{25}{32}$ |
| 11. | 40 penny nail penetrate a $2 \frac{1}{2}$ " dry joist? | $5 - 2$ |
| 12. | 60 penny nail penetrate a $3 \frac{1}{2}$ " dry frame? | $6 - 3$ |
| 13. | 5 penny nail penetrate a $5/8$ " dry pole? | $1 \frac{3}{4} - \frac{9}{16}$ |
| 14. | 9 penny nail penetrate a $3/4$ " green timber? | $2 \frac{3}{4} - \frac{11}{16}$ |
| 15. | 10 penny nail penetrate a 3" green rafter? | $3 - 2 \frac{9}{16}$ |
| 16. | 20 penny nail penetrate a $2 \frac{1}{2}$ " green pole? | $4 - 2 \frac{1}{16}$ |

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|---|--------------------------|
| 1. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 2. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 3. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 5. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 6. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 7. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 8. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 9. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 10. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 11. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 12. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 13. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 14. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 15. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 16. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |

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|-----|---|-----------------------------------|
| 1. | 2 penny nail penetrate a $\frac{3}{4}$ " dry board? | $1 - \frac{5}{8}$ |
| 2. | 10 penny nail penetrate a 1" green pole? | $3 - \frac{25}{32}$ |
| 3. | 5 penny nail penetrate a 3" green mast? | $1 \frac{3}{4} - 2 \frac{9}{16}$ |
| 4. | 11 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{2}$ " green beam? | $2 \frac{1}{2} - 1 \frac{9}{32}$ |
| 5. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{5}{8}$ " green joist? | $2 \frac{1}{2} - 1 \frac{19}{32}$ |
| 6. | 9 box gauge nail penetrate a 1 $\frac{1}{4}$ " dry mast? | $4 - 1$ |
| 7. | 2 penny nail penetrate a 2" green board? | $1 - 1 \frac{9}{16}$ |
| 8. | 3 penny nail penetrate a 3 $\frac{1}{2}$ " dry frame? | $1 \frac{1}{4} - 3$ |
| 9. | 2 common gauge nail penetrate a 2" dry rafter? | $6 - 1 \frac{1}{2}$ |
| 10. | 12 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{4}$ " dry frame? | $2 - 1$ |
| 11. | 9 common gauge nail penetrate a $\frac{5}{8}$ " green plank? | $3 - 1 \frac{19}{32}$ |
| 12. | 50 penny nail penetrate a $\frac{3}{4}$ " dry dimension? | $5 \frac{1}{2} - \frac{5}{8}$ |
| 13. | 8 penny nail penetrate a 4" dry pole? | $2 \frac{1}{2} - 3 \frac{1}{2}$ |
| 14. | 8 penny nail penetrate a 2 $\frac{1}{2}$ " green plank? | $2 \frac{1}{2} - 2 \frac{1}{16}$ |
| 15. | 12 penny nail penetrate a 3 $\frac{1}{2}$ " dry dimension? | $3 \frac{1}{4} - 3$ |
| 16. | 11 $\frac{1}{2}$ box gauge nail penetrate a 4" dry rafter? | $2 \frac{1}{2} - 3 \frac{1}{2}$ |

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|---|---|
| 1. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 3. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 4. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 5. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 6. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 7. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 8. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 9. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 10. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 11. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 12. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 13. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 14. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 15. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 16. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |

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|-----|---|----------------------------------|
| 1. | 8 penny nail penetrate a 1" green timber? | $2 \frac{1}{2} - 2\frac{25}{32}$ |
| 2. | 10 penny nail penetrate a $\frac{3}{8}$ " dry joist? | $3 - \frac{5}{16}$ |
| 3. | $10 \frac{1}{4}$ common gauge nail penetrate a 2" dry beam? | $2 \frac{1}{2} - 1 \frac{1}{2}$ |
| 4. | 6 common gauge nail penetrate a 3" dry beam? | $4 - 2 \frac{1}{2}$ |
| 5. | $14 \frac{1}{2}$ box gauge nail penetrate a 2" dry girder? | $1 \frac{1}{4} - 1 \frac{1}{2}$ |
| 6. | 4 penny nail penetrate a 4" dry plank? | $1 \frac{1}{2} - 3 \frac{1}{2}$ |
| 7. | 50 penny nail penetrate a $1 \frac{1}{2}$ " green mast? | $5 \frac{1}{2} - 1\frac{9}{32}$ |
| 8. | $10 \frac{1}{2}$ box gauge nail penetrate a 4" green timber? | $3 - 3\frac{9}{16}$ |
| 9. | $11 \frac{1}{2}$ common gauge nail penetrate a 2" green pole? | $2 - 1\frac{9}{16}$ |
| 10. | $11 \frac{1}{2}$ common gauge nail penetrate a $1 \frac{1}{2}$ " green joist? | $2 - 1\frac{9}{32}$ |
| 11. | $10 \frac{1}{4}$ common gauge nail penetrate a 2" green beam? | $2 \frac{1}{2} - 1\frac{9}{16}$ |
| 12. | 5 penny nail penetrate a 3" dry board? | $1 \frac{3}{4} - 2 \frac{1}{2}$ |
| 13. | $11 \frac{1}{2}$ common gauge nail penetrate a $2 \frac{1}{2}$ " green frame? | $2 - 2\frac{1}{16}$ |
| 14. | $12 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{2}$ " green timber? | $2 - 1\frac{9}{32}$ |
| 15. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green dimension? | $1 \frac{1}{2} - 1\frac{11}{16}$ |
| 16. | 60 penny nail penetrate a $2 \frac{1}{2}$ " green beam? | $6 - 2\frac{1}{16}$ |

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|--|---|
| 1. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 4. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 5. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 6. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 7. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 8. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 9. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 10. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 11. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 12. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 13. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 14. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 15. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |

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|-----|--|-------------|
| 1. | 12 ½ common gauge nail penetrate a 1" dry mast? | 1 ½ — ¾ |
| 2. | 40 penny nail penetrate a 3 ½" dry timber? | 5 — 3 |
| 3. | 40 penny nail penetrate a ½" dry joist? | 5 — 7/16 |
| 4. | 2 penny nail penetrate a 1 ¼" dry rafter? | 1 — 1 |
| 5. | 4 penny nail penetrate a 2" green joist? | 1 ½ — 19/16 |
| 6. | 10 penny nail penetrate a 1 ½" dry timber? | 3 — 1 ¼ |
| 7. | 4 common gauge nail penetrate a 2" green frame? | 5 — 19/16 |
| 8. | 8 common gauge nail penetrate a 3" dry frame? | 3 ½ — 2 ½ |
| 9. | 6 common gauge nail penetrate a 3" green girder? | 4 — 29/16 |
| 10. | 16 penny nail penetrate a 4 ½" green dimension? | 3 ½ — 41/16 |
| 11. | 9 penny nail penetrate a 2 ½" green board? | 2 ¾ — 21/16 |
| 12. | 60 penny nail penetrate a 5/8" dry beam? | 6 — 9/16 |
| 13. | 50 penny nail penetrate a 3" dry plank? | 5 ½ — 2 ½ |
| 14. | 20 penny nail penetrate a ½" green rafter? | 4 — 15/32 |
| 15. | 9 box gauge nail penetrate a 1 ½" green timber? | 4 — 19/32 |
| 16. | 7 penny nail penetrate a 3/8" dry mast? | 2 ¼ — 5/16 |

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|--|---|
| 1. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 2. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 3. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 4. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 5. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 6. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 8. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 9. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 10. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 11. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 12. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 13. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 14. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 15. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 16. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |

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|-----|---|-----------------------------------|
| 1. | 4 penny nail penetrate a 3/8" green plank? | $1 \frac{1}{2} - 1 \frac{11}{32}$ |
| 2. | 30 penny nail penetrate a 3/8" dry plank? | $4 \frac{1}{2} - \frac{5}{16}$ |
| 3. | 14 common gauge nail penetrate a 3" green girder? | $1 \frac{1}{4} - 2 \frac{9}{16}$ |
| 4. | 14 box gauge nail penetrate a 2" dry joist? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 5. | 14 common gauge nail penetrate a 1 1/4" dry timber? | $1 \frac{1}{4} - 1$ |
| 6. | 20 penny nail penetrate a 3" dry dimension? | $4 - 2 \frac{1}{2}$ |
| 7. | 2 penny nail penetrate a 2 1/2" dry timber? | $1 - 2$ |
| 8. | 15 common gauge nail penetrate a 4" green girder? | $1 - 3 \frac{9}{16}$ |
| 9. | 9 box gauge nail penetrate a 3" green mast? | $4 - 2 \frac{9}{16}$ |
| 10. | 12 penny nail penetrate a 2 1/2" dry joist? | $3 \frac{1}{4} - 2$ |
| 11. | 8 box gauge nail penetrate a 1 1/2" green mast? | $5 - 1 \frac{9}{32}$ |
| 12. | 2 penny nail penetrate a 1" green rafter? | $1 - 2 \frac{5}{32}$ |
| 13. | 12 penny nail penetrate a 2 1/2" dry frame? | $3 \frac{1}{4} - 2$ |
| 14. | 9 common gauge nail penetrate a 1 1/2" dry pole? | $3 - 1 \frac{1}{4}$ |
| 15. | 10 penny nail penetrate a 3" green plank? | $3 - 2 \frac{9}{16}$ |
| 16. | 12 1/2 box gauge nail penetrate a 3/4" green joist? | $2 - 1 \frac{1}{16}$ |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 3. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 4. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 5. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 6. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 7. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 8. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 9. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 10. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 11. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 14. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 15. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 16. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |

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|-----|--|--------------|
| 1. | 12 ½ common gauge nail penetrate a 3 ½" dry frame? | 1 ½ — 3 |
| 2. | 12 ½ common gauge nail penetrate a 1 ¼" green plank? | 1 ½ — 1 ⅓ |
| 3. | 50 penny nail penetrate a 1 ½" dry mast? | 5 ½ — 1 ¼ |
| 4. | 10 ¼ common gauge nail penetrate a 1 ½" green frame? | 2 ½ — 1 ⅞ |
| 5. | 20 penny nail penetrate a 1" green girder? | 4 — 25/32 |
| 6. | 8 penny nail penetrate a 3" green frame? | 2 ½ — 2 9/16 |
| 7. | 60 penny nail penetrate a 2 ½" green frame? | 6 — 2 1/16 |
| 8. | 11 ½ common gauge nail penetrate a 2" dry dimension? | 2 — 1 ½ |
| 9. | 5 penny nail penetrate a ½" dry rafter? | 1 ¾ — 7/16 |
| 10. | 50 penny nail penetrate a 1 ¼" green board? | 5 ½ — 1 ⅓ |
| 11. | 10 ½ box gauge nail penetrate a 3/8" green plank? | 3 — 11/32 |
| 12. | 9 box gauge nail penetrate a 1 ½" dry rafter? | 4 — 1 ¼ |
| 13. | 12 ½ box gauge nail penetrate a 3/8" dry rafter? | 2 — 5/16 |
| 14. | 7 penny nail penetrate a 4 ½" dry mast? | 2 ¼ — 4 |
| 15. | 2 penny nail penetrate a 2 ½" dry joist? | 1 — 2 |
| 16. | 9 common gauge nail penetrate a 1 ¼" dry joist? | 3 — 1 |

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|--|--------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 3. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 4. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 5. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 6. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 7. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 8. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 9. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 10. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 11. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 12. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 13. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 14. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 15. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 16. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |

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|-----|--|----------------------------------|
| 1. | 12 penny nail penetrate a 1" green dimension? | $3 \frac{1}{4} - \frac{25}{32}$ |
| 2. | 14 common gauge nail penetrate a $3 \frac{1}{2}$ " green dimension? | $1 \frac{1}{4} - 3 \frac{1}{16}$ |
| 3. | 14 box gauge nail penetrate a $\frac{3}{8}$ " dry dimension? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 4. | 3 penny nail penetrate a $1 \frac{1}{2}$ " green timber? | $1 \frac{1}{4} - 1 \frac{9}{32}$ |
| 5. | 20 penny nail penetrate a 4" dry dimension? | $4 - 3 \frac{1}{2}$ |
| 6. | $10 \frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry board? | $2 \frac{1}{2} - \frac{5}{16}$ |
| 7. | $11 \frac{1}{2}$ box gauge nail penetrate a $4 \frac{1}{2}$ " green mast? | $2 \frac{1}{2} - 4 \frac{1}{16}$ |
| 8. | 30 penny nail penetrate a $1 \frac{1}{4}$ " dry timber? | $4 \frac{1}{2} - 1$ |
| 9. | 12 penny nail penetrate a $\frac{3}{4}$ " dry joist? | $3 \frac{1}{4} - \frac{5}{8}$ |
| 10. | 8 box gauge nail penetrate a $2 \frac{1}{2}$ " green mast? | $5 - 2 \frac{1}{16}$ |
| 11. | 40 penny nail penetrate a 4" green pole? | $5 - 3 \frac{9}{16}$ |
| 12. | 8 box gauge nail penetrate a 3" green board? | $5 - 2 \frac{9}{16}$ |
| 13. | $11 \frac{1}{2}$ box gauge nail penetrate a 3" dry girder? | $2 \frac{1}{2} - 2 \frac{1}{2}$ |
| 14. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " dry timber? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 15. | 20 penny nail penetrate a $1 \frac{1}{4}$ " dry girder? | $4 - 1$ |
| 16. | $12 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green beam? | $2 - 1 \frac{1}{16}$ |

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|---|--------------------------|
| 1. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 2. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 3. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 5. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 6. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 7. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 8. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 9. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 10. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 11. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 12. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 13. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 14. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 15. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 16. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |

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|-----|---|---------------------------------|
| 1. | 14 box gauge nail penetrate a 1" green rafter? | $1 \frac{1}{2} - \frac{25}{32}$ |
| 2. | 11 $\frac{1}{2}$ box gauge nail penetrate a 4 $\frac{1}{2}$ " dry timber? | $2 \frac{1}{2} - 4$ |
| 3. | 11 $\frac{1}{2}$ box gauge nail penetrate a 5/8" green board? | $2 \frac{1}{2} - \frac{19}{32}$ |
| 4. | 40 penny nail penetrate a 3" dry rafter? | $5 - 2 \frac{1}{2}$ |
| 5. | 3 penny nail penetrate a 2 $\frac{1}{2}$ " dry beam? | $1 \frac{1}{4} - 2$ |
| 6. | 40 penny nail penetrate a 3" dry pole? | $5 - 2 \frac{1}{2}$ |
| 7. | 10 $\frac{1}{2}$ box gauge nail penetrate a 3/8" green rafter? | $3 - \frac{11}{32}$ |
| 8. | 12 $\frac{1}{2}$ common gauge nail penetrate a 3/8" dry timber? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 9. | 5 penny nail penetrate a 3/4" green joist? | $1 \frac{3}{4} - \frac{11}{16}$ |
| 10. | 9 box gauge nail penetrate a 2" dry mast? | $4 - 1 \frac{1}{2}$ |
| 11. | 4 common gauge nail penetrate a 2 $\frac{1}{2}$ " dry dimension? | $5 - 2$ |
| 12. | 11 $\frac{1}{2}$ common gauge nail penetrate a 2 $\frac{1}{2}$ " dry dimension? | $2 - 2$ |
| 13. | 9 penny nail penetrate a 1" dry board? | $2 \frac{3}{4} - \frac{3}{4}$ |
| 14. | 12 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{2}$ " green board? | $2 - \frac{19}{32}$ |
| 15. | 14 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry dimension? | $1 \frac{1}{4} - \frac{7}{16}$ |
| 16. | 6 common gauge nail penetrate a 3/4" green dimension? | $4 - \frac{11}{16}$ |

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|---|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 3. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 4. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 5. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 6. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 7. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 8. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 9. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 10. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 11. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 12. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 13. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 14. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 15. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 16. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |

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|-----|--|--|
| 1. | 2 penny nail penetrate a 3 1/2" green mast? | 1 — 3 ¹ / ₁₆ |
| 2. | 7 penny nail penetrate a 1" green plank? | 2 1/4 — 2 ⁵ / ₃₂ |
| 3. | 2 common gauge nail penetrate a 1" dry rafter? | 6 — 3/4 |
| 4. | 15 common gauge nail penetrate a 1" green beam? | 1 — 2 ⁵ / ₃₂ |
| 5. | 11 1/2 common gauge nail penetrate a 2" dry joist? | 2 — 1 1/2 |
| 6. | 10 penny nail penetrate a 1" dry joist? | 3 — 3/4 |
| 7. | 10 1/2 box gauge nail penetrate a 1 1/2" green mast? | 3 — 1 ⁹ / ₃₂ |
| 8. | 50 penny nail penetrate a 3/4" green plank? | 5 1/2 — 1 ¹ / ₁₆ |
| 9. | 8 box gauge nail penetrate a 3" dry beam? | 5 — 2 1/2 |
| 10. | 30 penny nail penetrate a 1" dry timber? | 4 1/2 — 3/4 |
| 11. | 7 penny nail penetrate a 1/2" dry joist? | 2 1/4 — 7/ ₁₆ |
| 12. | 16 penny nail penetrate a 5/8" dry joist? | 3 1/2 — 9/ ₁₆ |
| 13. | 60 penny nail penetrate a 2" green timber? | 6 — 1 ⁹ / ₁₆ |
| 14. | 8 penny nail penetrate a 3" dry timber? | 2 1/2 — 2 1/2 |
| 15. | 9 box gauge nail penetrate a 5/8" green plank? | 4 — 1 ⁹ / ₃₂ |
| 16. | 3 penny nail penetrate a 4 1/2" dry plank? | 1 1/4 — 4 |

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|--|--------------------------|
| 1. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 2. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 3. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 4. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 5. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 6. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 8. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 9. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 10. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 11. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 12. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 13. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 14. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 15. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 16. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |

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|-----|---|----------------------------------|
| 1. | 8 penny nail penetrate a 1" green pole? | $2 \frac{1}{2} - 2\frac{25}{32}$ |
| 2. | 5 penny nail penetrate a $2 \frac{1}{2}$ " green mast? | $1 \frac{3}{4} - 2\frac{1}{16}$ |
| 3. | 7 penny nail penetrate a $1 \frac{1}{2}$ " green board? | $2 \frac{1}{4} - 1\frac{9}{32}$ |
| 4. | $10 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry dimension? | $3 - 1$ |
| 5. | 14 box gauge nail penetrate a $\frac{3}{8}$ " green mast? | $1 \frac{1}{2} - 1\frac{11}{32}$ |
| 6. | $12 \frac{1}{2}$ common gauge nail penetrate a 3" dry plank? | $1 \frac{1}{2} - 2 \frac{1}{2}$ |
| 7. | 8 box gauge nail penetrate a $1 \frac{1}{2}$ " green joist? | $5 - 1\frac{9}{32}$ |
| 8. | 2 penny nail penetrate a $4 \frac{1}{2}$ " green beam? | $1 - 4\frac{1}{16}$ |
| 9. | $12 \frac{1}{2}$ box gauge nail penetrate a 4" green frame? | $2 - 3\frac{9}{16}$ |
| 10. | 30 penny nail penetrate a $\frac{3}{8}$ " green joist? | $4 \frac{1}{2} - 1\frac{11}{32}$ |
| 11. | 9 box gauge nail penetrate a 1" dry rafter? | $4 - \frac{3}{4}$ |
| 12. | 12 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $3 \frac{1}{4} - \frac{9}{16}$ |
| 13. | 4 penny nail penetrate a $\frac{3}{4}$ " dry rafter? | $1 \frac{1}{2} - \frac{5}{8}$ |
| 14. | 60 penny nail penetrate a $4 \frac{1}{2}$ " green plank? | $6 - 4\frac{1}{16}$ |
| 15. | $12 \frac{1}{2}$ common gauge nail penetrate a $3 \frac{1}{2}$ " green plank? | $1 \frac{1}{2} - 3\frac{1}{16}$ |
| 16. | 5 penny nail penetrate a $1 \frac{1}{4}$ " dry beam? | $1 \frac{3}{4} - 1$ |

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|--|--|
| 1. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 2. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 3. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 4. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 5. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 6. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 7. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 8. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 9. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 14. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 15. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |

165

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|-----|---|-------------------|
| 1. | 15 common gauge nail penetrate a $5/8$ " green timber? | $1 - 19/32$ |
| 2. | 30 penny nail penetrate a $3\ 1/2$ " dry board? | $4\ 1/2 - 3$ |
| 3. | 14 common gauge nail penetrate a 4" green dimension? | $1\ 1/4 - 39/16$ |
| 4. | 10 penny nail penetrate a $5/8$ " dry pole? | $3 - 9/16$ |
| 5. | 7 penny nail penetrate a $3/8$ " green board? | $2\ 1/4 - 11/32$ |
| 6. | 5 penny nail penetrate a 3" green rafter? | $1\ 3/4 - 29/16$ |
| 7. | $12\ 1/2$ common gauge nail penetrate a $1\ 1/4$ " green plank? | $1\ 1/2 - 11/32$ |
| 8. | 40 penny nail penetrate a $4\ 1/2$ " dry dimension? | $5 - 4$ |
| 9. | 2 common gauge nail penetrate a $1/2$ " dry dimension? | $6 - 7/16$ |
| 10. | 9 penny nail penetrate a 4" dry pole? | $2\ 3/4 - 3\ 1/2$ |
| 11. | 8 penny nail penetrate a 4" dry beam? | $2\ 1/2 - 3\ 1/2$ |
| 12. | 8 penny nail penetrate a 2" green dimension? | $2\ 1/2 - 19/16$ |
| 13. | 9 box gauge nail penetrate a 3" green rafter? | $4 - 29/16$ |
| 14. | $12\ 1/2$ box gauge nail penetrate a $3\ 1/2$ " green pole? | $2 - 31/16$ |
| 15. | 40 penny nail penetrate a $1\ 1/4$ " dry joist? | $5 - 1$ |
| 16. | 8 common gauge nail penetrate a $1\ 1/4$ " dry joist? | $3\ 1/2 - 1$ |

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|--|--|
| 1. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 2. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 3. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 4. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 5. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 8. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 9. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 11. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 12. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 13. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 14. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 15. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 16. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |

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|-----|---|----------------|
| 1. | 9 box gauge nail penetrate a 4 1/2" dry plank? | 4 — 4 |
| 2. | 14 1/2 box gauge nail penetrate a 1" green rafter? | 1 1/4 — 25/32 |
| 3. | 10 1/4 common gauge nail penetrate a 3/8" green girder? | 2 1/2 — 11/32 |
| 4. | 11 1/2 box gauge nail penetrate a 1/2" green joist? | 2 1/2 — 15/32 |
| 5. | 8 penny nail penetrate a 3/4" dry girder? | 2 1/2 — 5/8 |
| 6. | 5 penny nail penetrate a 4 1/2" green beam? | 1 3/4 — 4 1/16 |
| 7. | 3 penny nail penetrate a 4" green rafter? | 1 1/4 — 3 9/16 |
| 8. | 15 1/2 box gauge nail penetrate a 3/4" green joist? | 1 — 11/16 |
| 9. | 2 penny nail penetrate a 1" dry girder? | 1 — 3/4 |
| 10. | 6 penny nail penetrate a 1 1/2" dry frame? | 2 — 1 1/4 |
| 11. | 16 penny nail penetrate a 3/4" dry plank? | 3 1/2 — 5/8 |
| 12. | 14 common gauge nail penetrate a 4" dry pole? | 1 1/4 — 3 1/2 |
| 13. | 10 penny nail penetrate a 2 1/2" green beam? | 3 — 2 1/16 |
| 14. | 50 penny nail penetrate a 1 1/4" green mast? | 5 1/2 — 1 1/32 |
| 15. | 14 box gauge nail penetrate a 1 1/2" dry beam? | 1 1/2 — 1 1/4 |
| 16. | 7 penny nail penetrate a 5/8" green board? | 2 1/4 — 19/32 |

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|--|---|
| 1. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 2. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 3. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 4. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 5. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 6. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 7. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 8. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 9. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 10. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 11. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 14. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 15. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 16. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |

167

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|-----|---|--------------|
| 1. | 2 penny nail penetrate a 1 ½" dry joist? | 1 — 1 ¼ |
| 2. | 14 box gauge nail penetrate a 2" dry timber? | 1 ½ — 1 ½ |
| 3. | 4 common gauge nail penetrate a 4 ½" dry girder? | 5 — 4 |
| 4. | 6 penny nail penetrate a 2 ½" dry pole? | 2 — 2 |
| 5. | 30 penny nail penetrate a 1 ½" green pole? | 4 ½ — 1 9/32 |
| 6. | 5 common gauge nail penetrate a ½" dry girder? | 4 ½ — 7/16 |
| 7. | 4 penny nail penetrate a ½" dry joist? | 1 ½ — 7/16 |
| 8. | 3 common gauge nail penetrate a 3 ½" dry joist? | 5 ½ — 3 |
| 9. | 2 common gauge nail penetrate a 5/8" dry dimension? | 6 — 9/16 |
| 10. | 3 penny nail penetrate a 5/8" green board? | 1 ¼ — 19/32 |
| 11. | 11 ½ box gauge nail penetrate a 1 ½" green pole? | 2 ½ — 1 9/32 |
| 12. | 4 penny nail penetrate a 3/4" dry mast? | 1 ½ — 5/8 |
| 13. | 14 ½ box gauge nail penetrate a 1 ½" green rafter? | 1 ¼ — 1 9/32 |
| 14. | 30 penny nail penetrate a 3/4" green frame? | 4 ½ — 11/16 |
| 15. | 3 penny nail penetrate a 3 ½" dry beam? | 1 ¼ — 3 |
| 16. | 16 penny nail penetrate a 4" green pole? | 3 ½ — 3 9/16 |

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|--|-------------------------------|
| 1. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 2. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 3. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |
| 4. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 5. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 6. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 8. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 9. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 10. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 11. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 12. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 13. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 14. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 15. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 16. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |

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|-----|--|----------------------------------|
| 1. | 9 penny nail penetrate a $\frac{3}{4}$ " green frame? | $2 \frac{3}{4} - \frac{11}{16}$ |
| 2. | 12 penny nail penetrate a $\frac{3}{4}$ " green frame? | $3 \frac{1}{4} - \frac{11}{16}$ |
| 3. | 9 box gauge nail penetrate a $\frac{3}{8}$ " green timber? | $4 - \frac{11}{32}$ |
| 4. | 11 $\frac{1}{2}$ box gauge nail penetrate a $4 \frac{1}{2}$ " green plank? | $2 \frac{1}{2} - 4 \frac{1}{16}$ |
| 5. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " green mast? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 6. | 8 box gauge nail penetrate a $4 \frac{1}{2}$ " green rafter? | $5 - 4 \frac{1}{16}$ |
| 7. | 8 box gauge nail penetrate a $\frac{5}{8}$ " green rafter? | $5 - \frac{19}{32}$ |
| 8. | 14 box gauge nail penetrate a $2 \frac{1}{2}$ " dry beam? | $1 \frac{1}{2} - 2$ |
| 9. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry timber? | $2 - \frac{7}{16}$ |
| 10. | 40 penny nail penetrate a 3" dry board? | $5 - 2 \frac{1}{2}$ |
| 11. | 50 penny nail penetrate a $3 \frac{1}{2}$ " dry beam? | $5 \frac{1}{2} - 3$ |
| 12. | 20 penny nail penetrate a 2" green girder? | $4 - \frac{19}{16}$ |
| 13. | 5 penny nail penetrate a $\frac{3}{8}$ " green mast? | $1 \frac{3}{4} - \frac{11}{32}$ |
| 14. | 40 penny nail penetrate a $1 \frac{1}{2}$ " green pole? | $5 - \frac{19}{32}$ |
| 15. | 16 penny nail penetrate a $2 \frac{1}{2}$ " dry plank? | $3 \frac{1}{2} - 2$ |
| 16. | 8 box gauge nail penetrate a 2" green timber? | $5 - \frac{19}{16}$ |

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|--|---|
| 1. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 2. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 3. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 4. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 5. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 6. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 7. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 8. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 9. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 10. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 11. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 14. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 15. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 16. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |

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|-----|---|----------------|
| 1. | 7 penny nail penetrate a 3 1/2" green frame? | 2 1/4 — 3 1/16 |
| 2. | 16 penny nail penetrate a 3/8" dry beam? | 3 1/2 — 5/16 |
| 3. | 10 1/4 common gauge nail penetrate a 1/2" green mast? | 2 1/2 — 15/32 |
| 4. | 40 penny nail penetrate a 4" dry frame? | 5 — 3 1/2 |
| 5. | 20 penny nail penetrate a 2" green rafter? | 4 — 1 9/16 |
| 6. | 5 penny nail penetrate a 3/8" green rafter? | 1 3/4 — 11/32 |
| 7. | 8 common gauge nail penetrate a 1 1/4" dry rafter? | 3 1/2 — 1 |
| 8. | 20 penny nail penetrate a 3/4" dry mast? | 4 — 5/8 |
| 9. | 3 common gauge nail penetrate a 1 1/2" dry girder? | 5 1/2 — 1 1/4 |
| 10. | 12 penny nail penetrate a 3" dry timber? | 3 1/4 — 2 1/2 |
| 11. | 10 1/4 common gauge nail penetrate a 4" dry timber? | 2 1/2 — 3 1/2 |
| 12. | 9 common gauge nail penetrate a 4 1/2" green board? | 3 — 4 1/16 |
| 13. | 11 1/2 box gauge nail penetrate a 5/8" dry girder? | 2 1/2 — 9/16 |
| 14. | 8 penny nail penetrate a 4 1/2" green rafter? | 2 1/2 — 4 1/16 |
| 15. | 8 common gauge nail penetrate a 3/4" green girder? | 3 1/2 — 11/16 |
| 16. | 16 penny nail penetrate a 1 1/4" green rafter? | 3 1/2 — 1 1/32 |

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|---|---|
| 1. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 5. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 6. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 7. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 10. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 11. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 14. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 15. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 16. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |

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|-----|---|--------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 3" green joist? | 3 — 2 ⁹ / ₁₆ |
| 2. | 8 box gauge nail penetrate a ¾" green beam? | 5 — 1 ¹ / ₁₆ |
| 3. | 12 penny nail penetrate a 3 ½" dry mast? | 3 ¼ — 3 |
| 4. | 6 common gauge nail penetrate a 2" dry rafter? | 4 — 1 ½ |
| 5. | 11 ½ common gauge nail penetrate a 3/8" green pole? | 2 — 1 ¹ / ₃₂ |
| 6. | 14 common gauge nail penetrate a 3" green mast? | 1 ¼ — 2 ⁹ / ₁₆ |
| 7. | 15 ½ box gauge nail penetrate a 2 ½" green girder? | 1 — 2 ¹ / ₁₆ |
| 8. | 10 ½ box gauge nail penetrate a 2 ½" green board? | 3 — 2 ¹ / ₁₆ |
| 9. | 8 box gauge nail penetrate a 2 ½" dry joist? | 5 — 2 |
| 10. | 14 ½ box gauge nail penetrate a 1 ¼" dry frame? | 1 ¼ — 1 |
| 11. | 9 box gauge nail penetrate a 3" green mast? | 4 — 2 ⁹ / ₁₆ |
| 12. | 20 penny nail penetrate a 5/8" green plank? | 4 — 1 ⁹ / ₃₂ |
| 13. | 40 penny nail penetrate a 4 ½" green plank? | 5 — 4 ¹ / ₁₆ |
| 14. | 4 common gauge nail penetrate a 1 ¼" dry mast? | 5 — 1 |
| 15. | 6 penny nail penetrate a 1" green rafter? | 2 — 2 ⁵ / ₃₂ |
| 16. | 9 common gauge nail penetrate a 4 ½" dry plank? | 3 — 4 |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 3. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 4. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 5. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 6. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 7. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 8. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 9. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 10. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 11. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 14. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 15. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 16. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

171

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|-----|--|----------------|
| 1. | 3 penny nail penetrate a 4 1/2" green mast? | 1 1/4 — 4 1/16 |
| 2. | 3 penny nail penetrate a 2 1/2" green joist? | 1 1/4 — 2 1/16 |
| 3. | 15 1/2 box gauge nail penetrate a 1/2" green girder? | 1 — 15/32 |
| 4. | 60 penny nail penetrate a 4" green girder? | 6 — 3 9/16 |
| 5. | 11 1/2 common gauge nail penetrate a 1 1/2" dry rafter? | 2 — 1 1/4 |
| 6. | 50 penny nail penetrate a 3 1/2" dry timber? | 5 1/2 — 3 |
| 7. | 10 1/2 box gauge nail penetrate a 1" dry frame? | 3 — 3/4 |
| 8. | 3 penny nail penetrate a 3 1/2" dry girder? | 1 1/4 — 3 |
| 9. | 40 penny nail penetrate a 3/4" dry beam? | 5 — 5/8 |
| 10. | 60 penny nail penetrate a 1/2" dry frame? | 6 — 7/16 |
| 11. | 12 1/2 common gauge nail penetrate a 2 1/2" green dimension? | 1 1/2 — 2 1/16 |
| 12. | 20 penny nail penetrate a 3/4" green beam? | 4 — 11/16 |
| 13. | 30 penny nail penetrate a 3/4" dry board? | 4 1/2 — 5/8 |
| 14. | 7 penny nail penetrate a 5/8" dry timber? | 2 1/4 — 9/16 |
| 15. | 10 1/4 common gauge nail penetrate a 3/4" dry pole? | 2 1/2 — 5/8 |
| 16. | 3 common gauge nail penetrate a 3" green rafter? | 5 1/2 — 2 9/16 |

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|--|-------------------------------|
| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 4. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 5. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 6. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 8. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 9. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 10. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 11. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 12. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 13. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 14. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 15. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 16. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |

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|-----|--|-----------------------------------|
| 1. | 10 $\frac{1}{4}$ common gauge nail penetrate a 3 $\frac{1}{2}$ " dry mast? | 2 $\frac{1}{2}$ — 3 |
| 2. | 14 $\frac{1}{2}$ box gauge nail penetrate a 1" green girder? | 1 $\frac{1}{4}$ — $\frac{25}{32}$ |
| 3. | 5 penny nail penetrate a 4 $\frac{1}{2}$ " dry plank? | 1 $\frac{3}{4}$ — 4 |
| 4. | 10 $\frac{1}{2}$ box gauge nail penetrate a 3" green board? | 3 — $2\frac{9}{16}$ |
| 5. | 5 penny nail penetrate a 1 $\frac{1}{4}$ " dry beam? | 1 $\frac{3}{4}$ — 1 |
| 6. | 40 penny nail penetrate a $\frac{1}{2}$ " dry mast? | 5 — $\frac{7}{16}$ |
| 7. | 4 common gauge nail penetrate a 4 $\frac{1}{2}$ " green girder? | 5 — $4\frac{1}{16}$ |
| 8. | 10 penny nail penetrate a 3 $\frac{1}{2}$ " green timber? | 3 — $3\frac{1}{16}$ |
| 9. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry frame? | 2 — $\frac{5}{8}$ |
| 10. | 11 $\frac{1}{2}$ box gauge nail penetrate a 1" dry dimension? | 2 $\frac{1}{2}$ — $\frac{3}{4}$ |
| 11. | 12 penny nail penetrate a 1 $\frac{1}{2}$ " dry board? | 3 $\frac{1}{4}$ — 1 $\frac{1}{4}$ |
| 12. | 9 common gauge nail penetrate a 1 $\frac{1}{4}$ " dry beam? | 3 — 1 |
| 13. | 3 penny nail penetrate a $\frac{3}{4}$ " dry rafter? | 1 $\frac{1}{4}$ — $\frac{5}{8}$ |
| 14. | 3 common gauge nail penetrate a 4 $\frac{1}{2}$ " dry beam? | 5 $\frac{1}{2}$ — 4 |
| 15. | 6 common gauge nail penetrate a 1" green beam? | 4 — $\frac{25}{32}$ |
| 16. | 9 common gauge nail penetrate a $\frac{3}{8}$ " green girder? | 3 — $\frac{11}{32}$ |

1. pilot hole for a 20 gauge screw?	H $2\frac{1}{64} - \frac{1}{16} = \frac{17}{64}$
2. pilot hole for a 10 gauge screw?	30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$
3. pilot hole for a 11 gauge screw?	27 $1\frac{3}{64} - \frac{1}{16} = \frac{9}{64}$
4. countersink hole for a 1 gauge screw?	6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$
5. clearance hole for a 5 gauge screw?	30 $\frac{1}{8} = \frac{1}{8}$
6. clearance hole for a 3 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
7. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
8. countersink hole for a 3 gauge screw?	2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$
9. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
10. countersink hole for a 12 gauge screw?	S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$
11. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
12. pilot hole for a 3 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
13. countersink hole for a 10 gauge screw?	O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$
14. countersink hole for a 9 gauge screw?	N $1\frac{1}{64} + \frac{1}{8} = \frac{19}{64}$
15. pilot hole for a 6 gauge screw?	47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$
16. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$

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|-----|---|----------------------------------|
| 1. | 7 penny nail penetrate a 1" green beam? | $2 \frac{1}{4} - \frac{25}{32}$ |
| 2. | 60 penny nail penetrate a $\frac{3}{4}$ " green joist? | $6 - \frac{11}{16}$ |
| 3. | 2 common gauge nail penetrate a $2 \frac{1}{2}$ " green joist? | $6 - 2 \frac{1}{16}$ |
| 4. | 14 common gauge nail penetrate a $1 \frac{1}{4}$ " dry rafter? | $1 \frac{1}{4} - 1$ |
| 5. | 6 penny nail penetrate a 4" green frame? | $2 - 3 \frac{9}{16}$ |
| 6. | 8 box gauge nail penetrate a $1 \frac{1}{2}$ " green rafter? | $5 - 1 \frac{9}{32}$ |
| 7. | 3 penny nail penetrate a 4" dry mast? | $1 \frac{1}{4} - 3 \frac{1}{2}$ |
| 8. | 9 box gauge nail penetrate a 4" green beam? | $4 - 3 \frac{9}{16}$ |
| 9. | $10 \frac{1}{4}$ common gauge nail penetrate a $2 \frac{1}{2}$ " green board? | $2 \frac{1}{2} - 2 \frac{1}{16}$ |
| 10. | $12 \frac{1}{2}$ common gauge nail penetrate a 4" dry joist? | $1 \frac{1}{2} - 3 \frac{1}{2}$ |
| 11. | 3 common gauge nail penetrate a $4 \frac{1}{2}$ " dry frame? | $5 \frac{1}{2} - 4$ |
| 12. | 16 penny nail penetrate a $\frac{3}{4}$ " green plank? | $3 \frac{1}{2} - \frac{11}{16}$ |
| 13. | 6 penny nail penetrate a $2 \frac{1}{2}$ " green joist? | $2 - 2 \frac{1}{16}$ |
| 14. | 6 common gauge nail penetrate a $1 \frac{1}{2}$ " green timber? | $4 - 1 \frac{9}{32}$ |
| 15. | 10 penny nail penetrate a 1" green pole? | $3 - \frac{25}{32}$ |
| 16. | 10 penny nail penetrate a 4" dry plank? | $3 - 3 \frac{1}{2}$ |

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|--|-------------------------------|
| 1. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 2. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 3. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 4. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 5. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 7. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 8. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 9. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 10. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 11. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 12. countersink hole for a 18 gauge screw? | $27/64 - 19/64 + 1/8 = 27/64$ |
| 13. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 14. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 15. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 16. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |

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|-----|--|---------------------------------------|
| 1. | 12 ½ box gauge nail penetrate a 1" dry plank? | 2 — ¾ |
| 2. | 6 common gauge nail penetrate a 3" green board? | 4 — 2 ⁹ / ₁₆ |
| 3. | 9 common gauge nail penetrate a 1 ½" green joist? | 3 — 1 ⁹ / ₃₂ |
| 4. | 20 penny nail penetrate a 1 ¼" green plank? | 4 — 1 ¹ / ₃₂ |
| 5. | 7 penny nail penetrate a 2 ½" dry plank? | 2 ¼ — 2 |
| 6. | 8 box gauge nail penetrate a 2" green board? | 5 — 1 ⁹ / ₁₆ |
| 7. | 2 penny nail penetrate a 1" green rafter? | 1 — 2 ⁵ / ₃₂ |
| 8. | 9 penny nail penetrate a 4" green rafter? | 2 ¾ — 3 ⁹ / ₁₆ |
| 9. | 10 ½ box gauge nail penetrate a 4 ½" green board? | 3 — 4 ¹ / ₁₆ |
| 10. | 30 penny nail penetrate a 3/8" green frame? | 4 ½ — 1 ¹¹ / ₃₂ |
| 11. | 11 ½ box gauge nail penetrate a ½" dry rafter? | 2 ½ — 7 ¹ / ₁₆ |
| 12. | 11 ½ common gauge nail penetrate a 4" dry mast? | 2 — 3 ½ |
| 13. | 9 penny nail penetrate a 3" dry mast? | 2 ¾ — 2 ½ |
| 14. | 14 common gauge nail penetrate a 4 ½" green board? | 1 ¼ — 4 ¹ / ₁₆ |
| 15. | 20 penny nail penetrate a 4" green frame? | 4 — 3 ⁹ / ₁₆ |
| 16. | 5 penny nail penetrate a 2 ½" dry mast? | 1 ¾ — 2 |

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|--|---|
| 1. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 2. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 3. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 4. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 5. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 10. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 11. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 12. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 13. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 14. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 15. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 16. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |

175

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|-----|---|---------------------------------------|
| 1. | 12 ½ box gauge nail penetrate a 3 ½" green dimension? | 2 — 3 ¹ / ₁₆ |
| 2. | 9 common gauge nail penetrate a 3/8" dry plank? | 3 — 5 ¹ / ₁₆ |
| 3. | 60 penny nail penetrate a 3/4" green rafter? | 6 — 11 ¹ / ₁₆ |
| 4. | 60 penny nail penetrate a 1 ½" green joist? | 6 — 19 ¹ / ₃₂ |
| 5. | 7 penny nail penetrate a 2 ½" green board? | 2 ¼ — 2 ¹ / ₁₆ |
| 6. | 9 common gauge nail penetrate a 1 ¼" dry girder? | 3 — 1 |
| 7. | 30 penny nail penetrate a 2" green pole? | 4 ½ — 19 ¹ / ₁₆ |
| 8. | 10 penny nail penetrate a ½" dry beam? | 3 — 7 ¹ / ₁₆ |
| 9. | 14 box gauge nail penetrate a 3/8" dry mast? | 1 ½ — 5 ¹ / ₁₆ |
| 10. | 10 penny nail penetrate a 1 ½" dry girder? | 3 — 1 ¼ |
| 11. | 50 penny nail penetrate a 4" green girder? | 5 ½ — 39 ¹ / ₁₆ |
| 12. | 8 common gauge nail penetrate a 1 ½" dry plank? | 3 ½ — 1 ¼ |
| 13. | 7 penny nail penetrate a 5/8" green beam? | 2 ¼ — 19 ¹ / ₃₂ |
| 14. | 20 penny nail penetrate a 4 ½" green plank? | 4 — 4 ¹ / ₁₆ |
| 15. | 40 penny nail penetrate a 1 ½" dry timber? | 5 — 1 ¼ |
| 16. | 50 penny nail penetrate a 5/8" dry mast? | 5 ½ — 9 ¹ / ₁₆ |

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|--|---|
| 1. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 2. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 8. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 10. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 11. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 12. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 14. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 15. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 16. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |

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|-----|---|--------------|
| 1. | 10 ½ box gauge nail penetrate a 3" dry dimension? | 3 — 2 ½ |
| 2. | 8 box gauge nail penetrate a 5/8" green frame? | 5 — 19/32 |
| 3. | 4 penny nail penetrate a 5/8" green joist? | 1 ½ — 19/32 |
| 4. | 14 box gauge nail penetrate a 2" dry girder? | 1 ½ — 1 ½ |
| 5. | 12 penny nail penetrate a 4 ½" dry pole? | 3 ¼ — 4 |
| 6. | 40 penny nail penetrate a 5/8" dry girder? | 5 — 9/16 |
| 7. | 14 ½ box gauge nail penetrate a 1" green plank? | 1 ¼ — 25/32 |
| 8. | 4 penny nail penetrate a 2 ½" green girder? | 1 ½ — 2 1/16 |
| 9. | 5 penny nail penetrate a 1 ¼" dry pole? | 1 ¾ — 1 |
| 10. | 12 penny nail penetrate a 5/8" green rafter? | 3 ¼ — 19/32 |
| 11. | 16 penny nail penetrate a ½" dry plank? | 3 ½ — 7/16 |
| 12. | 8 box gauge nail penetrate a 1 ½" dry plank? | 5 — 1 ¼ |
| 13. | 12 penny nail penetrate a 3/4" dry pole? | 3 ¼ — 5/8 |
| 14. | 12 ½ common gauge nail penetrate a 3/4" green beam? | 1 ½ — 11/16 |
| 15. | 8 box gauge nail penetrate a 5/8" green board? | 5 — 19/32 |
| 16. | 10 ¼ common gauge nail penetrate a 2" dry girder? | 2 ½ — 1 ½ |

1. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
2. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$
3. pilot hole for a 9 gauge screw?	35 $11/64 - 1/16 = 7/64$
4. countersink hole for a 11 gauge screw?	Q $13/64 + 1/8 = 21/64$
5. clearance hole for a 10 gauge screw?	12 $3/16 = 3/16$
6. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$
7. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
8. clearance hole for a 24 gauge screw?	V $3/8 = 3/8$
9. countersink hole for a 5 gauge screw?	E $1/8 + 1/8 = 1/4$
10. clearance hole for a 11 gauge screw?	6 $13/64 = 13/64$
11. pilot hole for a 6 gauge screw?	47 $9/64 - 1/16 = 5/64$
12. countersink hole for a 18 gauge screw?	$27/64$ $19/64 + 1/8 = 27/64$
13. clearance hole for a 11 gauge screw?	6 $13/64 = 13/64$
14. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$
15. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
16. clearance hole for a 11 gauge screw?	6 $13/64 = 13/64$

177

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|-----|---|----------------|
| 1. | 10 penny nail penetrate a 1 1/4" green mast? | 3 — 1 1/32 |
| 2. | 30 penny nail penetrate a 1" dry frame? | 4 1/2 — 3/4 |
| 3. | 9 penny nail penetrate a 2 1/2" green girder? | 2 3/4 — 2 1/16 |
| 4. | 4 penny nail penetrate a 3" green dimension? | 1 1/2 — 2 9/16 |
| 5. | 8 box gauge nail penetrate a 2" dry mast? | 5 — 1 1/2 |
| 6. | 5 penny nail penetrate a 2 1/2" dry girder? | 1 3/4 — 2 |
| 7. | 12 1/2 box gauge nail penetrate a 3/8" dry beam? | 2 — 5/16 |
| 8. | 8 common gauge nail penetrate a 4 1/2" dry timber? | 3 1/2 — 4 |
| 9. | 12 penny nail penetrate a 1" green girder? | 3 1/4 — 25/32 |
| 10. | 11 1/2 common gauge nail penetrate a 4 1/2" green mast? | 2 — 4 1/16 |
| 11. | 9 common gauge nail penetrate a 5/8" green plank? | 3 — 19/32 |
| 12. | 3 common gauge nail penetrate a 3 1/2" dry girder? | 5 1/2 — 3 |
| 13. | 8 penny nail penetrate a 3/4" green board? | 2 1/2 — 11/16 |
| 14. | 11 1/2 box gauge nail penetrate a 1" green pole? | 2 1/2 — 25/32 |
| 15. | 15 1/2 box gauge nail penetrate a 4 1/2" dry rafter? | 1 — 4 |
| 16. | 6 common gauge nail penetrate a 5/8" dry plank? | 4 — 9/16 |

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|---|---|
| 1. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 2. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 3. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 4. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 5. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 6. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 7. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 8. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 9. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 10. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 11. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 12. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 13. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 14. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |

178

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|-----|---|--------------|
| 1. | 2 penny nail penetrate a 1 ½" dry pole? | 1 — 1 ¼ |
| 2. | 3 penny nail penetrate a 3" dry pole? | 1 ¼ — 2 ½ |
| 3. | 15 ½ box gauge nail penetrate a 4 ½" green joist? | 1 — 4 ⅓ |
| 4. | 4 penny nail penetrate a 3/8" dry mast? | 1 ½ — 5/16 |
| 5. | 2 common gauge nail penetrate a 4 ½" green girder? | 6 — 4 ⅓ |
| 6. | 16 penny nail penetrate a 1 ¼" green joist? | 3 ½ — 1 ⅓ |
| 7. | 11 ½ common gauge nail penetrate a ½" dry plank? | 2 — 7/16 |
| 8. | 50 penny nail penetrate a 3/4" dry joist? | 5 ½ — 5/8 |
| 9. | 12 ½ common gauge nail penetrate a 1 ¼" dry girder? | 1 ½ — 1 |
| 10. | 20 penny nail penetrate a 3" green frame? | 4 — 2 9/16 |
| 11. | 9 common gauge nail penetrate a 1" green mast? | 3 — 25/32 |
| 12. | 9 penny nail penetrate a 1 ¼" dry board? | 2 ¾ — 1 |
| 13. | 60 penny nail penetrate a 2 ½" dry beam? | 6 — 2 |
| 14. | 8 penny nail penetrate a 2 ½" green pole? | 2 ½ — 2 ⅓ |
| 15. | 8 box gauge nail penetrate a 1 ¼" green plank? | 5 — 1 ⅓ |
| 16. | 7 penny nail penetrate a 2" green pole? | 2 ¼ — 1 9/16 |

1. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
2. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
3. pilot hole for a 1 gauge screw?	78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$
4. pilot hole for a 8 gauge screw?	35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$
5. pilot hole for a 7 gauge screw?	41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$
6. clearance hole for a 24 gauge screw?	V $\frac{3}{8} = \frac{3}{8}$
7. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
8. countersink hole for a 18 gauge screw?	$\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$
9. pilot hole for a 11 gauge screw?	27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$
10. pilot hole for a 3 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
11. pilot hole for a 8 gauge screw?	35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$
12. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
13. clearance hole for a 6 gauge screw?	27 $\frac{9}{64} = \frac{9}{64}$
14. pilot hole for a 7 gauge screw?	41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$
15. clearance hole for a 5 gauge screw?	30 $\frac{1}{8} = \frac{1}{8}$
16. pilot hole for a 12 gauge screw?	22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$

179

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|-----|---|---------------------------------|
| 1. | 8 penny nail penetrate a 1" dry mast? | $2 \frac{1}{2} - \frac{3}{4}$ |
| 2. | 10 $\frac{1}{2}$ box gauge nail penetrate a 2" green rafter? | $3 - 1\frac{9}{16}$ |
| 3. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " green plank? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 4. | 8 penny nail penetrate a 4 $\frac{1}{2}$ " green frame? | $2 \frac{1}{2} - 4\frac{1}{16}$ |
| 5. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " dry joist? | $1 \frac{1}{2} - \frac{5}{8}$ |
| 6. | 15 common gauge nail penetrate a 2" green rafter? | $1 - 1\frac{9}{16}$ |
| 7. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry mast? | $2 - \frac{9}{16}$ |
| 8. | 14 box gauge nail penetrate a $\frac{1}{2}$ " green dimension? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 9. | 20 penny nail penetrate a 4 $\frac{1}{2}$ " dry frame? | $4 - 4$ |
| 10. | 11 $\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " dry rafter? | $2 - \frac{5}{16}$ |
| 11. | 2 penny nail penetrate a 3 $\frac{1}{2}$ " green board? | $1 - 3\frac{1}{16}$ |
| 12. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1 $\frac{1}{4}$ " dry pole? | $2 - 1$ |
| 13. | 14 box gauge nail penetrate a $\frac{5}{8}$ " dry girder? | $1 \frac{1}{2} - \frac{9}{16}$ |
| 14. | 60 penny nail penetrate a 1 $\frac{1}{2}$ " green rafter? | $6 - 1\frac{9}{32}$ |
| 15. | 40 penny nail penetrate a $\frac{3}{8}$ " green beam? | $5 - \frac{11}{32}$ |
| 16. | 14 common gauge nail penetrate a 3" green dimension? | $1 \frac{1}{4} - 2\frac{9}{16}$ |

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|---|---|
| 1. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 3. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 4. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 5. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 6. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 7. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 8. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 9. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 10. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 13. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 14. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 15. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 16. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |

180

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|-----|--|---------------------------------|
| 1. | 10 penny nail penetrate a 2" green timber? | $3 - 1\frac{9}{16}$ |
| 2. | 8 box gauge nail penetrate a $\frac{3}{8}$ " dry timber? | $5 - \frac{5}{16}$ |
| 3. | 3 common gauge nail penetrate a $3\frac{1}{2}$ " dry board? | $5\frac{1}{2} - 3$ |
| 4. | 20 penny nail penetrate a $\frac{3}{4}$ " dry board? | $4 - \frac{5}{8}$ |
| 5. | 8 penny nail penetrate a $\frac{3}{8}$ " green rafter? | $2\frac{1}{2} - 1\frac{11}{32}$ |
| 6. | 9 penny nail penetrate a $4\frac{1}{2}$ " green board? | $2\frac{3}{4} - 4\frac{1}{16}$ |
| 7. | $11\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{4}$ " dry timber? | $2\frac{1}{2} - 1$ |
| 8. | $11\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{2}$ " green mast? | $2\frac{1}{2} - 1\frac{9}{32}$ |
| 9. | 16 penny nail penetrate a $\frac{3}{4}$ " dry plank? | $3\frac{1}{2} - \frac{5}{8}$ |
| 10. | $12\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " green dimension? | $2 - 3\frac{1}{16}$ |
| 11. | 6 common gauge nail penetrate a $4\frac{1}{2}$ " green pole? | $4 - 4\frac{1}{16}$ |
| 12. | 10 penny nail penetrate a 2" green timber? | $3 - 1\frac{9}{16}$ |
| 13. | $10\frac{1}{2}$ box gauge nail penetrate a 2" green pole? | $3 - 1\frac{9}{16}$ |
| 14. | $11\frac{1}{2}$ common gauge nail penetrate a $4\frac{1}{2}$ " green board? | $2 - 4\frac{1}{16}$ |
| 15. | 7 penny nail penetrate a $2\frac{1}{2}$ " dry frame? | $2\frac{1}{4} - 2$ |
| 16. | 30 penny nail penetrate a $1\frac{1}{2}$ " dry girder? | $4\frac{1}{2} - 1\frac{1}{4}$ |

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|--|---|
| 1. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 2. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 3. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 6. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 7. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 8. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 9. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 10. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 11. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 12. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 13. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 14. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 15. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 16. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |

181

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|-----|---|---------------------------------|
| 1. | 8 penny nail penetrate a $3/4$ " green dimension? | $2 \frac{1}{2} - \frac{11}{16}$ |
| 2. | 4 penny nail penetrate a $\frac{1}{2}$ " green mast? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 3. | 2 penny nail penetrate a $4 \frac{1}{2}$ " green mast? | $1 - 4\frac{1}{16}$ |
| 4. | 8 box gauge nail penetrate a 3" dry mast? | $5 - 2 \frac{1}{2}$ |
| 5. | 10 penny nail penetrate a 3" dry plank? | $3 - 2 \frac{1}{2}$ |
| 6. | 2 common gauge nail penetrate a $3 \frac{1}{2}$ " dry frame? | $6 - 3$ |
| 7. | 3 penny nail penetrate a 3" green beam? | $1 \frac{1}{4} - 2\frac{9}{16}$ |
| 8. | 9 box gauge nail penetrate a 4" dry beam? | $4 - 3 \frac{1}{2}$ |
| 9. | 10 penny nail penetrate a 2" dry pole? | $3 - 1 \frac{1}{2}$ |
| 10. | 6 penny nail penetrate a $2 \frac{1}{2}$ " dry frame? | $2 - 2$ |
| 11. | 50 penny nail penetrate a $5/8$ " dry girder? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 12. | 4 common gauge nail penetrate a $5/8$ " green timber? | $5 - \frac{19}{32}$ |
| 13. | 2 penny nail penetrate a $3 \frac{1}{2}$ " green rafter? | $1 - 3\frac{1}{16}$ |
| 14. | $12 \frac{1}{2}$ common gauge nail penetrate a $2 \frac{1}{2}$ " dry dimension? | $1 \frac{1}{2} - 2$ |
| 15. | $15 \frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " green girder? | $1 - 2\frac{1}{16}$ |
| 16. | 4 penny nail penetrate a 3" dry frame? | $1 \frac{1}{2} - 2 \frac{1}{2}$ |

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|--|---|
| 1. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 2. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 3. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 4. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 5. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 6. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 7. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 8. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 9. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 11. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 14. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 15. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 16. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |

182

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|-----|---|---------------|
| 1. | 8 penny nail penetrate a 1 1/4" dry timber? | 2 1/2 — 1 |
| 2. | 4 penny nail penetrate a 1/2" dry joist? | 1 1/2 — 7/16 |
| 3. | 11 1/2 box gauge nail penetrate a 2" green mast? | 2 1/2 — 19/16 |
| 4. | 60 penny nail penetrate a 3 1/2" green rafter? | 6 — 31/16 |
| 5. | 9 penny nail penetrate a 1" green frame? | 2 3/4 — 25/32 |
| 6. | 12 penny nail penetrate a 2" green girder? | 3 1/4 — 19/16 |
| 7. | 10 penny nail penetrate a 4" green dimension? | 3 — 39/16 |
| 8. | 12 penny nail penetrate a 3/8" green mast? | 3 1/4 — 11/32 |
| 9. | 16 penny nail penetrate a 4" green girder? | 3 1/2 — 39/16 |
| 10. | 9 penny nail penetrate a 1/2" dry frame? | 2 3/4 — 7/16 |
| 11. | 12 penny nail penetrate a 4 1/2" green dimension? | 3 1/4 — 41/16 |
| 12. | 16 penny nail penetrate a 2 1/2" dry dimension? | 3 1/2 — 2 |
| 13. | 11 1/2 common gauge nail penetrate a 5/8" green girder? | 2 — 19/32 |
| 14. | 11 1/2 box gauge nail penetrate a 2" dry pole? | 2 1/2 — 1 1/2 |
| 15. | 9 common gauge nail penetrate a 1 1/4" green board? | 3 — 11/32 |
| 16. | 11 1/2 common gauge nail penetrate a 1 1/4" dry girder? | 2 — 1 |

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| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 3. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 4. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 5. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 6. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 8. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 9. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 10. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 11. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 12. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 13. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 14. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 15. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 16. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |

183

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|-----|--|-----------|
| 1. | 10 ½ box gauge nail penetrate a 3" dry pole? | 3 — 2 ½ |
| 2. | 4 penny nail penetrate a 1 ½" green rafter? | 1 ½ — 1 ⅞ |
| 3. | 9 box gauge nail penetrate a 5/8" green girder? | 4 — 1 ⅞ |
| 4. | 5 penny nail penetrate a 4" dry rafter? | 1 ¾ — 3 ½ |
| 5. | 16 penny nail penetrate a 2 ½" green joist? | 3 ½ — 2 ⅛ |
| 6. | 5 penny nail penetrate a 3/4" dry mast? | 1 ¾ — ⅝ |
| 7. | 3 penny nail penetrate a 1 ¼" green plank? | 1 ¼ — 1 ⅓ |
| 8. | 20 penny nail penetrate a 2 ½" green girder? | 4 — 2 ⅛ |
| 9. | 12 ½ common gauge nail penetrate a 5/8" green joist? | 1 ½ — 1 ⅞ |
| 10. | 5 penny nail penetrate a 1 ½" green joist? | 1 ¾ — 1 ⅞ |
| 11. | 7 penny nail penetrate a ½" green plank? | 2 ¼ — 1 ⅝ |
| 12. | 3 penny nail penetrate a 4" green dimension? | 1 ¼ — 3 ⅞ |
| 13. | 8 box gauge nail penetrate a 1 ¼" dry plank? | 5 — 1 |
| 14. | 6 penny nail penetrate a 1" green board? | 2 — 2 ⅝ |
| 15. | 8 penny nail penetrate a 2" green girder? | 2 ½ — 1 ⅞ |
| 16. | 2 penny nail penetrate a 2 ½" dry joist? | 1 — 2 |

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|---|-------------------------------|
| 1. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 2. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 3. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 4. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 5. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 6. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 7. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 8. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 9. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 10. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 11. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 12. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 13. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 15. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 16. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |

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|-----|---|------------------|
| 1. | 9 common gauge nail penetrate a $3/8$ " green frame? | $3 - 11/32$ |
| 2. | 11 $1/2$ box gauge nail penetrate a $4 1/2$ " green mast? | $2 1/2 - 4 1/16$ |
| 3. | 11 $1/2$ common gauge nail penetrate a $3 1/2$ " green plank? | $2 - 3 1/16$ |
| 4. | 60 penny nail penetrate a $3/8$ " green timber? | $6 - 11/32$ |
| 5. | 30 penny nail penetrate a 1" green pole? | $4 1/2 - 25/32$ |
| 6. | 11 $1/2$ box gauge nail penetrate a 1" green plank? | $2 1/2 - 25/32$ |
| 7. | 12 $1/2$ common gauge nail penetrate a $1/2$ " dry mast? | $1 1/2 - 7/16$ |
| 8. | 11 $1/2$ common gauge nail penetrate a $2 1/2$ " dry beam? | $2 - 2$ |
| 9. | 10 $1/4$ common gauge nail penetrate a 2" green plank? | $2 1/2 - 19/16$ |
| 10. | 5 penny nail penetrate a $5/8$ " green timber? | $1 3/4 - 19/32$ |
| 11. | 30 penny nail penetrate a 2" dry girder? | $4 1/2 - 1 1/2$ |
| 12. | 20 penny nail penetrate a $3 1/2$ " dry beam? | $4 - 3$ |
| 13. | 7 penny nail penetrate a $3/8$ " green plank? | $2 1/4 - 11/32$ |
| 14. | 10 $1/2$ box gauge nail penetrate a $1 1/2$ " green plank? | $3 - 19/32$ |
| 15. | 5 penny nail penetrate a $5/8$ " dry board? | $1 3/4 - 9/16$ |
| 16. | 5 penny nail penetrate a 3" dry dimension? | $1 3/4 - 2 1/2$ |

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|---|--|
| 1. pilot hole for a 20 gauge screw? | H $2\frac{1}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 2. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 3. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 4. pilot hole for a 8 gauge screw? | 35 $1\frac{1}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 5. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 6. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 7. pilot hole for a 20 gauge screw? | H $2\frac{1}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 8. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 9. countersink hole for a 16 gauge screw? | X $1\frac{7}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 10. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 14. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 15. pilot hole for a 8 gauge screw? | 35 $1\frac{1}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |

185

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|-----|---|---------------------------------------|
| 1. | 10 penny nail penetrate a 2 ½" green board? | 3 — 2 ¹ / ₁₆ |
| 2. | 2 penny nail penetrate a 3 ½" dry plank? | 1 — 3 |
| 3. | 10 ¼ common gauge nail penetrate a 2" dry mast? | 2 ½ — 1 ½ |
| 4. | 9 box gauge nail penetrate a 3 ½" green joist? | 4 — 3 ¹ / ₁₆ |
| 5. | 12 penny nail penetrate a 1 ¼" green timber? | 3 ¼ — 1 ¹ / ₃₂ |
| 6. | 40 penny nail penetrate a 2 ½" dry plank? | 5 — 2 |
| 7. | 12 ½ common gauge nail penetrate a ¾" green rafter? | 1 ½ — 1 ¹ / ₁₆ |
| 8. | 6 penny nail penetrate a 3/8" dry pole? | 2 — 5 ¹ / ₁₆ |
| 9. | 9 common gauge nail penetrate a 1" green joist? | 3 — 25 ¹ / ₃₂ |
| 10. | 40 penny nail penetrate a 2 ½" green rafter? | 5 — 2 ¹ / ₁₆ |
| 11. | 8 box gauge nail penetrate a 3 ½" dry beam? | 5 — 3 |
| 12. | 40 penny nail penetrate a ¾" green board? | 5 — 1 ¹ / ₁₆ |
| 13. | 9 common gauge nail penetrate a 3 ½" green timber? | 3 — 3 ¹ / ₁₆ |
| 14. | 6 common gauge nail penetrate a 3/8" dry pole? | 4 — 5 ¹ / ₁₆ |
| 15. | 50 penny nail penetrate a 1" green beam? | 5 ½ — 25 ¹ / ₃₂ |
| 16. | 12 ½ common gauge nail penetrate a 3" dry board? | 1 ½ — 2 ½ |

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|---|--|
| 1. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 2. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 5. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 6. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 7. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 8. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 9. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 10. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 11. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 12. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 14. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 15. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 16. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |

186

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|-----|---|--------------|
| 1. | 10 penny nail penetrate a 1 ½" dry timber? | 3 — 1 ¼ |
| 2. | 2 penny nail penetrate a ¾" green pole? | 1 — 11/16 |
| 3. | 10 penny nail penetrate a 3 ½" dry pole? | 3 — 3 |
| 4. | 12 ½ common gauge nail penetrate a 4" dry joist? | 1 ½ — 3 ½ |
| 5. | 8 penny nail penetrate a 2 ½" green rafter? | 2 ½ — 2 1/16 |
| 6. | 60 penny nail penetrate a 3/8" dry pole? | 6 — 5/16 |
| 7. | 15 common gauge nail penetrate a 2 ½" dry beam? | 1 — 2 |
| 8. | 20 penny nail penetrate a 3 ½" green board? | 4 — 3 1/16 |
| 9. | 4 common gauge nail penetrate a 4 ½" dry timber? | 5 — 4 |
| 10. | 8 box gauge nail penetrate a 4 ½" green rafter? | 5 — 4 1/16 |
| 11. | 8 box gauge nail penetrate a 3/8" green board? | 5 — 11/32 |
| 12. | 3 penny nail penetrate a 1 ¼" dry joist? | 1 ¼ — 1 |
| 13. | 11 ½ common gauge nail penetrate a 5/8" dry beam? | 2 — 9/16 |
| 14. | 9 box gauge nail penetrate a 2" dry girder? | 4 — 1 ½ |
| 15. | 9 common gauge nail penetrate a 2" green beam? | 3 — 1 9/16 |
| 16. | 8 box gauge nail penetrate a ¾" dry joist? | 5 — 5/8 |

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|--|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 3. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 4. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 5. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 6. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 7. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 8. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 9. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 10. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 11. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 12. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 13. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 14. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 15. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 16. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |

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|-----|---|---------------------------------|
| 1. | 9 penny nail penetrate a $\frac{1}{2}$ " dry frame? | $2\frac{3}{4} - \frac{7}{16}$ |
| 2. | 14 box gauge nail penetrate a 4" dry joist? | $1\frac{1}{2} - 3\frac{1}{2}$ |
| 3. | 7 penny nail penetrate a $1\frac{1}{2}$ " green frame? | $2\frac{1}{4} - 1\frac{9}{32}$ |
| 4. | 16 penny nail penetrate a 3" green beam? | $3\frac{1}{2} - 2\frac{9}{16}$ |
| 5. | 60 penny nail penetrate a 4" dry rafter? | $6 - 3\frac{1}{2}$ |
| 6. | 16 penny nail penetrate a $1\frac{1}{2}$ " dry girder? | $3\frac{1}{2} - 1\frac{1}{4}$ |
| 7. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " green rafter? | $2\frac{1}{2} - 1\frac{11}{32}$ |
| 8. | 8 box gauge nail penetrate a 1" green rafter? | $5 - 2\frac{25}{32}$ |
| 9. | 40 penny nail penetrate a 1" dry frame? | $5 - \frac{3}{4}$ |
| 10. | 6 penny nail penetrate a $3\frac{1}{2}$ " dry girder? | $2 - 3$ |
| 11. | 30 penny nail penetrate a $2\frac{1}{2}$ " green joist? | $4\frac{1}{2} - 2\frac{1}{16}$ |
| 12. | 40 penny nail penetrate a $\frac{5}{8}$ " green girder? | $5 - 1\frac{19}{32}$ |
| 13. | 8 box gauge nail penetrate a $3\frac{1}{2}$ " dry mast? | $5 - 3$ |
| 14. | 12 penny nail penetrate a $2\frac{1}{2}$ " dry rafter? | $3\frac{1}{4} - 2$ |
| 15. | 12 penny nail penetrate a $\frac{3}{4}$ " dry mast? | $3\frac{1}{4} - \frac{5}{8}$ |
| 16. | $10\frac{1}{4}$ common gauge nail penetrate a $3\frac{1}{2}$ " dry joist? | $2\frac{1}{2} - 3$ |

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|--|-------------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 3. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 4. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 5. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 6. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 7. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 8. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 9. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 10. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 11. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 12. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 13. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 14. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 15. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 16. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |

188

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|-----|---|----------------------------------|
| 1. | 6 penny nail penetrate a 4" dry frame? | $2 - 3 \frac{1}{2}$ |
| 2. | 16 penny nail penetrate a 2" dry beam? | $3 \frac{1}{2} - 1 \frac{1}{2}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{4}$ " green plank? | $3 - 1 \frac{1}{32}$ |
| 4. | 14 box gauge nail penetrate a 3 $\frac{1}{2}$ " green beam? | $1 \frac{1}{2} - 3 \frac{1}{16}$ |
| 5. | 6 penny nail penetrate a 3 $\frac{1}{2}$ " dry girder? | $2 - 3$ |
| 6. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{1}{2}$ " green dimension? | $2 \frac{1}{2} - \frac{15}{32}$ |
| 7. | 9 penny nail penetrate a 2 $\frac{1}{2}$ " dry mast? | $2 \frac{3}{4} - 2$ |
| 8. | 50 penny nail penetrate a 4" dry girder? | $5 \frac{1}{2} - 3 \frac{1}{2}$ |
| 9. | 11 $\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " green board? | $2 - \frac{11}{32}$ |
| 10. | 10 $\frac{1}{4}$ common gauge nail penetrate a 4 $\frac{1}{2}$ " dry mast? | $2 \frac{1}{2} - 4$ |
| 11. | 3 common gauge nail penetrate a 3 $\frac{1}{2}$ " dry pole? | $5 \frac{1}{2} - 3$ |
| 12. | 12 penny nail penetrate a 3" dry plank? | $3 \frac{1}{4} - 2 \frac{1}{2}$ |
| 13. | 9 penny nail penetrate a $\frac{1}{2}$ " dry frame? | $2 \frac{3}{4} - \frac{7}{16}$ |
| 14. | 16 penny nail penetrate a 1" dry pole? | $3 \frac{1}{2} - \frac{3}{4}$ |
| 15. | 14 box gauge nail penetrate a $\frac{3}{4}$ " green dimension? | $1 \frac{1}{2} - \frac{11}{16}$ |
| 16. | 9 penny nail penetrate a $\frac{3}{4}$ " green frame? | $2 \frac{3}{4} - \frac{11}{16}$ |

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|---|---|
| 1. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 2. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 3. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 4. countersink hole for a 18 gauge screw? | $\frac{27}{64} \quad \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 5. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 6. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 7. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 8. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 10. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 11. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 12. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 13. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 14. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 15. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |

189

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|-----|---|----------------|
| 1. | 14 box gauge nail penetrate a 3 1/2" green girder? | 1 1/2 — 3 1/16 |
| 2. | 5 common gauge nail penetrate a 1/2" dry rafter? | 4 1/2 — 7/16 |
| 3. | 16 penny nail penetrate a 3" green board? | 3 1/2 — 2 9/16 |
| 4. | 12 1/2 box gauge nail penetrate a 4 1/2" green plank? | 2 — 4 1/16 |
| 5. | 9 box gauge nail penetrate a 4 1/2" green girder? | 4 — 4 1/16 |
| 6. | 40 penny nail penetrate a 4 1/2" green pole? | 5 — 4 1/16 |
| 7. | 40 penny nail penetrate a 3 1/2" green timber? | 5 — 3 1/16 |
| 8. | 8 common gauge nail penetrate a 2 1/2" green frame? | 3 1/2 — 2 1/16 |
| 9. | 2 penny nail penetrate a 1" dry plank? | 1 — 3/4 |
| 10. | 14 common gauge nail penetrate a 3 1/2" dry girder? | 1 1/4 — 3 |
| 11. | 4 penny nail penetrate a 2 1/2" dry rafter? | 1 1/2 — 2 |
| 12. | 9 penny nail penetrate a 4 1/2" dry girder? | 2 3/4 — 4 |
| 13. | 2 penny nail penetrate a 3" green board? | 1 — 2 9/16 |
| 14. | 16 penny nail penetrate a 5/8" dry dimension? | 3 1/2 — 9/16 |
| 15. | 50 penny nail penetrate a 1 1/2" green timber? | 5 1/2 — 1 9/32 |
| 16. | 6 penny nail penetrate a 4 1/2" dry dimension? | 2 — 4 |

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|---|--------------------------|
| 1. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 2. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 3. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 4. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 5. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 6. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 7. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 8. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 9. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 10. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 11. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 12. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 13. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 14. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 15. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 16. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |

190

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|-----|--|---------------------------------|
| 1. | 9 common gauge nail penetrate a 1" dry rafter? | $3 - \frac{3}{4}$ |
| 2. | 12 $\frac{1}{2}$ common gauge nail penetrate a 2 $\frac{1}{2}$ " dry girder? | $1 \frac{1}{2} - 2$ |
| 3. | 2 penny nail penetrate a 3" green joist? | $1 - 2\frac{9}{16}$ |
| 4. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1" dry dimension? | $2 - \frac{3}{4}$ |
| 5. | 8 box gauge nail penetrate a $\frac{1}{2}$ " green board? | $5 - \frac{15}{32}$ |
| 6. | 6 penny nail penetrate a 1 $\frac{1}{4}$ " green timber? | $2 - 1\frac{1}{32}$ |
| 7. | 10 $\frac{1}{4}$ common gauge nail penetrate a 4" green joist? | $2 \frac{1}{2} - 3\frac{9}{16}$ |
| 8. | 20 penny nail penetrate a 4" dry frame? | $4 - 3 \frac{1}{2}$ |
| 9. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " green girder? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 10. | 3 penny nail penetrate a 2" green rafter? | $1 \frac{1}{4} - 1\frac{9}{16}$ |
| 11. | 20 penny nail penetrate a 3" dry rafter? | $4 - 2 \frac{1}{2}$ |
| 12. | 9 box gauge nail penetrate a 1 $\frac{1}{4}$ " green plank? | $4 - 1\frac{1}{32}$ |
| 13. | 9 common gauge nail penetrate a $\frac{5}{8}$ " green mast? | $3 - \frac{19}{32}$ |
| 14. | 10 $\frac{1}{4}$ common gauge nail penetrate a 4 $\frac{1}{2}$ " green timber? | $2 \frac{1}{2} - 4\frac{1}{16}$ |
| 15. | 3 common gauge nail penetrate a 4" dry joist? | $5 \frac{1}{2} - 3 \frac{1}{2}$ |
| 16. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green frame? | $3 - 1\frac{1}{16}$ |

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|---|-----------------------------|
| 1. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 2. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 3. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 4. countersink hole for a 24 gauge screw? | $1/2 \quad 3/8 + 1/8 = 1/2$ |
| 5. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 6. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 7. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 8. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 9. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 10. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 11. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 12. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 13. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 14. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 15. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 16. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |

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|-----|---|---------------------------------|
| 1. | 2 penny nail penetrate a 3" green joist? | $1 - 2\frac{9}{16}$ |
| 2. | 12 penny nail penetrate a $\frac{5}{8}$ " dry pole? | $3\frac{1}{4} - \frac{9}{16}$ |
| 3. | 40 penny nail penetrate a $\frac{3}{8}$ " green dimension? | $5 - 1\frac{11}{32}$ |
| 4. | 3 penny nail penetrate a $3\frac{1}{2}$ " dry girder? | $1\frac{1}{4} - 3$ |
| 5. | $11\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green beam? | $2\frac{1}{2} - 1\frac{19}{32}$ |
| 6. | 16 penny nail penetrate a $2\frac{1}{2}$ " green board? | $3\frac{1}{2} - 2\frac{1}{16}$ |
| 7. | 2 common gauge nail penetrate a 1" dry dimension? | $6 - \frac{3}{4}$ |
| 8. | 5 common gauge nail penetrate a $2\frac{1}{2}$ " green board? | $4\frac{1}{2} - 2\frac{1}{16}$ |
| 9. | $10\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " dry board? | $3 - 3$ |
| 10. | $11\frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " dry girder? | $2 - \frac{9}{16}$ |
| 11. | 9 penny nail penetrate a $\frac{3}{4}$ " dry frame? | $2\frac{3}{4} - \frac{5}{8}$ |
| 12. | 3 penny nail penetrate a 4" dry mast? | $1\frac{1}{4} - 3\frac{1}{2}$ |
| 13. | 2 penny nail penetrate a $\frac{1}{2}$ " green frame? | $1 - 1\frac{15}{32}$ |
| 14. | $10\frac{1}{2}$ box gauge nail penetrate a 4" green dimension? | $3 - 3\frac{9}{16}$ |
| 15. | 2 common gauge nail penetrate a $4\frac{1}{2}$ " green rafter? | $6 - 4\frac{1}{16}$ |
| 16. | 40 penny nail penetrate a $1\frac{1}{2}$ " green pole? | $5 - 1\frac{9}{32}$ |

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|--|-------------------------------|
| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 3. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 4. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 5. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 6. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 7. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 9. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 10. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 11. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 12. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 13. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 14. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 15. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 16. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |

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|-----|--|--------------------------------|
| 1. | 14 box gauge nail penetrate a $1\frac{1}{4}$ " dry beam? | $1\frac{1}{2} - 1$ |
| 2. | 20 penny nail penetrate a $\frac{5}{8}$ " green dimension? | $4 - \frac{19}{32}$ |
| 3. | 5 common gauge nail penetrate a 1" green pole? | $4\frac{1}{2} - \frac{25}{32}$ |
| 4. | 4 penny nail penetrate a $\frac{5}{8}$ " green dimension? | $1\frac{1}{2} - \frac{19}{32}$ |
| 5. | $10\frac{1}{2}$ box gauge nail penetrate a 1" dry girder? | $3 - \frac{3}{4}$ |
| 6. | $10\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{2}$ " dry plank? | $3 - 1\frac{1}{4}$ |
| 7. | 12 penny nail penetrate a $1\frac{1}{4}$ " dry rafter? | $3\frac{1}{4} - 1$ |
| 8. | 8 box gauge nail penetrate a $\frac{5}{8}$ " dry joist? | $5 - \frac{9}{16}$ |
| 9. | 16 penny nail penetrate a $2\frac{1}{2}$ " dry girder? | $3\frac{1}{2} - 2$ |
| 10. | 4 penny nail penetrate a $2\frac{1}{2}$ " green timber? | $1\frac{1}{2} - 2\frac{1}{16}$ |
| 11. | 8 penny nail penetrate a $\frac{1}{2}$ " dry mast? | $2\frac{1}{2} - \frac{7}{16}$ |
| 12. | 60 penny nail penetrate a $2\frac{1}{2}$ " dry pole? | $6 - 2$ |
| 13. | 10 penny nail penetrate a $3\frac{1}{2}$ " green beam? | $3 - 3\frac{1}{16}$ |
| 14. | 15 common gauge nail penetrate a 4" dry pole? | $1 - 3\frac{1}{2}$ |
| 15. | 3 penny nail penetrate a $1\frac{1}{2}$ " dry girder? | $1\frac{1}{4} - 1\frac{1}{4}$ |
| 16. | 15 common gauge nail penetrate a $2\frac{1}{2}$ " green joist? | $1 - 2\frac{1}{16}$ |

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|--|---|
| 1. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 2. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 3. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 4. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 5. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 6. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 7. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 8. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 9. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 10. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 11. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 12. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 13. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 14. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 15. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |

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|-----|---|--------------------------------------|
| 1. | 11 ½ box gauge nail penetrate a 2" green plank? | 2 ½ — 1 ⁹ / ₁₆ |
| 2. | 7 penny nail penetrate a ½" green beam? | 2 ¼ — 1 ⁵ / ₃₂ |
| 3. | 60 penny nail penetrate a 1 ¼" dry girder? | 6 — 1 |
| 4. | 5 common gauge nail penetrate a 2" dry rafter? | 4 ½ — 1 ½ |
| 5. | 8 box gauge nail penetrate a 4" dry plank? | 5 — 3 ½ |
| 6. | 40 penny nail penetrate a 5/8" dry plank? | 5 — 9/16 |
| 7. | 9 common gauge nail penetrate a 4 ½" dry girder? | 3 — 4 |
| 8. | 9 common gauge nail penetrate a 3/4" green frame? | 3 — 11/16 |
| 9. | 9 box gauge nail penetrate a 3/8" green plank? | 4 — 11/32 |
| 10. | 11 ½ common gauge nail penetrate a 5/8" green mast? | 2 — 19/32 |
| 11. | 10 penny nail penetrate a 2" green pole? | 3 — 19/16 |
| 12. | 5 penny nail penetrate a ½" green rafter? | 1 ¾ — 15/32 |
| 13. | 14 box gauge nail penetrate a 3/8" green rafter? | 1 ½ — 11/32 |
| 14. | 12 penny nail penetrate a 1 ½" dry timber? | 3 ¼ — 1 ¼ |
| 15. | 60 penny nail penetrate a 2" dry board? | 6 — 1 ½ |
| 16. | 14 box gauge nail penetrate a 2" dry plank? | 1 ½ — 1 ½ |

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|--|---|
| 1. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 2. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 3. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 4. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 7. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 9. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 10. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 14. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 15. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 16. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

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|-----|---|-------------------------------|
| 1. | 2 common gauge nail penetrate a $5/8$ " green pole? | $6 - 19/32$ |
| 2. | 8 box gauge nail penetrate a $2\frac{1}{2}$ " green board? | $5 - 2\frac{1}{16}$ |
| 3. | $10\frac{1}{4}$ common gauge nail penetrate a 2" dry plank? | $2\frac{1}{2} - 1\frac{1}{2}$ |
| 4. | $10\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry mast? | $3 - 7/16$ |
| 5. | 10 penny nail penetrate a $4\frac{1}{2}$ " green joist? | $3 - 4\frac{1}{16}$ |
| 6. | $10\frac{1}{2}$ box gauge nail penetrate a $3/8$ " dry girder? | $3 - 5/16$ |
| 7. | 6 common gauge nail penetrate a 4" green dimension? | $4 - 39/16$ |
| 8. | 9 common gauge nail penetrate a $1\frac{1}{2}$ " green beam? | $3 - 19/32$ |
| 9. | 20 penny nail penetrate a 2" dry rafter? | $4 - 1\frac{1}{2}$ |
| 10. | $11\frac{1}{2}$ box gauge nail penetrate a $5/8$ " dry rafter? | $2\frac{1}{2} - 9/16$ |
| 11. | 2 penny nail penetrate a 2" dry girder? | $1 - 1\frac{1}{2}$ |
| 12. | 8 box gauge nail penetrate a 3" green pole? | $5 - 29/16$ |
| 13. | 50 penny nail penetrate a 2" dry mast? | $5\frac{1}{2} - 1\frac{1}{2}$ |
| 14. | 9 penny nail penetrate a 1" dry timber? | $2\frac{3}{4} - 3/4$ |
| 15. | $12\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{2}$ " green pole? | $2 - 19/32$ |
| 16. | 9 box gauge nail penetrate a $4\frac{1}{2}$ " dry beam? | $4 - 4$ |

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|---|-----------------------------------|
| 1. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 2. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 3. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 4. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 5. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 6. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 7. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 8. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 9. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 10. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 11. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 12. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 13. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 14. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 15. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 16. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |

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|-----|--|--------------|
| 1. | 11 ½ common gauge nail penetrate a 2" dry beam? | 2 — 1 ½ |
| 2. | 8 penny nail penetrate a 2 ½" dry board? | 2 ½ — 2 |
| 3. | 30 penny nail penetrate a 4 ½" green beam? | 4 ½ — 4 ⅓ |
| 4. | 6 common gauge nail penetrate a 3/8" green mast? | 4 — 11/32 |
| 5. | 5 common gauge nail penetrate a 3/4" dry frame? | 4 ½ — 5/8 |
| 6. | 5 common gauge nail penetrate a 3" green girder? | 4 ½ — 2 9/16 |
| 7. | 14 ½ box gauge nail penetrate a 2 ½" green mast? | 1 ¼ — 2 1/16 |
| 8. | 30 penny nail penetrate a 3/8" dry mast? | 4 ½ — 5/16 |
| 9. | 14 box gauge nail penetrate a 5/8" dry plank? | 1 ½ — 9/16 |
| 10. | 10 ½ box gauge nail penetrate a 1 ½" green beam? | 3 — 1 9/32 |
| 11. | 50 penny nail penetrate a 3 ½" dry girder? | 5 ½ — 3 |
| 12. | 12 penny nail penetrate a 3" dry rafter? | 3 ¼ — 2 ½ |
| 13. | 14 common gauge nail penetrate a ½" green beam? | 1 ¼ — 15/32 |
| 14. | 7 penny nail penetrate a 1" green girder? | 2 ¼ — 25/32 |
| 15. | 7 penny nail penetrate a 3/4" green frame? | 2 ¼ — 11/16 |
| 16. | 50 penny nail penetrate a 2 ½" green pole? | 5 ½ — 2 1/16 |

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|--|---|
| 1. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 2. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 3. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 4. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 5. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 6. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 7. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 8. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 12. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 13. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 14. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 15. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 16. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |

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|-----|---|--------------------------------------|
| 1. | 6 common gauge nail penetrate a 4 ½" dry girder? | 4 — 4 |
| 2. | 3 penny nail penetrate a 3" green timber? | 1 ¼ — 2 ⁹ / ₁₆ |
| 3. | 10 penny nail penetrate a 1 ¼" green girder? | 3 — 1 ¹ / ₃₂ |
| 4. | 60 penny nail penetrate a 3 ½" green dimension? | 6 — 3 ¹ / ₁₆ |
| 5. | 10 ½ box gauge nail penetrate a 5/8" green board? | 3 — 1 ⁹ / ₃₂ |
| 6. | 7 penny nail penetrate a 3" dry frame? | 2 ¼ — 2 ½ |
| 7. | 12 ½ box gauge nail penetrate a ½" green pole? | 2 — 1 ⁵ / ₃₂ |
| 8. | 9 box gauge nail penetrate a 4" dry joist? | 4 — 3 ½ |
| 9. | 5 penny nail penetrate a 4 ½" dry joist? | 1 ¾ — 4 |
| 10. | 60 penny nail penetrate a 2" dry board? | 6 — 1 ½ |
| 11. | 14 ½ box gauge nail penetrate a 4 ½" dry board? | 1 ¼ — 4 |
| 12. | 3 common gauge nail penetrate a 4" dry mast? | 5 ½ — 3 ½ |
| 13. | 50 penny nail penetrate a 4 ½" green pole? | 5 ½ — 4 ¹ / ₁₆ |
| 14. | 60 penny nail penetrate a 4 ½" dry rafter? | 6 — 4 |
| 15. | 14 ½ box gauge nail penetrate a ½" dry board? | 1 ¼ — 7 ¹ / ₁₆ |
| 16. | 6 penny nail penetrate a 3/4" green timber? | 2 — 1 ¹ / ₁₆ |

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|---|--|
| 1. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 2. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 3. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 6. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 7. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 8. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 9. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 10. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 13. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 14. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 15. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 16. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |

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|-----|---|------------|
| 1. | 10 ½ box gauge nail penetrate a 1 ½" dry pole? | 3 — 1 ¼ |
| 2. | 60 penny nail penetrate a 1 ½" green dimension? | 6 — 1 ⅞ |
| 3. | 6 penny nail penetrate a ½" dry pole? | 2 — 7/16 |
| 4. | 12 ½ box gauge nail penetrate a 4" dry mast? | 2 — 3 ½ |
| 5. | 6 penny nail penetrate a 4" green board? | 2 — 3 ⅞ |
| 6. | 15 ½ box gauge nail penetrate a 4 ½" green beam? | 1 — 4 ⅞ |
| 7. | 9 box gauge nail penetrate a 4 ½" green board? | 4 — 4 ⅞ |
| 8. | 14 common gauge nail penetrate a 5/8" green girder? | 1 ¼ — 1 ⅞ |
| 9. | 5 common gauge nail penetrate a 3" green dimension? | 4 ½ — 2 ⅞ |
| 10. | 14 box gauge nail penetrate a 3/8" dry pole? | 1 ½ — 5/16 |
| 11. | 10 ¼ common gauge nail penetrate a 4 ½" green timber? | 2 ½ — 4 ⅞ |
| 12. | 40 penny nail penetrate a 4" dry joist? | 5 — 3 ½ |
| 13. | 30 penny nail penetrate a 4 ½" green timber? | 4 ½ — 4 ⅞ |
| 14. | 8 common gauge nail penetrate a 3/4" dry board? | 3 ½ — 5/8 |
| 15. | 15 common gauge nail penetrate a 2" green rafter? | 1 — 1 ⅞ |
| 16. | 11 ½ box gauge nail penetrate a 3/8" dry mast? | 2 ½ — 5/16 |

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|--|--------------------------|
| 1. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 2. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 3. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 4. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 5. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 6. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 7. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 8. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 9. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 10. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 11. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 12. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 13. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 14. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 15. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 16. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |

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|-----|--|---------------|
| 1. | 7 penny nail penetrate a 1 1/2" green dimension? | 2 1/4 — 19/32 |
| 2. | 9 box gauge nail penetrate a 3/8" dry joist? | 4 — 5/16 |
| 3. | 8 penny nail penetrate a 3/4" green mast? | 2 1/2 — 11/16 |
| 4. | 12 1/2 box gauge nail penetrate a 1" dry mast? | 2 — 3/4 |
| 5. | 9 common gauge nail penetrate a 3/4" dry frame? | 3 — 5/8 |
| 6. | 10 1/4 common gauge nail penetrate a 2" green girder? | 2 1/2 — 19/16 |
| 7. | 8 penny nail penetrate a 5/8" dry beam? | 2 1/2 — 9/16 |
| 8. | 8 box gauge nail penetrate a 4" green mast? | 5 — 39/16 |
| 9. | 60 penny nail penetrate a 3" dry dimension? | 6 — 2 1/2 |
| 10. | 40 penny nail penetrate a 2 1/2" green dimension? | 5 — 21/16 |
| 11. | 9 common gauge nail penetrate a 3 1/2" green mast? | 3 — 31/16 |
| 12. | 14 1/2 box gauge nail penetrate a 1 1/4" green timber? | 1 1/4 — 11/32 |
| 13. | 6 penny nail penetrate a 4 1/2" dry rafter? | 2 — 4 |
| 14. | 7 penny nail penetrate a 4 1/2" dry frame? | 2 1/4 — 4 |
| 15. | 7 penny nail penetrate a 1" green pole? | 2 1/4 — 25/32 |
| 16. | 8 box gauge nail penetrate a 3/8" dry board? | 5 — 5/16 |

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|---|---|
| 1. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 2. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 3. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 4. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 5. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 6. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 7. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 8. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 9. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 10. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 11. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 12. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 13. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 14. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 15. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 16. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |

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|-----|--|--------------------------------|
| 1. | 50 penny nail penetrate a $\frac{3}{4}$ " green rafter? | $5\frac{1}{2} - \frac{11}{16}$ |
| 2. | 3 penny nail penetrate a 4" dry rafter? | $1\frac{1}{4} - 3\frac{1}{2}$ |
| 3. | 10 penny nail penetrate a $2\frac{1}{2}$ " dry dimension? | $3 - 2$ |
| 4. | 14 box gauge nail penetrate a $\frac{1}{2}$ " green timber? | $1\frac{1}{2} - \frac{15}{32}$ |
| 5. | $10\frac{1}{4}$ common gauge nail penetrate a $4\frac{1}{2}$ " dry rafter? | $2\frac{1}{2} - 4$ |
| 6. | 9 penny nail penetrate a $1\frac{1}{4}$ " dry timber? | $2\frac{3}{4} - 1$ |
| 7. | 14 common gauge nail penetrate a $\frac{3}{4}$ " green frame? | $1\frac{1}{4} - \frac{11}{16}$ |
| 8. | 9 common gauge nail penetrate a 2" dry beam? | $3 - 1\frac{1}{2}$ |
| 9. | 9 penny nail penetrate a $\frac{3}{4}$ " dry frame? | $2\frac{3}{4} - \frac{5}{8}$ |
| 10. | 5 penny nail penetrate a $\frac{3}{8}$ " green girder? | $1\frac{3}{4} - \frac{11}{32}$ |
| 11. | 7 penny nail penetrate a $1\frac{1}{4}$ " dry plank? | $2\frac{1}{4} - 1$ |
| 12. | 2 common gauge nail penetrate a $3\frac{1}{2}$ " dry frame? | $6 - 3$ |
| 13. | $12\frac{1}{2}$ common gauge nail penetrate a $4\frac{1}{2}$ " green mast? | $1\frac{1}{2} - 4\frac{1}{16}$ |
| 14. | 6 penny nail penetrate a $\frac{1}{2}$ " dry joist? | $2 - \frac{7}{16}$ |
| 15. | 3 penny nail penetrate a 2" green board? | $1\frac{1}{4} - \frac{19}{16}$ |
| 16. | 2 penny nail penetrate a $\frac{5}{8}$ " green rafter? | $1 - \frac{19}{32}$ |

1. clearance hole for a 4 gauge screw? 35 $7/64 = 7/64$
2. clearance hole for a 3 gauge screw? 41 $3/32 = 3/32$
3. pilot hole for a 4 gauge screw? 55 $7/64 - 1/16 = 3/64$
4. countersink hole for a 10 gauge screw? O $3/16 + 1/8 = 5/16$
5. countersink hole for a 3 gauge screw? 2 $3/32 + 1/8 = 7/32$
6. pilot hole for a 11 gauge screw? 27 $13/64 - 1/16 = 9/64$
7. countersink hole for a 12 gauge screw? S $7/32 + 1/8 = 11/32$
8. countersink hole for a 5 gauge screw? E $1/8 + 1/8 = 1/4$
9. countersink hole for a 18 gauge screw? $27/64$ $19/64 + 1/8 = 27/64$
10. countersink hole for a 11 gauge screw? Q $13/64 + 1/8 = 21/64$
11. clearance hole for a 2 gauge screw? 41 $3/32 = 3/32$
12. countersink hole for a 11 gauge screw? Q $13/64 + 1/8 = 21/64$
13. pilot hole for a 10 gauge screw? 30 $3/16 - 1/16 = 1/8$
14. pilot hole for a 2 gauge screw? 67 $3/32 - 1/16 = 1/32$
15. countersink hole for a 1 gauge screw? 6 $5/64 + 1/8 = 13/64$
16. countersink hole for a 7 gauge screw? L $5/32 + 1/8 = 9/32$

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|-----|---|--------------------------------|
| 1. | 7 penny nail penetrate a $\frac{1}{2}$ " dry rafter? | $2\frac{1}{4} - \frac{7}{16}$ |
| 2. | 6 penny nail penetrate a 3" dry frame? | $2 - 2\frac{1}{2}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry board? | $3 - \frac{5}{16}$ |
| 4. | 3 penny nail penetrate a 1" dry rafter? | $1\frac{1}{4} - \frac{3}{4}$ |
| 5. | 15 $\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{4}$ " dry plank? | $1 - 1$ |
| 6. | 9 penny nail penetrate a $4\frac{1}{2}$ " green frame? | $2\frac{3}{4} - 4\frac{1}{16}$ |
| 7. | 4 penny nail penetrate a 4" dry frame? | $1\frac{1}{2} - 3\frac{1}{2}$ |
| 8. | 5 penny nail penetrate a 1" dry timber? | $1\frac{3}{4} - \frac{3}{4}$ |
| 9. | 4 common gauge nail penetrate a $\frac{3}{4}$ " green mast? | $5 - \frac{11}{16}$ |
| 10. | 3 penny nail penetrate a $4\frac{1}{2}$ " dry beam? | $1\frac{1}{4} - 4$ |
| 11. | 7 penny nail penetrate a 4" green board? | $2\frac{1}{4} - 3\frac{9}{16}$ |
| 12. | 9 common gauge nail penetrate a $2\frac{1}{2}$ " green dimension? | $3 - 2\frac{1}{16}$ |
| 13. | 3 penny nail penetrate a 3" dry plank? | $1\frac{1}{4} - 2\frac{1}{2}$ |
| 14. | 14 box gauge nail penetrate a $4\frac{1}{2}$ " green dimension? | $1\frac{1}{2} - 4\frac{1}{16}$ |
| 15. | 5 penny nail penetrate a $\frac{3}{4}$ " dry board? | $1\frac{3}{4} - \frac{5}{8}$ |
| 16. | 5 penny nail penetrate a $2\frac{1}{2}$ " dry pole? | $1\frac{3}{4} - 2$ |

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|--|---|
| 1. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 2. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 3. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 4. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 5. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 6. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 7. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 8. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 9. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 10. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 14. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 15. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 16. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |

201

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|-----|---|--------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 1 ½" green mast? | 3 — 1 ⁹ / ₃₂ |
| 2. | 11 ½ box gauge nail penetrate a 3" green dimension? | 2 ½ — 2 ⁹ / ₁₆ |
| 3. | 15 common gauge nail penetrate a 2 ½" green girder? | 1 — 2 ¹ / ₁₆ |
| 4. | 60 penny nail penetrate a 4 ½" dry frame? | 6 — 4 |
| 5. | 20 penny nail penetrate a 1 ¼" dry rafter? | 4 — 1 |
| 6. | 50 penny nail penetrate a 3 ½" green mast? | 5 ½ — 3 ¹ / ₁₆ |
| 7. | 14 box gauge nail penetrate a 2" dry rafter? | 1 ½ — 1 ½ |
| 8. | 10 ½ box gauge nail penetrate a 1 ½" green joist? | 3 — 1 ⁹ / ₃₂ |
| 9. | 6 penny nail penetrate a ½" dry plank? | 2 — ⁷ / ₁₆ |
| 10. | 50 penny nail penetrate a 1" green plank? | 5 ½ — ²⁵ / ₃₂ |
| 11. | 4 penny nail penetrate a 1" green timber? | 1 ½ — ²⁵ / ₃₂ |
| 12. | 11 ½ common gauge nail penetrate a 1" green timber? | 2 — ²⁵ / ₃₂ |
| 13. | 2 penny nail penetrate a 4" green plank? | 1 — 3 ⁹ / ₁₆ |
| 14. | 6 penny nail penetrate a 1 ¼" green pole? | 2 — 1 ¹ / ₃₂ |
| 15. | 7 penny nail penetrate a 3 ½" dry dimension? | 2 ¼ — 3 |
| 16. | 6 penny nail penetrate a 3" dry mast? | 2 — 2 ½ |

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|---|---|
| 1. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 2. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 3. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 4. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 5. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 6. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 7. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 8. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 9. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 10. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 11. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 12. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 13. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 14. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 16. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |

202

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|-----|---|---------------------------------|
| 1. | 8 common gauge nail penetrate a 3" dry beam? | $3 \frac{1}{2} - 2 \frac{1}{2}$ |
| 2. | 50 penny nail penetrate a $\frac{3}{8}$ " dry rafter? | $5 \frac{1}{2} - \frac{5}{16}$ |
| 3. | 8 box gauge nail penetrate a $\frac{3}{8}$ " dry pole? | $5 - \frac{5}{16}$ |
| 4. | 5 common gauge nail penetrate a $\frac{3}{4}$ " dry rafter? | $4 \frac{1}{2} - \frac{5}{8}$ |
| 5. | 20 penny nail penetrate a 3" green girder? | $4 - 2\frac{9}{16}$ |
| 6. | 14 box gauge nail penetrate a $4 \frac{1}{2}$ " dry rafter? | $1 \frac{1}{2} - 4$ |
| 7. | 14 box gauge nail penetrate a $\frac{1}{2}$ " green girder? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 8. | 4 common gauge nail penetrate a 3" dry board? | $5 - 2 \frac{1}{2}$ |
| 9. | $11 \frac{1}{2}$ box gauge nail penetrate a 4" green pole? | $2 \frac{1}{2} - 3\frac{9}{16}$ |
| 10. | 7 penny nail penetrate a $1 \frac{1}{4}$ " green girder? | $2 \frac{1}{4} - 1\frac{1}{32}$ |
| 11. | 12 penny nail penetrate a 4" green dimension? | $3 \frac{1}{4} - 3\frac{9}{16}$ |
| 12. | 14 box gauge nail penetrate a $3 \frac{1}{2}$ " green girder? | $1 \frac{1}{2} - 3\frac{1}{16}$ |
| 13. | 40 penny nail penetrate a 3" dry pole? | $5 - 2 \frac{1}{2}$ |
| 14. | 9 common gauge nail penetrate a $\frac{5}{8}$ " dry frame? | $3 - \frac{9}{16}$ |
| 15. | 14 common gauge nail penetrate a 4" green mast? | $1 \frac{1}{4} - 3\frac{9}{16}$ |
| 16. | 50 penny nail penetrate a $\frac{3}{8}$ " green pole? | $5 \frac{1}{2} - 1\frac{1}{32}$ |

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| 1. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 2. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 3. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 4. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 5. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 6. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 7. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 8. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 12. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 13. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 14. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 15. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 16. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |

203

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|-----|---|--------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 4" dry girder? | 3 — 3 ½ |
| 2. | 10 ½ box gauge nail penetrate a 1 ¼" dry dimension? | 3 — 1 |
| 3. | 2 penny nail penetrate a 1 ½" dry frame? | 1 — 1 ¼ |
| 4. | 10 ½ box gauge nail penetrate a 2" dry rafter? | 3 — 1 ½ |
| 5. | 12 ½ box gauge nail penetrate a 3" green pole? | 2 — 2 ⁹ / ₁₆ |
| 6. | 15 common gauge nail penetrate a 2" dry rafter? | 1 — 1 ½ |
| 7. | 11 ½ box gauge nail penetrate a ¾" dry timber? | 2 ½ — 5 ⁵ / ₈ |
| 8. | 40 penny nail penetrate a 3 ½" green dimension? | 5 — 3 ¹ / ₁₆ |
| 9. | 8 penny nail penetrate a 4 ½" green timber? | 2 ½ — 4 ¹ / ₁₆ |
| 10. | 40 penny nail penetrate a 4" dry plank? | 5 — 3 ½ |
| 11. | 10 ½ box gauge nail penetrate a 3" dry rafter? | 3 — 2 ½ |
| 12. | 5 penny nail penetrate a 4 ½" dry beam? | 1 ¾ — 4 |
| 13. | 8 penny nail penetrate a 4" dry girder? | 2 ½ — 3 ½ |
| 14. | 50 penny nail penetrate a 5/8" dry pole? | 5 ½ — 9 ⁹ / ₁₆ |
| 15. | 2 common gauge nail penetrate a 4" dry girder? | 6 — 3 ½ |
| 16. | 50 penny nail penetrate a 4 ½" green beam? | 5 ½ — 4 ¹ / ₁₆ |

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|---|--|
| 1. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 2. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 3. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 4. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 5. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 6. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 7. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 8. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 9. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 10. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 11. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 12. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 14. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 15. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 16. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |

204

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|-----|--|--------------|
| 1. | 14 ½ box gauge nail penetrate a 5/8" dry frame? | 1 ¼ — 9/16 |
| 2. | 9 box gauge nail penetrate a 4" green plank? | 4 — 39/16 |
| 3. | 10 penny nail penetrate a 4" green board? | 3 — 39/16 |
| 4. | 9 box gauge nail penetrate a 1 ½" dry pole? | 4 — 1 ¼ |
| 5. | 12 ½ common gauge nail penetrate a 1 ¼" green plank? | 1 ½ — 1 1/32 |
| 6. | 60 penny nail penetrate a 3/8" dry girder? | 6 — 5/16 |
| 7. | 20 penny nail penetrate a 4 ½" dry dimension? | 4 — 4 |
| 8. | 12 penny nail penetrate a 2" dry joist? | 3 ¼ — 1 ½ |
| 9. | 50 penny nail penetrate a 3/8" green dimension? | 5 ½ — 11/32 |
| 10. | 14 box gauge nail penetrate a 3 ½" dry plank? | 1 ½ — 3 |
| 11. | 12 ½ common gauge nail penetrate a 4" dry beam? | 1 ½ — 3 ½ |
| 12. | 2 common gauge nail penetrate a 2" dry timber? | 6 — 1 ½ |
| 13. | 9 box gauge nail penetrate a 2 ½" green joist? | 4 — 2 1/16 |
| 14. | 60 penny nail penetrate a 1 ½" dry rafter? | 6 — 1 ¼ |
| 15. | 20 penny nail penetrate a 3/4" dry mast? | 4 — 5/8 |
| 16. | 9 common gauge nail penetrate a 5/8" green frame? | 3 — 19/32 |

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|--|--|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 3. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 4. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 5. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 6. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 7. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 8. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 10. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 11. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 12. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 13. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 14. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 15. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 16. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |

205

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|-----|--|----------------------------------|
| 1. | 14 common gauge nail penetrate a 5/8" dry mast? | $1 \frac{1}{4} - \frac{9}{16}$ |
| 2. | 30 penny nail penetrate a 1/2" dry pole? | $4 \frac{1}{2} - \frac{7}{16}$ |
| 3. | 9 penny nail penetrate a 3 1/2" green beam? | $2 \frac{3}{4} - 3 \frac{1}{16}$ |
| 4. | 20 penny nail penetrate a 1/2" green pole? | $4 - \frac{15}{32}$ |
| 5. | 12 1/2 common gauge nail penetrate a 1" dry girder? | $1 \frac{1}{2} - \frac{3}{4}$ |
| 6. | 4 penny nail penetrate a 4" dry joist? | $1 \frac{1}{2} - 3 \frac{1}{2}$ |
| 7. | 12 1/2 box gauge nail penetrate a 1 1/4" dry pole? | $2 - 1$ |
| 8. | 20 penny nail penetrate a 5/8" dry plank? | $4 - \frac{9}{16}$ |
| 9. | 10 1/2 box gauge nail penetrate a 3" green dimension? | $3 - 2 \frac{9}{16}$ |
| 10. | 6 common gauge nail penetrate a 2 1/2" green dimension? | $4 - 2 \frac{1}{16}$ |
| 11. | 10 penny nail penetrate a 4" green girder? | $3 - 3 \frac{9}{16}$ |
| 12. | 8 common gauge nail penetrate a 1/2" green frame? | $3 \frac{1}{2} - \frac{15}{32}$ |
| 13. | 11 1/2 common gauge nail penetrate a 5/8" dry dimension? | $2 - \frac{9}{16}$ |
| 14. | 12 1/2 box gauge nail penetrate a 3/8" green rafter? | $2 - \frac{11}{32}$ |
| 15. | 12 1/2 box gauge nail penetrate a 1 1/4" green rafter? | $2 - 1 \frac{1}{32}$ |
| 16. | 16 penny nail penetrate a 2" green joist? | $3 \frac{1}{2} - 1 \frac{9}{16}$ |

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|--|---|
| 1. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 2. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 3. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 4. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 7. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 10. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 11. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 12. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 13. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 14. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 15. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 16. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |

206

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|-----|--|---------------------------------|
| 1. | 15 common gauge nail penetrate a 2" green mast? | $1 - 1\frac{9}{16}$ |
| 2. | 9 penny nail penetrate a $\frac{3}{8}$ " dry joist? | $2\frac{3}{4} - \frac{5}{16}$ |
| 3. | 12 penny nail penetrate a 4" green timber? | $3\frac{1}{4} - 3\frac{9}{16}$ |
| 4. | 14 common gauge nail penetrate a $\frac{3}{8}$ " dry pole? | $1\frac{1}{4} - \frac{5}{16}$ |
| 5. | 9 penny nail penetrate a $\frac{3}{8}$ " green girder? | $2\frac{3}{4} - 1\frac{11}{32}$ |
| 6. | 8 box gauge nail penetrate a 4" green joist? | $5 - 3\frac{9}{16}$ |
| 7. | 40 penny nail penetrate a 2" dry frame? | $5 - 1\frac{1}{2}$ |
| 8. | 50 penny nail penetrate a $2\frac{1}{2}$ " green girder? | $5\frac{1}{2} - 2\frac{1}{16}$ |
| 9. | 50 penny nail penetrate a 4" green plank? | $5\frac{1}{2} - 3\frac{9}{16}$ |
| 10. | $11\frac{1}{2}$ common gauge nail penetrate a $2\frac{1}{2}$ " green mast? | $2 - 2\frac{1}{16}$ |
| 11. | 15 common gauge nail penetrate a $\frac{3}{4}$ " green joist? | $1 - 1\frac{11}{16}$ |
| 12. | $15\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{2}$ " dry rafter? | $1 - 1\frac{1}{4}$ |
| 13. | 8 box gauge nail penetrate a $\frac{5}{8}$ " dry joist? | $5 - \frac{9}{16}$ |
| 14. | 2 common gauge nail penetrate a 3" dry rafter? | $6 - 2\frac{1}{2}$ |
| 15. | 9 penny nail penetrate a 1" green rafter? | $2\frac{3}{4} - 2\frac{25}{32}$ |
| 16. | 30 penny nail penetrate a $\frac{3}{8}$ " dry rafter? | $4\frac{1}{2} - \frac{5}{16}$ |

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|---|--|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 3. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 7. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 8. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 9. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 10. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 11. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 12. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 13. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 14. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 15. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 16. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |

207

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|-----|---|--------------------------------|
| 1. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{1}{2}$ " dry plank? | $2\frac{1}{2} - \frac{7}{16}$ |
| 2. | 10 $\frac{1}{4}$ common gauge nail penetrate a 2" dry joist? | $2\frac{1}{2} - 1\frac{1}{2}$ |
| 3. | 20 penny nail penetrate a 4 $\frac{1}{2}$ " green plank? | $4 - 4\frac{1}{16}$ |
| 4. | 60 penny nail penetrate a $\frac{3}{8}$ " green girder? | $6 - \frac{11}{32}$ |
| 5. | 40 penny nail penetrate a 2" green beam? | $5 - 1\frac{9}{16}$ |
| 6. | 16 penny nail penetrate a $\frac{3}{4}$ " green joist? | $3\frac{1}{2} - \frac{11}{16}$ |
| 7. | 60 penny nail penetrate a $\frac{3}{8}$ " green beam? | $6 - \frac{11}{32}$ |
| 8. | 12 penny nail penetrate a 2" green frame? | $3\frac{1}{4} - 1\frac{9}{16}$ |
| 9. | 7 penny nail penetrate a 2" dry timber? | $2\frac{1}{4} - 1\frac{1}{2}$ |
| 10. | 10 $\frac{1}{4}$ common gauge nail penetrate a 2" green dimension? | $2\frac{1}{2} - 1\frac{9}{16}$ |
| 11. | 8 penny nail penetrate a 4 $\frac{1}{2}$ " green joist? | $2\frac{1}{2} - 4\frac{1}{16}$ |
| 12. | 12 $\frac{1}{2}$ common gauge nail penetrate a 1 $\frac{1}{4}$ " green board? | $1\frac{1}{2} - 1\frac{1}{32}$ |
| 13. | 5 penny nail penetrate a $\frac{3}{4}$ " green pole? | $1\frac{3}{4} - \frac{11}{16}$ |
| 14. | 60 penny nail penetrate a $\frac{1}{2}$ " green mast? | $6 - \frac{15}{32}$ |
| 15. | 50 penny nail penetrate a $\frac{3}{4}$ " dry dimension? | $5\frac{1}{2} - \frac{5}{8}$ |
| 16. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green girder? | $3 - \frac{11}{16}$ |

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|--|---|
| 1. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 4. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 5. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 6. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 7. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 8. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 9. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 10. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 12. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 13. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 14. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 15. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 16. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |

208

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|-----|--|----------------------------------|
| 1. | 10 penny nail penetrate a 2" dry rafter? | $3 - 1 \frac{1}{2}$ |
| 2. | 12 penny nail penetrate a $\frac{3}{4}$ " green board? | $3 \frac{1}{4} - \frac{11}{16}$ |
| 3. | 12 penny nail penetrate a $2 \frac{1}{2}$ " dry pole? | $3 \frac{1}{4} - 2$ |
| 4. | 4 penny nail penetrate a 2" dry mast? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 5. | 9 common gauge nail penetrate a 2" green pole? | $3 - 1 \frac{9}{16}$ |
| 6. | $11 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green board? | $2 \frac{1}{2} - \frac{11}{32}$ |
| 7. | 30 penny nail penetrate a $\frac{3}{4}$ " green plank? | $4 \frac{1}{2} - \frac{11}{16}$ |
| 8. | 50 penny nail penetrate a 2" dry pole? | $5 \frac{1}{2} - 1 \frac{1}{2}$ |
| 9. | $10 \frac{1}{4}$ common gauge nail penetrate a $3 \frac{1}{2}$ " dry pole? | $2 \frac{1}{2} - 3$ |
| 10. | 4 penny nail penetrate a 3" green plank? | $1 \frac{1}{2} - 2 \frac{9}{16}$ |
| 11. | 9 common gauge nail penetrate a $1 \frac{1}{4}$ " dry mast? | $3 - 1$ |
| 12. | 2 penny nail penetrate a 2" green rafter? | $1 - 1 \frac{9}{16}$ |
| 13. | $15 \frac{1}{2}$ box gauge nail penetrate a $4 \frac{1}{2}$ " green dimension? | $1 - 4 \frac{1}{16}$ |
| 14. | 40 penny nail penetrate a 4" dry frame? | $5 - 3 \frac{1}{2}$ |
| 15. | 6 penny nail penetrate a 1" green pole? | $2 - \frac{25}{32}$ |
| 16. | 3 common gauge nail penetrate a $4 \frac{1}{2}$ " green mast? | $5 \frac{1}{2} - 4 \frac{1}{16}$ |

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|--|---|
| 1. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 4. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 5. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 6. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 7. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 8. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 12. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 13. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 14. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 15. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 16. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |

209

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|-----|--|---------------------------------|
| 1. | 9 penny nail penetrate a $\frac{1}{2}$ " dry board? | $2 \frac{3}{4} - \frac{7}{16}$ |
| 2. | 9 penny nail penetrate a $\frac{3}{8}$ " dry board? | $2 \frac{3}{4} - \frac{5}{16}$ |
| 3. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry pole? | $2 \frac{1}{2} - \frac{5}{16}$ |
| 4. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry girder? | $2 - \frac{5}{8}$ |
| 5. | 20 penny nail penetrate a $1 \frac{1}{4}$ " green frame? | $4 - 1\frac{1}{32}$ |
| 6. | 7 penny nail penetrate a 4" green mast? | $2 \frac{1}{4} - 3\frac{9}{16}$ |
| 7. | 6 penny nail penetrate a $\frac{5}{8}$ " green joist? | $2 - \frac{19}{32}$ |
| 8. | 12 penny nail penetrate a $\frac{5}{8}$ " dry plank? | $3 \frac{1}{4} - \frac{9}{16}$ |
| 9. | 20 penny nail penetrate a $3 \frac{1}{2}$ " green rafter? | $4 - 3\frac{1}{16}$ |
| 10. | 4 penny nail penetrate a 2" dry dimension? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 11. | 6 penny nail penetrate a $1 \frac{1}{4}$ " dry timber? | $2 - 1$ |
| 12. | 12 $\frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " dry rafter? | $2 - 2$ |
| 13. | 10 $\frac{1}{2}$ box gauge nail penetrate a 1" green frame? | $3 - \frac{25}{32}$ |
| 14. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green dimension? | $2 \frac{1}{2} - 1\frac{1}{16}$ |
| 15. | 30 penny nail penetrate a 3" dry rafter? | $4 \frac{1}{2} - 2 \frac{1}{2}$ |
| 16. | 10 $\frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry joist? | $3 - 1$ |

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| 1. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 2. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 3. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 4. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 5. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 6. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 7. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 8. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 9. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 11. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 12. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 13. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 14. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 15. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 16. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |

210

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|-----|---|--------------------------------|
| 1. | 8 penny nail penetrate a 3/8" dry mast? | $2\frac{1}{2} - \frac{5}{16}$ |
| 2. | 14 ½ box gauge nail penetrate a 3" dry mast? | $1\frac{1}{4} - 2\frac{1}{2}$ |
| 3. | 16 penny nail penetrate a 5/8" green dimension? | $3\frac{1}{2} - \frac{19}{32}$ |
| 4. | 5 common gauge nail penetrate a 1" dry joist? | $4\frac{1}{2} - \frac{3}{4}$ |
| 5. | 8 box gauge nail penetrate a 3 ½" dry dimension? | $5 - 3$ |
| 6. | 10 penny nail penetrate a 3" green dimension? | $3 - 2\frac{9}{16}$ |
| 7. | 60 penny nail penetrate a 3/4" dry rafter? | $6 - \frac{5}{8}$ |
| 8. | 20 penny nail penetrate a 3/8" dry dimension? | $4 - \frac{5}{16}$ |
| 9. | 10 ¼ common gauge nail penetrate a 2" dry rafter? | $2\frac{1}{2} - 1\frac{1}{2}$ |
| 10. | 5 common gauge nail penetrate a 5/8" dry plank? | $4\frac{1}{2} - \frac{9}{16}$ |
| 11. | 4 common gauge nail penetrate a 5/8" dry beam? | $5 - \frac{9}{16}$ |
| 12. | 11 ½ box gauge nail penetrate a 1 ½" dry dimension? | $2\frac{1}{2} - 1\frac{1}{4}$ |
| 13. | 11 ½ common gauge nail penetrate a 3/8" green rafter? | $2 - \frac{11}{32}$ |
| 14. | 10 penny nail penetrate a 2 ½" dry board? | $3 - 2$ |
| 15. | 30 penny nail penetrate a 2 ½" dry pole? | $4\frac{1}{2} - 2$ |
| 16. | 3 penny nail penetrate a 1" green timber? | $1\frac{1}{4} - \frac{25}{32}$ |

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|--|-------------------------------|
| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 3. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 4. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 5. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 6. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 7. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 8. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 9. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 10. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 11. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 12. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 13. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 14. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 15. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 16. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |

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|-----|---|-----------------------------------|
| 1. | 5 penny nail penetrate a 4" green timber? | $1 \frac{3}{4} - 3 \frac{9}{16}$ |
| 2. | 6 penny nail penetrate a 1 $\frac{1}{2}$ " green girder? | $2 - 1 \frac{9}{32}$ |
| 3. | 4 penny nail penetrate a 1" green mast? | $1 \frac{1}{2} - 2 \frac{5}{32}$ |
| 4. | 4 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $1 \frac{1}{2} - 1 \frac{15}{32}$ |
| 5. | 50 penny nail penetrate a 2" green frame? | $5 \frac{1}{2} - 1 \frac{9}{16}$ |
| 6. | 6 penny nail penetrate a $\frac{5}{8}$ " green pole? | $2 - 1 \frac{19}{32}$ |
| 7. | 7 penny nail penetrate a 1 $\frac{1}{2}$ " dry rafter? | $2 \frac{1}{4} - 1 \frac{1}{4}$ |
| 8. | 14 box gauge nail penetrate a 1 $\frac{1}{4}$ " dry pole? | $1 \frac{1}{2} - 1$ |
| 9. | 60 penny nail penetrate a 1 $\frac{1}{4}$ " dry mast? | $6 - 1$ |
| 10. | 5 common gauge nail penetrate a 4" dry pole? | $4 \frac{1}{2} - 3 \frac{1}{2}$ |
| 11. | 7 penny nail penetrate a 2" green beam? | $2 \frac{1}{4} - 1 \frac{9}{16}$ |
| 12. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green pole? | $2 \frac{1}{2} - 1 \frac{11}{32}$ |
| 13. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry mast? | $2 - \frac{5}{16}$ |
| 14. | 5 penny nail penetrate a 1" green joist? | $1 \frac{3}{4} - 2 \frac{25}{32}$ |
| 15. | 60 penny nail penetrate a 4 $\frac{1}{2}$ " dry rafter? | $6 - 4$ |
| 16. | 16 penny nail penetrate a 4" green beam? | $3 \frac{1}{2} - 3 \frac{9}{16}$ |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 3. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 4. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 7. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 9. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 10. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 11. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 12. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 13. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 14. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 15. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 16. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |

212

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|-----|---|--------------------------------|
| 1. | 30 penny nail penetrate a $\frac{3}{4}$ " green mast? | $4\frac{1}{2} - \frac{11}{16}$ |
| 2. | 5 penny nail penetrate a $\frac{1}{2}$ " dry girder? | $1\frac{3}{4} - \frac{7}{16}$ |
| 3. | $12\frac{1}{2}$ common gauge nail penetrate a $2\frac{1}{2}$ " dry board? | $1\frac{1}{2} - 2$ |
| 4. | 14 box gauge nail penetrate a $\frac{5}{8}$ " dry rafter? | $1\frac{1}{2} - \frac{9}{16}$ |
| 5. | 9 penny nail penetrate a $\frac{5}{8}$ " dry frame? | $2\frac{3}{4} - \frac{9}{16}$ |
| 6. | 14 common gauge nail penetrate a $1\frac{1}{2}$ " dry timber? | $1\frac{1}{4} - 1\frac{1}{4}$ |
| 7. | $12\frac{1}{2}$ box gauge nail penetrate a 4" dry mast? | $2 - 3\frac{1}{2}$ |
| 8. | 20 penny nail penetrate a $\frac{1}{2}$ " green pole? | $4 - \frac{15}{32}$ |
| 9. | 30 penny nail penetrate a $\frac{5}{8}$ " dry girder? | $4\frac{1}{2} - \frac{9}{16}$ |
| 10. | 5 penny nail penetrate a $\frac{1}{2}$ " dry board? | $1\frac{3}{4} - \frac{7}{16}$ |
| 11. | 6 penny nail penetrate a $4\frac{1}{2}$ " dry beam? | $2 - 4$ |
| 12. | $12\frac{1}{2}$ box gauge nail penetrate a 1" dry plank? | $2 - \frac{3}{4}$ |
| 13. | 14 common gauge nail penetrate a 2" green beam? | $1\frac{1}{4} - 1\frac{9}{16}$ |
| 14. | 3 penny nail penetrate a $1\frac{1}{4}$ " green rafter? | $1\frac{1}{4} - 1\frac{1}{32}$ |
| 15. | 8 penny nail penetrate a 3" dry timber? | $2\frac{1}{2} - 2\frac{1}{2}$ |
| 16. | 8 box gauge nail penetrate a $4\frac{1}{2}$ " green girder? | $5 - 4\frac{1}{16}$ |

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|--|---|
| 1. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 2. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 3. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 4. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 5. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 6. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 7. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 8. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 9. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 10. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 11. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 15. countersink hole for a 24 gauge screw? | $\frac{1}{2}$ $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 16. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |

213

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|-----|--|----------------|
| 1. | 5 penny nail penetrate a 3 1/2" dry board? | 1 3/4 — 3 |
| 2. | 11 1/2 common gauge nail penetrate a 5/8" dry mast? | 2 — 9/16 |
| 3. | 15 common gauge nail penetrate a 2" dry girder? | 1 — 1 1/2 |
| 4. | 14 box gauge nail penetrate a 3 1/2" green joist? | 1 1/2 — 3 1/16 |
| 5. | 10 penny nail penetrate a 5/8" dry rafter? | 3 — 9/16 |
| 6. | 3 penny nail penetrate a 1 1/2" green rafter? | 1 1/4 — 1 9/32 |
| 7. | 20 penny nail penetrate a 3/4" green joist? | 4 — 11/16 |
| 8. | 12 1/2 common gauge nail penetrate a 3/4" green frame? | 1 1/2 — 11/16 |
| 9. | 10 1/2 box gauge nail penetrate a 3/4" green joist? | 3 — 11/16 |
| 10. | 30 penny nail penetrate a 2 1/2" dry board? | 4 1/2 — 2 |
| 11. | 8 box gauge nail penetrate a 5/8" green frame? | 5 — 19/32 |
| 12. | 30 penny nail penetrate a 3/4" green rafter? | 4 1/2 — 11/16 |
| 13. | 11 1/2 box gauge nail penetrate a 1/2" dry girder? | 2 1/2 — 7/16 |
| 14. | 10 1/4 common gauge nail penetrate a 3" dry timber? | 2 1/2 — 2 1/2 |
| 15. | 14 box gauge nail penetrate a 1 1/4" green joist? | 1 1/2 — 1 1/32 |
| 16. | 10 1/2 box gauge nail penetrate a 5/8" dry mast? | 3 — 9/16 |

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|--|---|
| 1. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 2. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 3. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 4. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 5. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 6. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 7. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 8. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 9. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 10. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ N $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 11. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 12. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 13. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 14. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 15. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 16. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |

214

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|-----|---|-------------|
| 1. | 9 penny nail penetrate a 1 ½" dry rafter? | 2 ¾ — 1 ¼ |
| 2. | 12 ½ common gauge nail penetrate a 2" dry plank? | 1 ½ — 1 ½ |
| 3. | 60 penny nail penetrate a 1 ¼" dry dimension? | 6 — 1 |
| 4. | 10 ½ box gauge nail penetrate a 4" dry girder? | 3 — 3 ½ |
| 5. | 8 box gauge nail penetrate a 5/8" green pole? | 5 — 19/32 |
| 6. | 12 ½ common gauge nail penetrate a 3 ½" dry joist? | 1 ½ — 3 |
| 7. | 12 ½ box gauge nail penetrate a 3" dry pole? | 2 — 2 ½ |
| 8. | 4 penny nail penetrate a 1" green girder? | 1 ½ — 25/32 |
| 9. | 11 ½ common gauge nail penetrate a 1 ¼" green rafter? | 2 — 11/32 |
| 10. | 9 penny nail penetrate a 3/8" green pole? | 2 ¾ — 11/32 |
| 11. | 8 penny nail penetrate a 1 ½" dry rafter? | 2 ½ — 1 ¼ |
| 12. | 4 penny nail penetrate a ½" dry timber? | 1 ½ — 7/16 |
| 13. | 16 penny nail penetrate a 1" green frame? | 3 ½ — 25/32 |
| 14. | 3 penny nail penetrate a 3/4" green rafter? | 1 ¼ — 11/16 |
| 15. | 8 common gauge nail penetrate a 4" dry pole? | 3 ½ — 3 ½ |
| 16. | 9 box gauge nail penetrate a ½" dry plank? | 4 — 7/16 |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 3. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 4. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 5. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 6. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 7. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 8. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 10. countersink hole for a 24 gauge screw? | $\frac{1}{2} \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 11. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 12. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 13. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 14. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 15. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 16. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |

215

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|-----|--|-----------------------------------|
| 1. | 50 penny nail penetrate a 5/8" dry girder? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 2. | 8 penny nail penetrate a 4 1/2" green beam? | $2 \frac{1}{2} - 4 \frac{1}{16}$ |
| 3. | 4 penny nail penetrate a 4" green timber? | $1 \frac{1}{2} - 3 \frac{9}{16}$ |
| 4. | 16 penny nail penetrate a 1" green mast? | $3 \frac{1}{2} - 2 \frac{5}{32}$ |
| 5. | 8 box gauge nail penetrate a 2 1/2" green pole? | $5 - 2 \frac{1}{16}$ |
| 6. | 9 common gauge nail penetrate a 3/4" green beam? | $3 - 1 \frac{1}{16}$ |
| 7. | 12 1/2 common gauge nail penetrate a 1 1/4" green joist? | $1 \frac{1}{2} - 1 \frac{1}{32}$ |
| 8. | 6 penny nail penetrate a 3/8" green plank? | $2 - 1 \frac{1}{32}$ |
| 9. | 60 penny nail penetrate a 1/2" dry frame? | $6 - \frac{7}{16}$ |
| 10. | 14 common gauge nail penetrate a 1/2" green board? | $1 \frac{1}{4} - 1 \frac{15}{32}$ |
| 11. | 14 box gauge nail penetrate a 2" dry board? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 12. | 8 box gauge nail penetrate a 1 1/2" green beam? | $5 - 1 \frac{9}{32}$ |
| 13. | 9 penny nail penetrate a 1 1/4" dry plank? | $2 \frac{3}{4} - 1$ |
| 14. | 10 1/4 common gauge nail penetrate a 1 1/2" dry board? | $2 \frac{1}{2} - 1 \frac{1}{4}$ |
| 15. | 7 penny nail penetrate a 4 1/2" dry board? | $2 \frac{1}{4} - 4$ |
| 16. | 11 1/2 common gauge nail penetrate a 2 1/2" green pole? | $2 - 2 \frac{1}{16}$ |

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|--|--------------------------|
| 1. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 2. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 3. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 4. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 5. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 6. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 7. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 8. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 9. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 10. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 11. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 12. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 13. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 14. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 15. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 16. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |

216

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|-----|--|--------------|
| 1. | 10 ½ box gauge nail penetrate a 3" dry mast? | 3 — 2 ½ |
| 2. | 6 penny nail penetrate a 3" dry pole? | 2 — 2 ½ |
| 3. | 40 penny nail penetrate a 1" green board? | 5 — 25/32 |
| 4. | 8 penny nail penetrate a 1 ½" dry plank? | 2 ½ — 1 ¼ |
| 5. | 9 box gauge nail penetrate a 1" dry beam? | 4 — ¾ |
| 6. | 12 ½ common gauge nail penetrate a 1" dry timber? | 1 ½ — ¾ |
| 7. | 11 ½ common gauge nail penetrate a 5/8" dry mast? | 2 — 9/16 |
| 8. | 10 penny nail penetrate a 5/8" green girder? | 3 — 19/32 |
| 9. | 10 ¼ common gauge nail penetrate a ½" green mast? | 2 ½ — 15/32 |
| 10. | 9 common gauge nail penetrate a 5/8" dry girder? | 3 — 9/16 |
| 11. | 15 ½ box gauge nail penetrate a 1 ¼" dry board? | 1 — 1 |
| 12. | 2 common gauge nail penetrate a 3 ½" green board? | 6 — 3 1/16 |
| 13. | 16 penny nail penetrate a 4" dry plank? | 3 ½ — 3 ½ |
| 14. | 8 box gauge nail penetrate a 3/4" green dimension? | 5 — 11/16 |
| 15. | 40 penny nail penetrate a 2 ½" dry girder? | 5 — 2 |
| 16. | 16 penny nail penetrate a 2 ½" green rafter? | 3 ½ — 2 1/16 |

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|--|-------------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 3. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 4. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 5. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 6. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 7. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 8. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 9. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 10. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 11. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 12. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 13. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 15. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 16. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |

217

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|-----|---|------------------------------------|
| 1. | 2 penny nail penetrate a 4 ½" dry rafter? | 1 — 4 |
| 2. | 11 ½ box gauge nail penetrate a 1" dry board? | 2 ½ — ¾ |
| 3. | 10 ½ box gauge nail penetrate a 4" green joist? | 3 — 3 ⁹ / ₁₆ |
| 4. | 11 ½ common gauge nail penetrate a 5/8" green rafter? | 2 — 19/32 |
| 5. | 11 ½ common gauge nail penetrate a 1" dry beam? | 2 — ¾ |
| 6. | 30 penny nail penetrate a ½" dry dimension? | 4 ½ — 7/16 |
| 7. | 14 ½ box gauge nail penetrate a 2 ½" dry dimension? | 1 ¼ — 2 |
| 8. | 50 penny nail penetrate a 5/8" green frame? | 5 ½ — 19/32 |
| 9. | 6 penny nail penetrate a 4 ½" green timber? | 2 — 4 ¹ / ₁₆ |
| 10. | 11 ½ box gauge nail penetrate a 3/4" green girder? | 2 ½ — 11/16 |
| 11. | 6 common gauge nail penetrate a 2 ½" dry girder? | 4 — 2 |
| 12. | 4 common gauge nail penetrate a 1 ¼" green rafter? | 5 — 1 ¹ / ₃₂ |
| 13. | 3 penny nail penetrate a 3" dry board? | 1 ¼ — 2 ½ |
| 14. | 15 common gauge nail penetrate a 1 ¼" dry dimension? | 1 — 1 |
| 15. | 30 penny nail penetrate a 3/8" dry pole? | 4 ½ — 5/16 |
| 16. | 11 ½ box gauge nail penetrate a 2" dry joist? | 2 ½ — 1 ½ |

1.	clearance hole for a 9 gauge screw?	17	$11/64 = 11/64$
2.	countersink hole for a 2 gauge screw?	2	$3/32 + 1/8 = 7/32$
3.	pilot hole for a 2 gauge screw?	67	$3/32 - 1/16 = 1/32$
4.	pilot hole for a 20 gauge screw?	H	$21/64 - 1/16 = 17/64$
5.	pilot hole for a 3 gauge screw?	67	$3/32 - 1/16 = 1/32$
6.	countersink hole for a 14 gauge screw?	V	$1/4 + 1/8 = 3/8$
7.	clearance hole for a 16 gauge screw?	H	$17/64 = 17/64$
8.	pilot hole for a 6 gauge screw?	47	$9/64 - 1/16 = 5/64$
9.	clearance hole for a 8 gauge screw?	17	$11/64 = 11/64$
10.	pilot hole for a 7 gauge screw?	41	$5/32 - 1/16 = 3/32$
11.	pilot hole for a 8 gauge screw?	35	$11/64 - 1/16 = 7/64$
12.	countersink hole for a 6 gauge screw?	H	$9/64 + 1/8 = 17/64$
13.	countersink hole for a 4 gauge screw?	B	$7/64 + 1/8 = 15/64$
14.	countersink hole for a 4 gauge screw?	B	$7/64 + 1/8 = 15/64$
15.	clearance hole for a 8 gauge screw?	17	$11/64 = 11/64$
16.	clearance hole for a 3 gauge screw?	41	$3/32 = 3/32$

218

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|-----|---|------------------------------------|
| 1. | 9 penny nail penetrate a 4" green dimension? | $2 \frac{3}{4} - 3 \frac{9}{16}$ |
| 2. | 16 penny nail penetrate a $\frac{1}{2}$ " green timber? | $3 \frac{1}{2} - 15 \frac{15}{32}$ |
| 3. | 14 common gauge nail penetrate a $4 \frac{1}{2}$ " dry beam? | $1 \frac{1}{4} - 4$ |
| 4. | 9 box gauge nail penetrate a $4 \frac{1}{2}$ " dry rafter? | $4 - 4$ |
| 5. | 8 penny nail penetrate a $2 \frac{1}{2}$ " green dimension? | $2 \frac{1}{2} - 2 \frac{1}{16}$ |
| 6. | 15 common gauge nail penetrate a $\frac{5}{8}$ " dry mast? | $1 - \frac{9}{16}$ |
| 7. | 9 common gauge nail penetrate a 1" dry board? | $3 - \frac{3}{4}$ |
| 8. | 9 penny nail penetrate a $\frac{1}{2}$ " dry dimension? | $2 \frac{3}{4} - 7 \frac{7}{16}$ |
| 9. | 8 penny nail penetrate a 1" dry joist? | $2 \frac{1}{2} - \frac{3}{4}$ |
| 10. | 5 common gauge nail penetrate a $3 \frac{1}{2}$ " green rafter? | $4 \frac{1}{2} - 3 \frac{1}{16}$ |
| 11. | 4 penny nail penetrate a 1" green joist? | $1 \frac{1}{2} - 25 \frac{25}{32}$ |
| 12. | 16 penny nail penetrate a $\frac{1}{2}$ " green frame? | $3 \frac{1}{2} - 15 \frac{15}{32}$ |
| 13. | 14 box gauge nail penetrate a $3 \frac{1}{2}$ " green mast? | $1 \frac{1}{2} - 3 \frac{1}{16}$ |
| 14. | 7 penny nail penetrate a $3 \frac{1}{2}$ " green board? | $2 \frac{1}{4} - 3 \frac{1}{16}$ |
| 15. | 5 common gauge nail penetrate a $1 \frac{1}{2}$ " green board? | $4 \frac{1}{2} - 19 \frac{19}{32}$ |
| 16. | 60 penny nail penetrate a $2 \frac{1}{2}$ " dry plank? | $6 - 2$ |

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|---|-------------------------------|
| 1. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 2. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 3. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 4. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 5. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 6. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 7. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 8. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 9. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 10. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 11. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 12. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 13. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 14. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 15. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 16. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |

219

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|-----|--|----------------|
| 1. | 9 box gauge nail penetrate a 1 1/2" green joist? | 4 — 1 9/32 |
| 2. | 12 penny nail penetrate a 2 1/2" green girder? | 3 1/4 — 2 1/16 |
| 3. | 5 common gauge nail penetrate a 3/8" green joist? | 4 1/2 — 11/32 |
| 4. | 2 penny nail penetrate a 3" green beam? | 1 — 2 9/16 |
| 5. | 11 1/2 box gauge nail penetrate a 2 1/2" dry timber? | 2 1/2 — 2 |
| 6. | 60 penny nail penetrate a 2" dry beam? | 6 — 1 1/2 |
| 7. | 9 penny nail penetrate a 2 1/2" green board? | 2 3/4 — 2 1/16 |
| 8. | 10 1/4 common gauge nail penetrate a 1" dry pole? | 2 1/2 — 3/4 |
| 9. | 8 penny nail penetrate a 4" dry timber? | 2 1/2 — 3 1/2 |
| 10. | 50 penny nail penetrate a 3/4" dry plank? | 5 1/2 — 5/8 |
| 11. | 50 penny nail penetrate a 3 1/2" green girder? | 5 1/2 — 3 1/16 |
| 12. | 30 penny nail penetrate a 4 1/2" green dimension? | 4 1/2 — 4 1/16 |
| 13. | 12 penny nail penetrate a 3/8" green dimension? | 3 1/4 — 11/32 |
| 14. | 2 common gauge nail penetrate a 2" green frame? | 6 — 1 9/16 |
| 15. | 20 penny nail penetrate a 4 1/2" dry rafter? | 4 — 4 |
| 16. | 8 box gauge nail penetrate a 4" dry joist? | 5 — 3 1/2 |

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|--|-------------------------------|
| 1. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 2. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 3. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 4. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 5. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 6. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 7. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 8. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 9. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 10. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 11. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 12. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 13. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 14. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 15. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 16. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |

220

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|-----|---|---------------------------------|
| 1. | 5 common gauge nail penetrate a $5/8$ " dry mast? | $4\frac{1}{2} - \frac{9}{16}$ |
| 2. | 12 penny nail penetrate a $5/8$ " dry pole? | $3\frac{1}{4} - \frac{9}{16}$ |
| 3. | 7 penny nail penetrate a 4" dry pole? | $2\frac{1}{4} - 3\frac{1}{2}$ |
| 4. | 9 penny nail penetrate a $1/2$ " dry mast? | $2\frac{3}{4} - \frac{7}{16}$ |
| 5. | $10\frac{1}{2}$ box gauge nail penetrate a $3/4$ " dry girder? | $3 - \frac{5}{8}$ |
| 6. | $10\frac{1}{4}$ common gauge nail penetrate a 3" dry board? | $2\frac{1}{2} - 2\frac{1}{2}$ |
| 7. | 6 common gauge nail penetrate a 2" green mast? | $4 - 1\frac{9}{16}$ |
| 8. | 2 penny nail penetrate a $3/4$ " green joist? | $1 - 1\frac{11}{16}$ |
| 9. | 16 penny nail penetrate a 2" dry mast? | $3\frac{1}{2} - 1\frac{1}{2}$ |
| 10. | 20 penny nail penetrate a $2\frac{1}{2}$ " green beam? | $4 - 2\frac{1}{16}$ |
| 11. | 20 penny nail penetrate a $3/8$ " green pole? | $4 - 1\frac{11}{32}$ |
| 12. | 6 penny nail penetrate a $1/2$ " green dimension? | $2 - 1\frac{15}{32}$ |
| 13. | 14 common gauge nail penetrate a 1" green board? | $1\frac{1}{4} - 2\frac{25}{32}$ |
| 14. | 4 penny nail penetrate a $4\frac{1}{2}$ " dry frame? | $1\frac{1}{2} - 4$ |
| 15. | 50 penny nail penetrate a $4\frac{1}{2}$ " green dimension? | $5\frac{1}{2} - 4\frac{1}{16}$ |
| 16. | $10\frac{1}{2}$ box gauge nail penetrate a $1/2$ " dry dimension? | $3 - \frac{7}{16}$ |

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|--|---|
| 1. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 2. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 3. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 4. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 5. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 6. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 8. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 12. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 13. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 14. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 16. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

221

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|-----|--|---------------|
| 1. | 8 box gauge nail penetrate a 2 1/2" green plank? | 5 — 2 1/16 |
| 2. | 60 penny nail penetrate a 3/8" green board? | 6 — 11/32 |
| 3. | 10 penny nail penetrate a 5/8" green timber? | 3 — 19/32 |
| 4. | 4 penny nail penetrate a 2" dry girder? | 1 1/2 — 1 1/2 |
| 5. | 3 penny nail penetrate a 1/2" green rafter? | 1 1/4 — 15/32 |
| 6. | 50 penny nail penetrate a 5/8" green frame? | 5 1/2 — 19/32 |
| 7. | 4 common gauge nail penetrate a 1 1/2" green plank? | 5 — 19/32 |
| 8. | 12 1/2 box gauge nail penetrate a 5/8" dry joist? | 2 — 9/16 |
| 9. | 8 box gauge nail penetrate a 4 1/2" green joist? | 5 — 4 1/16 |
| 10. | 12 penny nail penetrate a 5/8" green dimension? | 3 1/4 — 19/32 |
| 11. | 10 1/2 box gauge nail penetrate a 1 1/4" dry girder? | 3 — 1 |
| 12. | 14 common gauge nail penetrate a 3/4" green dimension? | 1 1/4 — 11/16 |
| 13. | 9 box gauge nail penetrate a 1 1/2" green plank? | 4 — 19/32 |
| 14. | 4 penny nail penetrate a 3" green rafter? | 1 1/2 — 29/16 |
| 15. | 9 common gauge nail penetrate a 4" dry joist? | 3 — 3 1/2 |
| 16. | 10 1/2 box gauge nail penetrate a 5/8" dry dimension? | 3 — 9/16 |

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|---|--------------------------|
| 1. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 2. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 3. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 4. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 5. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 6. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 7. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 8. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 9. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 10. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 11. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 12. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 13. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 14. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 15. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 16. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |

222

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|-----|---|----------------|
| 1. | 10 penny nail penetrate a 2 1/2" dry dimension? | 3 — 2 |
| 2. | 9 penny nail penetrate a 2" dry beam? | 2 3/4 — 1 1/2 |
| 3. | 14 box gauge nail penetrate a 3" dry plank? | 1 1/2 — 2 1/2 |
| 4. | 15 1/2 box gauge nail penetrate a 5/8" dry frame? | 1 — 9/16 |
| 5. | 5 penny nail penetrate a 2" dry rafter? | 1 3/4 — 1 1/2 |
| 6. | 16 penny nail penetrate a 4 1/2" green rafter? | 3 1/2 — 4 1/16 |
| 7. | 50 penny nail penetrate a 1" green frame? | 5 1/2 — 25/32 |
| 8. | 8 penny nail penetrate a 1 1/2" green girder? | 2 1/2 — 19/32 |
| 9. | 60 penny nail penetrate a 4" green girder? | 6 — 39/16 |
| 10. | 30 penny nail penetrate a 5/8" green plank? | 4 1/2 — 19/32 |
| 11. | 10 penny nail penetrate a 4" dry mast? | 3 — 3 1/2 |
| 12. | 40 penny nail penetrate a 3 1/2" green beam? | 5 — 3 1/16 |
| 13. | 8 penny nail penetrate a 2" green joist? | 2 1/2 — 19/16 |
| 14. | 12 1/2 box gauge nail penetrate a 4 1/2" green dimension? | 2 — 4 1/16 |
| 15. | 5 penny nail penetrate a 3" green board? | 1 3/4 — 29/16 |
| 16. | 3 common gauge nail penetrate a 3/4" green beam? | 5 1/2 — 11/16 |

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|---|-------------------------|
| 1. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 2. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 3. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 4. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 5. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 6. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 7. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 8. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 9. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 10. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 11. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 12. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 13. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 14. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 15. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 16. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |

223

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|-----|---|----------------------------------|
| 1. | 60 penny nail penetrate a 4" dry timber? | $6 - 3 \frac{1}{2}$ |
| 2. | 60 penny nail penetrate a $1 \frac{1}{4}$ " dry rafter? | $6 - 1$ |
| 3. | 16 penny nail penetrate a 4" dry board? | $3 \frac{1}{2} - 3 \frac{1}{2}$ |
| 4. | 8 common gauge nail penetrate a 3" green joist? | $3 \frac{1}{2} - 2 \frac{9}{16}$ |
| 5. | 50 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 6. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry rafter? | $3 - \frac{7}{16}$ |
| 7. | 14 common gauge nail penetrate a 1" green beam? | $1 \frac{1}{4} - \frac{25}{32}$ |
| 8. | 8 box gauge nail penetrate a $1 \frac{1}{2}$ " dry plank? | $5 - 1 \frac{1}{4}$ |
| 9. | 6 penny nail penetrate a $4 \frac{1}{2}$ " dry timber? | $2 - 4$ |
| 10. | 8 box gauge nail penetrate a 4" dry frame? | $5 - 3 \frac{1}{2}$ |
| 11. | $11 \frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " dry board? | $2 - \frac{7}{16}$ |
| 12. | $10 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry girder? | $3 - 1$ |
| 13. | 15 common gauge nail penetrate a $\frac{5}{8}$ " dry plank? | $1 - \frac{9}{16}$ |
| 14. | 20 penny nail penetrate a $\frac{3}{8}$ " dry beam? | $4 - \frac{5}{16}$ |
| 15. | 8 box gauge nail penetrate a $1 \frac{1}{4}$ " dry timber? | $5 - 1$ |
| 16. | 9 box gauge nail penetrate a $\frac{5}{8}$ " dry pole? | $4 - \frac{9}{16}$ |

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|---|--------------------------|
| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 3. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 4. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 5. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 6. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 7. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 8. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 9. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 10. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 11. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 12. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 13. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 14. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 15. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 16. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |

224

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|-----|---|-----------------------------------|
| 1. | 2 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $1 - \frac{15}{32}$ |
| 2. | 16 penny nail penetrate a $4 \frac{1}{2}$ " green plank? | $3 \frac{1}{2} - 4 \frac{1}{16}$ |
| 3. | 14 common gauge nail penetrate a 2" green mast? | $1 \frac{1}{4} - 1 \frac{9}{16}$ |
| 4. | 40 penny nail penetrate a $1 \frac{1}{4}$ " green mast? | $5 - 1 \frac{1}{32}$ |
| 5. | 2 penny nail penetrate a $1 \frac{1}{2}$ " green dimension? | $1 - 1 \frac{9}{32}$ |
| 6. | $15 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry pole? | $1 - 1$ |
| 7. | 16 penny nail penetrate a $3 \frac{1}{2}$ " dry frame? | $3 \frac{1}{2} - 3$ |
| 8. | $11 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry board? | $2 \frac{1}{2} - 1$ |
| 9. | 5 common gauge nail penetrate a $3 \frac{1}{2}$ " green board? | $4 \frac{1}{2} - 3 \frac{1}{16}$ |
| 10. | 50 penny nail penetrate a $1 \frac{1}{2}$ " green board? | $5 \frac{1}{2} - 1 \frac{9}{32}$ |
| 11. | 20 penny nail penetrate a $\frac{3}{8}$ " green board? | $4 - 1 \frac{1}{32}$ |
| 12. | $10 \frac{1}{4}$ common gauge nail penetrate a 1" green board? | $2 \frac{1}{2} - 2 \frac{25}{32}$ |
| 13. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green rafter? | $3 - 1 \frac{19}{32}$ |
| 14. | 30 penny nail penetrate a $1 \frac{1}{4}$ " green frame? | $4 \frac{1}{2} - 1 \frac{1}{32}$ |
| 15. | 9 penny nail penetrate a $1 \frac{1}{4}$ " green plank? | $2 \frac{3}{4} - 1 \frac{1}{32}$ |
| 16. | 6 penny nail penetrate a 1" green girder? | $2 - 2 \frac{25}{32}$ |

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|--|---|
| 1. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 2. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 3. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 4. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 5. countersink hole for a 20 gauge screw? | $\frac{29}{64} \quad \frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 6. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 7. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 8. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 9. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 10. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 11. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 12. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 13. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 14. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 15. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 16. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |

225

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|-----|---|---------------|
| 1. | 4 penny nail penetrate a 1 1/4" dry pole? | 1 1/2 — 1 |
| 2. | 12 1/2 box gauge nail penetrate a 1 1/4" dry dimension? | 2 — 1 |
| 3. | 8 box gauge nail penetrate a 1 1/4" dry frame? | 5 — 1 |
| 4. | 14 box gauge nail penetrate a 3" dry board? | 1 1/2 — 2 1/2 |
| 5. | 8 penny nail penetrate a 3/8" green mast? | 2 1/2 — 11/32 |
| 6. | 5 common gauge nail penetrate a 3/4" green plank? | 4 1/2 — 11/16 |
| 7. | 11 1/2 common gauge nail penetrate a 3/4" dry plank? | 2 — 5/8 |
| 8. | 40 penny nail penetrate a 3/4" dry plank? | 5 — 5/8 |
| 9. | 4 common gauge nail penetrate a 2 1/2" dry pole? | 5 — 2 |
| 10. | 4 common gauge nail penetrate a 2 1/2" dry girder? | 5 — 2 |
| 11. | 60 penny nail penetrate a 4" green rafter? | 6 — 39/16 |
| 12. | 10 1/4 common gauge nail penetrate a 3/4" dry girder? | 2 1/2 — 5/8 |
| 13. | 3 penny nail penetrate a 3/8" dry plank? | 1 1/4 — 5/16 |
| 14. | 6 penny nail penetrate a 2 1/2" green frame? | 2 — 2 1/16 |
| 15. | 60 penny nail penetrate a 2 1/2" dry frame? | 6 — 2 |
| 16. | 4 penny nail penetrate a 3 1/2" dry timber? | 1 1/2 — 3 |

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|--|-------------------------------|
| 1. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 2. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 3. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 4. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 5. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 6. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 7. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 8. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 9. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 10. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 11. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 12. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 13. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 14. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 15. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 16. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |

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|-----|--|-------------------------------------|
| 1. | 10 $\frac{1}{4}$ common gauge nail penetrate a 1 $\frac{1}{4}$ " green frame? | 2 $\frac{1}{2}$ — 1 $\frac{1}{32}$ |
| 2. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " green rafter? | 2 $\frac{1}{2}$ — 1 $\frac{15}{32}$ |
| 3. | 40 penny nail penetrate a 3" dry girder? | 5 — 2 $\frac{1}{2}$ |
| 4. | 8 common gauge nail penetrate a 2" dry plank? | 3 $\frac{1}{2}$ — 1 $\frac{1}{2}$ |
| 5. | 12 $\frac{1}{2}$ common gauge nail penetrate a 1 $\frac{1}{2}$ " green girder? | 1 $\frac{1}{2}$ — 1 $\frac{9}{32}$ |
| 6. | 9 box gauge nail penetrate a $\frac{3}{4}$ " green pole? | 4 — 1 $\frac{11}{16}$ |
| 7. | 9 box gauge nail penetrate a 3" dry frame? | 4 — 2 $\frac{1}{2}$ |
| 8. | 3 common gauge nail penetrate a 4" green pole? | 5 $\frac{1}{2}$ — 3 $\frac{9}{16}$ |
| 9. | 10 $\frac{1}{4}$ common gauge nail penetrate a 4 $\frac{1}{2}$ " dry girder? | 2 $\frac{1}{2}$ — 4 |
| 10. | 12 $\frac{1}{2}$ common gauge nail penetrate a 1" green board? | 1 $\frac{1}{2}$ — 2 $\frac{25}{32}$ |
| 11. | 9 common gauge nail penetrate a $\frac{1}{2}$ " dry plank? | 3 — 7 $\frac{7}{16}$ |
| 12. | 9 box gauge nail penetrate a 1 $\frac{1}{2}$ " green frame? | 4 — 1 $\frac{9}{32}$ |
| 13. | 30 penny nail penetrate a 2" green beam? | 4 $\frac{1}{2}$ — 1 $\frac{9}{16}$ |
| 14. | 8 box gauge nail penetrate a 1" green pole? | 5 — 2 $\frac{25}{32}$ |
| 15. | 15 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry joist? | 1 — 5 $\frac{5}{16}$ |
| 16. | 12 $\frac{1}{2}$ common gauge nail penetrate a 3" green dimension? | 1 $\frac{1}{2}$ — 2 $\frac{9}{16}$ |

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|--|--------------------------|
| 1. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 2. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 4. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 5. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 6. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 7. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 8. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 9. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 10. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 11. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 12. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 13. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 14. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 15. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 16. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |

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|-----|---|--------------------------------------|
| 1. | 11 ½ box gauge nail penetrate a 2 ½" green board? | 2 ½ — 2 ¹ / ₁₆ |
| 2. | 10 ¼ common gauge nail penetrate a 5/8" dry frame? | 2 ½ — 9 ¹ / ₁₆ |
| 3. | 9 common gauge nail penetrate a 3 ½" dry girder? | 3 — 3 |
| 4. | 3 common gauge nail penetrate a 2 ½" green joist? | 5 ½ — 2 ¹ / ₁₆ |
| 5. | 60 penny nail penetrate a 3" green plank? | 6 — 2 ⁹ / ₁₆ |
| 6. | 9 box gauge nail penetrate a 3" dry plank? | 4 — 2 ½ |
| 7. | 15 common gauge nail penetrate a 2" green girder? | 1 — 1 ⁹ / ₁₆ |
| 8. | 10 ½ box gauge nail penetrate a ½" green board? | 3 — 15 ¹ / ₃₂ |
| 9. | 3 common gauge nail penetrate a 2 ½" green timber? | 5 ½ — 2 ¹ / ₁₆ |
| 10. | 2 penny nail penetrate a 2 ½" dry beam? | 1 — 2 |
| 11. | 50 penny nail penetrate a 4" green rafter? | 5 ½ — 3 ⁹ / ₁₆ |
| 12. | 9 penny nail penetrate a 2 ½" dry board? | 2 ¾ — 2 |
| 13. | 9 box gauge nail penetrate a 1 ½" dry mast? | 4 — 1 ¼ |
| 14. | 12 ½ box gauge nail penetrate a 3" green dimension? | 2 — 2 ⁹ / ₁₆ |
| 15. | 9 penny nail penetrate a 3/4" dry frame? | 2 ¾ — 5 ¹ / ₈ |
| 16. | 10 ¼ common gauge nail penetrate a 3" dry joist? | 2 ½ — 2 ½ |

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|--|-------------------------------|
| 1. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 2. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 3. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 4. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 5. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 6. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 8. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 9. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 10. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 11. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 12. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 13. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 14. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 15. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 16. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |

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|-----|--|------------------------------------|
| 1. | 3 penny nail penetrate a 3" green joist? | $1 \frac{1}{4} - 2 \frac{9}{16}$ |
| 2. | 16 penny nail penetrate a $\frac{1}{2}$ " dry frame? | $3 \frac{1}{2} - 7 \frac{7}{16}$ |
| 3. | 10 penny nail penetrate a $\frac{5}{8}$ " green timber? | $3 - 19 \frac{19}{32}$ |
| 4. | 8 box gauge nail penetrate a 2" dry board? | $5 - 1 \frac{1}{2}$ |
| 5. | 4 common gauge nail penetrate a 4" green girder? | $5 - 3 \frac{9}{16}$ |
| 6. | $12 \frac{1}{2}$ common gauge nail penetrate a $2 \frac{1}{2}$ " green timber? | $1 \frac{1}{2} - 2 \frac{1}{16}$ |
| 7. | 3 common gauge nail penetrate a 4" green pole? | $5 \frac{1}{2} - 3 \frac{9}{16}$ |
| 8. | 30 penny nail penetrate a $\frac{3}{8}$ " dry mast? | $4 \frac{1}{2} - 5 \frac{1}{16}$ |
| 9. | $10 \frac{1}{4}$ common gauge nail penetrate a $3 \frac{1}{2}$ " green beam? | $2 \frac{1}{2} - 3 \frac{1}{16}$ |
| 10. | $11 \frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " green frame? | $2 \frac{1}{2} - 3 \frac{1}{16}$ |
| 11. | 2 penny nail penetrate a 4" dry board? | $1 - 3 \frac{1}{2}$ |
| 12. | 5 penny nail penetrate a 1" green dimension? | $1 \frac{3}{4} - 25 \frac{25}{32}$ |
| 13. | 3 penny nail penetrate a $\frac{1}{2}$ " dry board? | $1 \frac{1}{4} - 7 \frac{7}{16}$ |
| 14. | $10 \frac{1}{4}$ common gauge nail penetrate a 2" dry beam? | $2 \frac{1}{2} - 1 \frac{1}{2}$ |
| 15. | 4 penny nail penetrate a $\frac{3}{4}$ " green board? | $1 \frac{1}{2} - 11 \frac{11}{16}$ |
| 16. | $10 \frac{1}{2}$ box gauge nail penetrate a 2" green joist? | $3 - 19 \frac{19}{16}$ |

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| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 3. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 4. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 5. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 6. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 7. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 8. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 9. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 10. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 11. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 12. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 13. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 14. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 15. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 16. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |

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|-----|---|---|
| 1. | 8 box gauge nail penetrate a 2 1/2" green beam? | 5 — 2 ¹ / ₁₆ |
| 2. | 11 1/2 box gauge nail penetrate a 3/4" dry joist? | 2 1/2 — 5/8 |
| 3. | 2 penny nail penetrate a 3 1/2" green rafter? | 1 — 3 ¹ / ₁₆ |
| 4. | 40 penny nail penetrate a 3" dry frame? | 5 — 2 1/2 |
| 5. | 14 box gauge nail penetrate a 3/4" green girder? | 1 1/2 — 11 ¹ / ₁₆ |
| 6. | 3 penny nail penetrate a 3" dry beam? | 1 1/4 — 2 1/2 |
| 7. | 9 common gauge nail penetrate a 4 1/2" green girder? | 3 — 4 ¹ / ₁₆ |
| 8. | 11 1/2 box gauge nail penetrate a 1" dry rafter? | 2 1/2 — 3/4 |
| 9. | 12 1/2 common gauge nail penetrate a 3/4" dry timber? | 1 1/2 — 5/8 |
| 10. | 10 1/4 common gauge nail penetrate a 1" dry pole? | 2 1/2 — 3/4 |
| 11. | 16 penny nail penetrate a 1 1/4" green dimension? | 3 1/2 — 1 ¹ / ₃₂ |
| 12. | 9 common gauge nail penetrate a 4 1/2" green plank? | 3 — 4 ¹ / ₁₆ |
| 13. | 14 box gauge nail penetrate a 3 1/2" dry mast? | 1 1/2 — 3 |
| 14. | 5 penny nail penetrate a 2 1/2" green girder? | 1 3/4 — 2 ¹ / ₁₆ |
| 15. | 60 penny nail penetrate a 4 1/2" dry frame? | 6 — 4 |
| 16. | 6 penny nail penetrate a 4" green joist? | 2 — 3 ⁹ / ₁₆ |

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| 1. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 2. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 4. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 5. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 7. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 8. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 9. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 10. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 11. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 12. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 13. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 14. countersink hole for a 18 gauge screw? | $27/64 - 19/64 + 1/8 = 27/64$ |
| 15. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 16. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |

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|-----|--|---------------|
| 1. | 30 penny nail penetrate a 1 1/2" dry beam? | 4 1/2 — 1 1/4 |
| 2. | 4 penny nail penetrate a 3/8" green frame? | 1 1/2 — 11/32 |
| 3. | 3 penny nail penetrate a 3/4" green pole? | 1 1/4 — 11/16 |
| 4. | 11 1/2 common gauge nail penetrate a 1 1/2" green joist? | 2 — 19/32 |
| 5. | 8 box gauge nail penetrate a 3" dry mast? | 5 — 2 1/2 |
| 6. | 2 penny nail penetrate a 1/2" dry timber? | 1 — 7/16 |
| 7. | 20 penny nail penetrate a 1 1/4" green timber? | 4 — 11/32 |
| 8. | 40 penny nail penetrate a 4" green dimension? | 5 — 39/16 |
| 9. | 7 penny nail penetrate a 4 1/2" dry girder? | 2 1/4 — 4 |
| 10. | 50 penny nail penetrate a 2 1/2" dry dimension? | 5 1/2 — 2 |
| 11. | 2 penny nail penetrate a 4 1/2" green dimension? | 1 — 41/16 |
| 12. | 40 penny nail penetrate a 4" dry frame? | 5 — 3 1/2 |
| 13. | 5 common gauge nail penetrate a 1 1/2" green mast? | 4 1/2 — 19/32 |
| 14. | 8 box gauge nail penetrate a 3 1/2" green beam? | 5 — 31/16 |
| 15. | 8 box gauge nail penetrate a 5/8" green timber? | 5 — 19/32 |
| 16. | 20 penny nail penetrate a 1 1/4" green plank? | 4 — 11/32 |

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|---|--------------------------|
| 1. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 2. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 3. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 5. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 6. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 7. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 9. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 10. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 11. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 12. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 13. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 14. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 15. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 16. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |

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|-----|---|----------------|
| 1. | 10 penny nail penetrate a 2 1/2" green rafter? | 3 — 2 1/16 |
| 2. | 10 1/2 box gauge nail penetrate a 5/8" green board? | 3 — 19/32 |
| 3. | 9 box gauge nail penetrate a 4 1/2" green dimension? | 4 — 4 1/16 |
| 4. | 9 box gauge nail penetrate a 3 1/2" green timber? | 4 — 3 1/16 |
| 5. | 6 penny nail penetrate a 3/8" green girder? | 2 — 11/32 |
| 6. | 30 penny nail penetrate a 1" dry beam? | 4 1/2 — 3/4 |
| 7. | 9 common gauge nail penetrate a 4 1/2" dry frame? | 3 — 4 |
| 8. | 11 1/2 box gauge nail penetrate a 1 1/2" green joist? | 2 1/2 — 19/32 |
| 9. | 9 penny nail penetrate a 3/4" dry rafter? | 2 3/4 — 5/8 |
| 10. | 5 penny nail penetrate a 1 1/4" green rafter? | 1 3/4 — 1 1/32 |
| 11. | 6 common gauge nail penetrate a 2 1/2" green mast? | 4 — 2 1/16 |
| 12. | 11 1/2 common gauge nail penetrate a 5/8" dry board? | 2 — 9/16 |
| 13. | 10 1/2 box gauge nail penetrate a 1" dry timber? | 3 — 3/4 |
| 14. | 10 penny nail penetrate a 3 1/2" green girder? | 3 — 3 1/16 |
| 15. | 8 penny nail penetrate a 1/2" green mast? | 2 1/2 — 15/32 |
| 16. | 30 penny nail penetrate a 3 1/2" dry timber? | 4 1/2 — 3 |

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| 1. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 2. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 3. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 4. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 5. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 6. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 7. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 9. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 10. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 11. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 12. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 13. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 14. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 15. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 16. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |

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|-----|--|----------------|
| 1. | 8 box gauge nail penetrate a 4 1/2" green mast? | 5 — 4 1/16 |
| 2. | 15 1/2 box gauge nail penetrate a 3" green plank? | 1 — 2 9/16 |
| 3. | 9 penny nail penetrate a 2 1/2" green dimension? | 2 3/4 — 2 1/16 |
| 4. | 7 penny nail penetrate a 4 1/2" dry timber? | 2 1/4 — 4 |
| 5. | 14 1/2 box gauge nail penetrate a 2 1/2" green beam? | 1 1/4 — 2 1/16 |
| 6. | 8 penny nail penetrate a 3 1/2" green joist? | 2 1/2 — 3 1/16 |
| 7. | 12 1/2 common gauge nail penetrate a 1/2" green joist? | 1 1/2 — 15/32 |
| 8. | 12 penny nail penetrate a 1 1/2" dry joist? | 3 1/4 — 1 1/4 |
| 9. | 8 penny nail penetrate a 1/2" green rafter? | 2 1/2 — 15/32 |
| 10. | 15 common gauge nail penetrate a 2" green joist? | 1 — 1 9/16 |
| 11. | 10 penny nail penetrate a 3 1/2" dry frame? | 3 — 3 |
| 12. | 60 penny nail penetrate a 1 1/2" green girder? | 6 — 1 9/32 |
| 13. | 3 penny nail penetrate a 2" green pole? | 1 1/4 — 1 9/16 |
| 14. | 60 penny nail penetrate a 3/8" green mast? | 6 — 11/32 |
| 15. | 12 1/2 common gauge nail penetrate a 3/8" green beam? | 1 1/2 — 11/32 |
| 16. | 10 1/2 box gauge nail penetrate a 4 1/2" dry pole? | 3 — 4 |

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|--|--------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 3. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 4. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 5. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 6. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 7. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 8. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 9. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 10. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 11. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 12. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 13. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 14. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 15. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 16. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |

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| 1. | 60 penny nail penetrate a 3 ½" dry plank? | 6 — 3 |
| 2. | 8 penny nail penetrate a 1 ¼" dry plank? | 2 ½ — 1 |
| 3. | 14 box gauge nail penetrate a 3 ½" green rafter? | 1 ½ — 3 ¹ / ₁₆ |
| 4. | 6 penny nail penetrate a ½" green pole? | 2 — 15 ¹⁵ / ₃₂ |
| 5. | 11 ½ common gauge nail penetrate a 3 ½" dry frame? | 2 — 3 |
| 6. | 9 penny nail penetrate a 1 ½" dry rafter? | 2 ³ / ₄ — 1 ¹ / ₄ |
| 7. | 12 ½ common gauge nail penetrate a 3" dry board? | 1 ½ — 2 ½ |
| 8. | 10 ½ box gauge nail penetrate a 2" green dimension? | 3 — 1 ⁹ / ₁₆ |
| 9. | 14 box gauge nail penetrate a 3/8" dry dimension? | 1 ½ — 5 ⁵ / ₁₆ |
| 10. | 8 penny nail penetrate a 3/4" green pole? | 2 ½ — 11 ¹¹ / ₁₆ |
| 11. | 5 penny nail penetrate a 1" green beam? | 1 ³ / ₄ — 25 ²⁵ / ₃₂ |
| 12. | 16 penny nail penetrate a 4" green board? | 3 ½ — 3 ⁹ / ₁₆ |
| 13. | 4 penny nail penetrate a 4 ½" dry dimension? | 1 ½ — 4 |
| 14. | 60 penny nail penetrate a 1 ½" dry frame? | 6 — 1 ¹ / ₄ |
| 15. | 9 penny nail penetrate a ½" green beam? | 2 ³ / ₄ — 15 ¹⁵ / ₃₂ |
| 16. | 16 penny nail penetrate a ½" dry rafter? | 3 ½ — 7 ⁷ / ₁₆ |

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|---|---|
| 1. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 2. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 3. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 4. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 5. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 6. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 7. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 8. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 9. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 10. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 11. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 12. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 13. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 14. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 15. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 16. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |

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|-----|--|----------------------------------|
| 1. | 6 common gauge nail penetrate a $\frac{3}{4}$ " dry timber? | $4 - \frac{5}{8}$ |
| 2. | 5 penny nail penetrate a 4" green joist? | $1 \frac{3}{4} - 3 \frac{9}{16}$ |
| 3. | 6 penny nail penetrate a $\frac{3}{4}$ " dry joist? | $2 - \frac{5}{8}$ |
| 4. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry plank? | $2 - \frac{5}{16}$ |
| 5. | 6 common gauge nail penetrate a 2" dry timber? | $4 - 1 \frac{1}{2}$ |
| 6. | 20 penny nail penetrate a 3" green frame? | $4 - 2 \frac{9}{16}$ |
| 7. | 7 penny nail penetrate a $\frac{5}{8}$ " dry joist? | $2 \frac{1}{4} - \frac{9}{16}$ |
| 8. | 9 common gauge nail penetrate a $1 \frac{1}{4}$ " green mast? | $3 - 1 \frac{1}{32}$ |
| 9. | 5 penny nail penetrate a $2 \frac{1}{2}$ " dry rafter? | $1 \frac{3}{4} - 2$ |
| 10. | 5 common gauge nail penetrate a $4 \frac{1}{2}$ " dry beam? | $4 \frac{1}{2} - 4$ |
| 11. | 9 penny nail penetrate a $3 \frac{1}{2}$ " dry frame? | $2 \frac{3}{4} - 3$ |
| 12. | 5 common gauge nail penetrate a $\frac{5}{8}$ " green pole? | $4 \frac{1}{2} - 1 \frac{9}{32}$ |
| 13. | 5 penny nail penetrate a 3" dry beam? | $1 \frac{3}{4} - 2 \frac{1}{2}$ |
| 14. | 14 common gauge nail penetrate a $3 \frac{1}{2}$ " dry frame? | $1 \frac{1}{4} - 3$ |
| 15. | 8 box gauge nail penetrate a $1 \frac{1}{2}$ " green timber? | $5 - 1 \frac{9}{32}$ |
| 16. | 20 penny nail penetrate a $3 \frac{1}{2}$ " green board? | $4 - 3 \frac{1}{16}$ |

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|---|-------------------------------|
| 1. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 2. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 3. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 4. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 5. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 6. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 7. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 8. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 9. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 10. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 11. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 12. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 13. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 15. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 16. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |

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| 1. | 12 penny nail penetrate a 2 ½" green timber? | 3 ¼ — 2 ¹ / ₁₆ |
| 2. | 6 penny nail penetrate a 1" dry timber? | 2 — ¾ |
| 3. | 10 ¼ common gauge nail penetrate a 5/8" dry mast? | 2 ½ — 9 ¹ / ₁₆ |
| 4. | 11 ½ common gauge nail penetrate a 1 ½" dry board? | 2 — 1 ¼ |
| 5. | 7 penny nail penetrate a 3/4" dry plank? | 2 ¼ — 5 ¹ / ₈ |
| 6. | 3 penny nail penetrate a 3/4" dry board? | 1 ¼ — 5 ¹ / ₈ |
| 7. | 14 box gauge nail penetrate a 4" green beam? | 1 ½ — 3 ⁹ / ₁₆ |
| 8. | 10 ½ box gauge nail penetrate a 3" green board? | 3 — 2 ⁹ / ₁₆ |
| 9. | 8 penny nail penetrate a 4 ½" dry beam? | 2 ½ — 4 |
| 10. | 10 ¼ common gauge nail penetrate a 5/8" green dimension? | 2 ½ — 1 ⁹ / ₃₂ |
| 11. | 15 common gauge nail penetrate a 3/4" green mast? | 1 — 1 ¹ / ₁₆ |
| 12. | 9 penny nail penetrate a 1" dry mast? | 2 ¾ — ¾ |
| 13. | 3 penny nail penetrate a 3 ½" green board? | 1 ¼ — 3 ¹ / ₁₆ |
| 14. | 10 ¼ common gauge nail penetrate a 4" green pole? | 2 ½ — 3 ⁹ / ₁₆ |
| 15. | 10 ½ box gauge nail penetrate a 1 ½" green girder? | 3 — 1 ⁹ / ₃₂ |
| 16. | 8 penny nail penetrate a 5/8" dry dimension? | 2 ½ — 9 ¹ / ₁₆ |

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|--|--------------------------|
| 1. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 2. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 3. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 4. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 5. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 6. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 7. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 8. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 9. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 10. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 11. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 12. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 13. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 14. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 15. clearance hole for a 2 gauge screw? | 41 $3/32 = 3/32$ |
| 16. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |

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|-----|---|---------------------------------|
| 1. | 3 common gauge nail penetrate a 1" green rafter? | $5 \frac{1}{2} - 25/32$ |
| 2. | 9 box gauge nail penetrate a 1 $\frac{1}{4}$ " green board? | $4 - 11/32$ |
| 3. | 10 $\frac{1}{4}$ common gauge nail penetrate a 2 $\frac{1}{2}$ " green frame? | $2 \frac{1}{2} - 21/16$ |
| 4. | 5 penny nail penetrate a 2" dry girder? | $1 \frac{3}{4} - 1 \frac{1}{2}$ |
| 5. | 3 penny nail penetrate a 2" dry girder? | $1 \frac{1}{4} - 1 \frac{1}{2}$ |
| 6. | 40 penny nail penetrate a 2 $\frac{1}{2}$ " green rafter? | $5 - 21/16$ |
| 7. | 60 penny nail penetrate a 2 $\frac{1}{2}$ " dry joist? | $6 - 2$ |
| 8. | 50 penny nail penetrate a 2" green timber? | $5 \frac{1}{2} - 19/16$ |
| 9. | 40 penny nail penetrate a 4 $\frac{1}{2}$ " green rafter? | $5 - 41/16$ |
| 10. | 14 $\frac{1}{2}$ box gauge nail penetrate a 3/4" green girder? | $1 \frac{1}{4} - 11/16$ |
| 11. | 7 penny nail penetrate a 3 $\frac{1}{2}$ " dry rafter? | $2 \frac{1}{4} - 3$ |
| 12. | 8 box gauge nail penetrate a 3/8" dry frame? | $5 - 5/16$ |
| 13. | 10 $\frac{1}{2}$ box gauge nail penetrate a 5/8" green dimension? | $3 - 19/32$ |
| 14. | 9 common gauge nail penetrate a $\frac{1}{2}$ " dry frame? | $3 - 7/16$ |
| 15. | 20 penny nail penetrate a 3 $\frac{1}{2}$ " green dimension? | $4 - 31/16$ |
| 16. | 14 box gauge nail penetrate a 4 $\frac{1}{2}$ " green pole? | $1 \frac{1}{2} - 41/16$ |

1. countersink hole for a 16 gauge screw?	X $17/64 + 1/8 = 25/64$
2. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
3. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
4. countersink hole for a 12 gauge screw?	S $7/32 + 1/8 = 11/32$
5. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
6. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
7. clearance hole for a 18 gauge screw?	N $19/64 = 19/64$
8. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$
9. countersink hole for a 9 gauge screw?	N $11/64 + 1/8 = 19/64$
10. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
11. pilot hole for a 14 gauge screw?	12 $1/4 - 1/16 = 3/16$
12. pilot hole for a 24 gauge screw?	O $3/8 - 1/16 = 5/16$
13. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
14. pilot hole for a 14 gauge screw?	12 $1/4 - 1/16 = 3/16$
15. clearance hole for a 12 gauge screw?	2 $7/32 = 7/32$
16. pilot hole for a 7 gauge screw?	41 $5/32 - 1/16 = 3/32$

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|-----|--|-----------------------------------|
| 1. | 10 penny nail penetrate a 4" dry board? | $3 - 3 \frac{1}{2}$ |
| 2. | 9 common gauge nail penetrate a $3 \frac{1}{2}$ " dry girder? | $3 - 3$ |
| 3. | 14 $\frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " dry joist? | $1 \frac{1}{4} - 3$ |
| 4. | 20 penny nail penetrate a 2" dry dimension? | $4 - 1 \frac{1}{2}$ |
| 5. | 10 penny nail penetrate a 1" dry rafter? | $3 - \frac{3}{4}$ |
| 6. | 8 penny nail penetrate a $1 \frac{1}{2}$ " green girder? | $2 \frac{1}{2} - 1 \frac{9}{32}$ |
| 7. | 5 penny nail penetrate a $\frac{5}{8}$ " dry joist? | $1 \frac{3}{4} - \frac{9}{16}$ |
| 8. | 16 penny nail penetrate a $\frac{3}{8}$ " green girder? | $3 \frac{1}{2} - 1 \frac{11}{32}$ |
| 9. | 14 box gauge nail penetrate a 1" dry rafter? | $1 \frac{1}{2} - \frac{3}{4}$ |
| 10. | 14 common gauge nail penetrate a $3 \frac{1}{2}$ " dry board? | $1 \frac{1}{4} - 3$ |
| 11. | 2 penny nail penetrate a $\frac{5}{8}$ " green girder? | $1 - 1 \frac{19}{32}$ |
| 12. | 12 penny nail penetrate a 1" green pole? | $3 \frac{1}{4} - 2 \frac{25}{32}$ |
| 13. | 20 penny nail penetrate a $3 \frac{1}{2}$ " green pole? | $4 - 3 \frac{1}{16}$ |
| 14. | 20 penny nail penetrate a $\frac{5}{8}$ " green girder? | $4 - 1 \frac{19}{32}$ |
| 15. | 50 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 16. | 12 penny nail penetrate a 1" dry mast? | $3 \frac{1}{4} - \frac{3}{4}$ |

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|--|-------------------------------|
| 1. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 2. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 3. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 5. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 6. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 7. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 8. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 9. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 10. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 11. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 12. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 13. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 14. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 15. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 16. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |

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|-----|---|----------------|
| 1. | 2 penny nail penetrate a 1 1/2" green frame? | 1 — 19/32 |
| 2. | 3 penny nail penetrate a 4 1/2" green plank? | 1 1/4 — 4 1/16 |
| 3. | 30 penny nail penetrate a 1" dry joist? | 4 1/2 — 3/4 |
| 4. | 5 penny nail penetrate a 1/2" green plank? | 1 3/4 — 15/32 |
| 5. | 6 common gauge nail penetrate a 3/8" dry girder? | 4 — 5/16 |
| 6. | 10 1/4 common gauge nail penetrate a 1 1/4" green beam? | 2 1/2 — 1 1/32 |
| 7. | 4 penny nail penetrate a 1 1/2" dry girder? | 1 1/2 — 1 1/4 |
| 8. | 10 1/4 common gauge nail penetrate a 3" dry mast? | 2 1/2 — 2 1/2 |
| 9. | 50 penny nail penetrate a 4 1/2" dry frame? | 5 1/2 — 4 |
| 10. | 5 penny nail penetrate a 5/8" dry mast? | 1 3/4 — 9/16 |
| 11. | 6 common gauge nail penetrate a 1 1/4" green joist? | 4 — 1 1/32 |
| 12. | 40 penny nail penetrate a 4 1/2" dry joist? | 5 — 4 |
| 13. | 10 penny nail penetrate a 2" green dimension? | 3 — 19/16 |
| 14. | 2 penny nail penetrate a 2 1/2" green rafter? | 1 — 2 1/16 |
| 15. | 2 common gauge nail penetrate a 3/4" dry board? | 6 — 5/8 |
| 16. | 2 penny nail penetrate a 3" dry frame? | 1 — 2 1/2 |

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|--|--|
| 1. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 2. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 3. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 4. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 5. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 6. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 7. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 8. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 9. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 10. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 11. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 12. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 13. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 14. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 15. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 16. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |

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|-----|---|--------------------------------------|
| 1. | 11 ½ common gauge nail penetrate a 2 ½" green mast? | 2 — 2 ¹ / ₁₆ |
| 2. | 16 penny nail penetrate a 4 ½" dry dimension? | 3 ½ — 4 |
| 3. | 4 penny nail penetrate a 1" green dimension? | 1 ½ — 2 ⁵ / ₃₂ |
| 4. | 6 common gauge nail penetrate a 4 ½" green rafter? | 4 — 4 ¹ / ₁₆ |
| 5. | 5 penny nail penetrate a 5/8" green mast? | 1 ¾ — 1 ⁹ / ₃₂ |
| 6. | 6 penny nail penetrate a 3 ½" dry board? | 2 — 3 |
| 7. | 9 box gauge nail penetrate a 3/8" dry girder? | 4 — 5 ¹ / ₁₆ |
| 8. | 8 penny nail penetrate a 3 ½" dry girder? | 2 ½ — 3 |
| 9. | 20 penny nail penetrate a 1 ¼" green plank? | 4 — 1 ¹ / ₃₂ |
| 10. | 60 penny nail penetrate a 2 ½" green rafter? | 6 — 2 ¹ / ₁₆ |
| 11. | 40 penny nail penetrate a 4" green mast? | 5 — 3 ⁹ / ₁₆ |
| 12. | 11 ½ box gauge nail penetrate a 4" dry mast? | 2 ½ — 3 ½ |
| 13. | 12 penny nail penetrate a ½" dry plank? | 3 ¼ — 7 ¹ / ₁₆ |
| 14. | 4 common gauge nail penetrate a 1" green beam? | 5 — 2 ⁵ / ₃₂ |
| 15. | 8 box gauge nail penetrate a 3" green timber? | 5 — 2 ⁹ / ₁₆ |
| 16. | 10 penny nail penetrate a 3/8" green mast? | 3 — 1 ¹ / ₃₂ |

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| 1. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 2. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 3. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 4. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 5. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 6. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 10. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 13. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 14. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 15. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 16. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |

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|-----|---|----------------------------------|
| 1. | 8 common gauge nail penetrate a 1" dry beam? | $3 \frac{1}{2} - \frac{3}{4}$ |
| 2. | 10 penny nail penetrate a $\frac{3}{8}$ " dry joist? | $3 - \frac{5}{16}$ |
| 3. | 6 penny nail penetrate a $1 \frac{1}{2}$ " green girder? | $2 - 1\frac{9}{32}$ |
| 4. | 10 penny nail penetrate a $\frac{1}{2}$ " green board? | $3 - 1\frac{15}{32}$ |
| 5. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{3}{8}$ " green board? | $1 \frac{1}{2} - 1\frac{11}{32}$ |
| 6. | 10 penny nail penetrate a $\frac{3}{8}$ " green girder? | $3 - 1\frac{11}{32}$ |
| 7. | 9 penny nail penetrate a $2 \frac{1}{2}$ " green board? | $2 \frac{3}{4} - 2\frac{1}{16}$ |
| 8. | 9 penny nail penetrate a $1 \frac{1}{4}$ " green board? | $2 \frac{3}{4} - 1\frac{1}{32}$ |
| 9. | $12 \frac{1}{2}$ box gauge nail penetrate a 1" green rafter? | $2 - 2\frac{25}{32}$ |
| 10. | 50 penny nail penetrate a $4 \frac{1}{2}$ " green mast? | $5 \frac{1}{2} - 4\frac{1}{16}$ |
| 11. | 12 penny nail penetrate a $1 \frac{1}{2}$ " green mast? | $3 \frac{1}{4} - 1\frac{9}{32}$ |
| 12. | 8 penny nail penetrate a 3" green dimension? | $2 \frac{1}{2} - 2\frac{9}{16}$ |
| 13. | $11 \frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " dry frame? | $2 \frac{1}{2} - 3$ |
| 14. | 10 penny nail penetrate a $3 \frac{1}{2}$ " dry plank? | $3 - 3$ |
| 15. | $10 \frac{1}{2}$ box gauge nail penetrate a 3" green timber? | $3 - 2\frac{9}{16}$ |
| 16. | 5 penny nail penetrate a 2" green beam? | $1 \frac{3}{4} - 1\frac{9}{16}$ |

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| 1. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 2. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 3. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 4. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 5. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 6. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 7. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 8. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 9. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 10. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 11. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 12. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 13. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 15. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 16. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |

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|-----|---|--------------------------------------|
| 1. | 11 ½ box gauge nail penetrate a 3" green mast? | 2 ½ — 2 ⁹ / ₁₆ |
| 2. | 20 penny nail penetrate a 5/8" dry frame? | 4 — 9/16 |
| 3. | 20 penny nail penetrate a 1 ¼" green beam? | 4 — 1 ¹ / ₃₂ |
| 4. | 12 ½ common gauge nail penetrate a 4 ½" green board? | 1 ½ — 4 ¹ / ₁₆ |
| 5. | 8 common gauge nail penetrate a 3 ½" green beam? | 3 ½ — 3 ¹ / ₁₆ |
| 6. | 16 penny nail penetrate a 3 ½" dry dimension? | 3 ½ — 3 |
| 7. | 30 penny nail penetrate a 1 ¼" dry joist? | 4 ½ — 1 |
| 8. | 14 common gauge nail penetrate a 4 ½" dry frame? | 1 ¼ — 4 |
| 9. | 8 box gauge nail penetrate a 3/8" dry frame? | 5 — 5/16 |
| 10. | 3 penny nail penetrate a 3" green timber? | 1 ¼ — 2 ⁹ / ₁₆ |
| 11. | 9 penny nail penetrate a 3/4" dry beam? | 2 ¾ — 5/8 |
| 12. | 6 penny nail penetrate a 2 ½" green frame? | 2 — 2 ¹ / ₁₆ |
| 13. | 16 penny nail penetrate a 4" dry mast? | 3 ½ — 3 ½ |
| 14. | 10 ½ box gauge nail penetrate a 1 ¼" green dimension? | 3 — 1 ¹ / ₃₂ |
| 15. | 2 common gauge nail penetrate a 3/8" green timber? | 6 — 1 ¹ / ₃₂ |
| 16. | 15 ½ box gauge nail penetrate a 3/8" dry mast? | 1 — 5/16 |

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| 1. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 2. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 3. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 4. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 5. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 6. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 7. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 8. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 12. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 13. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 14. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 15. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 16. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |

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|-----|--|------------------------------------|
| 1. | 20 penny nail penetrate a 3 ½" dry girder? | 4 — 3 |
| 2. | 30 penny nail penetrate a 1 ½" dry beam? | 4 ½ — 1 ¼ |
| 3. | 10 penny nail penetrate a ½" green mast? | 3 — 15/32 |
| 4. | 7 penny nail penetrate a 4 ½" dry board? | 2 ¼ — 4 |
| 5. | 14 ½ box gauge nail penetrate a 1 ¼" green girder? | 1 ¼ — 1 1/32 |
| 6. | 4 penny nail penetrate a 3" dry plank? | 1 ½ — 2 ½ |
| 7. | 20 penny nail penetrate a 2 ½" dry frame? | 4 — 2 |
| 8. | 2 penny nail penetrate a 3" green mast? | 1 — 2 ⁹ / ₁₆ |
| 9. | 7 penny nail penetrate a 2" dry rafter? | 2 ¼ — 1 ½ |
| 10. | 12 ½ box gauge nail penetrate a 4 ½" green frame? | 2 — 4 ¹ / ₁₆ |
| 11. | 9 box gauge nail penetrate a 3" green mast? | 4 — 2 ⁹ / ₁₆ |
| 12. | 9 penny nail penetrate a 3 ½" dry frame? | 2 ¾ — 3 |
| 13. | 8 penny nail penetrate a 5/8" dry beam? | 2 ½ — 9/16 |
| 14. | 3 penny nail penetrate a 1 ½" dry mast? | 1 ¼ — 1 ¼ |
| 15. | 50 penny nail penetrate a 1" dry pole? | 5 ½ — ¾ |
| 16. | 10 ½ box gauge nail penetrate a 2" green pole? | 3 — 1 ⁹ / ₁₆ |

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|--|-------------------------------|
| 1. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 2. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 3. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 4. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 5. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 6. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 7. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 9. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 10. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 11. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 12. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 13. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 14. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 15. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 16. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |

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| 1. | 6 penny nail penetrate a $\frac{3}{4}$ " dry dimension? | $2 - \frac{5}{8}$ |
| 2. | 12 $\frac{1}{2}$ common gauge nail penetrate a 4" dry rafter? | $1 \frac{1}{2} - 3 \frac{1}{2}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a 4" dry timber? | $3 - 3 \frac{1}{2}$ |
| 4. | 50 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 5. | 8 common gauge nail penetrate a $3 \frac{1}{2}$ " green beam? | $3 \frac{1}{2} - 3 \frac{1}{16}$ |
| 6. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry frame? | $2 - \frac{9}{16}$ |
| 7. | 30 penny nail penetrate a $\frac{3}{8}$ " dry rafter? | $4 \frac{1}{2} - \frac{5}{16}$ |
| 8. | 9 penny nail penetrate a $\frac{3}{8}$ " green frame? | $2 \frac{3}{4} - \frac{11}{32}$ |
| 9. | 9 box gauge nail penetrate a 3" dry mast? | $4 - 2 \frac{1}{2}$ |
| 10. | 7 penny nail penetrate a $2 \frac{1}{2}$ " green beam? | $2 \frac{1}{4} - 2 \frac{1}{16}$ |
| 11. | 2 penny nail penetrate a 1" dry pole? | $1 - \frac{3}{4}$ |
| 12. | 5 penny nail penetrate a 3" green plank? | $1 \frac{3}{4} - 2 \frac{9}{16}$ |
| 13. | 12 $\frac{1}{2}$ common gauge nail penetrate a $1 \frac{1}{4}$ " green timber? | $1 \frac{1}{2} - 1 \frac{1}{32}$ |
| 14. | 2 penny nail penetrate a $\frac{3}{4}$ " dry girder? | $1 - \frac{5}{8}$ |
| 15. | 2 common gauge nail penetrate a 1" green plank? | $6 - \frac{25}{32}$ |
| 16. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry girder? | $3 - \frac{9}{16}$ |

1. countersink hole for a 7 gauge screw?	L $5/32 + 1/8 = 9/32$
2. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
3. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
4. countersink hole for a 6 gauge screw?	H $9/64 + 1/8 = 17/64$
5. pilot hole for a 11 gauge screw?	27 $13/64 - 1/16 = 9/64$
6. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
7. countersink hole for a 5 gauge screw?	E $1/8 + 1/8 = 1/4$
8. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
9. pilot hole for a 7 gauge screw?	41 $5/32 - 1/16 = 3/32$
10. clearance hole for a 7 gauge screw?	22 $5/32 = 5/32$
11. countersink hole for a 6 gauge screw?	H $9/64 + 1/8 = 17/64$
12. pilot hole for a 11 gauge screw?	27 $13/64 - 1/16 = 9/64$
13. clearance hole for a 14 gauge screw?	E $1/4 = 1/4$
14. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
15. pilot hole for a 5 gauge screw?	52 $1/8 - 1/16 = 1/16$
16. pilot hole for a 16 gauge screw?	6 $17/64 - 1/16 = 13/64$

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|-----|---|---------------------------------|
| 1. | 10 penny nail penetrate a 1" dry board? | $3 - \frac{3}{4}$ |
| 2. | 50 penny nail penetrate a 2" green frame? | $5 \frac{1}{2} - 1\frac{9}{16}$ |
| 3. | 8 common gauge nail penetrate a 1" dry timber? | $3 \frac{1}{2} - \frac{3}{4}$ |
| 4. | 10 $\frac{1}{4}$ common gauge nail penetrate a 4" green beam? | $2 \frac{1}{2} - 3\frac{9}{16}$ |
| 5. | 50 penny nail penetrate a 2 $\frac{1}{2}$ " green joist? | $5 \frac{1}{2} - 2\frac{1}{16}$ |
| 6. | 8 box gauge nail penetrate a 4 $\frac{1}{2}$ " dry mast? | $5 - 4$ |
| 7. | 11 $\frac{1}{2}$ box gauge nail penetrate a 3" green timber? | $2 \frac{1}{2} - 2\frac{9}{16}$ |
| 8. | 20 penny nail penetrate a 1 $\frac{1}{2}$ " dry girder? | $4 - 1 \frac{1}{4}$ |
| 9. | 9 common gauge nail penetrate a 4 $\frac{1}{2}$ " dry beam? | $3 - 4$ |
| 10. | 10 $\frac{1}{4}$ common gauge nail penetrate a 3 $\frac{1}{2}$ " dry joist? | $2 \frac{1}{2} - 3$ |
| 11. | 8 penny nail penetrate a 1" green frame? | $2 \frac{1}{2} - \frac{25}{32}$ |
| 12. | 40 penny nail penetrate a 3 $\frac{1}{2}$ " green pole? | $5 - 3\frac{1}{16}$ |
| 13. | 4 penny nail penetrate a 2" dry timber? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 14. | 8 common gauge nail penetrate a 2 $\frac{1}{2}$ " dry board? | $3 \frac{1}{2} - 2$ |
| 15. | 20 penny nail penetrate a 3 $\frac{1}{2}$ " dry girder? | $4 - 3$ |
| 16. | 40 penny nail penetrate a 3 $\frac{1}{2}$ " dry mast? | $5 - 3$ |

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|--|-------------------------------|
| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 3. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 4. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 5. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 6. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 7. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 8. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 9. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 10. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 11. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 12. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 13. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 14. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 15. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 16. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |

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|-----|--|----------------------------------|
| 1. | 14 box gauge nail penetrate a 2" dry timber? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 2. | 2 penny nail penetrate a $\frac{1}{2}$ " green pole? | $1 - \frac{15}{32}$ |
| 3. | 5 penny nail penetrate a $1 \frac{1}{4}$ " green beam? | $1 \frac{3}{4} - 1 \frac{1}{32}$ |
| 4. | 8 penny nail penetrate a $\frac{5}{8}$ " green joist? | $2 \frac{1}{2} - \frac{19}{32}$ |
| 5. | 10 penny nail penetrate a 3" dry girder? | $3 - 2 \frac{1}{2}$ |
| 6. | $12 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " green frame? | $2 - \frac{11}{16}$ |
| 7. | 2 common gauge nail penetrate a $4 \frac{1}{2}$ " dry frame? | $6 - 4$ |
| 8. | 60 penny nail penetrate a $4 \frac{1}{2}$ " green pole? | $6 - 4 \frac{1}{16}$ |
| 9. | 50 penny nail penetrate a $\frac{5}{8}$ " green dimension? | $5 \frac{1}{2} - \frac{19}{32}$ |
| 10. | 3 penny nail penetrate a $4 \frac{1}{2}$ " green timber? | $1 \frac{1}{4} - 4 \frac{1}{16}$ |
| 11. | $11 \frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry frame? | $2 \frac{1}{2} - \frac{9}{16}$ |
| 12. | 9 common gauge nail penetrate a $\frac{3}{8}$ " green pole? | $3 - \frac{11}{32}$ |
| 13. | $10 \frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " green frame? | $3 - 2 \frac{1}{16}$ |
| 14. | 16 penny nail penetrate a $\frac{1}{2}$ " green board? | $3 \frac{1}{2} - \frac{15}{32}$ |
| 15. | 8 box gauge nail penetrate a 4" dry frame? | $5 - 3 \frac{1}{2}$ |
| 16. | 4 penny nail penetrate a $2 \frac{1}{2}$ " dry joist? | $1 \frac{1}{2} - 2$ |

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|--|---|
| 1. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 3. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 4. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 5. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 6. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 7. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 8. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 9. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 10. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 11. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 12. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 13. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 14. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 15. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 16. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |

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|-----|---|---------------------------------------|
| 1. | 12 ½ box gauge nail penetrate a 2 ½" green dimension? | 2 — 2 ¹ / ₁₆ |
| 2. | 3 common gauge nail penetrate a 2 ½" dry mast? | 5 ½ — 2 |
| 3. | 5 common gauge nail penetrate a ½" green timber? | 4 ½ — 15 ¹ / ₃₂ |
| 4. | 5 common gauge nail penetrate a 5/8" dry frame? | 4 ½ — 9 ¹ / ₁₆ |
| 5. | 9 common gauge nail penetrate a 1" green rafter? | 3 — 25 ¹ / ₃₂ |
| 6. | 9 box gauge nail penetrate a 4 ½" green dimension? | 4 — 4 ¹ / ₁₆ |
| 7. | 30 penny nail penetrate a 1" green plank? | 4 ½ — 25 ¹ / ₃₂ |
| 8. | 16 penny nail penetrate a 2 ½" dry plank? | 3 ½ — 2 |
| 9. | 12 ½ common gauge nail penetrate a 2" green beam? | 1 ½ — 19 ¹ / ₁₆ |
| 10. | 10 ½ box gauge nail penetrate a 3" green girder? | 3 — 29 ¹ / ₁₆ |
| 11. | 8 box gauge nail penetrate a 3/8" green frame? | 5 — 11 ¹ / ₃₂ |
| 12. | 14 box gauge nail penetrate a 4" dry board? | 1 ½ — 3 ½ |
| 13. | 40 penny nail penetrate a ½" green joist? | 5 — 15 ¹ / ₃₂ |
| 14. | 10 penny nail penetrate a 1 ½" dry joist? | 3 — 1 ¼ |
| 15. | 3 common gauge nail penetrate a 4 ½" dry plank? | 5 ½ — 4 |
| 16. | 8 common gauge nail penetrate a ½" green dimension? | 3 ½ — 15 ¹ / ₃₂ |

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|---|--------------------------|
| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 3. countersink hole for a 7 gauge screw? | L $5/32 + 1/8 = 9/32$ |
| 4. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 5. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 6. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 7. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 8. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 9. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 10. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 11. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 12. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 13. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 14. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 15. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 16. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |

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|-----|--|---------------------------------|
| 1. | 2 common gauge nail penetrate a $\frac{1}{2}$ " green dimension? | $6 - \frac{15}{32}$ |
| 2. | 14 $\frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry plank? | $1 \frac{1}{4} - 1$ |
| 3. | 9 penny nail penetrate a $\frac{3}{4}$ " dry timber? | $2 \frac{3}{4} - \frac{5}{8}$ |
| 4. | 3 penny nail penetrate a $4 \frac{1}{2}$ " green board? | $1 \frac{1}{4} - 4\frac{1}{16}$ |
| 5. | 50 penny nail penetrate a $\frac{5}{8}$ " dry board? | $5 \frac{1}{2} - \frac{9}{16}$ |
| 6. | 20 penny nail penetrate a 3" green beam? | $4 - 2\frac{9}{16}$ |
| 7. | 12 $\frac{1}{2}$ box gauge nail penetrate a 4" green beam? | $2 - 3\frac{9}{16}$ |
| 8. | 4 penny nail penetrate a $\frac{3}{8}$ " dry plank? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 9. | 2 penny nail penetrate a $\frac{5}{8}$ " dry joist? | $1 - \frac{9}{16}$ |
| 10. | 12 penny nail penetrate a $1 \frac{1}{2}$ " green pole? | $3 \frac{1}{4} - 1\frac{9}{32}$ |
| 11. | 15 common gauge nail penetrate a 1" dry joist? | $1 - \frac{3}{4}$ |
| 12. | 11 $\frac{1}{2}$ box gauge nail penetrate a 4" green girder? | $2 \frac{1}{2} - 3\frac{9}{16}$ |
| 13. | 10 penny nail penetrate a $1 \frac{1}{2}$ " dry plank? | $3 - 1 \frac{1}{4}$ |
| 14. | 12 $\frac{1}{2}$ box gauge nail penetrate a 2" green beam? | $2 - 1\frac{9}{16}$ |
| 15. | 10 penny nail penetrate a $1 \frac{1}{4}$ " dry timber? | $3 - 1$ |
| 16. | 10 $\frac{1}{4}$ common gauge nail penetrate a $2 \frac{1}{2}$ " dry beam? | $2 \frac{1}{2} - 2$ |

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|--|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 3. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 4. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 5. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 6. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 7. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 8. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 9. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 10. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 11. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 12. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 13. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 14. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 15. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 16. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |

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|-----|---|----------------------------------|
| 1. | 4 common gauge nail penetrate a 2" dry mast? | $5 - 1 \frac{1}{2}$ |
| 2. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry dimension? | $3 - \frac{7}{16}$ |
| 3. | 16 penny nail penetrate a 3 $\frac{1}{2}$ " green beam? | $3 \frac{1}{2} - 3 \frac{1}{16}$ |
| 4. | 12 $\frac{1}{2}$ common gauge nail penetrate a $\frac{1}{2}$ " green rafter? | $1 \frac{1}{2} - \frac{15}{32}$ |
| 5. | 40 penny nail penetrate a $\frac{3}{4}$ " green dimension? | $5 - \frac{11}{16}$ |
| 6. | 8 box gauge nail penetrate a $\frac{3}{4}$ " dry pole? | $5 - \frac{5}{8}$ |
| 7. | 8 box gauge nail penetrate a $\frac{1}{2}$ " green joist? | $5 - \frac{15}{32}$ |
| 8. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green dimension? | $2 \frac{1}{2} - \frac{11}{32}$ |
| 9. | 7 penny nail penetrate a $\frac{1}{2}$ " green frame? | $2 \frac{1}{4} - \frac{15}{32}$ |
| 10. | 5 penny nail penetrate a 3" green board? | $1 \frac{3}{4} - 2 \frac{9}{16}$ |
| 11. | 3 penny nail penetrate a 4" green timber? | $1 \frac{1}{4} - 3 \frac{9}{16}$ |
| 12. | 9 common gauge nail penetrate a 1 $\frac{1}{4}$ " green board? | $3 - 1 \frac{1}{32}$ |
| 13. | 12 $\frac{1}{2}$ common gauge nail penetrate a 4 $\frac{1}{2}$ " green board? | $1 \frac{1}{2} - 4 \frac{1}{16}$ |
| 14. | 8 penny nail penetrate a 2 $\frac{1}{2}$ " green plank? | $2 \frac{1}{2} - 2 \frac{1}{16}$ |
| 15. | 10 penny nail penetrate a $\frac{5}{8}$ " dry frame? | $3 - \frac{9}{16}$ |
| 16. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green dimension? | $2 \frac{1}{2} - \frac{19}{32}$ |

1. clearance hole for a 20 gauge screw?	Q $21/64 = 21/64$
2. countersink hole for a 3 gauge screw?	2 $3/32 + 1/8 = 7/32$
3. clearance hole for a 24 gauge screw?	V $3/8 = 3/8$
4. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
5. countersink hole for a 18 gauge screw?	$27/64$ $19/64 + 1/8 = 27/64$
6. countersink hole for a 16 gauge screw?	X $17/64 + 1/8 = 25/64$
7. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
8. countersink hole for a 14 gauge screw?	V $1/4 + 1/8 = 3/8$
9. countersink hole for a 10 gauge screw?	O $3/16 + 1/8 = 5/16$
10. pilot hole for a 9 gauge screw?	35 $11/64 - 1/16 = 7/64$
11. pilot hole for a 9 gauge screw?	35 $11/64 - 1/16 = 7/64$
12. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
13. countersink hole for a 14 gauge screw?	V $1/4 + 1/8 = 3/8$
14. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
15. pilot hole for a 7 gauge screw?	41 $5/32 - 1/16 = 3/32$
16. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$

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| 1. | 3 common gauge nail penetrate a $\frac{3}{8}$ " dry board? | $5\frac{1}{2} - \frac{5}{16}$ |
| 2. | 5 penny nail penetrate a $2\frac{1}{2}$ " dry frame? | $1\frac{3}{4} - 2$ |
| 3. | 4 penny nail penetrate a 3" dry dimension? | $1\frac{1}{2} - 2\frac{1}{2}$ |
| 4. | 16 penny nail penetrate a $\frac{3}{8}$ " dry pole? | $3\frac{1}{2} - \frac{5}{16}$ |
| 5. | 30 penny nail penetrate a 2" dry joist? | $4\frac{1}{2} - 1\frac{1}{2}$ |
| 6. | 6 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $2 - \frac{15}{32}$ |
| 7. | 50 penny nail penetrate a $3\frac{1}{2}$ " green board? | $5\frac{1}{2} - 3\frac{1}{16}$ |
| 8. | 12 penny nail penetrate a 3" dry plank? | $3\frac{1}{4} - 2\frac{1}{2}$ |
| 9. | $12\frac{1}{2}$ box gauge nail penetrate a 4" green rafter? | $2 - 3\frac{9}{16}$ |
| 10. | $11\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry plank? | $2\frac{1}{2} - \frac{5}{16}$ |
| 11. | $10\frac{1}{4}$ common gauge nail penetrate a $1\frac{1}{4}$ " dry joist? | $2\frac{1}{2} - 1$ |
| 12. | 9 common gauge nail penetrate a 1" green joist? | $3 - \frac{25}{32}$ |
| 13. | 2 common gauge nail penetrate a $\frac{1}{2}$ " green joist? | $6 - \frac{15}{32}$ |
| 14. | 3 common gauge nail penetrate a $1\frac{1}{4}$ " dry timber? | $5\frac{1}{2} - 1$ |
| 15. | $10\frac{1}{2}$ box gauge nail penetrate a 1" green frame? | $3 - \frac{25}{32}$ |
| 16. | $11\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green frame? | $2 - \frac{11}{16}$ |

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|---|--|
| 1. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 3. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 4. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 5. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 6. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 7. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 8. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 9. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 10. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 12. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 13. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 14. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 15. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 16. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |

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|-----|---|---------------------------------|
| 1. | 14 common gauge nail penetrate a $\frac{1}{2}$ " green timber? | $1 \frac{1}{4} - \frac{15}{32}$ |
| 2. | 60 penny nail penetrate a 2" dry frame? | $6 - 1 \frac{1}{2}$ |
| 3. | 3 penny nail penetrate a $2 \frac{1}{2}$ " dry frame? | $1 \frac{1}{4} - 2$ |
| 4. | 9 box gauge nail penetrate a $1 \frac{1}{2}$ " green mast? | $4 - \frac{19}{32}$ |
| 5. | 14 box gauge nail penetrate a $\frac{3}{8}$ " dry timber? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 6. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " green joist? | $1 \frac{1}{2} - \frac{19}{32}$ |
| 7. | 8 box gauge nail penetrate a 4" dry dimension? | $5 - 3 \frac{1}{2}$ |
| 8. | 3 common gauge nail penetrate a 3" dry joist? | $5 \frac{1}{2} - 2 \frac{1}{2}$ |
| 9. | 60 penny nail penetrate a $4 \frac{1}{2}$ " green pole? | $6 - 4 \frac{1}{16}$ |
| 10. | $12 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry beam? | $2 - 1$ |
| 11. | 3 penny nail penetrate a 1" dry dimension? | $1 \frac{1}{4} - \frac{3}{4}$ |
| 12. | 8 box gauge nail penetrate a $\frac{1}{2}$ " dry board? | $5 - \frac{7}{16}$ |
| 13. | 6 penny nail penetrate a 1" green mast? | $2 - \frac{25}{32}$ |
| 14. | 6 penny nail penetrate a $3 \frac{1}{2}$ " dry rafter? | $2 - 3$ |
| 15. | 50 penny nail penetrate a 1" dry plank? | $5 \frac{1}{2} - \frac{3}{4}$ |
| 16. | 60 penny nail penetrate a 4" green rafter? | $6 - 3 \frac{9}{16}$ |

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|--|-------------------------------|
| 1. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 2. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 3. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 4. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 5. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 6. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 7. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 8. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 9. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 10. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 11. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 12. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 13. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 14. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 15. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 16. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |

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|-----|--|--------------|
| 1. | 12 ½ common gauge nail penetrate a ¾" green pole? | 1 ½ — 11/16 |
| 2. | 6 common gauge nail penetrate a 2" dry frame? | 4 — 1 ½ |
| 3. | 10 ½ box gauge nail penetrate a ½" dry plank? | 3 — 7/16 |
| 4. | 10 ¼ common gauge nail penetrate a 3 ½" green plank? | 2 ½ — 3 1/16 |
| 5. | 10 ¼ common gauge nail penetrate a 4" dry pole? | 2 ½ — 3 ½ |
| 6. | 6 penny nail penetrate a 1 ¼" dry rafter? | 2 — 1 |
| 7. | 2 penny nail penetrate a 4" dry pole? | 1 — 3 ½ |
| 8. | 30 penny nail penetrate a ½" green beam? | 4 ½ — 15/32 |
| 9. | 50 penny nail penetrate a 4 ½" dry frame? | 5 ½ — 4 |
| 10. | 12 ½ box gauge nail penetrate a 1" green board? | 2 — 25/32 |
| 11. | 5 common gauge nail penetrate a 3 ½" dry girder? | 4 ½ — 3 |
| 12. | 10 ½ box gauge nail penetrate a 3" green rafter? | 3 — 29/16 |
| 13. | 11 ½ common gauge nail penetrate a ½" dry pole? | 2 — 7/16 |
| 14. | 4 penny nail penetrate a 3" dry girder? | 1 ½ — 2 ½ |
| 15. | 8 box gauge nail penetrate a 2" green plank? | 5 — 19/16 |
| 16. | 16 penny nail penetrate a 4" dry joist? | 3 ½ — 3 ½ |

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|--|---|
| 1. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 2. countersink hole for a 24 gauge screw? | $\frac{1}{2} \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 3. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 6. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 7. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 8. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 11. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 12. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 13. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 14. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 16. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |

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|-----|---|----------------|
| 1. | 9 penny nail penetrate a 1 1/2" green beam? | 2 3/4 — 1 9/32 |
| 2. | 7 penny nail penetrate a 4" dry timber? | 2 1/4 — 3 1/2 |
| 3. | 14 1/2 box gauge nail penetrate a 3 1/2" dry frame? | 1 1/4 — 3 |
| 4. | 40 penny nail penetrate a 2 1/2" green pole? | 5 — 2 1/16 |
| 5. | 5 penny nail penetrate a 3 1/2" dry dimension? | 1 3/4 — 3 |
| 6. | 4 penny nail penetrate a 2 1/2" dry dimension? | 1 1/2 — 2 |
| 7. | 6 penny nail penetrate a 3/4" green frame? | 2 — 11/16 |
| 8. | 10 penny nail penetrate a 1" dry beam? | 3 — 3/4 |
| 9. | 8 box gauge nail penetrate a 1" green plank? | 5 — 25/32 |
| 10. | 15 common gauge nail penetrate a 2" dry dimension? | 1 — 1 1/2 |
| 11. | 60 penny nail penetrate a 3" dry rafter? | 6 — 2 1/2 |
| 12. | 4 penny nail penetrate a 1 1/2" dry frame? | 1 1/2 — 1 1/4 |
| 13. | 10 penny nail penetrate a 4" green frame? | 3 — 3 9/16 |
| 14. | 10 1/2 box gauge nail penetrate a 1" dry dimension? | 3 — 3/4 |
| 15. | 9 box gauge nail penetrate a 1 1/2" green timber? | 4 — 1 9/32 |
| 16. | 14 common gauge nail penetrate a 3/8" dry girder? | 1 1/4 — 5/16 |

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| 1. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 2. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 3. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 4. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 5. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 6. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 7. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 8. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 9. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 10. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 11. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 12. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 13. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 14. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 15. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 16. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |

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|-----|--|--------------------------------|
| 1. | 10 penny nail penetrate a $2\frac{1}{2}$ " dry timber? | $3 - 2$ |
| 2. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{5}{8}$ " dry rafter? | $2\frac{1}{2} - \frac{9}{16}$ |
| 3. | 12 penny nail penetrate a $4\frac{1}{2}$ " dry plank? | $3\frac{1}{4} - 4$ |
| 4. | 9 box gauge nail penetrate a $4\frac{1}{2}$ " dry dimension? | $4 - 4$ |
| 5. | 2 penny nail penetrate a $1\frac{1}{4}$ " dry board? | $1 - 1$ |
| 6. | $10\frac{1}{2}$ box gauge nail penetrate a 4" dry dimension? | $3 - 3\frac{1}{2}$ |
| 7. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry beam? | $2\frac{1}{2} - \frac{5}{16}$ |
| 8. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " green board? | $2\frac{1}{2} - \frac{11}{32}$ |
| 9. | 5 penny nail penetrate a $4\frac{1}{2}$ " green mast? | $1\frac{3}{4} - 4\frac{1}{16}$ |
| 10. | 5 penny nail penetrate a $2\frac{1}{2}$ " green joist? | $1\frac{3}{4} - 2\frac{1}{16}$ |
| 11. | $10\frac{1}{4}$ common gauge nail penetrate a $\frac{5}{8}$ " green beam? | $2\frac{1}{2} - \frac{19}{32}$ |
| 12. | 8 penny nail penetrate a 1" dry mast? | $2\frac{1}{2} - \frac{3}{4}$ |
| 13. | $10\frac{1}{4}$ common gauge nail penetrate a 2" dry girder? | $2\frac{1}{2} - 1\frac{1}{2}$ |
| 14. | 6 penny nail penetrate a $4\frac{1}{2}$ " green mast? | $2 - 4\frac{1}{16}$ |
| 15. | $10\frac{1}{2}$ box gauge nail penetrate a 2" dry beam? | $3 - 1\frac{1}{2}$ |
| 16. | 7 penny nail penetrate a $\frac{5}{8}$ " green girder? | $2\frac{1}{4} - \frac{19}{32}$ |

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| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 3. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 4. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 5. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 6. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 7. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 8. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 9. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 10. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 11. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 12. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 13. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 14. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 15. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 16. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |

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|-----|--|---------------|
| 1. | 30 penny nail penetrate a 2 1/2" dry rafter? | 4 1/2 — 2 |
| 2. | 50 penny nail penetrate a 3" dry timber? | 5 1/2 — 2 1/2 |
| 3. | 4 common gauge nail penetrate a 3/8" green timber? | 5 — 11/32 |
| 4. | 5 common gauge nail penetrate a 1/2" green girder? | 4 1/2 — 15/32 |
| 5. | 8 penny nail penetrate a 1" green pole? | 2 1/2 — 25/32 |
| 6. | 9 box gauge nail penetrate a 1 1/4" dry beam? | 4 — 1 |
| 7. | 2 penny nail penetrate a 1/2" green girder? | 1 — 15/32 |
| 8. | 12 1/2 box gauge nail penetrate a 1 1/4" dry rafter? | 2 — 1 |
| 9. | 10 1/4 common gauge nail penetrate a 3 1/2" dry joist? | 2 1/2 — 3 |
| 10. | 11 1/2 box gauge nail penetrate a 1 1/4" green plank? | 2 1/2 — 11/32 |
| 11. | 3 penny nail penetrate a 1/2" dry girder? | 1 1/4 — 7/16 |
| 12. | 50 penny nail penetrate a 1 1/4" green girder? | 5 1/2 — 11/32 |
| 13. | 4 common gauge nail penetrate a 3/8" green frame? | 5 — 11/32 |
| 14. | 8 penny nail penetrate a 1/2" green board? | 2 1/2 — 15/32 |
| 15. | 50 penny nail penetrate a 5/8" green board? | 5 1/2 — 19/32 |
| 16. | 9 box gauge nail penetrate a 3" dry mast? | 4 — 2 1/2 |

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| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 3. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 4. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 5. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 6. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 7. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 8. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 9. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 10. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 11. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 12. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 13. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 14. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 15. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 16. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

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|-----|---|----------------------------------|
| 1. | 8 box gauge nail penetrate a 1" green frame? | $5 - \frac{25}{32}$ |
| 2. | 4 common gauge nail penetrate a 2" dry mast? | $5 - 1 \frac{1}{2}$ |
| 3. | 60 penny nail penetrate a $\frac{3}{4}$ " dry plank? | $6 - \frac{5}{8}$ |
| 4. | 9 box gauge nail penetrate a $\frac{3}{4}$ " green pole? | $4 - \frac{11}{16}$ |
| 5. | 14 box gauge nail penetrate a 4" green rafter? | $1 \frac{1}{2} - 3 \frac{9}{16}$ |
| 6. | 9 box gauge nail penetrate a 3" green joist? | $4 - 2 \frac{9}{16}$ |
| 7. | 20 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $4 - \frac{15}{32}$ |
| 8. | 3 penny nail penetrate a $\frac{3}{4}$ " green joist? | $1 \frac{1}{4} - \frac{11}{16}$ |
| 9. | 60 penny nail penetrate a 1" green rafter? | $6 - \frac{25}{32}$ |
| 10. | 7 penny nail penetrate a $\frac{3}{8}$ " dry timber? | $2 \frac{1}{4} - \frac{5}{16}$ |
| 11. | 2 penny nail penetrate a $3 \frac{1}{2}$ " dry plank? | $1 - 3$ |
| 12. | 20 penny nail penetrate a $1 \frac{1}{2}$ " dry beam? | $4 - 1 \frac{1}{4}$ |
| 13. | 10 penny nail penetrate a $\frac{5}{8}$ " dry girder? | $3 - \frac{9}{16}$ |
| 14. | 3 penny nail penetrate a $\frac{3}{4}$ " dry plank? | $1 \frac{1}{4} - \frac{5}{8}$ |
| 15. | 4 penny nail penetrate a 1" dry mast? | $1 \frac{1}{2} - \frac{3}{4}$ |
| 16. | 9 common gauge nail penetrate a $4 \frac{1}{2}$ " dry timber? | $3 - 4$ |

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| 1. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 2. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 3. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 4. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 5. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 6. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 7. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 8. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 9. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 10. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 11. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 13. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 14. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 15. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 16. countersink hole for a 20 gauge screw? | $\frac{29}{64} - \frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |

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| 1. | 12 ½ common gauge nail penetrate a 3/8" green dimension? | 1 ½ — 11/32 |
| 2. | 5 penny nail penetrate a 2 ½" green timber? | 1 ¾ — 2 1/16 |
| 3. | 11 ½ common gauge nail penetrate a ½" dry joist? | 2 — 7/16 |
| 4. | 4 common gauge nail penetrate a 1" dry beam? | 5 — ¾ |
| 5. | 8 penny nail penetrate a 3/4" dry joist? | 2 ½ — 5/8 |
| 6. | 60 penny nail penetrate a 5/8" dry joist? | 6 — 9/16 |
| 7. | 15 common gauge nail penetrate a 3" dry joist? | 1 — 2 ½ |
| 8. | 10 ½ box gauge nail penetrate a 4 ½" dry timber? | 3 — 4 |
| 9. | 14 box gauge nail penetrate a 3/8" dry dimension? | 1 ½ — 5/16 |
| 10. | 9 penny nail penetrate a 1 ½" dry mast? | 2 ¾ — 1 ¼ |
| 11. | 14 box gauge nail penetrate a 3" dry joist? | 1 ½ — 2 ½ |
| 12. | 9 common gauge nail penetrate a 3 ½" green mast? | 3 — 3 1/16 |
| 13. | 3 common gauge nail penetrate a 1 ¼" dry timber? | 5 ½ — 1 |
| 14. | 12 ½ common gauge nail penetrate a 1" green beam? | 1 ½ — 25/32 |
| 15. | 4 common gauge nail penetrate a 3 ½" dry plank? | 5 — 3 |
| 16. | 20 penny nail penetrate a 1 ¼" green pole? | 4 — 1 1/32 |

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| 1. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 2. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 3. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 4. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 5. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 6. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 7. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 8. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 9. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 10. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 11. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 12. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 13. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 14. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 15. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 16. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |

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| 1. | 50 penny nail penetrate a $1\frac{1}{4}$ " dry beam? | $5\frac{1}{2} - 1$ |
| 2. | 6 common gauge nail penetrate a $\frac{1}{2}$ " green joist? | $4 - \frac{15}{32}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " green girder? | $3 - 3\frac{1}{16}$ |
| 4. | 20 penny nail penetrate a $\frac{1}{2}$ " dry board? | $4 - \frac{7}{16}$ |
| 5. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green frame? | $3 - \frac{19}{32}$ |
| 6. | 5 penny nail penetrate a 4" green board? | $1\frac{3}{4} - 3\frac{9}{16}$ |
| 7. | 5 common gauge nail penetrate a $4\frac{1}{2}$ " dry dimension? | $4\frac{1}{2} - 4$ |
| 8. | 6 penny nail penetrate a $\frac{3}{4}$ " dry mast? | $2 - \frac{5}{8}$ |
| 9. | 8 box gauge nail penetrate a $1\frac{1}{2}$ " green pole? | $5 - 1\frac{9}{32}$ |
| 10. | 15 common gauge nail penetrate a $\frac{5}{8}$ " green girder? | $1 - \frac{19}{32}$ |
| 11. | 15 common gauge nail penetrate a 3" green girder? | $1 - 2\frac{9}{16}$ |
| 12. | 14 common gauge nail penetrate a 2" dry mast? | $1\frac{1}{4} - 1\frac{1}{2}$ |
| 13. | 7 penny nail penetrate a $\frac{3}{4}$ " dry girder? | $2\frac{1}{4} - \frac{5}{8}$ |
| 14. | 30 penny nail penetrate a $\frac{3}{8}$ " dry rafter? | $4\frac{1}{2} - \frac{5}{16}$ |
| 15. | 11 $\frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " green timber? | $2 - \frac{19}{32}$ |
| 16. | 2 penny nail penetrate a 4" green joist? | $1 - 3\frac{9}{16}$ |

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| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 3. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 4. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 5. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 10. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 11. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 12. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 13. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 14. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 15. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 16. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |

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| 1. | 4 penny nail penetrate a 2 1/2" dry girder? | 1 1/2 — 2 |
| 2. | 2 penny nail penetrate a 3" dry rafter? | 1 — 2 1/2 |
| 3. | 10 1/4 common gauge nail penetrate a 5/8" dry plank? | 2 1/2 — 9/16 |
| 4. | 50 penny nail penetrate a 3 1/2" green board? | 5 1/2 — 3 1/16 |
| 5. | 2 common gauge nail penetrate a 3/8" green plank? | 6 — 11/32 |
| 6. | 8 penny nail penetrate a 4" dry plank? | 2 1/2 — 3 1/2 |
| 7. | 60 penny nail penetrate a 1 1/2" green mast? | 6 — 19/32 |
| 8. | 8 box gauge nail penetrate a 2" green beam? | 5 — 19/16 |
| 9. | 12 1/2 box gauge nail penetrate a 3/4" green timber? | 2 — 11/16 |
| 10. | 12 1/2 box gauge nail penetrate a 3" green beam? | 2 — 29/16 |
| 11. | 50 penny nail penetrate a 5/8" dry plank? | 5 1/2 — 9/16 |
| 12. | 10 penny nail penetrate a 5/8" green joist? | 3 — 19/32 |
| 13. | 8 box gauge nail penetrate a 4 1/2" dry frame? | 5 — 4 |
| 14. | 3 common gauge nail penetrate a 2" green pole? | 5 1/2 — 19/16 |
| 15. | 12 penny nail penetrate a 1 1/4" dry girder? | 3 1/4 — 1 |
| 16. | 30 penny nail penetrate a 3 1/2" dry dimension? | 4 1/2 — 3 |

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|---|--|
| 1. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 2. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 3. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 4. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 5. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 6. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 7. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 8. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 9. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 10. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 11. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 12. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 13. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 14. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 15. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 16. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |

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|-----|---|--------------------------------|
| 1. | 5 common gauge nail penetrate a $2\frac{1}{2}$ " dry dimension? | $4\frac{1}{2} - 2$ |
| 2. | 40 penny nail penetrate a 1" green timber? | $5 - \frac{25}{32}$ |
| 3. | 11 $\frac{1}{2}$ common gauge nail penetrate a $4\frac{1}{2}$ " green timber? | $2 - 4\frac{1}{16}$ |
| 4. | 12 $\frac{1}{2}$ box gauge nail penetrate a $3\frac{1}{2}$ " green timber? | $2 - 3\frac{1}{16}$ |
| 5. | 9 common gauge nail penetrate a 4" green dimension? | $3 - 3\frac{9}{16}$ |
| 6. | 5 common gauge nail penetrate a $\frac{5}{8}$ " green mast? | $4\frac{1}{2} - \frac{19}{32}$ |
| 7. | 30 penny nail penetrate a $4\frac{1}{2}$ " dry board? | $4\frac{1}{2} - 4$ |
| 8. | 11 $\frac{1}{2}$ common gauge nail penetrate a $\frac{3}{4}$ " green joist? | $2 - \frac{11}{16}$ |
| 9. | 6 penny nail penetrate a 4" dry rafter? | $2 - 3\frac{1}{2}$ |
| 10. | 60 penny nail penetrate a $\frac{3}{4}$ " green rafter? | $6 - \frac{11}{16}$ |
| 11. | 9 common gauge nail penetrate a $\frac{1}{2}$ " dry beam? | $3 - \frac{7}{16}$ |
| 12. | 30 penny nail penetrate a 2" dry board? | $4\frac{1}{2} - 1\frac{1}{2}$ |
| 13. | 9 penny nail penetrate a 4" dry rafter? | $2\frac{3}{4} - 3\frac{1}{2}$ |
| 14. | 10 penny nail penetrate a $\frac{3}{8}$ " green plank? | $3 - \frac{11}{32}$ |
| 15. | 7 penny nail penetrate a $\frac{3}{4}$ " dry frame? | $2\frac{1}{4} - \frac{5}{8}$ |
| 16. | 8 common gauge nail penetrate a $\frac{1}{2}$ " green frame? | $3\frac{1}{2} - \frac{15}{32}$ |

1. pilot hole for a 7 gauge screw?	41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$
2. clearance hole for a 20 gauge screw?	Q $\frac{21}{64} = \frac{21}{64}$
3. clearance hole for a 3 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
4. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
5. pilot hole for a 1 gauge screw?	78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$
6. countersink hole for a 5 gauge screw?	E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$
7. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
8. countersink hole for a 16 gauge screw?	X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$
9. clearance hole for a 9 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$
10. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
11. clearance hole for a 3 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
12. clearance hole for a 4 gauge screw?	35 $\frac{7}{64} = \frac{7}{64}$
13. clearance hole for a 18 gauge screw?	N $\frac{19}{64} = \frac{19}{64}$
14. clearance hole for a 1 gauge screw?	47 $\frac{5}{64} = \frac{5}{64}$
15. clearance hole for a 10 gauge screw?	12 $\frac{3}{16} = \frac{3}{16}$
16. countersink hole for a 8 gauge screw?	N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$

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| 1. | 12 ½ box gauge nail penetrate a 3/8" green dimension? | 2 — 11/32 |
| 2. | 40 penny nail penetrate a 4 ½" dry board? | 5 — 4 |
| 3. | 4 common gauge nail penetrate a ½" green rafter? | 5 — 15/32 |
| 4. | 15 ½ box gauge nail penetrate a 3 ½" dry girder? | 1 — 3 |
| 5. | 11 ½ common gauge nail penetrate a ½" dry pole? | 2 — 7/16 |
| 6. | 11 ½ common gauge nail penetrate a 3" green dimension? | 2 — 29/16 |
| 7. | 3 common gauge nail penetrate a 3 ½" green timber? | 5 ½ — 31/16 |
| 8. | 7 penny nail penetrate a 3 ½" dry board? | 2 ¼ — 3 |
| 9. | 5 penny nail penetrate a 3 ½" dry timber? | 1 ¾ — 3 |
| 10. | 3 penny nail penetrate a 1 ¼" green girder? | 1 ¼ — 11/32 |
| 11. | 7 penny nail penetrate a 4" dry rafter? | 2 ¼ — 3 ½ |
| 12. | 11 ½ common gauge nail penetrate a 5/8" dry beam? | 2 — 9/16 |
| 13. | 40 penny nail penetrate a 2 ½" dry frame? | 5 — 2 |
| 14. | 9 box gauge nail penetrate a 3/4" dry timber? | 4 — 5/8 |
| 15. | 9 common gauge nail penetrate a ½" dry board? | 3 — 7/16 |
| 16. | 2 penny nail penetrate a 4 ½" dry plank? | 1 — 4 |

1. countersink hole for a 1 gauge screw?	6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$
2. clearance hole for a 3 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
3. pilot hole for a 1 gauge screw?	78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$
4. clearance hole for a 6 gauge screw?	27 $\frac{9}{64} = \frac{9}{64}$
5. pilot hole for a 11 gauge screw?	27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$
6. countersink hole for a 14 gauge screw?	V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$
7. clearance hole for a 20 gauge screw?	Q $\frac{21}{64} = \frac{21}{64}$
8. clearance hole for a 7 gauge screw?	22 $\frac{5}{32} = \frac{5}{32}$
9. pilot hole for a 14 gauge screw?	12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$
10. pilot hole for a 7 gauge screw?	41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$
11. countersink hole for a 1 gauge screw?	6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$
12. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
13. countersink hole for a 10 gauge screw?	O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$
14. clearance hole for a 8 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$
15. clearance hole for a 9 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$
16. clearance hole for a 8 gauge screw?	17 $\frac{11}{64} = \frac{11}{64}$

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|-----|--|----------------------------------|
| 1. | 50 penny nail penetrate a $\frac{1}{2}$ " green rafter? | $5 \frac{1}{2} - \frac{15}{32}$ |
| 2. | 4 penny nail penetrate a 3" green joist? | $1 \frac{1}{2} - 2 \frac{9}{16}$ |
| 3. | 12 $\frac{1}{2}$ common gauge nail penetrate a 1" dry rafter? | $1 \frac{1}{2} - \frac{3}{4}$ |
| 4. | 6 penny nail penetrate a $\frac{5}{8}$ " dry pole? | $2 - \frac{9}{16}$ |
| 5. | 12 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{2}$ " green rafter? | $2 - 1 \frac{9}{32}$ |
| 6. | 11 $\frac{1}{2}$ box gauge nail penetrate a 4 $\frac{1}{2}$ " dry board? | $2 \frac{1}{2} - 4$ |
| 7. | 16 penny nail penetrate a $\frac{5}{8}$ " green pole? | $3 \frac{1}{2} - \frac{19}{32}$ |
| 8. | 4 penny nail penetrate a 1 $\frac{1}{4}$ " green frame? | $1 \frac{1}{2} - 1 \frac{1}{32}$ |
| 9. | 14 box gauge nail penetrate a $\frac{5}{8}$ " green girder? | $1 \frac{1}{2} - \frac{19}{32}$ |
| 10. | 9 penny nail penetrate a $\frac{1}{2}$ " dry board? | $2 \frac{3}{4} - \frac{7}{16}$ |
| 11. | 7 penny nail penetrate a 1 $\frac{1}{4}$ " green beam? | $2 \frac{1}{4} - 1 \frac{1}{32}$ |
| 12. | 6 common gauge nail penetrate a 1 $\frac{1}{2}$ " green board? | $4 - 1 \frac{9}{32}$ |
| 13. | 6 common gauge nail penetrate a 1" dry plank? | $4 - \frac{3}{4}$ |
| 14. | 12 $\frac{1}{2}$ common gauge nail penetrate a 3 $\frac{1}{2}$ " green rafter? | $1 \frac{1}{2} - 3 \frac{1}{16}$ |
| 15. | 14 common gauge nail penetrate a 4 $\frac{1}{2}$ " green joist? | $1 \frac{1}{4} - 4 \frac{1}{16}$ |
| 16. | 2 common gauge nail penetrate a 2" green board? | $6 - 1 \frac{9}{16}$ |

1. pilot hole for a 2 gauge screw?	67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$
2. clearance hole for a 2 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
3. clearance hole for a 12 gauge screw?	2 $\frac{7}{32} = \frac{7}{32}$
4. countersink hole for a 14 gauge screw?	V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$
5. pilot hole for a 12 gauge screw?	22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$
6. countersink hole for a 7 gauge screw?	L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$
7. pilot hole for a 24 gauge screw?	O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$
8. pilot hole for a 4 gauge screw?	55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$
9. countersink hole for a 1 gauge screw?	6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$
10. countersink hole for a 10 gauge screw?	O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$
11. clearance hole for a 16 gauge screw?	H $\frac{17}{64} = \frac{17}{64}$
12. countersink hole for a 2 gauge screw?	2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$
13. clearance hole for a 24 gauge screw?	V $\frac{3}{8} = \frac{3}{8}$
14. clearance hole for a 6 gauge screw?	27 $\frac{9}{64} = \frac{9}{64}$
15. clearance hole for a 3 gauge screw?	41 $\frac{3}{32} = \frac{3}{32}$
16. countersink hole for a 16 gauge screw?	X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$

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|-----|--|---------------------------------|
| 1. | 3 penny nail penetrate a $\frac{1}{2}$ " green girder? | $1 \frac{1}{4} - \frac{15}{32}$ |
| 2. | 2 penny nail penetrate a $3 \frac{1}{2}$ " dry board? | $1 - 3$ |
| 3. | 15 common gauge nail penetrate a $\frac{5}{8}$ " green board? | $1 - \frac{19}{32}$ |
| 4. | 9 common gauge nail penetrate a $\frac{5}{8}$ " green frame? | $3 - \frac{19}{32}$ |
| 5. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " green pole? | $2 - \frac{15}{32}$ |
| 6. | 9 box gauge nail penetrate a $4 \frac{1}{2}$ " green joist? | $4 - 4\frac{1}{16}$ |
| 7. | 9 common gauge nail penetrate a 2" dry joist? | $3 - 1 \frac{1}{2}$ |
| 8. | 50 penny nail penetrate a $1 \frac{1}{2}$ " dry plank? | $5 \frac{1}{2} - 1 \frac{1}{4}$ |
| 9. | 40 penny nail penetrate a 2" dry beam? | $5 - 1 \frac{1}{2}$ |
| 10. | 8 box gauge nail penetrate a $\frac{3}{8}$ " green mast? | $5 - \frac{11}{32}$ |
| 11. | 16 penny nail penetrate a 1" green board? | $3 \frac{1}{2} - \frac{25}{32}$ |
| 12. | 8 box gauge nail penetrate a 1" green plank? | $5 - \frac{25}{32}$ |
| 13. | 50 penny nail penetrate a $\frac{1}{2}$ " dry beam? | $5 \frac{1}{2} - \frac{7}{16}$ |
| 14. | 15 $\frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " green plank? | $1 - 1\frac{1}{32}$ |
| 15. | 11 $\frac{1}{2}$ common gauge nail penetrate a $3 \frac{1}{2}$ " green pole? | $2 - 3\frac{1}{16}$ |
| 16. | 20 penny nail penetrate a 1" dry timber? | $4 - \frac{3}{4}$ |

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|---|--|
| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 3. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 4. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 5. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 6. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 7. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 8. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 10. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 11. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 12. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 13. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 14. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 15. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 16. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |

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|-----|--|--------------|
| 1. | 15 ½ box gauge nail penetrate a 3/4" dry beam? | 1 — 5/8 |
| 2. | 5 penny nail penetrate a 5/8" dry board? | 1 ¾ — 9/16 |
| 3. | 15 common gauge nail penetrate a 2" dry board? | 1 — 1 ½ |
| 4. | 11 ½ box gauge nail penetrate a ½" dry joist? | 2 ½ — 7/16 |
| 5. | 8 box gauge nail penetrate a 1 ¼" dry joist? | 5 — 1 |
| 6. | 5 penny nail penetrate a 2 ½" dry timber? | 1 ¾ — 2 |
| 7. | 14 ½ box gauge nail penetrate a 1" green beam? | 1 ¼ — 25/32 |
| 8. | 14 ½ box gauge nail penetrate a 1" dry board? | 1 ¼ — ¾ |
| 9. | 5 penny nail penetrate a 1" dry pole? | 1 ¾ — ¾ |
| 10. | 10 penny nail penetrate a 1" green girder? | 3 — 25/32 |
| 11. | 4 penny nail penetrate a 3/8" green dimension? | 1 ½ — 11/32 |
| 12. | 2 common gauge nail penetrate a ½" green timber? | 6 — 15/32 |
| 13. | 10 ¼ common gauge nail penetrate a ½" dry dimension? | 2 ½ — 7/16 |
| 14. | 14 common gauge nail penetrate a 3 ½" green rafter? | 1 ¼ — 3 1/16 |
| 15. | 15 common gauge nail penetrate a 4 ½" dry plank? | 1 — 4 |
| 16. | 14 common gauge nail penetrate a 4" dry frame? | 1 ¼ — 3 ½ |

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|--|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 3. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 4. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 5. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 6. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 7. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 9. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 10. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 11. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 12. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 13. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 14. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 15. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 16. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |

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| 1. | 6 common gauge nail penetrate a 4 ½" dry pole? | 4 — 4 |
| 2. | 10 ¼ common gauge nail penetrate a 1 ¼" green pole? | 2 ½ — 1 ⅓ |
| 3. | 12 penny nail penetrate a ¾" dry plank? | 3 ¼ — ⅝ |
| 4. | 60 penny nail penetrate a 1" dry pole? | 6 — ¾ |
| 5. | 15 common gauge nail penetrate a 1" green mast? | 1 — ⅔ |
| 6. | 10 ½ box gauge nail penetrate a 2" green rafter? | 3 — 1 ⅙ |
| 7. | 20 penny nail penetrate a 1" green joist? | 4 — ⅔ |
| 8. | 10 penny nail penetrate a 1" green girder? | 3 — ⅔ |
| 9. | 4 penny nail penetrate a ⅝" green pole? | 1 ½ — 1 ⅓ |
| 10. | 40 penny nail penetrate a 2 ½" green plank? | 5 — 2 ⅙ |
| 11. | 40 penny nail penetrate a 3" green beam? | 5 — 2 ⅙ |
| 12. | 5 common gauge nail penetrate a 1" green timber? | 4 ½ — ⅔ |
| 13. | 9 box gauge nail penetrate a 4 ½" green plank? | 4 — 4 ⅙ |
| 14. | 9 common gauge nail penetrate a ⅝" green board? | 3 — 1 ⅓ |
| 15. | 3 penny nail penetrate a 4 ½" dry plank? | 1 ¼ — 4 |
| 16. | 15 ½ box gauge nail penetrate a 2 ½" dry pole? | 1 — 2 |

1. pilot hole for a 24 gauge screw? O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$
2. clearance hole for a 5 gauge screw? 30 $\frac{1}{8} = \frac{1}{8}$
3. countersink hole for a 18 gauge screw? $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$
4. countersink hole for a 18 gauge screw? $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$
5. clearance hole for a 6 gauge screw? 27 $\frac{9}{64} = \frac{9}{64}$
6. pilot hole for a 20 gauge screw? H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$
7. pilot hole for a 12 gauge screw? 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$
8. countersink hole for a 24 gauge screw? $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$
9. countersink hole for a 1 gauge screw? 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$
10. pilot hole for a 5 gauge screw? 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$
11. countersink hole for a 7 gauge screw? L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$
12. clearance hole for a 8 gauge screw? 17 $\frac{11}{64} = \frac{11}{64}$
13. clearance hole for a 14 gauge screw? E $\frac{1}{4} = \frac{1}{4}$
14. countersink hole for a 7 gauge screw? L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$
15. clearance hole for a 11 gauge screw? 6 $\frac{13}{64} = \frac{13}{64}$
16. pilot hole for a 14 gauge screw? 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$

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|-----|--|-----------------------------------|
| 1. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " green timber? | 2 $\frac{1}{2}$ — $\frac{11}{32}$ |
| 2. | 14 box gauge nail penetrate a 3 $\frac{1}{2}$ " green joist? | 1 $\frac{1}{2}$ — $3\frac{1}{16}$ |
| 3. | 10 $\frac{1}{2}$ box gauge nail penetrate a 4" dry frame? | 3 — 3 $\frac{1}{2}$ |
| 4. | 60 penny nail penetrate a 2 $\frac{1}{2}$ " green mast? | 6 — $2\frac{1}{16}$ |
| 5. | 20 penny nail penetrate a 3" green mast? | 4 — $2\frac{9}{16}$ |
| 6. | 10 penny nail penetrate a $\frac{1}{2}$ " green girder? | 3 — $\frac{15}{32}$ |
| 7. | 12 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{4}$ " dry board? | 2 — 1 |
| 8. | 2 penny nail penetrate a 3" green mast? | 1 — $2\frac{9}{16}$ |
| 9. | 6 common gauge nail penetrate a 3 $\frac{1}{2}$ " green beam? | 4 — $3\frac{1}{16}$ |
| 10. | 2 penny nail penetrate a 3 $\frac{1}{2}$ " green rafter? | 1 — $3\frac{1}{16}$ |
| 11. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green frame? | 2 — $\frac{19}{32}$ |
| 12. | 6 penny nail penetrate a 2 $\frac{1}{2}$ " green pole? | 2 — $2\frac{1}{16}$ |
| 13. | 8 penny nail penetrate a 2" dry pole? | 2 $\frac{1}{2}$ — 1 $\frac{1}{2}$ |
| 14. | 5 common gauge nail penetrate a 4" green plank? | 4 $\frac{1}{2}$ — $3\frac{9}{16}$ |
| 15. | 8 box gauge nail penetrate a 1" green beam? | 5 — $\frac{25}{32}$ |
| 16. | 5 common gauge nail penetrate a 2" dry rafter? | 4 $\frac{1}{2}$ — 1 $\frac{1}{2}$ |

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|--|-----------------------------------|
| 1. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 2. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 4. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 5. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 6. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 7. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 8. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 9. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 10. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 11. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 12. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 13. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 14. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 15. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 16. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |

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|-----|---|---------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 2" green timber? | 3 — 1 ⁹ / ₁₆ |
| 2. | 2 penny nail penetrate a 3 ½" green frame? | 1 — 3 ¹ / ₁₆ |
| 3. | 14 box gauge nail penetrate a 1 ¼" green dimension? | 1 ½ — 1 ¹ / ₃₂ |
| 4. | 4 penny nail penetrate a 1 ½" dry board? | 1 ½ — 1 ¼ |
| 5. | 10 ½ box gauge nail penetrate a 2" dry mast? | 3 — 1 ½ |
| 6. | 10 ½ box gauge nail penetrate a 5/8" green beam? | 3 — 1 ⁹ / ₃₂ |
| 7. | 7 penny nail penetrate a 3/8" green board? | 2 ¼ — 1 ¹¹ / ₃₂ |
| 8. | 5 penny nail penetrate a ½" dry plank? | 1 ¾ — 7/ ₁₆ |
| 9. | 14 box gauge nail penetrate a 1 ½" green rafter? | 1 ½ — 1 ⁹ / ₃₂ |
| 10. | 50 penny nail penetrate a 3/8" green timber? | 5 ½ — 1 ¹¹ / ₃₂ |
| 11. | 8 box gauge nail penetrate a 2" dry plank? | 5 — 1 ½ |
| 12. | 8 box gauge nail penetrate a 5/8" dry pole? | 5 — 9/ ₁₆ |
| 13. | 11 ½ box gauge nail penetrate a ½" dry plank? | 2 ½ — 7/ ₁₆ |
| 14. | 8 box gauge nail penetrate a 2 ½" dry board? | 5 — 2 |
| 15. | 15 ½ box gauge nail penetrate a 5/8" dry joist? | 1 — 9/ ₁₆ |
| 16. | 50 penny nail penetrate a 1" green girder? | 5 ½ — 2 ⁵ / ₃₂ |

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|--|---|
| 1. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 2. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 3. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 4. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 5. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 6. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 7. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 8. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 9. clearance hole for a 6 gauge screw? | 27 $\frac{9}{64} = \frac{9}{64}$ |
| 10. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 11. countersink hole for a 24 gauge screw? | $\frac{1}{2} \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 12. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 13. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 14. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 15. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 16. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |

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|-----|--|---------------------------------|
| 1. | 8 penny nail penetrate a $\frac{1}{2}$ " green timber? | $2 \frac{1}{2} - \frac{15}{32}$ |
| 2. | 50 penny nail penetrate a 2" green plank? | $5 \frac{1}{2} - \frac{19}{16}$ |
| 3. | 10 penny nail penetrate a 2" dry pole? | $3 - 1 \frac{1}{2}$ |
| 4. | 20 penny nail penetrate a $\frac{1}{2}$ " green plank? | $4 - \frac{15}{32}$ |
| 5. | 6 common gauge nail penetrate a $1 \frac{1}{2}$ " green pole? | $4 - \frac{19}{32}$ |
| 6. | 40 penny nail penetrate a $\frac{1}{2}$ " dry mast? | $5 - \frac{7}{16}$ |
| 7. | 20 penny nail penetrate a 4" green girder? | $4 - \frac{39}{16}$ |
| 8. | 50 penny nail penetrate a $\frac{3}{8}$ " dry joist? | $5 \frac{1}{2} - \frac{5}{16}$ |
| 9. | 9 common gauge nail penetrate a $\frac{1}{2}$ " green plank? | $3 - \frac{15}{32}$ |
| 10. | 4 penny nail penetrate a 2" dry pole? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 11. | 9 box gauge nail penetrate a $\frac{5}{8}$ " green beam? | $4 - \frac{19}{32}$ |
| 12. | $12 \frac{1}{2}$ common gauge nail penetrate a $\frac{5}{8}$ " dry mast? | $1 \frac{1}{2} - \frac{9}{16}$ |
| 13. | $14 \frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry board? | $1 \frac{1}{4} - \frac{5}{16}$ |
| 14. | 7 penny nail penetrate a $3 \frac{1}{2}$ " green frame? | $2 \frac{1}{4} - \frac{31}{16}$ |
| 15. | $11 \frac{1}{2}$ common gauge nail penetrate a 2" green pole? | $2 - \frac{19}{16}$ |
| 16. | 9 box gauge nail penetrate a $1 \frac{1}{2}$ " dry plank? | $4 - 1 \frac{1}{4}$ |

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|--|--------------------------|
| 1. clearance hole for a 10 gauge screw? | 12 $3/16 = 3/16$ |
| 2. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 3. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 4. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 5. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 6. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 8. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 9. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 10. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 11. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 12. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 13. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 14. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 15. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 16. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |

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|-----|---|--------------|
| 1. | 10 ½ box gauge nail penetrate a 3" dry frame? | 3 — 2 ½ |
| 2. | 40 penny nail penetrate a ½" green plank? | 5 — 15/32 |
| 3. | 6 common gauge nail penetrate a 3/4" dry rafter? | 4 — 5/8 |
| 4. | 10 ½ box gauge nail penetrate a 3 ½" green rafter? | 3 — 3 1/16 |
| 5. | 10 ¼ common gauge nail penetrate a 1 ¼" green beam? | 2 ½ — 1 1/32 |
| 6. | 16 penny nail penetrate a 4 ½" green girder? | 3 ½ — 4 1/16 |
| 7. | 14 common gauge nail penetrate a 3" green pole? | 1 ¼ — 2 9/16 |
| 8. | 50 penny nail penetrate a 3" green dimension? | 5 ½ — 2 9/16 |
| 9. | 12 penny nail penetrate a 4 ½" green pole? | 3 ¼ — 4 1/16 |
| 10. | 12 penny nail penetrate a 3 ½" green rafter? | 3 ¼ — 3 1/16 |
| 11. | 10 ¼ common gauge nail penetrate a 3 ½" green timber? | 2 ½ — 3 1/16 |
| 12. | 8 box gauge nail penetrate a 1" dry girder? | 5 — ¾ |
| 13. | 2 common gauge nail penetrate a ½" dry mast? | 6 — 7/16 |
| 14. | 8 box gauge nail penetrate a 3/4" green rafter? | 5 — 11/16 |
| 15. | 5 penny nail penetrate a 1 ¼" dry mast? | 1 ¾ — 1 |
| 16. | 8 box gauge nail penetrate a 3/8" dry rafter? | 5 — 5/16 |

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|--|---|
| 1. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 2. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 3. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 4. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 5. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 6. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 7. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 8. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 9. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 10. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 14. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 15. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 16. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |

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|-----|--|---------------------------------|
| 1. | 8 common gauge nail penetrate a 1" green dimension? | $3 \frac{1}{2} - \frac{25}{32}$ |
| 2. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green rafter? | $2 \frac{1}{2} - \frac{19}{32}$ |
| 3. | 11 $\frac{1}{2}$ common gauge nail penetrate a $3 \frac{1}{2}$ " dry girder? | $2 - 3$ |
| 4. | 2 penny nail penetrate a $1 \frac{1}{2}$ " green joist? | $1 - \frac{19}{32}$ |
| 5. | 12 $\frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " dry joist? | $2 - 2$ |
| 6. | 30 penny nail penetrate a $\frac{3}{4}$ " dry girder? | $4 \frac{1}{2} - \frac{5}{8}$ |
| 7. | 3 common gauge nail penetrate a 1" green timber? | $5 \frac{1}{2} - \frac{25}{32}$ |
| 8. | 11 $\frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " green dimension? | $2 \frac{1}{2} - 3\frac{1}{16}$ |
| 9. | 3 penny nail penetrate a $\frac{5}{8}$ " dry frame? | $1 \frac{1}{4} - \frac{9}{16}$ |
| 10. | 15 $\frac{1}{2}$ box gauge nail penetrate a $3 \frac{1}{2}$ " green joist? | $1 - 3\frac{1}{16}$ |
| 11. | 9 box gauge nail penetrate a $1 \frac{1}{2}$ " dry beam? | $4 - 1 \frac{1}{4}$ |
| 12. | 15 common gauge nail penetrate a $4 \frac{1}{2}$ " dry mast? | $1 - 4$ |
| 13. | 20 penny nail penetrate a $\frac{3}{4}$ " green plank? | $4 - \frac{11}{16}$ |
| 14. | 8 penny nail penetrate a 3" green timber? | $2 \frac{1}{2} - 2\frac{9}{16}$ |
| 15. | 5 penny nail penetrate a 3" dry mast? | $1 \frac{3}{4} - 2 \frac{1}{2}$ |
| 16. | 4 penny nail penetrate a $4 \frac{1}{2}$ " green board? | $1 \frac{1}{2} - 4\frac{1}{16}$ |

1. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
2. countersink hole for a 5 gauge screw?	E $1/8 + 1/8 = 1/4$
3. countersink hole for a 4 gauge screw?	B $7/64 + 1/8 = 15/64$
4. clearance hole for a 9 gauge screw?	17 $11/64 = 11/64$
5. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
6. countersink hole for a 3 gauge screw?	2 $3/32 + 1/8 = 7/32$
7. countersink hole for a 24 gauge screw?	$1/2$ $3/8 + 1/8 = 1/2$
8. countersink hole for a 16 gauge screw?	X $17/64 + 1/8 = 25/64$
9. pilot hole for a 3 gauge screw?	67 $3/32 - 1/16 = 1/32$
10. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
11. clearance hole for a 10 gauge screw?	12 $3/16 = 3/16$
12. pilot hole for a 24 gauge screw?	O $3/8 - 1/16 = 5/16$
13. pilot hole for a 11 gauge screw?	27 $13/64 - 1/16 = 9/64$
14. pilot hole for a 14 gauge screw?	12 $1/4 - 1/16 = 3/16$
15. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
16. pilot hole for a 7 gauge screw?	41 $5/32 - 1/16 = 3/32$

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|-----|---|---------------------------------------|
| 1. | 15 ½ box gauge nail penetrate a 4" dry board? | 1 — 3 ½ |
| 2. | 40 penny nail penetrate a 3" green board? | 5 — 2 ⁹ / ₁₆ |
| 3. | 9 penny nail penetrate a 2 ½" dry frame? | 2 ¾ — 2 |
| 4. | 2 common gauge nail penetrate a ¾" dry beam? | 6 — ⅝ |
| 5. | 3 penny nail penetrate a 1 ¼" dry rafter? | 1 ¼ — 1 |
| 6. | 10 penny nail penetrate a 3" green timber? | 3 — 2 ⁹ / ₁₆ |
| 7. | 60 penny nail penetrate a 3 ½" green beam? | 6 — 3 ¹ / ₁₆ |
| 8. | 9 penny nail penetrate a ½" green girder? | 2 ¾ — 15 ¹ / ₃₂ |
| 9. | 3 common gauge nail penetrate a 5/8" dry mast? | 5 ½ — 9 ¹ / ₁₆ |
| 10. | 8 penny nail penetrate a 4 ½" green dimension? | 2 ½ — 4 ¹ / ₁₆ |
| 11. | 50 penny nail penetrate a ¾" dry beam? | 5 ½ — ⅝ |
| 12. | 14 box gauge nail penetrate a 5/8" green rafter? | 1 ½ — 19 ¹ / ₃₂ |
| 13. | 4 common gauge nail penetrate a ½" green dimension? | 5 — 15 ¹ / ₃₂ |
| 14. | 16 penny nail penetrate a ¾" green timber? | 3 ½ — 11 ¹ / ₁₆ |
| 15. | 11 ½ box gauge nail penetrate a 3" green rafter? | 2 ½ — 2 ⁹ / ₁₆ |
| 16. | 10 ¼ common gauge nail penetrate a 2 ½" green beam? | 2 ½ — 2 ¹ / ₁₆ |

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| 1. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 2. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |
| 3. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 4. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 5. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 6. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 7. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 8. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 9. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 10. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 11. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 14. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 15. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 16. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |

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|-----|--|-----------|
| 1. | 50 penny nail penetrate a 1 ½" dry frame? | 5 ½ — 1 ¼ |
| 2. | 4 penny nail penetrate a 2 ½" green frame? | 1 ½ — 2 ⅓ |
| 3. | 9 penny nail penetrate a 1 ¼" dry pole? | 2 ¾ — 1 |
| 4. | 10 ½ box gauge nail penetrate a 4" dry pole? | 3 — 3 ½ |
| 5. | 9 common gauge nail penetrate a 2" green mast? | 3 — 1 ⅙ |
| 6. | 4 common gauge nail penetrate a 4 ½" dry girder? | 5 — 4 |
| 7. | 14 box gauge nail penetrate a ¾" dry dimension? | 1 ½ — ⅝ |
| 8. | 40 penny nail penetrate a ⅜" dry plank? | 5 — ⅕ |
| 9. | 2 penny nail penetrate a 4 ½" green joist? | 1 — 4 ⅓ |
| 10. | 7 penny nail penetrate a 4" green board? | 2 ¼ — 3 ⅓ |
| 11. | 15 common gauge nail penetrate a 3" green pole? | 1 — 2 ⅓ |
| 12. | 11 ½ common gauge nail penetrate a 1" green frame? | 2 — ⅔ |
| 13. | 10 penny nail penetrate a 1" green girder? | 3 — ⅔ |
| 14. | 50 penny nail penetrate a 1 ½" green timber? | 5 ½ — 1 ⅓ |
| 15. | 8 common gauge nail penetrate a 1 ½" green board? | 3 ½ — 1 ⅓ |
| 16. | 10 penny nail penetrate a 2" dry joist? | 3 — 1 ½ |

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|--|-----------------------------------|
| 1. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 2. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 3. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 4. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 5. pilot hole for a 24 gauge screw? | O $3/8 - 1/16 = 5/16$ |
| 6. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 7. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 8. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 9. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 10. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 11. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 12. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 13. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 14. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 15. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 16. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |

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|-----|--|--------------|
| 1. | 16 penny nail penetrate a 1 ½" dry mast? | 3 ½ — 1 ¼ |
| 2. | 10 ½ box gauge nail penetrate a 2" dry plank? | 3 — 1 ½ |
| 3. | 30 penny nail penetrate a 4" dry girder? | 4 ½ — 3 ½ |
| 4. | 9 box gauge nail penetrate a 3/8" green joist? | 4 — 11/32 |
| 5. | 14 box gauge nail penetrate a 2 ½" green joist? | 1 ½ — 2 1/16 |
| 6. | 10 ½ box gauge nail penetrate a 2 ½" green board? | 3 — 2 1/16 |
| 7. | 9 box gauge nail penetrate a 1 ¼" dry girder? | 4 — 1 |
| 8. | 10 ¼ common gauge nail penetrate a 2" dry rafter? | 2 ½ — 1 ½ |
| 9. | 50 penny nail penetrate a 5/8" green timber? | 5 ½ — 19/32 |
| 10. | 30 penny nail penetrate a 4" dry pole? | 4 ½ — 3 ½ |
| 11. | 20 penny nail penetrate a 3/4" dry plank? | 4 — 5/8 |
| 12. | 2 common gauge nail penetrate a ½" dry frame? | 6 — 7/16 |
| 13. | 5 penny nail penetrate a 4" dry pole? | 1 ¾ — 3 ½ |
| 14. | 8 penny nail penetrate a 2" green plank? | 2 ½ — 1 9/16 |
| 15. | 14 box gauge nail penetrate a 2" green dimension? | 1 ½ — 1 9/16 |
| 16. | 12 ½ common gauge nail penetrate a 2" green board? | 1 ½ — 1 9/16 |

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|--|---|
| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 3. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 4. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 5. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 6. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 7. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 8. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 9. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 10. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 11. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 12. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 13. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 14. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 15. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |
| 16. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |

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|-----|--|--------------------------------|
| 1. | 8 box gauge nail penetrate a $1\frac{1}{4}$ " green board? | $5 - 1\frac{1}{32}$ |
| 2. | 50 penny nail penetrate a $4\frac{1}{2}$ " green pole? | $5\frac{1}{2} - 4\frac{1}{16}$ |
| 3. | 8 penny nail penetrate a $3\frac{1}{2}$ " dry joist? | $2\frac{1}{2} - 3$ |
| 4. | 3 penny nail penetrate a $\frac{5}{8}$ " green timber? | $1\frac{1}{4} - \frac{19}{32}$ |
| 5. | 50 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $5\frac{1}{2} - \frac{9}{16}$ |
| 6. | 3 penny nail penetrate a $3\frac{1}{2}$ " green beam? | $1\frac{1}{4} - 3\frac{1}{16}$ |
| 7. | 10 penny nail penetrate a 4" dry pole? | $3 - 3\frac{1}{2}$ |
| 8. | $10\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green rafter? | $3 - \frac{11}{32}$ |
| 9. | 3 common gauge nail penetrate a 1" green plank? | $5\frac{1}{2} - \frac{25}{32}$ |
| 10. | 16 penny nail penetrate a 1" dry mast? | $3\frac{1}{2} - \frac{3}{4}$ |
| 11. | 12 penny nail penetrate a 4" green timber? | $3\frac{1}{4} - 3\frac{9}{16}$ |
| 12. | 5 penny nail penetrate a $\frac{3}{8}$ " green beam? | $1\frac{3}{4} - \frac{11}{32}$ |
| 13. | 2 penny nail penetrate a 1" green dimension? | $1 - \frac{25}{32}$ |
| 14. | 2 common gauge nail penetrate a $\frac{3}{8}$ " green timber? | $6 - \frac{11}{32}$ |
| 15. | 8 box gauge nail penetrate a $\frac{3}{8}$ " dry joist? | $5 - \frac{5}{16}$ |
| 16. | 2 penny nail penetrate a $\frac{3}{4}$ " green timber? | $1 - \frac{11}{16}$ |

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|---|---|
| 1. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 2. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 3. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 4. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 5. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 6. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 7. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 8. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 10. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 11. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 12. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 13. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 16. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |

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|-----|--|----------------------------------|
| 1. | 60 penny nail penetrate a 3" dry dimension? | $6 - 2 \frac{1}{2}$ |
| 2. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " green frame? | $2 - \frac{19}{32}$ |
| 3. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1 $\frac{1}{2}$ " green mast? | $2 - \frac{19}{32}$ |
| 4. | 2 penny nail penetrate a 4 $\frac{1}{2}$ " dry board? | $1 - 4$ |
| 5. | 16 penny nail penetrate a 1 $\frac{1}{4}$ " dry beam? | $3 \frac{1}{2} - 1$ |
| 6. | 9 box gauge nail penetrate a $\frac{3}{4}$ " dry dimension? | $4 - \frac{5}{8}$ |
| 7. | 10 $\frac{1}{4}$ common gauge nail penetrate a 1" green frame? | $2 \frac{1}{2} - \frac{25}{32}$ |
| 8. | 40 penny nail penetrate a 1" green pole? | $5 - \frac{25}{32}$ |
| 9. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1" dry plank? | $2 - \frac{3}{4}$ |
| 10. | 9 box gauge nail penetrate a 1 $\frac{1}{4}$ " green beam? | $4 - 1 \frac{1}{32}$ |
| 11. | 50 penny nail penetrate a 4" dry dimension? | $5 \frac{1}{2} - 3 \frac{1}{2}$ |
| 12. | 8 common gauge nail penetrate a $\frac{3}{8}$ " dry girder? | $3 \frac{1}{2} - \frac{5}{16}$ |
| 13. | 6 penny nail penetrate a 1" green pole? | $2 - \frac{25}{32}$ |
| 14. | 8 common gauge nail penetrate a 1" dry mast? | $3 \frac{1}{2} - \frac{3}{4}$ |
| 15. | 6 penny nail penetrate a 4" green dimension? | $2 - 3 \frac{9}{16}$ |
| 16. | 5 common gauge nail penetrate a 4 $\frac{1}{2}$ " green pole? | $4 \frac{1}{2} - 4 \frac{1}{16}$ |

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|---|---|
| 1. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 2. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 3. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 4. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 5. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 6. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 7. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 8. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 9. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 10. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 11. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 12. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 13. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |
| 14. clearance hole for a 2 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 15. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 16. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |

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|-----|---|-------------|
| 1. | 10 ¼ common gauge nail penetrate a 5/8" dry timber? | 2 ½ — 9/16 |
| 2. | 12 ½ box gauge nail penetrate a 4" green timber? | 2 — 39/16 |
| 3. | 12 ½ box gauge nail penetrate a 1 ½" dry board? | 2 — 1 ¼ |
| 4. | 10 ½ box gauge nail penetrate a 1" dry frame? | 3 — ¾ |
| 5. | 8 box gauge nail penetrate a 3 ½" dry plank? | 5 — 3 |
| 6. | 9 penny nail penetrate a 4 ½" dry timber? | 2 ¾ — 4 |
| 7. | 50 penny nail penetrate a 3/8" dry joist? | 5 ½ — 5/16 |
| 8. | 3 penny nail penetrate a 5/8" dry girder? | 1 ¼ — 9/16 |
| 9. | 11 ½ common gauge nail penetrate a 2" dry mast? | 2 — 1 ½ |
| 10. | 20 penny nail penetrate a 3/4" green mast? | 4 — 11/16 |
| 11. | 10 ½ box gauge nail penetrate a 3/8" green board? | 3 — 11/32 |
| 12. | 3 penny nail penetrate a 5/8" green beam? | 1 ¼ — 19/32 |
| 13. | 8 penny nail penetrate a 4" green timber? | 2 ½ — 39/16 |
| 14. | 10 ½ box gauge nail penetrate a 5/8" dry frame? | 3 — 9/16 |
| 15. | 60 penny nail penetrate a 5/8" green dimension? | 6 — 19/32 |
| 16. | 10 ½ box gauge nail penetrate a 2 ½" dry mast? | 3 — 2 |

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|---|--------------------------|
| 1. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 2. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 3. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 4. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 5. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 6. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 7. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 8. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 9. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 10. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 11. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 12. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 13. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 15. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 16. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |

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|-----|--|--------------------------------|
| 1. | 8 box gauge nail penetrate a 2" green mast? | $5 - 1\frac{9}{16}$ |
| 2. | 50 penny nail penetrate a $\frac{1}{2}$ " dry dimension? | $5\frac{1}{2} - \frac{7}{16}$ |
| 3. | 8 box gauge nail penetrate a $\frac{5}{8}$ " green rafter? | $5 - 1\frac{9}{32}$ |
| 4. | 2 common gauge nail penetrate a $1\frac{1}{2}$ " green dimension? | $6 - 1\frac{9}{32}$ |
| 5. | 40 penny nail penetrate a $1\frac{1}{4}$ " green girder? | $5 - 1\frac{1}{32}$ |
| 6. | 20 penny nail penetrate a $\frac{3}{4}$ " dry rafter? | $4 - \frac{5}{8}$ |
| 7. | 10 penny nail penetrate a $1\frac{1}{4}$ " dry frame? | $3 - 1$ |
| 8. | 9 common gauge nail penetrate a 3" dry rafter? | $3 - 2\frac{1}{2}$ |
| 9. | $11\frac{1}{2}$ common gauge nail penetrate a 3" green frame? | $2 - 2\frac{9}{16}$ |
| 10. | 60 penny nail penetrate a $1\frac{1}{4}$ " dry pole? | $6 - 1$ |
| 11. | 50 penny nail penetrate a $\frac{1}{2}$ " dry plank? | $5\frac{1}{2} - \frac{7}{16}$ |
| 12. | 14 box gauge nail penetrate a $2\frac{1}{2}$ " dry dimension? | $1\frac{1}{2} - 2$ |
| 13. | 8 penny nail penetrate a 2" dry pole? | $2\frac{1}{2} - 1\frac{1}{2}$ |
| 14. | 14 common gauge nail penetrate a 2" dry mast? | $1\frac{1}{4} - 1\frac{1}{2}$ |
| 15. | 14 box gauge nail penetrate a $4\frac{1}{2}$ " green rafter? | $1\frac{1}{2} - 4\frac{1}{16}$ |
| 16. | $11\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry beam? | $2\frac{1}{2} - \frac{7}{16}$ |

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|--|---|
| 1. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 2. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 3. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 4. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 5. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 6. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 7. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 8. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 10. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 13. countersink hole for a 9 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 15. countersink hole for a 20 gauge screw? | $\frac{29}{64} - \frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 16. clearance hole for a 5 gauge screw? | 30 $\frac{1}{8} = \frac{1}{8}$ |

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|-----|---|---------------------------------|
| 1. | 8 box gauge nail penetrate a $1\frac{1}{4}$ " dry board? | $5 - 1$ |
| 2. | 7 penny nail penetrate a 4" dry timber? | $2\frac{1}{4} - 3\frac{1}{2}$ |
| 3. | 16 penny nail penetrate a $3\frac{1}{2}$ " green girder? | $3\frac{1}{2} - 3\frac{1}{16}$ |
| 4. | $11\frac{1}{2}$ box gauge nail penetrate a 1" green pole? | $2\frac{1}{2} - 2\frac{25}{32}$ |
| 5. | 8 common gauge nail penetrate a $\frac{3}{8}$ " dry rafter? | $3\frac{1}{2} - \frac{5}{16}$ |
| 6. | 12 penny nail penetrate a 3" dry rafter? | $3\frac{1}{4} - 2\frac{1}{2}$ |
| 7. | 20 penny nail penetrate a $1\frac{1}{2}$ " green rafter? | $4 - 1\frac{9}{32}$ |
| 8. | $10\frac{1}{2}$ box gauge nail penetrate a 3" dry joist? | $3 - 2\frac{1}{2}$ |
| 9. | 30 penny nail penetrate a $\frac{5}{8}$ " dry frame? | $4\frac{1}{2} - \frac{9}{16}$ |
| 10. | 50 penny nail penetrate a $2\frac{1}{2}$ " green girder? | $5\frac{1}{2} - 2\frac{1}{16}$ |
| 11. | $10\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{2}$ " dry pole? | $3 - 1\frac{1}{4}$ |
| 12. | 8 box gauge nail penetrate a 2" green joist? | $5 - 1\frac{9}{16}$ |
| 13. | 16 penny nail penetrate a $\frac{5}{8}$ " dry rafter? | $3\frac{1}{2} - \frac{9}{16}$ |
| 14. | 5 common gauge nail penetrate a $1\frac{1}{2}$ " dry plank? | $4\frac{1}{2} - 1\frac{1}{4}$ |
| 15. | 50 penny nail penetrate a $\frac{3}{4}$ " green rafter? | $5\frac{1}{2} - 1\frac{1}{16}$ |
| 16. | 5 penny nail penetrate a 1" dry timber? | $1\frac{3}{4} - \frac{3}{4}$ |

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|--|--------------------------|
| 1. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 2. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 3. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 4. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 5. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 6. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 7. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 8. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 9. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |
| 10. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 11. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 12. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 13. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 14. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 15. clearance hole for a 20 gauge screw? | Q $21/64 = 21/64$ |
| 16. countersink hole for a 24 gauge screw? | $1/2$ $3/8 + 1/8 = 1/2$ |

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|-----|--|------------------------|
| 1. | 7 penny nail penetrate a $4\frac{1}{2}$ " dry beam? | $2\frac{1}{4} - 4$ |
| 2. | 2 penny nail penetrate a 1" green joist? | $1 - 25/32$ |
| 3. | 9 box gauge nail penetrate a $3\frac{1}{2}$ " green girder? | $4 - 31/16$ |
| 4. | 16 penny nail penetrate a $5/8$ " dry pole? | $3\frac{1}{2} - 9/16$ |
| 5. | $12\frac{1}{2}$ common gauge nail penetrate a $4\frac{1}{2}$ " green rafter? | $1\frac{1}{2} - 41/16$ |
| 6. | 50 penny nail penetrate a $3\frac{1}{2}$ " dry beam? | $5\frac{1}{2} - 3$ |
| 7. | 2 common gauge nail penetrate a 4" green frame? | $6 - 39/16$ |
| 8. | 20 penny nail penetrate a $3/4$ " green rafter? | $4 - 11/16$ |
| 9. | 30 penny nail penetrate a $1\frac{1}{4}$ " green mast? | $4\frac{1}{2} - 11/32$ |
| 10. | 60 penny nail penetrate a $3\frac{1}{2}$ " dry plank? | $6 - 3$ |
| 11. | 14 box gauge nail penetrate a $3/4$ " dry pole? | $1\frac{1}{2} - 5/8$ |
| 12. | $12\frac{1}{2}$ box gauge nail penetrate a $1\frac{1}{4}$ " dry joist? | $2 - 1$ |
| 13. | 6 common gauge nail penetrate a $3\frac{1}{2}$ " green dimension? | $4 - 31/16$ |
| 14. | 40 penny nail penetrate a $4\frac{1}{2}$ " green rafter? | $5 - 41/16$ |
| 15. | 7 penny nail penetrate a $4\frac{1}{2}$ " green joist? | $2\frac{1}{4} - 41/16$ |
| 16. | 9 common gauge nail penetrate a $4\frac{1}{2}$ " green pole? | $3 - 41/16$ |

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|--|--------------------------|
| 1. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 2. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 3. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 4. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 5. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 6. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 7. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 8. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 9. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 10. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 11. countersink hole for a 16 gauge screw? | X $17/64 + 1/8 = 25/64$ |
| 12. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 13. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 14. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 15. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 16. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |

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|-----|---|----------------|
| 1. | 6 penny nail penetrate a 2 1/2" dry mast? | 2 — 2 |
| 2. | 8 penny nail penetrate a 1 1/2" green mast? | 2 1/2 — 1 9/32 |
| 3. | 12 1/2 box gauge nail penetrate a 1/2" green girder? | 2 — 15/32 |
| 4. | 15 common gauge nail penetrate a 1/2" green plank? | 1 — 15/32 |
| 5. | 6 penny nail penetrate a 1/2" dry pole? | 2 — 7/16 |
| 6. | 2 common gauge nail penetrate a 2" dry beam? | 6 — 1 1/2 |
| 7. | 3 common gauge nail penetrate a 3/4" dry joist? | 5 1/2 — 5/8 |
| 8. | 14 box gauge nail penetrate a 4 1/2" green mast? | 1 1/2 — 4 1/16 |
| 9. | 9 penny nail penetrate a 1/2" dry rafter? | 2 3/4 — 7/16 |
| 10. | 10 1/2 box gauge nail penetrate a 3 1/2" dry timber? | 3 — 3 |
| 11. | 14 box gauge nail penetrate a 1/2" green plank? | 1 1/2 — 15/32 |
| 12. | 10 1/2 box gauge nail penetrate a 1/2" green dimension? | 3 — 15/32 |
| 13. | 8 penny nail penetrate a 4" green board? | 2 1/2 — 3 9/16 |
| 14. | 12 penny nail penetrate a 2" dry rafter? | 3 1/4 — 1 1/2 |
| 15. | 30 penny nail penetrate a 4 1/2" dry beam? | 4 1/2 — 4 |
| 16. | 14 box gauge nail penetrate a 2" green timber? | 1 1/2 — 1 9/16 |

1. pilot hole for a 9 gauge screw?	35 $11/64 - 1/16 = 7/64$
2. pilot hole for a 5 gauge screw?	52 $1/8 - 1/16 = 1/16$
3. pilot hole for a 3 gauge screw?	67 $3/32 - 1/16 = 1/32$
4. clearance hole for a 6 gauge screw?	27 $9/64 = 9/64$
5. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
6. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
7. pilot hole for a 10 gauge screw?	30 $3/16 - 1/16 = 1/8$
8. clearance hole for a 4 gauge screw?	35 $7/64 = 7/64$
9. countersink hole for a 2 gauge screw?	2 $3/32 + 1/8 = 7/32$
10. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
11. pilot hole for a 11 gauge screw?	27 $13/64 - 1/16 = 9/64$
12. pilot hole for a 2 gauge screw?	67 $3/32 - 1/16 = 1/32$
13. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
14. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
15. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
16. clearance hole for a 6 gauge screw?	27 $9/64 = 9/64$

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|-----|--|--------------------------------------|
| 1. | 10 ½ box gauge nail penetrate a 2" green board? | 3 — 1 ⁹ / ₁₆ |
| 2. | 3 common gauge nail penetrate a 2" green joist? | 5 ½ — 1 ⁹ / ₁₆ |
| 3. | 8 common gauge nail penetrate a 2" dry joist? | 3 ½ — 1 ½ |
| 4. | 5 penny nail penetrate a 3" dry mast? | 1 ¾ — 2 ½ |
| 5. | 9 common gauge nail penetrate a 3 ½" dry timber? | 3 — 3 |
| 6. | 15 common gauge nail penetrate a 4" green rafter? | 1 — 3 ⁹ / ₁₆ |
| 7. | 11 ½ box gauge nail penetrate a 1 ½" green rafter? | 2 ½ — 1 ⁹ / ₃₂ |
| 8. | 11 ½ common gauge nail penetrate a 3/8" dry plank? | 2 — 5 ¹ / ₁₆ |
| 9. | 14 ½ box gauge nail penetrate a 4" dry rafter? | 1 ¼ — 3 ½ |
| 10. | 5 penny nail penetrate a 3 ½" green mast? | 1 ¾ — 3 ¹ / ₁₆ |
| 11. | 14 common gauge nail penetrate a 3/8" dry mast? | 1 ¼ — 5 ¹ / ₁₆ |
| 12. | 12 ½ common gauge nail penetrate a 5/8" dry joist? | 1 ½ — 9 ¹ / ₁₆ |
| 13. | 14 box gauge nail penetrate a 4" dry frame? | 1 ½ — 3 ½ |
| 14. | 12 penny nail penetrate a 1 ¼" dry board? | 3 ¼ — 1 |
| 15. | 8 box gauge nail penetrate a 4" dry girder? | 5 — 3 ½ |
| 16. | 8 penny nail penetrate a 4" green mast? | 2 ½ — 3 ⁹ / ₁₆ |

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| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. clearance hole for a 24 gauge screw? | V $\frac{3}{8} = \frac{3}{8}$ |
| 3. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 4. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 5. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 6. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 7. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 8. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 10. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 11. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 12. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 13. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 14. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 15. pilot hole for a 4 gauge screw? | 55 $\frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 16. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |

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|-----|---|--------------------------------------|
| 1. | 11 ½ box gauge nail penetrate a 2 ½" dry dimension? | 2 ½ — 2 |
| 2. | 4 common gauge nail penetrate a 2 ½" green mast? | 5 — 2 ¹ / ₁₆ |
| 3. | 8 box gauge nail penetrate a 2 ½" dry timber? | 5 — 2 |
| 4. | 8 penny nail penetrate a 4 ½" green board? | 2 ½ — 4 ¹ / ₁₆ |
| 5. | 12 ½ common gauge nail penetrate a 4" dry pole? | 1 ½ — 3 ½ |
| 6. | 20 penny nail penetrate a ¾" green joist? | 4 — 1 ¹ / ₁₆ |
| 7. | 11 ½ box gauge nail penetrate a ¾" green timber? | 2 ½ — 1 ¹ / ₁₆ |
| 8. | 12 ½ box gauge nail penetrate a 2 ½" dry girder? | 2 — 2 |
| 9. | 11 ½ box gauge nail penetrate a 1" green beam? | 2 ½ — 2 ⁵ / ₃₂ |
| 10. | 11 ½ common gauge nail penetrate a 3 ½" green beam? | 2 — 3 ¹ / ₁₆ |
| 11. | 8 penny nail penetrate a ½" green board? | 2 ½ — 1 ⁵ / ₃₂ |
| 12. | 8 penny nail penetrate a 2 ½" dry timber? | 2 ½ — 2 |
| 13. | 4 penny nail penetrate a 1" dry timber? | 1 ½ — ¾ |
| 14. | 8 common gauge nail penetrate a 2" dry timber? | 3 ½ — 1 ½ |
| 15. | 14 box gauge nail penetrate a ¾" dry mast? | 1 ½ — ⅝ |
| 16. | 40 penny nail penetrate a ½" dry girder? | 5 — 7 ¹ / ₁₆ |

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|---|---|
| 1. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 2. countersink hole for a 24 gauge screw? | $\frac{1}{2} \quad \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 3. countersink hole for a 14 gauge screw? | $V \quad \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 4. clearance hole for a 24 gauge screw? | $V \quad \frac{3}{8} = \frac{3}{8}$ |
| 5. clearance hole for a 4 gauge screw? | $35 \quad \frac{7}{64} = \frac{7}{64}$ |
| 6. clearance hole for a 7 gauge screw? | $22 \quad \frac{5}{32} = \frac{5}{32}$ |
| 7. countersink hole for a 1 gauge screw? | $6 \quad \frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 8. clearance hole for a 2 gauge screw? | $41 \quad \frac{3}{32} = \frac{3}{32}$ |
| 9. clearance hole for a 14 gauge screw? | $E \quad \frac{1}{4} = \frac{1}{4}$ |
| 10. countersink hole for a 8 gauge screw? | $N \quad \frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. countersink hole for a 3 gauge screw? | $2 \quad \frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. pilot hole for a 16 gauge screw? | $6 \quad \frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 13. countersink hole for a 9 gauge screw? | $N \quad \frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 14. pilot hole for a 12 gauge screw? | $22 \quad \frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 15. pilot hole for a 4 gauge screw? | $55 \quad \frac{7}{64} - \frac{1}{16} = \frac{3}{64}$ |
| 16. clearance hole for a 18 gauge screw? | $N \quad \frac{19}{64} = \frac{19}{64}$ |

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|-----|---|---------------------------------------|
| 1. | 11 ½ box gauge nail penetrate a 4 ½" green dimension? | 2 ½ — 4 ¹ / ₁₆ |
| 2. | 9 penny nail penetrate a 3" green mast? | 2 ¾ — 2 ⁹ / ₁₆ |
| 3. | 16 penny nail penetrate a 3/8" green frame? | 3 ½ — 11 ¹ / ₃₂ |
| 4. | 10 ¼ common gauge nail penetrate a 1 ¼" dry board? | 2 ½ — 1 |
| 5. | 10 ¼ common gauge nail penetrate a 4" green rafter? | 2 ½ — 3 ⁹ / ₁₆ |
| 6. | 9 common gauge nail penetrate a 1 ¼" dry frame? | 3 — 1 |
| 7. | 6 penny nail penetrate a 1 ½" green pole? | 2 — 1 ⁹ / ₃₂ |
| 8. | 12 ½ box gauge nail penetrate a 4" dry plank? | 2 — 3 ½ |
| 9. | 11 ½ common gauge nail penetrate a 5/8" green frame? | 2 — 1 ⁹ / ₃₂ |
| 10. | 3 penny nail penetrate a ½" dry board? | 1 ¼ — 7 ¹ / ₁₆ |
| 11. | 10 ½ box gauge nail penetrate a 3 ½" dry girder? | 3 — 3 |
| 12. | 9 box gauge nail penetrate a 5/8" green timber? | 4 — 1 ⁹ / ₃₂ |
| 13. | 2 penny nail penetrate a 4" green timber? | 1 — 3 ⁹ / ₁₆ |
| 14. | 7 penny nail penetrate a 3/4" dry beam? | 2 ¼ — 5 ¹ / ₈ |
| 15. | 15 common gauge nail penetrate a 1 ¼" green joist? | 1 — 1 ¹ / ₃₂ |
| 16. | 12 penny nail penetrate a 3/4" green rafter? | 3 ¼ — 11 ¹ / ₁₆ |

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|--|-------------------------------|
| 1. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 2. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 3. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 4. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 5. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 6. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 7. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 8. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 9. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 10. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 11. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 12. countersink hole for a 5 gauge screw? | E $1/8 + 1/8 = 1/4$ |
| 13. clearance hole for a 14 gauge screw? | E $1/4 = 1/4$ |
| 14. countersink hole for a 18 gauge screw? | $27/64 - 19/64 + 1/8 = 27/64$ |
| 15. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 16. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |

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|-----|---|----------------------------------|
| 1. | 8 common gauge nail penetrate a 1" green beam? | $3 \frac{1}{2} - \frac{25}{32}$ |
| 2. | 16 penny nail penetrate a $1 \frac{1}{4}$ " green plank? | $3 \frac{1}{2} - 1 \frac{1}{32}$ |
| 3. | 10 penny nail penetrate a $3 \frac{1}{2}$ " dry timber? | $3 - 3$ |
| 4. | 7 penny nail penetrate a $\frac{5}{8}$ " dry timber? | $2 \frac{1}{4} - \frac{9}{16}$ |
| 5. | 2 common gauge nail penetrate a $4 \frac{1}{2}$ " green girder? | $6 - 4 \frac{1}{16}$ |
| 6. | 3 penny nail penetrate a $2 \frac{1}{2}$ " green board? | $1 \frac{1}{4} - 2 \frac{1}{16}$ |
| 7. | $15 \frac{1}{2}$ box gauge nail penetrate a 4" green frame? | $1 - 3 \frac{9}{16}$ |
| 8. | 5 penny nail penetrate a $\frac{3}{8}$ " dry pole? | $1 \frac{3}{4} - \frac{5}{16}$ |
| 9. | 6 common gauge nail penetrate a $1 \frac{1}{2}$ " dry frame? | $4 - 1 \frac{1}{4}$ |
| 10. | $12 \frac{1}{2}$ common gauge nail penetrate a 2" dry pole? | $1 \frac{1}{2} - 1 \frac{1}{2}$ |
| 11. | $10 \frac{1}{2}$ box gauge nail penetrate a 3" green beam? | $3 - 2 \frac{9}{16}$ |
| 12. | 12 penny nail penetrate a 2" dry beam? | $3 \frac{1}{4} - 1 \frac{1}{2}$ |
| 13. | 40 penny nail penetrate a 2" dry plank? | $5 - 1 \frac{1}{2}$ |
| 14. | 3 penny nail penetrate a $\frac{3}{4}$ " green pole? | $1 \frac{1}{4} - \frac{11}{16}$ |
| 15. | $10 \frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{4}$ " dry girder? | $3 - 1$ |
| 16. | $10 \frac{1}{4}$ common gauge nail penetrate a 4" green plank? | $2 \frac{1}{2} - 3 \frac{9}{16}$ |

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|--|-------------------------------|
| 1. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 2. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 3. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 4. clearance hole for a 11 gauge screw? | 6 $13/64 = 13/64$ |
| 5. pilot hole for a 6 gauge screw? | 47 $9/64 - 1/16 = 5/64$ |
| 6. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 7. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 8. countersink hole for a 12 gauge screw? | S $7/32 + 1/8 = 11/32$ |
| 9. clearance hole for a 18 gauge screw? | N $19/64 = 19/64$ |
| 10. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 11. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 12. pilot hole for a 16 gauge screw? | 6 $17/64 - 1/16 = 13/64$ |
| 13. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 14. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 15. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 16. clearance hole for a 12 gauge screw? | 2 $7/32 = 7/32$ |

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|-----|---|--------------------------------|
| 1. | 8 box gauge nail penetrate a 3" dry board? | $5 - 2\frac{1}{2}$ |
| 2. | 2 common gauge nail penetrate a 1" dry plank? | $6 - \frac{3}{4}$ |
| 3. | 16 penny nail penetrate a 3" green board? | $3\frac{1}{2} - 2\frac{9}{16}$ |
| 4. | $10\frac{1}{4}$ common gauge nail penetrate a $1\frac{1}{2}$ " dry beam? | $2\frac{1}{2} - 1\frac{1}{4}$ |
| 5. | 14 box gauge nail penetrate a $4\frac{1}{2}$ " dry frame? | $1\frac{1}{2} - 4$ |
| 6. | 8 box gauge nail penetrate a $4\frac{1}{2}$ " dry frame? | $5 - 4$ |
| 7. | $10\frac{1}{4}$ common gauge nail penetrate a $1\frac{1}{4}$ " green beam? | $2\frac{1}{2} - 1\frac{1}{32}$ |
| 8. | $11\frac{1}{2}$ common gauge nail penetrate a 3" green girder? | $2 - 2\frac{9}{16}$ |
| 9. | $11\frac{1}{2}$ box gauge nail penetrate a $2\frac{1}{2}$ " green mast? | $2\frac{1}{2} - 2\frac{1}{16}$ |
| 10. | 50 penny nail penetrate a $1\frac{1}{4}$ " dry frame? | $5\frac{1}{2} - 1$ |
| 11. | 30 penny nail penetrate a $1\frac{1}{4}$ " dry joist? | $4\frac{1}{2} - 1$ |
| 12. | $12\frac{1}{2}$ common gauge nail penetrate a $4\frac{1}{2}$ " green plank? | $1\frac{1}{2} - 4\frac{1}{16}$ |
| 13. | 8 box gauge nail penetrate a $\frac{3}{8}$ " dry mast? | $5 - \frac{5}{16}$ |
| 14. | 4 penny nail penetrate a $2\frac{1}{2}$ " green rafter? | $1\frac{1}{2} - 2\frac{1}{16}$ |
| 15. | 5 common gauge nail penetrate a 2" green rafter? | $4\frac{1}{2} - 1\frac{9}{16}$ |
| 16. | 8 penny nail penetrate a $3\frac{1}{2}$ " green board? | $2\frac{1}{2} - 3\frac{1}{16}$ |

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|--|--------------------------|
| 1. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 2. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 3. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 4. countersink hole for a 14 gauge screw? | V $1/4 + 1/8 = 3/8$ |
| 5. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 6. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 7. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 8. pilot hole for a 9 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 9. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 10. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 11. clearance hole for a 8 gauge screw? | 17 $11/64 = 11/64$ |
| 12. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 13. countersink hole for a 24 gauge screw? | $1/2 - 3/8 + 1/8 = 1/2$ |
| 14. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 15. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 16. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |

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|-----|--|--------------------------------|
| 1. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{4}$ " dry girder? | $2\frac{1}{2} - \frac{5}{8}$ |
| 2. | 5 penny nail penetrate a 1" green rafter? | $1\frac{3}{4} - \frac{25}{32}$ |
| 3. | 6 penny nail penetrate a $2\frac{1}{2}$ " green beam? | $2 - 2\frac{1}{16}$ |
| 4. | 10 $\frac{1}{4}$ common gauge nail penetrate a $\frac{3}{8}$ " dry board? | $2\frac{1}{2} - \frac{5}{16}$ |
| 5. | 50 penny nail penetrate a $3\frac{1}{2}$ " green pole? | $5\frac{1}{2} - 3\frac{1}{16}$ |
| 6. | 12 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{4}$ " dry plank? | $2 - \frac{5}{8}$ |
| 7. | 9 common gauge nail penetrate a $1\frac{1}{2}$ " green pole? | $3 - 1\frac{9}{32}$ |
| 8. | 9 common gauge nail penetrate a $\frac{3}{8}$ " dry girder? | $3 - \frac{5}{16}$ |
| 9. | 6 penny nail penetrate a $4\frac{1}{2}$ " dry rafter? | $2 - 4$ |
| 10. | 7 penny nail penetrate a $1\frac{1}{2}$ " green board? | $2\frac{1}{4} - 1\frac{9}{32}$ |
| 11. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{1}{2}$ " dry board? | $3 - \frac{7}{16}$ |
| 12. | 6 penny nail penetrate a $4\frac{1}{2}$ " green plank? | $2 - 4\frac{1}{16}$ |
| 13. | 6 common gauge nail penetrate a $\frac{3}{4}$ " green frame? | $4 - 1\frac{1}{16}$ |
| 14. | 11 $\frac{1}{2}$ common gauge nail penetrate a $2\frac{1}{2}$ " dry mast? | $2 - 2$ |
| 15. | 15 common gauge nail penetrate a $1\frac{1}{2}$ " dry dimension? | $1 - 1\frac{1}{4}$ |
| 16. | 2 penny nail penetrate a 1" dry beam? | $1 - \frac{3}{4}$ |

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|--|-----------------------------------|
| 1. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 2. countersink hole for a 24 gauge screw? | $1/2 \quad 3/8 + 1/8 = 1/2$ |
| 3. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 4. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 5. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 6. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |
| 7. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 8. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 9. countersink hole for a 1 gauge screw? | 6 $5/64 + 1/8 = 13/64$ |
| 10. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 11. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 12. countersink hole for a 10 gauge screw? | O $3/16 + 1/8 = 5/16$ |
| 13. pilot hole for a 7 gauge screw? | 41 $5/32 - 1/16 = 3/32$ |
| 14. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 15. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 16. clearance hole for a 9 gauge screw? | 17 $11/64 = 11/64$ |

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|-----|---|----------------------------------|
| 1. | 9 penny nail penetrate a $5/8$ " green board? | $2 \frac{3}{4} - 19/32$ |
| 2. | 50 penny nail penetrate a $3 \frac{1}{2}$ " green beam? | $5 \frac{1}{2} - 3 \frac{1}{16}$ |
| 3. | 6 penny nail penetrate a $4 \frac{1}{2}$ " dry pole? | $2 - 4$ |
| 4. | $12 \frac{1}{2}$ box gauge nail penetrate a 2" green pole? | $2 - 19/16$ |
| 5. | $10 \frac{1}{4}$ common gauge nail penetrate a 3" dry rafter? | $2 \frac{1}{2} - 2 \frac{1}{2}$ |
| 6. | 4 penny nail penetrate a 2" green timber? | $1 \frac{1}{2} - 19/16$ |
| 7. | $11 \frac{1}{2}$ box gauge nail penetrate a 2" green plank? | $2 \frac{1}{2} - 19/16$ |
| 8. | 5 penny nail penetrate a $2 \frac{1}{2}$ " green pole? | $1 \frac{3}{4} - 2 \frac{1}{16}$ |
| 9. | 8 box gauge nail penetrate a 2" green mast? | $5 - 19/16$ |
| 10. | $12 \frac{1}{2}$ common gauge nail penetrate a $1 \frac{1}{2}$ " dry frame? | $1 \frac{1}{2} - 1 \frac{1}{4}$ |
| 11. | 3 penny nail penetrate a $3/4$ " dry beam? | $1 \frac{1}{4} - 5/8$ |
| 12. | 9 common gauge nail penetrate a 4" dry mast? | $3 - 3 \frac{1}{2}$ |
| 13. | $10 \frac{1}{2}$ box gauge nail penetrate a 3" dry dimension? | $3 - 2 \frac{1}{2}$ |
| 14. | 60 penny nail penetrate a 1" dry dimension? | $6 - 3/4$ |
| 15. | 16 penny nail penetrate a $3/4$ " green pole? | $3 \frac{1}{2} - 11/16$ |
| 16. | 8 box gauge nail penetrate a 1" green mast? | $5 - 25/32$ |

1. pilot hole for a 16 gauge screw?	6 $17/64 - 1/16 = 13/64$
2. pilot hole for a 12 gauge screw?	22 $7/32 - 1/16 = 5/32$
3. pilot hole for a 5 gauge screw?	52 $1/8 - 1/16 = 1/16$
4. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$
5. clearance hole for a 2 gauge screw?	41 $3/32 = 3/32$
6. pilot hole for a 10 gauge screw?	30 $3/16 - 1/16 = 1/8$
7. countersink hole for a 5 gauge screw?	E $1/8 + 1/8 = 1/4$
8. clearance hole for a 1 gauge screw?	47 $5/64 = 5/64$
9. clearance hole for a 14 gauge screw?	E $1/4 = 1/4$
10. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
11. pilot hole for a 20 gauge screw?	H $21/64 - 1/16 = 17/64$
12. pilot hole for a 3 gauge screw?	67 $3/32 - 1/16 = 1/32$
13. countersink hole for a 9 gauge screw?	N $11/64 + 1/8 = 19/64$
14. pilot hole for a 8 gauge screw?	35 $11/64 - 1/16 = 7/64$
15. pilot hole for a 4 gauge screw?	55 $7/64 - 1/16 = 3/64$
16. countersink hole for a 12 gauge screw?	S $7/32 + 1/8 = 11/32$

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| 1. | 8 common gauge nail penetrate a 2" dry girder? | $3 \frac{1}{2} - 1 \frac{1}{2}$ |
| 2. | 50 penny nail penetrate a $1 \frac{1}{4}$ " dry girder? | $5 \frac{1}{2} - 1$ |
| 3. | 60 penny nail penetrate a $2 \frac{1}{2}$ " dry dimension? | $6 - 2$ |
| 4. | 30 penny nail penetrate a $2 \frac{1}{2}$ " green board? | $4 \frac{1}{2} - 2 \frac{1}{16}$ |
| 5. | $12 \frac{1}{2}$ box gauge nail penetrate a 3" green timber? | $2 - 2 \frac{9}{16}$ |
| 6. | 3 penny nail penetrate a $1 \frac{1}{4}$ " dry girder? | $1 \frac{1}{4} - 1$ |
| 7. | 4 penny nail penetrate a 4" dry beam? | $1 \frac{1}{2} - 3 \frac{1}{2}$ |
| 8. | 30 penny nail penetrate a 1" green beam? | $4 \frac{1}{2} - 2 \frac{5}{32}$ |
| 9. | 5 penny nail penetrate a $\frac{5}{8}$ " dry dimension? | $1 \frac{3}{4} - \frac{9}{16}$ |
| 10. | 3 penny nail penetrate a 4" dry frame? | $1 \frac{1}{4} - 3 \frac{1}{2}$ |
| 11. | 16 penny nail penetrate a $1 \frac{1}{2}$ " green frame? | $3 \frac{1}{2} - 1 \frac{9}{32}$ |
| 12. | $14 \frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry dimension? | $1 \frac{1}{4} - \frac{9}{16}$ |
| 13. | 7 penny nail penetrate a 4" dry pole? | $2 \frac{1}{4} - 3 \frac{1}{2}$ |
| 14. | 14 common gauge nail penetrate a $3 \frac{1}{2}$ " dry frame? | $1 \frac{1}{4} - 3$ |
| 15. | 60 penny nail penetrate a $4 \frac{1}{2}$ " green mast? | $6 - 4 \frac{1}{16}$ |
| 16. | 9 box gauge nail penetrate a $\frac{1}{2}$ " green beam? | $4 - 1 \frac{15}{32}$ |

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| 1. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 2. pilot hole for a 10 gauge screw? | 30 $3/16 - 1/16 = 1/8$ |
| 3. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 4. clearance hole for a 24 gauge screw? | V $3/8 = 3/8$ |
| 5. pilot hole for a 3 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 6. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 7. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 8. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 9. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 10. countersink hole for a 20 gauge screw? | $29/64$ $21/64 + 1/8 = 29/64$ |
| 11. countersink hole for a 2 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 12. countersink hole for a 18 gauge screw? | $27/64$ $19/64 + 1/8 = 27/64$ |
| 13. pilot hole for a 14 gauge screw? | 12 $1/4 - 1/16 = 3/16$ |
| 14. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 15. countersink hole for a 11 gauge screw? | Q $13/64 + 1/8 = 21/64$ |
| 16. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |

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|-----|--|----------------|
| 1. | 9 penny nail penetrate a 3 1/2" green frame? | 2 3/4 — 3 1/16 |
| 2. | 12 1/2 box gauge nail penetrate a 1 1/4" dry joist? | 2 — 1 |
| 3. | 15 1/2 box gauge nail penetrate a 5/8" green plank? | 1 — 19/32 |
| 4. | 11 1/2 box gauge nail penetrate a 2" dry plank? | 2 1/2 — 1 1/2 |
| 5. | 8 penny nail penetrate a 4" dry joist? | 2 1/2 — 3 1/2 |
| 6. | 2 common gauge nail penetrate a 3/4" green girder? | 6 — 11/16 |
| 7. | 8 box gauge nail penetrate a 3" green joist? | 5 — 29/16 |
| 8. | 3 penny nail penetrate a 3/4" green plank? | 1 1/4 — 11/16 |
| 9. | 16 penny nail penetrate a 1 1/4" green board? | 3 1/2 — 1 1/32 |
| 10. | 11 1/2 box gauge nail penetrate a 1 1/4" dry mast? | 2 1/2 — 1 |
| 11. | 8 box gauge nail penetrate a 3/8" dry pole? | 5 — 5/16 |
| 12. | 9 common gauge nail penetrate a 1 1/4" dry plank? | 3 — 1 |
| 13. | 12 penny nail penetrate a 3/4" green pole? | 3 1/4 — 11/16 |
| 14. | 12 1/2 common gauge nail penetrate a 2 1/2" green frame? | 1 1/2 — 2 1/16 |
| 15. | 12 1/2 common gauge nail penetrate a 1 1/2" dry board? | 1 1/2 — 1 1/4 |
| 16. | 30 penny nail penetrate a 3 1/2" dry pole? | 4 1/2 — 3 |

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|---|-----------------------------------|
| 1. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 2. pilot hole for a 6 gauge screw? | $47 \quad 9/64 - 1/16 = 5/64$ |
| 3. countersink hole for a 6 gauge screw? | $H \quad 9/64 + 1/8 = 17/64$ |
| 4. countersink hole for a 14 gauge screw? | $V \quad 1/4 + 1/8 = 3/8$ |
| 5. countersink hole for a 14 gauge screw? | $V \quad 1/4 + 1/8 = 3/8$ |
| 6. clearance hole for a 1 gauge screw? | $47 \quad 5/64 = 5/64$ |
| 7. pilot hole for a 1 gauge screw? | $78 \quad 5/64 - 1/16 = 1/64$ |
| 8. pilot hole for a 5 gauge screw? | $52 \quad 1/8 - 1/16 = 1/16$ |
| 9. countersink hole for a 8 gauge screw? | $N \quad 11/64 + 1/8 = 19/64$ |
| 10. countersink hole for a 7 gauge screw? | $L \quad 5/32 + 1/8 = 9/32$ |
| 11. pilot hole for a 3 gauge screw? | $67 \quad 3/32 - 1/16 = 1/32$ |
| 12. pilot hole for a 20 gauge screw? | $H \quad 21/64 - 1/16 = 17/64$ |
| 13. countersink hole for a 8 gauge screw? | $N \quad 11/64 + 1/8 = 19/64$ |
| 14. pilot hole for a 7 gauge screw? | $41 \quad 5/32 - 1/16 = 3/32$ |
| 15. clearance hole for a 11 gauge screw? | $6 \quad 13/64 = 13/64$ |
| 16. clearance hole for a 24 gauge screw? | $V \quad 3/8 = 3/8$ |

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|-----|--|--------------------------------|
| 1. | 6 penny nail penetrate a 2" green board? | $2 - 1\frac{9}{16}$ |
| 2. | 10 $\frac{1}{2}$ box gauge nail penetrate a 1 $\frac{1}{4}$ " green plank? | $3 - 1\frac{1}{32}$ |
| 3. | 8 common gauge nail penetrate a 4" dry rafter? | $3\frac{1}{2} - 3\frac{1}{2}$ |
| 4. | 15 common gauge nail penetrate a 4" green plank? | $1 - 3\frac{9}{16}$ |
| 5. | 6 penny nail penetrate a $\frac{5}{8}$ " green board? | $2 - 1\frac{9}{32}$ |
| 6. | 40 penny nail penetrate a $\frac{1}{2}$ " dry board? | $5 - \frac{7}{16}$ |
| 7. | 5 common gauge nail penetrate a 2 $\frac{1}{2}$ " green plank? | $4\frac{1}{2} - 2\frac{1}{16}$ |
| 8. | 3 common gauge nail penetrate a 3 $\frac{1}{2}$ " green plank? | $5\frac{1}{2} - 3\frac{1}{16}$ |
| 9. | 40 penny nail penetrate a 3 $\frac{1}{2}$ " green plank? | $5 - 3\frac{1}{16}$ |
| 10. | 2 penny nail penetrate a $\frac{1}{2}$ " dry joist? | $1 - \frac{7}{16}$ |
| 11. | 11 $\frac{1}{2}$ common gauge nail penetrate a 2 $\frac{1}{2}$ " dry rafter? | $2 - 2$ |
| 12. | 15 common gauge nail penetrate a 1 $\frac{1}{4}$ " dry joist? | $1 - 1$ |
| 13. | 11 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " dry plank? | $2\frac{1}{2} - \frac{5}{16}$ |
| 14. | 40 penny nail penetrate a 4 $\frac{1}{2}$ " dry mast? | $5 - 4$ |
| 15. | 12 $\frac{1}{2}$ box gauge nail penetrate a 2" dry girder? | $2 - 1\frac{1}{2}$ |
| 16. | 40 penny nail penetrate a 3" green mast? | $5 - 2\frac{9}{16}$ |

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|--|---|
| 1. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 2. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 3. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 4. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 5. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 6. clearance hole for a 4 gauge screw? | 35 $\frac{7}{64} = \frac{7}{64}$ |
| 7. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 8. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 9. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 10. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 11. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 12. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 13. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 14. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 15. pilot hole for a 3 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 16. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |

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|-----|---|----------------------------------|
| 1. | 30 penny nail penetrate a 3" dry plank? | $4 \frac{1}{2} - 2 \frac{1}{2}$ |
| 2. | 40 penny nail penetrate a $\frac{3}{8}$ " dry joist? | $5 - \frac{5}{16}$ |
| 3. | 14 box gauge nail penetrate a $1 \frac{1}{4}$ " green joist? | $1 \frac{1}{2} - 1 \frac{1}{32}$ |
| 4. | $10 \frac{1}{2}$ box gauge nail penetrate a 4" green board? | $3 - 3 \frac{9}{16}$ |
| 5. | $12 \frac{1}{2}$ common gauge nail penetrate a 4" green beam? | $1 \frac{1}{2} - 3 \frac{9}{16}$ |
| 6. | 14 common gauge nail penetrate a $\frac{5}{8}$ " dry dimension? | $1 \frac{1}{4} - \frac{9}{16}$ |
| 7. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry beam? | $3 - \frac{9}{16}$ |
| 8. | 7 penny nail penetrate a 2" dry plank? | $2 \frac{1}{4} - 1 \frac{1}{2}$ |
| 9. | $15 \frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " green rafter? | $1 - 2 \frac{1}{16}$ |
| 10. | 6 common gauge nail penetrate a $1 \frac{1}{2}$ " dry girder? | $4 - 1 \frac{1}{4}$ |
| 11. | $10 \frac{1}{2}$ box gauge nail penetrate a 2" dry board? | $3 - 1 \frac{1}{2}$ |
| 12. | 6 penny nail penetrate a $3 \frac{1}{2}$ " green plank? | $2 - 3 \frac{1}{16}$ |
| 13. | 8 penny nail penetrate a $4 \frac{1}{2}$ " dry dimension? | $2 \frac{1}{2} - 4$ |
| 14. | 50 penny nail penetrate a $3 \frac{1}{2}$ " green timber? | $5 \frac{1}{2} - 3 \frac{1}{16}$ |
| 15. | 9 penny nail penetrate a 3" dry mast? | $2 \frac{3}{4} - 2 \frac{1}{2}$ |
| 16. | 60 penny nail penetrate a 4" green board? | $6 - 3 \frac{9}{16}$ |

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|---|---|
| 1. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 2. countersink hole for a 24 gauge screw? | $\frac{1}{2} - \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ |
| 3. pilot hole for a 18 gauge screw? | B $\frac{19}{64} - \frac{1}{16} = \frac{15}{64}$ |
| 4. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 5. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 6. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 7. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 8. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 9. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 10. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 11. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 12. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 13. pilot hole for a 7 gauge screw? | 41 $\frac{5}{32} - \frac{1}{16} = \frac{3}{32}$ |
| 14. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 15. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 16. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |

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|-----|--|---------------------------------|
| 1. | 50 penny nail penetrate a $\frac{5}{8}$ " green board? | $5 \frac{1}{2} - \frac{19}{32}$ |
| 2. | 40 penny nail penetrate a 4" green timber? | $5 - 3\frac{9}{16}$ |
| 3. | 11 $\frac{1}{2}$ common gauge nail penetrate a 1" dry board? | $2 - \frac{3}{4}$ |
| 4. | 3 penny nail penetrate a $3 \frac{1}{2}$ " green pole? | $1 \frac{1}{4} - 3\frac{1}{16}$ |
| 5. | 12 $\frac{1}{2}$ box gauge nail penetrate a 4" dry plank? | $2 - 3 \frac{1}{2}$ |
| 6. | 10 penny nail penetrate a 1" green mast? | $3 - \frac{25}{32}$ |
| 7. | 8 common gauge nail penetrate a 2" dry mast? | $3 \frac{1}{2} - 1 \frac{1}{2}$ |
| 8. | 9 box gauge nail penetrate a $1 \frac{1}{2}$ " green plank? | $4 - 1\frac{9}{32}$ |
| 9. | 11 $\frac{1}{2}$ box gauge nail penetrate a $1 \frac{1}{2}$ " green plank? | $2 \frac{1}{2} - 1\frac{9}{32}$ |
| 10. | 3 common gauge nail penetrate a $4 \frac{1}{2}$ " green timber? | $5 \frac{1}{2} - 4\frac{1}{16}$ |
| 11. | 15 $\frac{1}{2}$ box gauge nail penetrate a $4 \frac{1}{2}$ " green beam? | $1 - 4\frac{1}{16}$ |
| 12. | 16 penny nail penetrate a $\frac{3}{8}$ " dry beam? | $3 \frac{1}{2} - \frac{5}{16}$ |
| 13. | 12 $\frac{1}{2}$ box gauge nail penetrate a $2 \frac{1}{2}$ " dry timber? | $2 - 2$ |
| 14. | 16 penny nail penetrate a $\frac{1}{2}$ " green dimension? | $3 \frac{1}{2} - \frac{15}{32}$ |
| 15. | 5 penny nail penetrate a 3" green plank? | $1 \frac{3}{4} - 2\frac{9}{16}$ |
| 16. | 10 $\frac{1}{2}$ box gauge nail penetrate a $\frac{3}{8}$ " green beam? | $3 - 1\frac{1}{32}$ |

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|---|---|
| 1. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 2. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 3. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 4. clearance hole for a 18 gauge screw? | N $\frac{19}{64} = \frac{19}{64}$ |
| 5. pilot hole for a 9 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 6. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 7. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 8. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 9. countersink hole for a 18 gauge screw? | $\frac{27}{64} - \frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 10. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 11. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 12. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 13. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 14. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 15. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 16. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |

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|-----|--|-----------|
| 1. | 12 ½ common gauge nail penetrate a 1 ¼" green pole? | 1 ½ — 1⅓ |
| 2. | 5 penny nail penetrate a 4" dry beam? | 1 ¾ — 3 ½ |
| 3. | 7 penny nail penetrate a 2 ½" dry mast? | 2 ¼ — 2 |
| 4. | 14 box gauge nail penetrate a 4" green beam? | 1 ½ — 3⅞ |
| 5. | 50 penny nail penetrate a 1" dry board? | 5 ½ — ¾ |
| 6. | 10 ½ box gauge nail penetrate a 1 ½" green timber? | 3 — 1⅞ |
| 7. | 10 ¼ common gauge nail penetrate a 1 ½" green pole? | 2 ½ — 1⅞ |
| 8. | 16 penny nail penetrate a 4" dry mast? | 3 ½ — 3 ½ |
| 9. | 2 penny nail penetrate a 5/8" dry girder? | 1 — ⅞ |
| 10. | 11 ½ common gauge nail penetrate a 3 ½" green dimension? | 2 — 3⅓ |
| 11. | 5 penny nail penetrate a 3" dry dimension? | 1 ¾ — 2 ½ |
| 12. | 5 penny nail penetrate a 3" green beam? | 1 ¾ — 2⅞ |
| 13. | 14 box gauge nail penetrate a 1 ¼" dry pole? | 1 ½ — 1 |
| 14. | 4 penny nail penetrate a 4" dry joist? | 1 ½ — 3 ½ |
| 15. | 30 penny nail penetrate a 3 ½" dry timber? | 4 ½ — 3 |
| 16. | 8 box gauge nail penetrate a ½" green joist? | 5 — 1⅝ |

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|--|--|
| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. countersink hole for a 16 gauge screw? | X $\frac{17}{64} + \frac{1}{8} = \frac{25}{64}$ |
| 3. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 4. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 5. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 6. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 7. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 8. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 9. pilot hole for a 8 gauge screw? | 35 $\frac{11}{64} - \frac{1}{16} = \frac{7}{64}$ |
| 10. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 11. countersink hole for a 7 gauge screw? | L $\frac{5}{32} + \frac{1}{8} = \frac{9}{32}$ |
| 12. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 13. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 14. clearance hole for a 9 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 15. pilot hole for a 6 gauge screw? | 47 $\frac{9}{64} - \frac{1}{16} = \frac{5}{64}$ |
| 16. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |

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|-----|---|----------------------------------|
| 1. | 5 penny nail penetrate a $\frac{3}{4}$ " green plank? | $1 \frac{3}{4} - \frac{11}{16}$ |
| 2. | 7 penny nail penetrate a 2" dry girder? | $2 \frac{1}{4} - 1 \frac{1}{2}$ |
| 3. | 8 penny nail penetrate a 3" green beam? | $2 \frac{1}{2} - 2 \frac{9}{16}$ |
| 4. | 9 penny nail penetrate a $2 \frac{1}{2}$ " dry plank? | $2 \frac{3}{4} - 2$ |
| 5. | $10 \frac{1}{2}$ box gauge nail penetrate a $\frac{5}{8}$ " dry rafter? | $3 - \frac{9}{16}$ |
| 6. | 3 penny nail penetrate a $\frac{5}{8}$ " green dimension? | $1 \frac{1}{4} - \frac{19}{32}$ |
| 7. | 6 penny nail penetrate a $\frac{3}{8}$ " dry frame? | $2 - \frac{5}{16}$ |
| 8. | 8 common gauge nail penetrate a 1" dry dimension? | $3 \frac{1}{2} - \frac{3}{4}$ |
| 9. | 8 box gauge nail penetrate a $\frac{3}{4}$ " dry girder? | $5 - \frac{5}{8}$ |
| 10. | $11 \frac{1}{2}$ box gauge nail penetrate a 2" green girder? | $2 \frac{1}{2} - 1 \frac{9}{16}$ |
| 11. | 30 penny nail penetrate a $1 \frac{1}{4}$ " dry dimension? | $4 \frac{1}{2} - 1$ |
| 12. | 14 box gauge nail penetrate a 4" green pole? | $1 \frac{1}{2} - 3 \frac{9}{16}$ |
| 13. | 4 penny nail penetrate a $\frac{3}{8}$ " dry beam? | $1 \frac{1}{2} - \frac{5}{16}$ |
| 14. | 6 common gauge nail penetrate a $1 \frac{1}{2}$ " dry pole? | $4 - 1 \frac{1}{4}$ |
| 15. | 2 penny nail penetrate a 4" green beam? | $1 - 3 \frac{9}{16}$ |
| 16. | 2 common gauge nail penetrate a $\frac{5}{8}$ " dry dimension? | $6 - \frac{9}{16}$ |

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| 1. clearance hole for a 14 gauge screw? | E $\frac{1}{4} = \frac{1}{4}$ |
| 2. countersink hole for a 10 gauge screw? | O $\frac{3}{16} + \frac{1}{8} = \frac{5}{16}$ |
| 3. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 4. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 5. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 6. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 7. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 8. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 9. clearance hole for a 7 gauge screw? | 22 $\frac{5}{32} = \frac{5}{32}$ |
| 10. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 11. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 12. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 13. countersink hole for a 18 gauge screw? | $\frac{27}{64}$ $\frac{19}{64} + \frac{1}{8} = \frac{27}{64}$ |
| 14. countersink hole for a 3 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 15. pilot hole for a 16 gauge screw? | 6 $\frac{17}{64} - \frac{1}{16} = \frac{13}{64}$ |
| 16. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |

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|-----|--|--------------------------------------|
| 1. | 10 ¼ common gauge nail penetrate a 2" green board? | 2 ½ — 1 ⁹ / ₁₆ |
| 2. | 50 penny nail penetrate a 1 ¼" green plank? | 5 ½ — 1 ¹ / ₃₂ |
| 3. | 50 penny nail penetrate a 3" dry joist? | 5 ½ — 2 ½ |
| 4. | 8 penny nail penetrate a 3/4" dry plank? | 2 ½ — 5/8 |
| 5. | 16 penny nail penetrate a ½" dry girder? | 3 ½ — 7/16 |
| 6. | 9 common gauge nail penetrate a 3/4" green girder? | 3 — 11/16 |
| 7. | 10 penny nail penetrate a 1 ½" dry rafter? | 3 — 1 ¼ |
| 8. | 50 penny nail penetrate a ½" dry joist? | 5 ½ — 7/16 |
| 9. | 8 box gauge nail penetrate a 3" dry beam? | 5 — 2 ½ |
| 10. | 11 ½ box gauge nail penetrate a ½" green timber? | 2 ½ — 15/32 |
| 11. | 10 ¼ common gauge nail penetrate a 1 ½" green dimension? | 2 ½ — 1 ⁹ / ₃₂ |
| 12. | 4 penny nail penetrate a 2 ½" green frame? | 1 ½ — 2 ¹ / ₁₆ |
| 13. | 12 ½ common gauge nail penetrate a 2" dry pole? | 1 ½ — 1 ½ |
| 14. | 6 penny nail penetrate a 3/8" green board? | 2 — 11/32 |
| 15. | 6 penny nail penetrate a ½" dry girder? | 2 — 7/16 |
| 16. | 11 ½ box gauge nail penetrate a 1 ¼" green beam? | 2 ½ — 1 ¹ / ₃₂ |

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|--|---|
| 1. pilot hole for a 24 gauge screw? | O $\frac{3}{8} - \frac{1}{16} = \frac{5}{16}$ |
| 2. countersink hole for a 4 gauge screw? | B $\frac{7}{64} + \frac{1}{8} = \frac{15}{64}$ |
| 3. countersink hole for a 5 gauge screw? | E $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ |
| 4. pilot hole for a 14 gauge screw? | 12 $\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$ |
| 5. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 6. pilot hole for a 5 gauge screw? | 52 $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$ |
| 7. countersink hole for a 12 gauge screw? | S $\frac{7}{32} + \frac{1}{8} = \frac{11}{32}$ |
| 8. clearance hole for a 3 gauge screw? | 41 $\frac{3}{32} = \frac{3}{32}$ |
| 9. pilot hole for a 1 gauge screw? | 78 $\frac{5}{64} - \frac{1}{16} = \frac{1}{64}$ |
| 10. countersink hole for a 8 gauge screw? | N $\frac{11}{64} + \frac{1}{8} = \frac{19}{64}$ |
| 11. clearance hole for a 1 gauge screw? | 47 $\frac{5}{64} = \frac{5}{64}$ |
| 12. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 13. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 14. countersink hole for a 14 gauge screw? | V $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ |
| 15. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 16. clearance hole for a 20 gauge screw? | Q $\frac{21}{64} = \frac{21}{64}$ |

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|-----|---|----------------|
| 1. | 3 common gauge nail penetrate a 3 1/2" green plank? | 5 1/2 — 3 1/16 |
| 2. | 11 1/2 box gauge nail penetrate a 4" green dimension? | 2 1/2 — 3 9/16 |
| 3. | 10 1/2 box gauge nail penetrate a 3" green mast? | 3 — 2 9/16 |
| 4. | 15 1/2 box gauge nail penetrate a 3" dry frame? | 1 — 2 1/2 |
| 5. | 9 common gauge nail penetrate a 3/4" green joist? | 3 — 11/16 |
| 6. | 4 penny nail penetrate a 2" dry pole? | 1 1/2 — 1 1/2 |
| 7. | 3 penny nail penetrate a 3/8" green girder? | 1 1/4 — 11/32 |
| 8. | 10 1/2 box gauge nail penetrate a 4" green timber? | 3 — 3 9/16 |
| 9. | 12 1/2 common gauge nail penetrate a 4" dry frame? | 1 1/2 — 3 1/2 |
| 10. | 10 penny nail penetrate a 1" green joist? | 3 — 25/32 |
| 11. | 12 1/2 common gauge nail penetrate a 1 1/4" dry girder? | 1 1/2 — 1 |
| 12. | 6 common gauge nail penetrate a 4" dry frame? | 4 — 3 1/2 |
| 13. | 10 1/2 box gauge nail penetrate a 2 1/2" green board? | 3 — 2 1/16 |
| 14. | 3 penny nail penetrate a 4 1/2" green board? | 1 1/4 — 4 1/16 |
| 15. | 16 penny nail penetrate a 3/8" green frame? | 3 1/2 — 11/32 |
| 16. | 60 penny nail penetrate a 1" green timber? | 6 — 25/32 |

1. countersink hole for a 8 gauge screw?	N $11/64 + 1/8 = 19/64$
2. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
3. clearance hole for a 9 gauge screw?	17 $11/64 = 11/64$
4. clearance hole for a 11 gauge screw?	6 $13/64 = 13/64$
5. clearance hole for a 6 gauge screw?	27 $9/64 = 9/64$
6. pilot hole for a 3 gauge screw?	67 $3/32 - 1/16 = 1/32$
7. pilot hole for a 24 gauge screw?	O $3/8 - 1/16 = 5/16$
8. pilot hole for a 8 gauge screw?	35 $11/64 - 1/16 = 7/64$
9. clearance hole for a 5 gauge screw?	30 $1/8 = 1/8$
10. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$
11. clearance hole for a 3 gauge screw?	41 $3/32 = 3/32$
12. clearance hole for a 10 gauge screw?	12 $3/16 = 3/16$
13. pilot hole for a 8 gauge screw?	35 $11/64 - 1/16 = 7/64$
14. pilot hole for a 1 gauge screw?	78 $5/64 - 1/16 = 1/64$
15. clearance hole for a 5 gauge screw?	30 $1/8 = 1/8$
16. countersink hole for a 1 gauge screw?	6 $5/64 + 1/8 = 13/64$

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| 1. | 12 penny nail penetrate a 2 1/2" dry dimension? | 3 1/4 — 2 |
| 2. | 8 box gauge nail penetrate a 5/8" green frame? | 5 — 19/32 |
| 3. | 5 penny nail penetrate a 2" dry mast? | 1 3/4 — 1 1/2 |
| 4. | 15 1/2 box gauge nail penetrate a 1 1/2" green joist? | 1 — 19/32 |
| 5. | 3 penny nail penetrate a 2 1/2" green timber? | 1 1/4 — 2 1/16 |
| 6. | 12 1/2 common gauge nail penetrate a 1/2" green mast? | 1 1/2 — 15/32 |
| 7. | 10 1/4 common gauge nail penetrate a 1 1/2" green mast? | 2 1/2 — 19/32 |
| 8. | 5 penny nail penetrate a 2 1/2" dry plank? | 1 3/4 — 2 |
| 9. | 8 penny nail penetrate a 1" dry girder? | 2 1/2 — 3/4 |
| 10. | 14 1/2 box gauge nail penetrate a 4 1/2" green mast? | 1 1/4 — 4 1/16 |
| 11. | 9 common gauge nail penetrate a 1 1/2" green board? | 3 — 19/32 |
| 12. | 2 penny nail penetrate a 3/8" dry mast? | 1 — 5/16 |
| 13. | 12 1/2 box gauge nail penetrate a 1/2" green rafter? | 2 — 15/32 |
| 14. | 5 penny nail penetrate a 1 1/2" green board? | 1 3/4 — 19/32 |
| 15. | 3 penny nail penetrate a 3" green pole? | 1 1/4 — 29/16 |
| 16. | 7 penny nail penetrate a 1" green girder? | 2 1/4 — 25/32 |

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| 1. countersink hole for a 20 gauge screw? | $29/64 \quad 21/64 + 1/8 = 29/64$ |
| 2. pilot hole for a 18 gauge screw? | B $19/64 - 1/16 = 15/64$ |
| 3. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 4. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 5. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 6. clearance hole for a 5 gauge screw? | 30 $1/8 = 1/8$ |
| 7. countersink hole for a 3 gauge screw? | 2 $3/32 + 1/8 = 7/32$ |
| 8. clearance hole for a 6 gauge screw? | 27 $9/64 = 9/64$ |
| 9. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 10. clearance hole for a 3 gauge screw? | 41 $3/32 = 3/32$ |
| 11. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |
| 12. countersink hole for a 18 gauge screw? | $27/64 \quad 19/64 + 1/8 = 27/64$ |
| 13. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 14. pilot hole for a 12 gauge screw? | 22 $7/32 - 1/16 = 5/32$ |
| 15. countersink hole for a 8 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 16. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |

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| 1. | 12 ½ box gauge nail penetrate a 3" dry beam? | 2 — 2 ½ |
| 2. | 11 ½ box gauge nail penetrate a 4" green plank? | 2 ½ — 3 ⁹ / ₁₆ |
| 3. | 11 ½ box gauge nail penetrate a 3 ½" green frame? | 2 ½ — 3 ¹ / ₁₆ |
| 4. | 3 penny nail penetrate a 1 ¼" dry joist? | 1 ¼ — 1 |
| 5. | 8 penny nail penetrate a 5/8" dry timber? | 2 ½ — ⁹ / ₁₆ |
| 6. | 20 penny nail penetrate a 2 ½" green mast? | 4 — 2 ¹ / ₁₆ |
| 7. | 16 penny nail penetrate a ½" dry board? | 3 ½ — ⁷ / ₁₆ |
| 8. | 60 penny nail penetrate a 4" dry frame? | 6 — 3 ½ |
| 9. | 7 penny nail penetrate a 1" dry timber? | 2 ¼ — ³ / ₄ |
| 10. | 4 common gauge nail penetrate a 1" green dimension? | 5 — ²⁵ / ₃₂ |
| 11. | 9 penny nail penetrate a 3" green timber? | 2 ³ / ₄ — 2 ⁹ / ₁₆ |
| 12. | 60 penny nail penetrate a 3/8" green plank? | 6 — ¹¹ / ₃₂ |
| 13. | 5 penny nail penetrate a 2 ½" dry rafter? | 1 ³ / ₄ — 2 |
| 14. | 9 penny nail penetrate a 3 ½" dry girder? | 2 ³ / ₄ — 3 |
| 15. | 9 common gauge nail penetrate a 1 ¼" green rafter? | 3 — 1 ¹ / ₃₂ |
| 16. | 9 penny nail penetrate a 3/8" green pole? | 2 ³ / ₄ — ¹¹ / ₃₂ |

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| 1. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 2. clearance hole for a 11 gauge screw? | 6 $\frac{13}{64} = \frac{13}{64}$ |
| 3. clearance hole for a 12 gauge screw? | 2 $\frac{7}{32} = \frac{7}{32}$ |
| 4. countersink hole for a 1 gauge screw? | 6 $\frac{5}{64} + \frac{1}{8} = \frac{13}{64}$ |
| 5. clearance hole for a 10 gauge screw? | 12 $\frac{3}{16} = \frac{3}{16}$ |
| 6. pilot hole for a 2 gauge screw? | 67 $\frac{3}{32} - \frac{1}{16} = \frac{1}{32}$ |
| 7. clearance hole for a 16 gauge screw? | H $\frac{17}{64} = \frac{17}{64}$ |
| 8. clearance hole for a 8 gauge screw? | 17 $\frac{11}{64} = \frac{11}{64}$ |
| 9. pilot hole for a 20 gauge screw? | H $\frac{21}{64} - \frac{1}{16} = \frac{17}{64}$ |
| 10. pilot hole for a 10 gauge screw? | 30 $\frac{3}{16} - \frac{1}{16} = \frac{1}{8}$ |
| 11. pilot hole for a 12 gauge screw? | 22 $\frac{7}{32} - \frac{1}{16} = \frac{5}{32}$ |
| 12. countersink hole for a 2 gauge screw? | 2 $\frac{3}{32} + \frac{1}{8} = \frac{7}{32}$ |
| 13. countersink hole for a 6 gauge screw? | H $\frac{9}{64} + \frac{1}{8} = \frac{17}{64}$ |
| 14. countersink hole for a 20 gauge screw? | $\frac{29}{64}$ $\frac{21}{64} + \frac{1}{8} = \frac{29}{64}$ |
| 15. pilot hole for a 11 gauge screw? | 27 $\frac{13}{64} - \frac{1}{16} = \frac{9}{64}$ |
| 16. countersink hole for a 11 gauge screw? | Q $\frac{13}{64} + \frac{1}{8} = \frac{21}{64}$ |

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| 1. 9 penny nail penetrate a $\frac{1}{2}$ " green frame? | $2 \frac{3}{4} - \frac{15}{32}$ |
| 2. 12 penny nail penetrate a 2" green plank? | $3 \frac{1}{4} - \frac{19}{16}$ |
| 3. 5 penny nail penetrate a 2" dry timber? | $1 \frac{3}{4} - 1 \frac{1}{2}$ |
| 4. 14 box gauge nail penetrate a $1 \frac{1}{2}$ " green board? | $1 \frac{1}{2} - \frac{19}{32}$ |
| 5. 9 penny nail penetrate a $\frac{3}{8}$ " green rafter? | $2 \frac{3}{4} - \frac{11}{32}$ |
| 6. 15 common gauge nail penetrate a 3" green pole? | $1 - \frac{29}{16}$ |
| 7. 7 penny nail penetrate a 1" green timber? | $2 \frac{1}{4} - \frac{25}{32}$ |
| 8. 12 penny nail penetrate a $4 \frac{1}{2}$ " green beam? | $3 \frac{1}{4} - 4 \frac{1}{16}$ |
| 9. 6 penny nail penetrate a $1 \frac{1}{2}$ " green frame? | $2 - \frac{19}{32}$ |
| 10. 40 penny nail penetrate a $2 \frac{1}{2}$ " green girder? | $5 - 2 \frac{1}{16}$ |
| 11. $12 \frac{1}{2}$ box gauge nail penetrate a 2" dry timber? | $2 - 1 \frac{1}{2}$ |
| 12. 3 common gauge nail penetrate a 2" green dimension? | $5 \frac{1}{2} - \frac{19}{16}$ |
| 13. 6 penny nail penetrate a $\frac{3}{4}$ " green timber? | $2 - \frac{11}{16}$ |
| 14. 9 box gauge nail penetrate a 3" dry joist? | $4 - 2 \frac{1}{2}$ |
| 15. $12 \frac{1}{2}$ common gauge nail penetrate a 2" green frame? | $1 \frac{1}{2} - \frac{19}{16}$ |
| 16. 3 common gauge nail penetrate a 2" green mast? | $5 \frac{1}{2} - \frac{19}{16}$ |

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| 1. clearance hole for a 16 gauge screw? | H $17/64 = 17/64$ |
| 2. countersink hole for a 9 gauge screw? | N $11/64 + 1/8 = 19/64$ |
| 3. pilot hole for a 1 gauge screw? | 78 $5/64 - 1/16 = 1/64$ |
| 4. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 5. pilot hole for a 2 gauge screw? | 67 $3/32 - 1/16 = 1/32$ |
| 6. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 7. pilot hole for a 8 gauge screw? | 35 $11/64 - 1/16 = 7/64$ |
| 8. countersink hole for a 6 gauge screw? | H $9/64 + 1/8 = 17/64$ |
| 9. pilot hole for a 11 gauge screw? | 27 $13/64 - 1/16 = 9/64$ |
| 10. clearance hole for a 1 gauge screw? | 47 $5/64 = 5/64$ |
| 11. countersink hole for a 4 gauge screw? | B $7/64 + 1/8 = 15/64$ |
| 12. clearance hole for a 7 gauge screw? | 22 $5/32 = 5/32$ |
| 13. pilot hole for a 5 gauge screw? | 52 $1/8 - 1/16 = 1/16$ |
| 14. pilot hole for a 4 gauge screw? | 55 $7/64 - 1/16 = 3/64$ |
| 15. pilot hole for a 20 gauge screw? | H $21/64 - 1/16 = 17/64$ |
| 16. clearance hole for a 4 gauge screw? | 35 $7/64 = 7/64$ |

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|-----|---|-----------------|
| 1. | 5 penny nail penetrate a 3 1/2" green beam? | 1 3/4 — 3 1/16 |
| 2. | 11 1/2 box gauge nail penetrate a 3/8" dry girder? | 2 1/2 — 5/16 |
| 3. | 10 1/2 box gauge nail penetrate a 1 1/4" dry rafter? | 3 — 1 |
| 4. | 20 penny nail penetrate a 3/8" dry frame? | 4 — 5/16 |
| 5. | 12 1/2 box gauge nail penetrate a 1 1/2" green beam? | 2 — 1 9/32 |
| 6. | 14 box gauge nail penetrate a 3" green board? | 1 1/2 — 2 9/16 |
| 7. | 20 penny nail penetrate a 2" green pole? | 4 — 1 9/16 |
| 8. | 50 penny nail penetrate a 3/8" green dimension? | 5 1/2 — 11 1/32 |
| 9. | 7 penny nail penetrate a 1/2" dry plank? | 2 1/4 — 7/16 |
| 10. | 8 penny nail penetrate a 1 1/4" dry frame? | 2 1/2 — 1 |
| 11. | 2 penny nail penetrate a 1" dry frame? | 1 — 3/4 |
| 12. | 15 1/2 box gauge nail penetrate a 3 1/2" green joist? | 1 — 3 1/16 |
| 13. | 2 penny nail penetrate a 3/4" green plank? | 1 — 11/16 |
| 14. | 10 penny nail penetrate a 1 1/4" dry plank? | 3 — 1 |
| 15. | 12 1/2 box gauge nail penetrate a 3" dry girder? | 2 — 2 1/2 |
| 16. | 4 common gauge nail penetrate a 1 1/2" green girder? | 5 — 1 9/32 |